
Chapter
Six

Approximating Functions

6.0	Introduction
6.1	Polynomial Interpolation
6.2	Divided Differences
6.3	Hermite Interpolation
6.4	Spline Interpolation
6.5	<i>B</i> -Splines: Basic Theory
6.6	<i>B</i> -Splines: Applications
6.7	Taylor Series
6.8	Best Approximation: Least-Squares Theory
6.9	Best Approximation: Chebyshev Theory
6.10	Interpolation in Higher Dimensions
6.11	Continued Fractions
6.12	Trigonometric Interpolation
6.13	Fast Fourier Transform
6.14	Adaptive Approximation

6.0

Introduction

In this chapter, the problem of representing functions within a computer is discussed. Several different subproblems will be considered. They differ according to the type of function being represented, whether it is known at relatively few points or at many (or all) points. The representation chosen (a polynomial, a spline function, a continued fraction, and so on) also determines the nature of the theory. We begin with the case of polynomials, which is the oldest and simplest.

6.1

Polynomial Interpolation

In this section, we solve the following problem: We are given a table of $n + 1$ data points (x_i, y_i) :

x	x_0	x_1	x_2	$\cdots$	x_n
y	y_0	y_1	y_2	$\cdots$	y_n

and we seek a polynomial p of lowest possible degree for which

$$p(x_i) = y_i \qquad (0 \leq i \leq n)$$

Such a polynomial is said to **interpolate** the data. Here is the theorem that governs this problem.

THEOREM 1 Theorem on Polynomial Interpolation

If $x_0, x_1, \dots, x_n$ are distinct real numbers, then for arbitrary values $y_0, y_1, \dots, y_n$, there is a unique polynomial p_n of degree at most n such that

$$p_n(x_i) = y_i \quad (0 \leq i \leq n)$$

Proof Let us prove the uniqueness or *unicity* first. Suppose there were two such polynomials, p_n and q_n . Then the polynomial $p_n - q_n$ would have the property $(p_n - q_n)(x_i) = 0$ for $0 \leq i \leq n$. Since the degree of $p_n - q_n$ can be at most n , this polynomial can have at most n zeros if it is not the 0 polynomial. Since the x_i are distinct, $p_n - q_n$ has $n + 1$ zeros; it must therefore be 0. Hence, $p_n \equiv q_n$.

For the existence part of the theorem, we proceed inductively. For $n = 0$, the existence is obvious since a constant function p_0 (polynomial of degree ≤ 0) can be chosen so that $p_0(x_0) = y_0$. Now suppose that we have obtained a polynomial p_{k-1} of degree $\leq k - 1$ with $p_{k-1}(x_i) = y_i$ for $0 \leq i \leq k - 1$. We try to construct p_k in the form

$$p_k(x) = p_{k-1}(x) + c(x - x_0)(x - x_1) \cdots (x - x_{k-1}) \quad (1)$$

Note that this is unquestionably a polynomial of degree at most k . Furthermore, p_k interpolates the data that p_{k-1} interpolates because

$$p_k(x_i) = p_{k-1}(x_i) = y_i \quad (0 \leq i \leq k - 1)$$

Now we determine the unknown coefficient c from the condition $p_k(x_k) = y_k$. This leads to the equation

$$p_{k-1}(x_k) + c(x_k - x_0)(x_k - x_1) \cdots (x_k - x_{k-1}) = y_k \quad (2)$$

Equation (2) can certainly be solved for c because the factors multiplying c are not 0. (Why?) ■

Newton Form of the Interpolation Polynomial

Before we attempt to write an algorithm that carries out the recursive process in this proof, we need to make some observations. First, the polynomials $p_0, p_1, \dots, p_n$ constructed in the proof have the property that each p_k is obtained simply by adding a single term to p_{k-1} . Thus, at the end of the process, p_n will be a sum of terms, and each $p_0, p_1, \dots, p_{n-1}$ will be clearly visible in the expression for p_n . Each p_k has the form

$$p_k(x) = c_0 + c_1(x - x_0) + c_2(x - x_0)(x - x_1) + \cdots + c_k(x - x_0) \cdots (x - x_{k-1}) \quad (3)$$

The compact form of this is

$$p_k(x) = \sum_{i=0}^k c_i \prod_{j=0}^{i-1} (x - x_j) \tag{4}$$

Here the convention has been adopted that $\prod_{j=0}^m (x - x_j) = 1$ whenever $m < 0$. The first few cases of Equation (4) are

$$\begin{aligned} p_0(x) &= c_0 \\ p_1(x) &= c_0 + c_1(x - x_0) \\ p_2(x) &= c_0 + c_1(x - x_0) + c_2(x - x_0)(x - x_1) \end{aligned}$$

and so on. These polynomials are called the **interpolation polynomials in Newton's form**.

To evaluate $p_k(x)$, assuming that the coefficients $c_0, c_1, \dots, c_k$ are known, an efficient method called **nested multiplication** or **Horner's algorithm** is used. This can be explained most easily for an arbitrary expression of the form

$$u = \sum_{i=0}^k c_i \prod_{j=0}^{i-1} d_j = c_0 + c_1 d_0 + c_2 d_0 d_1 + \dots + c_k d_0 d_1 \dots d_{k-1} \tag{5}$$

The idea is to write it in the form

$$u = (\dots((c_k d_{k-1} + c_{k-1}) d_{k-2} + c_{k-2}) d_{k-3} + \dots + c_1) d_0 + c_0 \tag{6}$$

The algorithm to compute u can be developed as follows: We denote the quantities in the parentheses by $u_k, u_{k-1}, \dots, u_0$, starting with the innermost parentheses. Thus,

$$\begin{aligned} u_k &\leftarrow c_k \\ u_{k-1} &\leftarrow u_k d_{k-1} + c_{k-1} \\ u_{k-2} &\leftarrow u_{k-1} d_{k-2} + c_{k-2} \\ &\vdots \\ u_0 &\leftarrow u_1 d_0 + c_0 \end{aligned}$$

Since only u_0 is wanted, we can write algorithmically

```

u ← c_k
for i = k - 1 to 0 step - 1 do
    u ← u d_i + c_i
end do
    
```

Going back to the polynomial given by Equation (3) or (4), we use the following algorithm to obtain $u = p_k(t)$ for a prescribed value of t :

```
u ← ck
for i = k - 1 to 0 step - 1 do
    u ← (t - xi)u + ci
end do
```

Now we can write an algorithm for computing the coefficients c_i in Equation (4). Notice that in Equation (2) the coefficient c is what we later have labeled c_k . The formula for c_k is then

$$c_k = \frac{y_k - p_{k-1}(x_k)}{(x_k - x_0)(x_k - x_1) \cdots (x_k - x_{k-1})}$$

An algorithm to compute $c_0, c_1, \dots, c_n$ from the table of values for $x_0, x_1, \dots, x_n$ and $y_0, y_1, \dots, y_n$ follows:

```
c0 ← y0
for k = 1 to n do
    d ← xk - xk-1
    u ← ck-1
    for i = k - 2 to 0 step - 1 do
        u ← u(xk - xi) + ci
        d ← d(xk - xi)
    end do
    ck ← (yk - u)/d
end do
```

The inner loop computes $p_{k-1}(x_k)$ and $(x_k - x_0)(x_k - x_1) \cdots (x_k - x_{k-1})$, the former by a straightforward modification of the algorithm discussed previously.

The algorithm just given serves a didactic purpose only. It illustrates the important process of translating a constructive existence proof into an algorithm suitable for a computer program. However, a more efficient procedure exists that achieves the same result. This alternative method uses **divided differences** to compute the coefficients $c_0, c_1, \dots, c_k$ in Equation (4). This method will be presented in Section 6.2 (p. 329).

The preceding algorithm was programmed and given a simple test as follows: We define the polynomial

$$p_3(x) = 4x^3 + 35x^2 - 84x - 954 \tag{7}$$

Four values of this function are given here:

x	5	-7	-6	0
y	1	-23	-54	-954

(8)

This set of values was given as input to the computer routine, and it correctly computed the coefficients $c_0 = 1, c_1 = 2, c_2 = 3, c_3 = 4$. These are the coefficients in the Newton form of the polynomial in Equation (7); namely,

$$p_3(x) = 1 + 2(x - 5) + 3(x - 5)(x + 7) + 4(x - 5)(x + 7)(x + 6)$$

Lagrange Form of the Interpolation Polynomial

We now present an alternative form for the interpolating polynomial p associated with a table of data points (x_i, y_i) for $0 \leq i \leq n$. It is important to understand that there is one and only one interpolating polynomial of degree $\leq n$ associated with the data (assuming, of course, that the $n + 1$ abscissas x_i are *distinct*). However, the possibility certainly exists for expressing this polynomial in different forms and for arriving at it by different algorithms.

The alternative method will express p in the form

$$p(x) = y_0 \ell_0(x) + y_1 \ell_1(x) + \cdots + y_n \ell_n(x) = \sum_{k=0}^n y_k \ell_k(x) \quad (9)$$

Here $\ell_0, \ell_1, \dots, \ell_n$ are polynomials that depend on the nodes $x_0, x_1, \dots, x_n$ but *not* on the *ordinates* $y_0, y_1, \dots, y_n$. Since all the ordinates could be 0 except for a 1 occupying the i th position, we see that

$$\delta_{ij} = p_n(x_j) = \sum_{k=0}^n y_k \ell_k(x_j) = \sum_{k=0}^n \delta_{ki} \ell_k(x_j) = \ell_i(x_j)$$

(Recall that the **Kronecker delta** is defined by $\delta_{ki} = 1$ if $k = i$ and $\delta_{ki} = 0$ if $k \neq i$.) We can easily arrive at a set of polynomials having this property.

Let us consider ℓ_0 . It is to be a polynomial of degree n that takes the value 0 at $x_1, x_2, \dots, x_n$ and the value 1 at x_0 . Clearly, ℓ_0 must be of the form

$$\ell_0(x) = c(x - x_1)(x - x_2) \cdots (x - x_n) = c \prod_{j=1}^n (x - x_j)$$

The value of c is obtained by putting $x = x_0$, so that

$$1 = c \prod_{j=1}^n (x_0 - x_j)$$

and

$$c = \prod_{j=1}^n (x_0 - x_j)^{-1}$$

Hence, we have

$$\ell_0(x) = \prod_{j=1}^n \frac{x - x_j}{x_0 - x_j}$$

Each ℓ_i is obtained by similar reasoning, and the general formula is then

$$\ell_i(x) = \prod_{\substack{j=0 \\ j \neq i}}^n \frac{x - x_j}{x_i - x_j} \quad (0 \leq i \leq n) \quad (10)$$

For the set of nodes $x_0, x_1, \dots, x_n$, these polynomials are known as the **cardinal functions**. With the cardinal polynomials in hand, Equation (9) gives the **Lagrange form** of the interpolating polynomials.

EXAMPLE 1 What are the cardinal functions and Lagrange form of the interpolating polynomial for the data in Table (8)?

Solution The nodes are 5, −7, −6, 0. Hence, the cardinal functions are

$$\begin{aligned}\ell_0(x) &= \frac{(x+7)(x+6)x}{(5+7)(5+6)5} = \frac{1}{660} \times (x+6)(x+7) \\ \ell_1(x) &= \frac{(x-5)(x+6)x}{(-7-5)(-7+6)(-7)} = \frac{-1}{84} \times (x-5)(x+6) \\ \ell_2(x) &= \frac{(x-5)(x+7)x}{(-6-5)(-6+7)(-6)} = \frac{-1}{66} \times (x-5)(x+7) \\ \ell_3(x) &= \frac{(x-5)(x+7)(x+6)}{(0-5)(0+7)(0+6)} = \frac{-1}{210} (x-5)(x+6)(x+7)\end{aligned}$$

The interpolating polynomial is

$$p_3(x) = \ell_0(x) - 23\ell_1(x) - 54\ell_2(x) - 954\ell_3(x)$$

This may not look like the same p_3 as in Equation (7), but it must be identical to it as a function even though its form is quite different. ■

EXAMPLE 2 Find an interpolation formula for the two-point table

x	x_0	x_1
y	y_0	y_1

Solution The Lagrange interpolation formula gives us

$$p(x) = y_0 \left(\frac{x-x_1}{x_0-x_1} \right) + y_1 \left(\frac{x-x_0}{x_1-x_0} \right)$$

Still other algorithms for polynomial interpolation have various advantages and disadvantages. Since there is one and only one polynomial of degree $\leq n$ that takes prescribed values at $n+1$ given (distinct) points, these algorithms produce the same polynomial in different forms. For example, we can require our polynomial to be expressed in powers of x :

$$p(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n \tag{11}$$

The interpolation conditions, $p(x_i) = y_i$ for $0 \leq i \leq n$, lead to a system of $n+1$ linear equations for determining $a_0, a_1, \dots, a_n$. This system has the form

$$\begin{bmatrix} 1 & x_0 & x_0^2 & \cdots & x_0^n \\ 1 & x_1 & x_1^2 & \cdots & x_1^n \\ 1 & x_2 & x_2^2 & \cdots & x_2^n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^n \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} = \begin{bmatrix} y_0 \\ y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} \tag{12}$$

The coefficient matrix in Equation (12) is called a **Vandermonde matrix**. It is non-singular because the system has a unique solution for any choice of $y_0, y_1, \dots, y_n$ (Theorem 1). The determinant of the Vandermonde matrix is therefore nonzero for distinct nodes $x_0, x_1, \dots, x_n$. A formula for it is given in Problem 6.1.34 (p. 327). However, the Vandermonde matrix is often *ill conditioned*, and the coefficients a_i may therefore be inaccurately determined by solving System (12). (See Gautschi [1984].) Furthermore, the amount of work involved to obtain the polynomial in Equation (11) is excessive. Therefore, this approach is *not* recommended.

If we let $x = x_i$ for $0 \leq i \leq n$ in the Lagrange polynomial (9), we obtain the following $n + 1$ linear equations:

$$\begin{bmatrix} \ell_0(x_0) & \ell_1(x_0) & \cdots & \ell_n(x_0) \\ \ell_0(x_1) & \ell_1(x_1) & \cdots & \ell_n(x_1) \\ \vdots & \vdots & \ddots & \vdots \\ \ell_0(x_n) & \ell_1(x_n) & \cdots & \ell_n(x_n) \end{bmatrix} \begin{bmatrix} y_0 \\ y_1 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} p(x_0) \\ p(x_1) \\ \vdots \\ p(x_n) \end{bmatrix}$$

The coefficient matrix reduces to the identity matrix I , and the solution is $y_i = p(x_i)$ for $0 \leq i \leq n$, as it should be. Since only one interpolating polynomial can exist for the $n + 1$ distinct points $x_0, x_1, \dots, x_n$, we obtain an identity satisfied by any polynomial of degree at most n :

$$p(x) = \sum_{i=0}^n p(x_i)\ell_i(x)$$

For numerical work, it is probably best to use the Newton form of the interpolation polynomial. This can be combined with the divided difference algorithm (discussed in Section 6.2, p. 331) for computing the required coefficients. On the other hand, the Lagrange form of the interpolating polynomial can be written down at once, since the coefficients in the Lagrange formula are the given y_i . This fact will be useful in Chapter 7 when quadrature formulas are constructed.

A situation where the Lagrange form would have the advantage over the Newton form would be if one had a single set of fixed x_i nodes with many different y_i values associated with them, such as from experimental data. Here the cardinal functions would remain the same for each case.

Another advantage of the Newton form is that if more data points are added to the interpolation problem, the coefficients already computed will not have to be changed. Thus in Equation (3), c_0 depends on the point (x_0, y_0) . Then c_1 depends on the two points (x_0, y_0) and (x_1, y_1) , and so on. The Newton form easily accommodates additional data to be interpolated.

The efficient algorithm of nested multiplication applies to the Newton form and to the form in Equation (11). This is a clear advantage over the Lagrange form. However, there are algorithms for efficiently evaluating the Lagrange form. See Werner [1984] and the references given there. See also Problems 6.1.31–32 (p. 326).

The Error in Polynomial Interpolation

We present now some theorems that pertain to the discrepancy between a function and a polynomial interpolant to it.

THEOREM 2 Theorem on Polynomial Interpolation Error

Let f be a function in $C^{n+1}[a, b]$, and let p be the polynomial of degree at most n that interpolates the function f at $n + 1$ distinct points $x_0, x_1, \dots, x_n$ in the interval $[a, b]$. To each x in $[a, b]$ there corresponds a point ξ_x in (a, b) such that

$$f(x) - p(x) = \frac{1}{(n + 1)!} f^{(n+1)}(\xi_x) \prod_{i=0}^n (x - x_i) \tag{13}$$

Proof If x is one of the nodes of interpolation x_i , the assertion is obviously true since both sides of Equation (13) reduce to 0. So, let x be any point other than a node. Put

$$w(t) \equiv \prod_{i=0}^n (t - x_i) \quad \phi \equiv f - p - \lambda w$$

where λ is the real number that makes $\phi(x) = 0$. (Remember, x is fixed.) Thus,

$$\lambda = \frac{f(x) - p(x)}{w(x)}$$

Now $\phi \in C^{n+1}[a, b]$, and ϕ vanishes at the $n + 2$ points $x, x_0, x_1, \dots, x_n$. By Rolle's Theorem, ϕ' has at least $n + 1$ distinct zeros in (a, b) . Similarly, ϕ'' has at least n distinct zeros in (a, b) . If this argument is repeated, we conclude eventually that $\phi^{(n+1)}$ has at least one zero, say ξ_x , in (a, b) . Now

$$\phi^{(n+1)} = f^{(n+1)} - p^{(n+1)} - \lambda w^{(n+1)} = f^{(n+1)} - (n + 1)! \lambda$$

Hence, we have

$$0 = \phi^{(n+1)}(\xi_x) = f^{(n+1)}(\xi_x) - (n + 1)! \lambda = f^{(n+1)}(\xi_x) - (n + 1)! \frac{f(x) - p(x)}{w(x)}$$

This is Equation (13) in disguise. ■

EXAMPLE 3 If the function $f(x) = \sin x$ is approximated by a polynomial of degree 9 that interpolates f at ten points in the interval $[0, 1]$, how large is the error on this interval?

Solution To answer this question, we use Equation (13), in Theorem 2. Obviously, we have $|f^{(10)}(\xi_x)| \leq 1$ and $\prod_{i=0}^9 |x - x_i| \leq 1$. So, for all x in $[0, 1]$,

$$|\sin x - p(x)| \leq \frac{1}{10!} < 2.8 \times 10^{-7} \tag{14} \tag{15} \tag{16} \tag{17} \tag{18} \tag{19} \tag{20} \tag{21} \tag{22} \tag{23} \tag{24} \tag{25} \tag{26} \tag{27} \tag{28} \tag{29} \tag{30} \tag{31} \tag{32} \tag{33} \tag{34} \tag{35} \tag{36} \tag{37} \tag{38} \tag{39} \tag{40} \tag{41} \tag{42} \tag{43} \tag{44} \tag{45} \tag{46} \tag{47} \tag{48} \tag{49} \tag{50} \tag{51} \tag{52} \tag{53} \tag{54} \tag{55} \tag{56} \tag{57} \tag{58} \tag{59} \tag{60} \tag{61} \tag{62} \tag{63} \tag{64} \tag{65} \tag{66} \tag{67} \tag{68} \tag{69} \tag{70} \tag{71} \tag{72} \tag{73} \tag{74} \tag{75} \tag{76} \tag{77} \tag{78} \tag{79} \tag{80} \tag{81} \tag{82} \tag{83} \tag{84} \tag{85} \tag{86} \tag{87} \tag{88} \tag{89} \tag{90} \tag{91} \tag{92} \tag{93} \tag{94} \tag{95} \tag{96} \tag{97} \tag{98} \tag{99} \tag{100} \tag{101} \tag{102} \tag{103} \tag{104} \tag{105} \tag{106} \tag{107} \tag{108} \tag{109} \tag{110} \tag{111} \tag{112} \tag{113} \tag{114} \tag{115} 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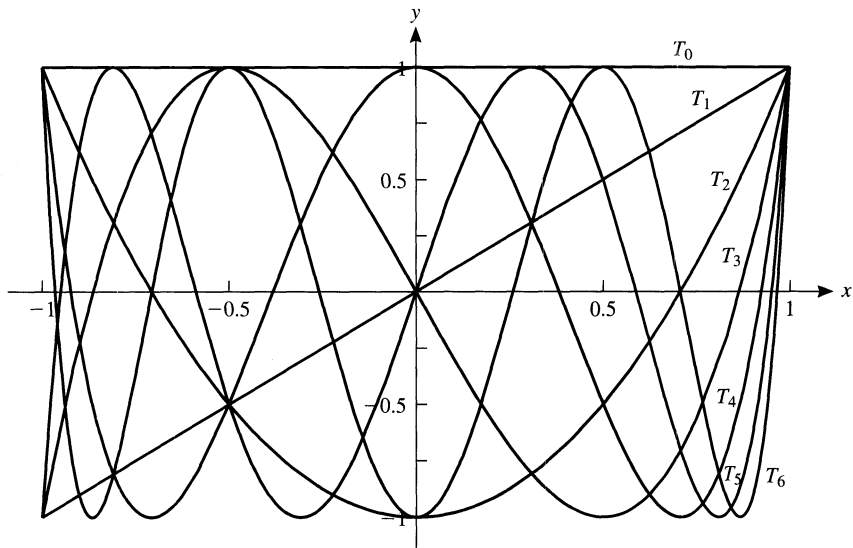


FIGURE 6.1
Chebyshev polynomials
 $T_n(x)$ for $0 \leq n \leq 6$

The Chebyshev polynomials (of the first kind) are defined recursively as follows:

$$\begin{cases} T_0(x) = 1 & T_1(x) = x \\ T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x) & (n \geq 1) \end{cases}$$

The explicit forms of the next few T_n are readily calculated:

$$\begin{aligned} T_2(x) &= 2x^2 - 1 \\ T_3(x) &= 4x^3 - 3x \\ T_4(x) &= 8x^4 - 8x^2 + 1 \\ T_5(x) &= 16x^5 - 20x^3 + 5x \\ T_6(x) &= 32x^6 - 48x^4 + 18x^2 - 1 \end{aligned}$$

Figure 6.1 show the graphs of these first few Chebyshev polynomials.

These polynomials arose when Chebyshev (Чебышев) was studying the motion of linkages in a steam locomotive. Since then they have become a primary ingredient in applied mathematics. The Chebyshev polynomials have many marvelous properties. (See Rivlin [1990].)

THEOREM 3 Theorem on Chebyshev Polynomials

For x in the interval $[-1, 1]$, the Chebyshev polynomials have this closed-form expression:

$$T_n(x) = \cos(n \cos^{-1} x) \quad (n \geq 0)$$

Proof Recall the addition formula for the cosine:

$$\cos(A + B) = \cos A \cos B - \sin A \sin B$$

From this we obtain

$$\begin{aligned}\cos(n+1)\theta &= \cos\theta \cos n\theta - \sin\theta \sin n\theta \\ \cos(n-1)\theta &= \cos\theta \cos n\theta + \sin\theta \sin n\theta\end{aligned}$$

On adding these two equations and rearranging, we obtain

$$\cos(n+1)\theta = 2\cos\theta \cos n\theta - \cos(n-1)\theta \tag{14}$$

Now let $\theta = \cos^{-1}x$ and $x = \cos\theta$. Equation (14) shows that the functions f_n defined by

$$f_n(x) = \cos(n \cos^{-1}x)$$

obey this system of equations:

$$\begin{cases} f_0(x) = 1 & f_1(x) = x \\ f_{n+1}(x) = 2xf_n(x) - f_{n-1}(x) & (n \geq 1) \end{cases}$$

It follows that $f_n = T_n$ for all n . ■

From the formula in Theorem 3, we obtain further properties of the Chebyshev polynomials, such as

$$|T_n(x)| \leq 1 \quad (-1 \leq x \leq 1)$$

$$T_n\left(\cos \frac{j\pi}{n}\right) = (-1)^j \quad (0 \leq j \leq n)$$

$$T_n\left(\cos \frac{2j-1}{2n}\pi\right) = 0 \quad (1 \leq j \leq n)$$

A **monic polynomial** is one in which the term of highest degree has a coefficient of unity. From the definition of the Chebyshev polynomials, we see that in $T_n(x)$ the term of highest degree is $2^{n-1}x^n$ for $n > 0$. Therefore, $2^{1-n}T_n$ is a monic polynomial for $n > 0$.

THEOREM 4 Theorem on Monic Polynomials

If p is a monic polynomial of degree n , then

$$\|p\|_\infty = \max_{-1 \leq x \leq 1} |p(x)| \geq 2^{1-n}$$

Proof We proceed by contradiction. Suppose that

$$|p(x)| < 2^{1-n} \quad (|x| \leq 1)$$

Let $q = 2^{1-n}T_n$ and $x_i = \cos(i\pi/n)$. As noted above, q is a monic polynomial of degree n . Then

$$(-1)^i p(x_i) \leq |p(x_i)| < 2^{1-n} = (-1)^i q(x_i)$$

Consequently,

$$(-1)^i [q(x_i) - p(x_i)] > 0 \quad (0 \leq i \leq n)$$

This shows that the polynomial $q - p$ oscillates in sign $n + 1$ times on the interval $[-1, 1]$. It therefore must have at least n roots in $(-1, 1)$. But this is not possible because $q - p$ has degree at most $n - 1$. (Remember that q and p are monic, and the term x^n will not appear in $q - p$.) ■

Choosing the Nodes

In Theorem 2, assume that the interpolation nodes are in the interval $[-1, 1]$. If x is in this same interval, then ξ_x also will be in that interval. Hence, we can deduce that

$$\max_{|x| \leq 1} |f(x) - p(x)| \leq \frac{1}{(n + 1)!} \max_{|x| \leq 1} |f^{(n+1)}(x)| \max_{|x| \leq 1} \left| \prod_{i=0}^n (x - x_i) \right|$$

By Theorem 4, we have (for any set of nodes)

$$\max_{|x| \leq 1} \left| \prod_{i=0}^n (x - x_i) \right| \geq 2^{-n}$$

This minimum value will be attained if $\prod_{i=0}^n (x - x_i)$ is the monic multiple of T_{n+1} , that is, $2^{-n} T_{n+1}$. The nodes then will be the roots of T_{n+1} . These are

$$x_i = \cos\left(\frac{2i + 1}{2n + 2} \pi\right) \quad (0 \leq i \leq n)$$

These considerations establish the next result.

THEOREM 5 Theorem on Interpolation Error, Chebyshev Nodes

If the nodes x_i are the roots of the Chebyshev polynomial T_{n+1} , then the error formula in Theorem 2 yields (for $|x| \leq 1$)

$$|f(x) - p(x)| \leq \frac{1}{2^n (n + 1)!} \max_{|t| \leq 1} |f^{(n+1)}(t)|$$

The Convergence of Interpolating Polynomials

If a continuous function f is prescribed on an interval $[a, b]$, and if interpolating polynomials p_n of higher and higher degree (with evenly spaced nodes) are constructed for f , then the natural expectation is that these polynomials will converge to f uniformly on $[a, b]$. That is, we expect the quantity

$$\|f - p_n\|_\infty = \max_{a \leq x \leq b} |f(x) - p_n(x)|$$

to converge to 0 as $n \rightarrow \infty$. We have seen examples, such as $f(x) = \sin x$, in which this is true. But it must be recognized that $\sin x$ is far from being a typical continuous function. It is of class C^∞ on the real line and, as a function of a complex variable, it is an **entire function**, which means that it has no singularities at all in (the finite part of) the complex plane.

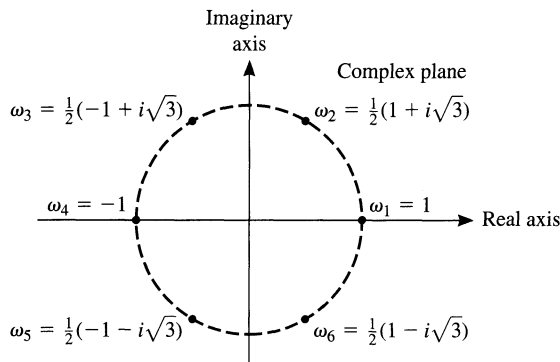


FIGURE 6.2
The 6 sixth roots of unity

The surprising state of affairs is that for most continuous functions, the quantity $\|f - p_n\|_\infty$ will not converge to 0. The first example of this was exhibited by Meray in 1884. He took as nodes the n **nth roots of unity** in the complex plane. These are points evenly spaced on the unit circle $|z| = 1$, as shown in Figure 6.2 for $n = 6$.

We denote these nodes by $\omega_1, \omega_2, \dots, \omega_n$. Now consider the problem of interpolating the function $f(z) = 1/z$ at the nodes $\omega_1, \omega_2, \dots, \omega_n$. A unique polynomial p_{n-1} of degree $n - 1$ solves this problem. Not surprisingly, it is $p_{n-1}(z) = z^{n-1}$ because

$$p_{n-1}(\omega_j) = \omega_j^{n-1} = \frac{1}{\omega_j} \omega_j^n = \frac{1}{\omega_j} = f(\omega_j) \quad (1 \leq j \leq n)$$

Let us measure the discrepancy between f and p_{n-1} on the unit circle. We have

$$\begin{aligned} \|f - p_{n-1}\|_\infty &= \max_{|z|=1} |f(z) - p_{n-1}(z)| = \max_{|z|=1} |z^{-1} - z^{n-1}| \\ &= \max_{|z|=1} \frac{1}{|z|} |1 - z^n| = 2 \end{aligned}$$

(As z traverses the circle $|z| = 1$, the same is true of z^n , and when $z^n = -1$ we obtain 2 in the preceding calculation.) Thus, when $n \rightarrow \infty$, p_n and f in this example are always two units apart.

Another example, this time in the real domain, was given by Runge in 1901. This is the function $f(x) = (x^2 + 1)^{-1}$ on the interval $[-5, 5]$. If interpolating polynomials p_n are constructed for this function using equally spaced nodes in $[-5, 5]$, it is found that the sequence $\|f - p_n\|_\infty$ is not bounded. A computer experiment to see this behavior is outlined in the next section. Even for modest values of n (say $n = 15$), we see wild oscillations in the polynomial p_n . This phenomenon has become a standard example in textbooks and papers and is called the **Runge function** or **Runge example**. See, for example, Epperson [1987].

The proof that interpolation at equally spaced nodes in the Runge function produces unbounded polynomials is in the book on interpolation by Steffensen [1950] first published in 1927. Further background requires complex variable treatment, and for that we recommend the book by Davis [1982] on approximation and interpolation. The Runge function seems like a benign one from the viewpoint of real

variables, but in the complex plane it has a singularity nearby, on the imaginary axis. This is the source of the trouble. The analysis is rather hard and beyond the scope of this book. The positive result is Theorem 7 below, which shows that an interpolation polynomial p_n converges uniformly to f under certain conditions. However, the negative result is Theorem 6 below, which shows that the interpolants to f are unbounded (and hence cannot converge uniformly to f). The difference is in the quantifiers in these theorems!

Subsequent work by other mathematicians, such as Faber, disclosed the following very general result in 1914:

THEOREM 6 Faber’s Theorem

For any prescribed system of nodes

$$a \leq x_0^{(n)} < x_1^{(n)} < \cdots < x_n^{(n)} \leq b \qquad (n \geq 0) \tag{15}$$

there exists a continuous function f on $[a, b]$ such that the interpolating polynomials for f using these nodes fail to converge uniformly to f .

This matter is a little more subtle than it seems at first because we have the following positive result:

THEOREM 7 Theorem on Convergence of Interpolants

If f is a continuous function on $[a, b]$, then there is a system of nodes as in Equation (15) such that the polynomials p_n of interpolation to f at these nodes satisfy $\lim_{n \rightarrow \infty} \|f - p_n\|_\infty = 0$.

This theorem is obtained by joining two other powerful theorems: the **Weierstrass Approximation Theorem** and the **Chebyshev Alternation Theorem**. We turn now to the first of these two results. (The second appears in Section 6.9, p. 416.)

THEOREM 8 Weierstrass Approximation Theorem

If f is continuous on $[a, b]$ and if $\varepsilon > 0$, then there is a polynomial p satisfying $|f(x) - p(x)| \leq \varepsilon$ on the interval $[a, b]$.

Proof It suffices to prove this theorem for the special interval $[0, 1]$, since the change of variable $x = a + t(b - a)$ can be used to go back and forth between $[a, b]$ and $[0, 1]$. Of course, a polynomial of a linear function of t is a polynomial in t of the same degree. If $f \in C[0, 1]$, the sequence of Bernstein polynomials $B_n f$, as explained later, converges uniformly to f . ■

Now, let us discuss the **Bernstein polynomials**, introduced by Serge Bernstein in 1912. They are given by

$$(B_n f)(x) = \sum_{k=0}^n f\left(\frac{k}{n}\right) g_{nk}(x) \quad \text{where} \quad g_{nk}(x) = \binom{n}{k} x^k (1-x)^{n-k}$$

In the definition of g_{nk} , the term $\binom{n}{k}$ is a **binomial coefficient**, defined by

$$\binom{n}{k} = \begin{cases} \frac{n!}{k!(n-k)!} & 0 \leq k \leq n \\ 0 & \text{otherwise} \end{cases}$$

Why do we care about the Bernstein polynomials? We use them primarily to give an elementary proof of the Weierstrass Theorem. However, Bernstein polynomials are used also in computer-aided design. (Recently, there has been a tendency to replace them by B -splines in this application.)

It is essential to interpret B_n as a linear operator that maps elements of $C[0, 1]$ into other elements. The linearity of B_n is expressed by the equation

$$B_n(af + bg) = aB_nf + bB_ng \quad (a, b \in \mathbb{R}; \quad f, g \in C[0, 1])$$

which is easily verified. Another important property of these operators is that they are *positive*. This means that if $f \geq 0$, then $B_nf \geq 0$. The truth of this follows at once from the fact that $g_{nk} \geq 0$ on the interval $[0, 1]$. (We regard all functions as being defined only on the interval $[0, 1]$.)

The properties of linearity and positivity together are extremely powerful, as indicated in the following theorem.

THEOREM 9 Bohman-Korovkin Theorem

Let L_n ($n \geq 1$) be a sequence of positive linear operators defined on $C[a, b]$ and taking values in the same space. If $\|L_nf - f\|_\infty \rightarrow 0$ for the three functions $f(x) = 1, x, x^2$, then the same is true for all $f \in C[a, b]$.

Proof In the space $C[a, b]$, we can take absolute values, denoting by $|f|$ that function whose value at x is $|f(x)|$. Notice also that if L is a positive linear operator, we have this implication:

$$f \geq g \Rightarrow f - g \geq 0 \Rightarrow L(f - g) \geq 0 \Rightarrow Lf - Lg \geq 0 \Rightarrow Lf \geq Lg$$

A further property is that $L(|f|) \geq |Lf|$, for $|f| \geq f$ and $|f| \geq -f$, from which $L(|f|) \geq Lf$ and $L(|f|) \geq -Lf$. Now set $h_k(x) = x^k$ for $k = 0, 1, 2$. Also define functions α_n, β_n , and γ_n by putting

$$\alpha_n = L_nh_0 - h_0 \quad \beta_n = L_nh_1 - h_1 \quad \gamma_n = L_nh_2 - h_2$$

The hypothesis of the theorem asserts that

$$\|\alpha_n\|_\infty \rightarrow 0 \quad \|\beta_n\|_\infty \rightarrow 0 \quad \|\gamma_n\|_\infty \rightarrow 0$$

(Here and for the remainder of the proof we are using the infinity norm.) For the main part of the proof, let f be any element of $C[a, b]$. Let $\varepsilon > 0$. We want to prove

that an integer m exists such that

$$n \geq m \Rightarrow \|L_n f - f\|_\infty < 3\varepsilon$$

Since f is continuous on a compact interval, it is uniformly continuous. Consequently, a positive δ exists such that for all x and y in $[a, b]$,

$$|x - y| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon$$

Observe that with $c = 2\|f\|_\infty/\delta^2$, we have

$$|x - y| \geq \delta \Rightarrow |f(x) - f(y)| \leq 2\|f\|_\infty \leq 2\|f\|_\infty \frac{(x - y)^2}{\delta^2} = c(x - y)^2$$

Thus, for all x and y in $[a, b]$, we have

$$|f(x) - f(y)| \leq \varepsilon + c(x - y)^2$$

This inequality can be written in the form

$$|f - f(y)h_0| \leq \varepsilon h_0 + c[h_2 - 2yh_1 + y^2h_0]$$

To verify this, simply substitute x and use the definitions of h_k . From remarks made at the beginning of the proof, we infer that

$$|L_n f - f(y)L_n h_0| \leq \varepsilon L_n h_0 + c[L_n h_2 - 2yL_n h_1 + y^2 L_n h_0]$$

This is an inequality among functions, and we are at liberty to substitute y into it:

$$\begin{aligned} & |(L_n f)(y) - f(y)(L_n h_0)(y)| \\ & \leq \varepsilon (L_n h_0)(y) + c[(L_n h_2)(y) - 2y(L_n h_1)(y) + y^2(L_n h_0)(y)] \\ & = \varepsilon[1 + \alpha_n(y)] + c[y^2 + \gamma_n(y) - 2y(y + \beta_n(y)) + y^2(1 + \alpha_n(y))] \\ & = \varepsilon + \varepsilon\alpha_n + c\gamma_n(y) - 2cy\beta_n(y) + cy^2\alpha_n(y) \\ & \leq \varepsilon + \varepsilon\|\alpha_n\|_\infty + c\|\gamma_n\|_\infty + 2c\|h_1\|_\infty\|\beta_n\|_\infty + c\|h_2\|_\infty\|\alpha_n\|_\infty \end{aligned}$$

Select m so that the final right-hand side of this inequality is less than 2ε whenever $n \geq m$. Then for $n \geq m$, we have

$$\|L_n - f \cdot L_n h_0\|_\infty \leq 2\varepsilon$$

Finally, we write

$$\begin{aligned} \|L_n f - f\|_\infty & \leq \|L_n f - f \cdot L_n h_0\|_\infty + \|f \cdot L_n h_0 - f \cdot h_0\|_\infty \\ & \leq 2\varepsilon + \|f\|_\infty \|\alpha_n\|_\infty \end{aligned}$$

If necessary, increase m so that for $n \geq m$ we have $\|f\|_\infty \|\alpha_n\|_\infty < \varepsilon$. Then the last term above is no greater than 3ε . ■

In the proof of the Weierstrass Theorem, a remaining detail is to show that $B_n h_k \rightarrow h_k$ for $k = 0, 1, 2$. Here $h_k(x) = x^k$. For h_0 , we use the Binomial Theorem to write

$$(B_n h_0)(x) = \sum_{k=0}^n g_{nk}(x) = \sum_{k=0}^n \binom{n}{k} x^k (1-x)^{n-k} = (x + (1-x))^n = 1$$

For h_1 , we have, with the help of Problem 6.1.35 (p. 327),

$$\begin{aligned} (B_n h_1)(x) &= \sum_{k=0}^n \left[\frac{k}{n} \right] \binom{n}{k} x^k (1-x)^{n-k} = \sum_{k=1}^n \binom{n-1}{k-1} x^k (1-x)^{n-k} \\ &= x \sum_{k=0}^{n-1} \binom{n-1}{k} x^k (1-x)^{n-1-k} = x \end{aligned}$$

Lastly, for $k = 2$, we have, using Problem 6.1.35 (p. 327) twice,

$$\begin{aligned} (B_n h_2)(x) &= \sum_{k=0}^n \left[\frac{k}{n} \right]^2 \binom{n}{k} x^k (1-x)^{n-k} = \sum_{k=1}^n \left[\frac{k}{n} \right] \binom{n-1}{k-1} x^k (1-x)^{n-k} \\ &= \sum_{k=1}^n \left[\frac{n-1}{n} \frac{k-1}{n-1} + \frac{1}{n} \right] \binom{n-1}{k-1} x^k (1-x)^{n-k} \\ &= \frac{n-1}{n} x^2 \sum_{k=2}^n \binom{n-2}{k-2} x^{k-2} (1-x)^{n-k} + \frac{x}{n} \\ &= \frac{n-1}{n} x^2 + \frac{x}{n} \rightarrow x^2 \end{aligned}$$

PROBLEMS 6.1

1. Find the polynomials of least degree that interpolate these sets of data:

a.

x	3	7
y	5	-1

b.

x	7	1	2
y	146	2	1

c.

x	3	7	1	2
y	10	146	2	1

d.

x	3	7	1	2
y	12	146	2	1

e.

x	1.5	2.7	3.1	-2.1	-6.6	11.0
y	0.0	0.0	0.0	1.0	0.0	0.0

2. The polynomial p of degree $\leq n$ that interpolates a given function f at $n + 1$ prescribed nodes is uniquely defined. Hence, there is a mapping $f \mapsto p$. Denote this mapping by L and show that

$$Lf = \sum_{i=0}^n f(x_i) \ell_i$$

Show that L is **linear**; that is, $L(af + bg) = aLf + bLg$.

3. (Continuation) Refer to the preceding problem and define another mapping, G , by the formula

$$Gf = \sum_{i=0}^n f(x_i) \ell_i^2$$

Prove that Gf is a polynomial of degree at most $2n$, that Gf interpolates f at the nodes, and that Gf is nonnegative whenever f is nonnegative.

4. (Continuation) Prove that the mapping, L , in Problem 6.1.2 (above) has the property that $Lq = q$ for every polynomial q of degree at most n .
5. (Continuation) Prove that $\sum_{i=0}^n \ell_i(x) = 1$ for all x .
6. (Continuation) Prove that if p is a polynomial of degree $\leq n$ that interpolates the function f at $x_0, x_1, \dots, x_n$ (distinct points), then

$$f(x) - p(x) = \sum_{i=0}^n [f(x) - f(x_i)] \ell_i(x)$$

7. Prove that the algorithm for computing the coefficients c_i in the Newton form of the interpolating polynomial involves n^2 long operations (multiplications and divisions).
8. Discuss the problem of determining a polynomial of degree at most 2 for which $p(0)$, $p(1)$, and $p'(\xi)$ are prescribed, ξ being any preassigned point.
9. Prove that if g interpolates the function f at $x_0, x_1, \dots, x_{n-1}$ and if h interpolates f at $x_1, x_2, \dots, x_n$, then the function

$$g(x) + \frac{x_0 - x}{x_n - x_0} [g(x) - h(x)]$$

interpolates f at $x_0, x_1, \dots, x_n$. Notice that h and g need not be polynomials.

10. Prove that the coefficient of x^n in the polynomial p of Equation (9) is

$$\sum_{i=0}^n y_i \prod_{\substack{j=0 \\ j \neq i}}^n (x_i - x_j)^{-1}$$

11. Prove that for any polynomial q of degree $\leq n - 1$,

$$\sum_{i=0}^n q(x_i) \prod_{\substack{j=0 \\ j \neq i}}^n (x_i - x_j)^{-1} = 0$$

12. Determine whether the algorithm

```
x ← anbn
for i = 1 to n do
  x ← (x + an-i)bi
end do
```

computes

$$x = \sum_{i=0}^n a_i \prod_{j=0}^i b_j$$

13. Prove that if we take *any* set of 23 nodes in the interval $[-1, 1]$ and interpolate the function $f(x) = \cosh x$ with a polynomial p of degree 22, then the relative error $|p(x) - f(x)|/|f(x)|$ is no greater than 5×10^{-16} on $[-1, 1]$.
14. Let p be a polynomial of degree $\leq n - 1$ that interpolates the function $f(x) = \sinh x$ at any set of n nodes in the interval $[-1, 1]$, subject only to the condition that one of the nodes is 0. Prove that the error obeys this inequality on $[-1, 1]$:

$$|p(x) - f(x)| \leq \frac{2^n}{n!} |f(x)|$$

15. What is the final value of v in the algorithm shown?

```
v ← ci-1
for j = i to n do
    v ← vx + cj
end do
```

What is the number of additions and subtractions involved in this algorithm?

16. Write an efficient algorithm for evaluating

$$u = \sum_{i=1}^n \prod_{j=1}^i d_j$$

17. Suppose that p is a polynomial of degree greater than n that interpolates f at $n + 1$ nodes. What can you discover about $f(x) - p(x)$?
18. Prove or disprove: If n is a divisor of m , then each zero of T_n is a zero of T_m .
19. Find a polynomial that assumes these values:

x	1	2	0	3
y	3	2	-4	5

20. Prove that for $x \geq 1$, $T_n(x) = \cosh(n \cosh^{-1} x)$. *Hint:* Imitate the proof of Theorem 3 on Chebyshev polynomials.
21. Write the Lagrange and Newton interpolating polynomials for these data:

x	2	0	3
$f(x)$	11	7	28

22. Find the Lagrange and Newton forms of the interpolating polynomial for these data:

x	-2	0	1
$f(x)$	0	1	-1

Write both polynomials in the form $a + bx + cx^2$ to verify that they are identical as functions.

23. Consider the data

x	$-\sqrt{\frac{3}{5}}$	0	$\sqrt{\frac{3}{5}}$
$f(x)$	$f\left(-\sqrt{\frac{3}{5}}\right)$	$f(0)$	$f\left(\sqrt{\frac{3}{5}}\right)$

- What are the Newton interpolation polynomial and the Lagrange interpolation polynomial for these data?
24. The formula for the leading coefficient in T_n is 2^{n-1} . What is the formula for the coefficient of x^{n-2} ? What about x^{n-1} ?
25. Find the interpolating polynomial for the table

x	1	3	2	6
y	-2	-22	-1	-37

26. The equation $x - 9^{-x} = 0$ has a solution in $[0, 1]$. Find the interpolation polynomial on $x_0 = 0, x_1 = 0.5, x_2 = 1$ for the function on the left side of the equation. By setting the interpolation polynomial equal to 0 and solving the equation, find an approximate solution to the equation.
27. If we interpolate the function $f(x) = e^{x-1}$ with a polynomial p of degree 12 using 13 nodes in $[-1, 1]$, what is a good upper bound for $|f(x) - p(x)|$ on $[-1, 1]$?
28. Let p_k be the polynomial of degree $\leq k$ such that $p_k(x_i) = y_i$ for $0 \leq i \leq k$. Prove that $p_k = p_{k-1}$ if and only if $p_{k-1}(x_k) = y_k$.
29. Devise a method for solving an equation $f(x) = 0$ that gives the correct root in $n + 1$ steps if f^{-1} is a polynomial of degree n in a neighborhood of the root sought. Here f^{-1} is an inverse function: $f^{-1}(f(x)) = x$.
30. Prove the following: If g is a function (not necessarily a polynomial) that interpolates a function f at nodes $x_0, x_1, \dots, x_{n-1}$, and if h is a function such that $h(x_i) = \delta_{in}$ ($0 \leq i \leq n$), then for some c the function $g + ch$ interpolates f at $x_0, x_1, \dots, x_n$.
31. Refer to the Lagrange interpolation process and define

$$w_i = \prod_{\substack{j=0 \\ j \neq i}}^n (x_i - x_j)^{-1}$$

Show that if x is not a node, then the interpolating polynomial can be evaluated by the formula

$$p(x) = \left[\sum_{i=0}^n y_i w_i (x - x_i)^{-1} \right] \bigg/ \left[\sum_{i=0}^n w_i (x - x_i)^{-1} \right]$$

- This is called the **barycentric form** of the Lagrange interpolation process.
32. (Continuation) Show that the evaluation of p in the preceding problem is stable in the sense that if the w_i are incorrectly computed, we still have the interpolation property: $\lim_{x \rightarrow x_k} p(x) = y_k$ ($0 \leq k \leq n$).
33. Let E be an $(n + 1)$ -dimensional vector space of functions defined on a domain D . Let $x_0, x_1, \dots, x_n$ be distinct points in D . Show that the interpolation problem

$$f(x_i) = y_i \quad (0 \leq i \leq n) \quad f \in E$$

has a unique solution for any choice of ordinates y_i if and only if no element of E other than 0 vanishes at all the points $x_0, x_1, \dots, x_n$.

34. Prove that

$$\det \begin{vmatrix} 1 & x_0 & x_0^2 & \cdots & x_0^n \\ 1 & x_1 & x_1^2 & \cdots & x_1^n \\ 1 & x_2 & x_2^2 & \cdots & x_2^n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^n \end{vmatrix} = \prod_{0 \leq j < k \leq n} (x_k - x_j) \\ = (x_n - x_0)(x_n - x_1)(x_n - x_2) \cdots (x_n - x_{n-1}) \cdots \\ (x_3 - x_0)(x_3 - x_1)(x_3 - x_2)(x_2 - x_0) \cdots \\ (x_2 - x_1)(x_1 - x_0)$$

35. Prove that

$$\left[\begin{matrix} k \\ n \end{matrix} \right] \binom{n}{k} = \binom{n-1}{k-1}$$

36. The functions $1/(1+x^2)$ and e^{-x^2} have a similar appearance. Do they behave similarly in the interpolation process for equally spaced nodes?
37. The first U.S. postage stamp was issued in 1885, with the cost to mail a letter set at 2 cents. In 1917, the cost was raised to 3 cents but then was returned to 2 cents in 1919. In 1932, it was upped to 3 cents again, where it remained for 26 years. Then a series of increases took place as follows: 1958 = 4 cents, 1963 = 5 cents, 1968 = 6 cents, 1971 = 8 cents, 1974 = 10 cents, 1978 = 15 cents, 1981 = 18 cents in March and 20 cents in October, 1985 = 22 cents, 1988 = 25 cents, 1991 = 29 cents, 1995 = 32 cents, 1999 = 33 cents, and 2001 = 34 cents. Determine the Newton interpolation polynomial for these data. Based on this, when will it cost \$1 to mail a letter? When will it cost \$10?

6.2 Divided Differences

In the preceding section, we discussed the problem of *interpolating* a function by a polynomial. We return to that problem now. Let f be a function whose values are known or computable at a set of points (*nodes*) $x_0, x_1, \dots, x_n$. We assume in this section that these points are *distinct*, but they need not be *ordered* on the real line. We know that there exists a unique polynomial p of degree at most n that interpolates f at the $n+1$ nodes:

$$p(x_i) = f(x_i) \quad (0 \leq i \leq n) \tag{1}$$

Of course, the polynomial p can be constructed as a linear combination of the basic polynomials $1, x, x^2, \dots, x^n$. As discussed in the previous section, this basis is not recommended, and we prefer to use a basis appropriate to the Newton form of the interpolating polynomial:

$$\begin{aligned}
 q_0(x) &= 1 \\
 q_1(x) &= (x - x_0) \\
 q_2(x) &= (x - x_0)(x - x_1) \\
 q_3(x) &= (x - x_0)(x - x_1)(x - x_2) \\
 &\vdots \\
 q_n(x) &= (x - x_0)(x - x_1)(x - x_2) \cdots (x - x_{n-1})
 \end{aligned}$$

These lead to the Newton form

$$p(x) = \sum_{j=0}^n c_j q_j(x)$$

The interpolation conditions give rise to a linear system of equations for the determination of the unknown coefficients, c_j :

$$\sum_{j=0}^n c_j q_j(x_i) = f(x_i) \quad (0 \leq i \leq n) \quad (2)$$

In this system of equations, the coefficient matrix is an $(n+1) \times (n+1)$ matrix A whose elements are

$$a_{ij} = q_j(x_i) \quad (0 \leq i, j \leq n) \quad (3)$$

The matrix A is lower triangular because

$$\begin{aligned}
 q_j(x) &= \prod_{k=0}^{j-1} (x - x_k) \\
 q_j(x_i) &= \prod_{k=0}^{j-1} (x_i - x_k) = 0 \quad \text{if } i \leq j-1
 \end{aligned} \quad (4)$$

For example, consider the case of three nodes with

$$\begin{aligned}
 p_2(x) &= c_0 q_0(x) + c_1 q_1(x) + c_2 q_2(x) \\
 &= c_0 + c_1(x - x_0) + c_2(x - x_0)(x - x_1)
 \end{aligned}$$

Setting $x = x_0$, $x = x_1$, and $x = x_2$, we have a lower triangular system

$$\begin{bmatrix} 1 & 0 & 0 \\ 1 & (x_1 - x_0) & 0 \\ 1 & (x_2 - x_0) & (x_2 - x_0)(x_2 - x_1) \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} f(x_0) \\ f(x_1) \\ f(x_2) \end{bmatrix} \quad (5)$$

In solving Equation (2) for $c_0, c_1, \dots, c_n$, we can start at the top and work down, computing the coefficients, c_j , in the order given by their subscripts. In this process, we see that c_0 depends on only $f(x_0)$, that c_1 depends on $f(x_0)$ and $f(x_1)$, and so on. Thus, c_n depends on f at $x_0, x_1, \dots, x_n$. The notation

$$c_n = f[x_0, x_1, \dots, x_n] \quad (6)$$

was adopted many years ago to signify this dependency. We are therefore *defining* the symbol $f[x_0, x_1, \dots, x_n]$ to be the coefficient of q_n when $\sum_{k=0}^n c_k q_k$ interpolates f at $x_0, x_1, \dots, x_n$. Since

$$q_n(x) = (x - x_0)(x - x_1) \cdots (x - x_{n-1}) = x^n + \text{lower-order terms}$$

we can say also that $f[x_0, x_1, \dots, x_n]$ is the coefficient of x^n in the polynomial of degree at most n that interpolates f at $x_0, x_1, \dots, x_n$. In all of the preceding description, n can take on arbitrary values. The expressions $f[x_0, x_1, \dots, x_n]$ are called **divided differences** of f .

Explicit formulas for the first few divided differences will now be given. First, $f[x_0]$ is the coefficient of x^0 in the polynomial of degree 0 interpolating f at x_0 . Thus, we must have

$$f[x_0] = f(x_0) \quad (7)$$

The quantity $f[x_0, x_1]$ is the coefficient of x in the polynomial of degree at most 1 interpolating f at x_0 and x_1 . Since that polynomial is

$$p(x) = f(x_0) + \frac{f(x_1) - f(x_0)}{x_1 - x_0}(x - x_0) \quad (8)$$

we see that the coefficient of $q_1(x)$ is

$$f[x_0, x_1] = \frac{f(x_1) - f(x_0)}{x_1 - x_0} \quad (9)$$

This gives a hint as to why the term **divided difference** was adopted. A divided difference table with the following form can be displayed:

$$\begin{array}{ccc} x_0 & f(x_0) & f[x_0, x_1] \\ x_1 & f(x_1) & \end{array}$$

and the interpolation polynomial is easily formed from it:

$$p(x) = f(x_0) + f[x_0, x_1](x - x_0)$$

Formulas (7) and (9) can also be obtained by solving for c_0 and c_1 in System (5) because $c_0 = f[x_0]$ and $c_1 = f[x_0, x_1]$, in accordance with Equation (6). Equation (1) allows us to write the Newton interpolating polynomial in the form

$$p(x) = \sum_{k=0}^n c_k q_k(x) = \sum_{k=0}^n f[x_0, x_1, \dots, x_k] \prod_{j=0}^{k-1} (x - x_j) \quad (10)$$

The correctness of Equation (10) is proved by the following reasoning: If we *truncate* the sum in Equation (10) at the value $k = m$, we obtain $\sum_{k=0}^m c_k q_k(x)$. We know that this is the polynomial of degree at most m that interpolates f at $x_0, x_1, \dots, x_m$. Hence, $c_m = f[x_0, x_1, \dots, x_m]$. This is true for all m in the range $0 \leq m \leq n$.

Higher-Order Divided Differences

For computing the higher-order divided differences, the following theorem can be used.

THEOREM 1
Theorem on Higher-Order Divided Differences

Divided differences satisfy the equation

$$f[x_0, x_1, \dots, x_n] = \frac{f[x_1, x_2, \dots, x_n] - f[x_0, x_1, \dots, x_{n-1}]}{x_n - x_0} \tag{11}$$

Proof First, let p_k denote the polynomial of degree at most k that interpolates f at the nodes $x_0, x_1, \dots, x_k$. We shall be needing p_n and p_{n-1} . Let q denote the polynomial of degree at most $n - 1$ that interpolates f at $x_1, x_2, \dots, x_n$. Then we have

$$p_n(x) = q(x) + \frac{x - x_n}{x_n - x_0} [q(x) - p_{n-1}(x)] \tag{12}$$

This equation is proved by noting first that on both sides of the equation there appears a polynomial of degree at most n . Then we verify that the values of these polynomials on the right and the left are the same at the points $x_0, x_1, \dots, x_n$. Hence, the polynomials must be identical. Now let us examine the coefficient of x^n on the left and right sides of Equation (12). These coefficients must be equal, and so we arrive at Equation (11).

Theorem 1 gives us these particular formulas:

$$\begin{aligned} f[x_0, x_1] &= \frac{f[x_1] - f[x_0]}{x_1 - x_0} \\ f[x_0, x_1, x_2] &= \frac{f[x_1, x_2] - f[x_0, x_1]}{x_2 - x_0} \\ &\vdots \end{aligned}$$

In these formulas, $x_0, x_1, x_2, \dots$ can be interpreted as independent variables. Because of that, we also have equations such as

$$f[x_i, x_{i+1}, \dots, x_{i+j}] = \frac{f[x_{i+1}, x_{i+2}, \dots, x_{i+j}] - f[x_i, x_{i+1}, \dots, x_{i+j-1}]}{x_{i+j} - x_i} \tag{13}$$

Here $f[x_i]$, $f[x_i, x_{i+1}]$, $f[x_i, x_{i+1}, x_{i+2}]$, $f[x_i, x_{i+1}, x_{i+2}, x_{i+3}]$, and so on are differences of order 0, 1, 2, 3, and so forth, respectively.

If a table of function values $(x_i, f(x_i))$ is given, we can construct from it a table of divided differences. This is customarily laid out in the following form, where differences of orders 0, 1, 2, and 3 are shown in each successive column:

x_0	$f[x_0]$	$f[x_0, x_1]$	$f[x_0, x_1, x_2]$	$f[x_0, x_1, x_2, x_3]$
x_1	$f[x_1]$	$f[x_1, x_2]$	$f[x_1, x_2, x_3]$	
x_2	$f[x_2]$	$f[x_2, x_3]$		
x_3	$f[x_3]$			

The information to the left of the vertical line is given, and the quantities on the right are to be computed. Formula (11) is used to do this. The recursive nature of Formula (11) dictates the triangular form of the divided difference table. For example, the given data do not allow us to compute $f[x_3, x_4]$, $f[x_2, x_3, x_4]$, and so on.

By comparing Equations (10) and (11), we see that the coefficients required in the Newton interpolating polynomial occupy the top row in the divided difference table.

EXAMPLE 1 Compute a divided difference table for these function values:

x	3	1	5	6
$f(x)$	1	-3	2	4

(14)

Solution We arrange the given table vertically and compute divided differences by the use of Formula (11), arriving at

3	1	2	$-\frac{3}{8}$	$\frac{7}{40}$
1	-3	$\frac{5}{4}$	$\frac{3}{20}$	
5		2	2	
6	4			

■

EXAMPLE 2 Find the Newton interpolating polynomial for the function values in Table (14).

Solution $p(x) = 1 + 2(x - 3) - \frac{3}{8}(x - 3)(x - 1) + \frac{7}{40}(x - 3)(x - 1)(x - 5)$ ■

Algorithm for Divided Differences

An algorithm for computing a divided difference table can be very efficient and is recommended as the best means for producing an interpolating polynomial. Let us change the notation so that our divided difference table has the entries shown here:

x_0	c_{00}	c_{01}	c_{02}	c_{03}	$\cdots$	$c_{0,n-1}$	$c_{0,n}$
x_1	c_{10}	c_{11}	c_{12}	c_{13}	$\cdots$	$c_{1,n-1}$	
x_2	c_{20}	c_{21}	c_{22}	c_{23}	$\ddots$		
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$			
$\vdots$	$\vdots$	$\vdots$	$\ddots$				
x_{n-1}	$c_{n-1,0}$	$c_{n-1,1}$					
x_n	c_{n0}						

Again, the vertical line separates the data (on the left) from the entries to be computed. It is clear that we have set

$$c_{ij} = f[x_i, x_{i+1}, \dots, x_{i+j}]$$

An algorithm is obtained from a direct translation of Equation (13):

```
for j = 1 to n do
  for i = 0 to n - j do
     $c_{ij} \leftarrow (c_{i+1,j-1} - c_{i,j-1}) / (x_{i+j} - x_i)$ 
  end do
end do
```

In this algorithm, the numbers c_{i0} (which are input) are the values of the function f at the points x_i . They are also the values that the interpolating polynomial will have at those points. The interpolating polynomial, of course, is

$$\begin{aligned} p(x) &= c_{00} + c_{01}(x - x_0) + c_{02}(x - x_0)(x - x_1) + \cdots \\ &\quad + c_{0n}(x - x_0)(x - x_1) \cdots (x - x_{n-1}) \\ &= \sum_{i=0}^n c_{0i} \prod_{j=0}^{i-1} (x - x_j) \end{aligned}$$

If the divided difference algorithm is being used only to compute the coefficients in the Newton interpolating polynomial, then another algorithm can be designed that uses less storage space in the computer. A singly subscripted variable can be used. Let it be denoted by $d = [d_0, d_1, \dots, d_n]$. At the beginning, we put the given function values $f(x_0), f(x_1), \dots, f(x_n)$ into d . Observe that d_0 is already the first of the desired coefficients for the Newton interpolating polynomial. Next, we compute the first column of divided differences, putting them in positions $d_1, d_2, \dots, d_n$. After this has been done, d_0 will still have its original value, but d_1 will have become the next of the required coefficients in the polynomial. We continue this pattern, being careful to store the new divided differences in the *bottom part* of the d -vector so as not to disturb the *top part*, where the final values are gradually being created. Here is the algorithm:

```

for  $i = 0$  to  $n$  do
     $d_i \leftarrow f(x_i)$ 
end do
for  $j = 1$  to  $n$  do
    for  $i = n$  to  $j$  step  $-1$  do
         $d_i \leftarrow (d_i - d_{i-1}) / (x_i - x_{i-j})$ 
    end do
end do
    
```

At the conclusion of this algorithm, the vector d contains the coefficients of the polynomial:

$$p(x) = \sum_{i=0}^n d_i \prod_{j=0}^{i-1} (x - x_j)$$

In Problem 6.2.10 (p. 336), the efficiency of this procedure is compared to that of the procedure discussed in Section 6.1 (p. 311).

Divided Difference Properties

We conclude this section with some of the nice properties of divided differences.

THEOREM 2 Theorem on Permutations in Divided Differences

The divided difference is a symmetric function of its arguments. Thus, if $(z_0, z_1, \dots, z_n)$ is a permutation of $(x_0, x_1, \dots, x_n)$, then

$$f[z_0, z_1, \dots, z_n] = f[x_0, x_1, \dots, x_n] \quad (15)$$

Proof The divided difference on the left side of Equation (15) is the coefficient of x^n in the polynomial of degree at most n that interpolates f at the points $z_0, z_1, \dots, z_n$. The divided difference on the right is the coefficient of x^n in the polynomial of degree at most n that interpolates f at the points $x_0, x_1, \dots, x_n$. These two polynomials are, of course, the same. ■

THEOREM 3 Theorem on Error in Newton Interpolation

Let p be the polynomial of degree at most n that interpolates a function f at a set of $n + 1$ distinct nodes, $x_0, x_1, \dots, x_n$. If t is a point different from the nodes, then

$$f(t) - p(t) = f[x_0, x_1, \dots, x_n, t] \prod_{j=0}^n (t - x_j) \quad (16)$$

Proof First, let q be the polynomial of degree at most $n + 1$ that interpolates f at the nodes $x_0, x_1, \dots, x_n, t$. We know that q is obtained from p by adding one term. In fact,

$$q(x) = p(x) + f[x_0, x_1, \dots, x_n, t] \prod_{j=0}^n (x - x_j)$$

Since $q(t) = f(t)$, we obtain at once (by letting $x = t$)

$$f(t) = p(t) + f[x_0, x_1, \dots, x_n, t] \prod_{j=0}^n (t - x_j) \quad \blacksquare$$

THEOREM 4 Theorem on Derivatives and Divided Differences

If f is n times continuously differentiable on $[a, b]$ and if $x_0, x_1, \dots, x_n$ are distinct points in $[a, b]$, then there exists a point ξ in (a, b) such that

$$f[x_0, x_1, \dots, x_n] = \frac{1}{n!} f^{(n)}(\xi) \quad (17)$$

Proof First, let p be the polynomial of degree at most $n - 1$ that interpolates f at the nodes $x_0, x_1, \dots, x_{n-1}$. By Theorem 2 in Section 6.1 (p. 315), there exists a point ξ in (a, b) such that

$$f(x_n) - p(x_n) = \frac{1}{n!} f^{(n)}(\xi) \prod_{j=0}^{n-1} (x_n - x_j) \quad (18)$$

By Theorem 3 in this section, we have

$$f(x_n) - p(x_n) = f[x_0, x_1, \dots, x_n] \prod_{j=0}^{n-1} (x_n - x_j) \quad (19)$$

By comparing Equations (18) and (19), we deduce Equation (17). ■

Hermite-Genocchi Formula

A formula for divided differences, called the **Hermite-Genocchi** formula, is needed in many situations. It asserts that the divided difference $f[x_0, x_1, \dots, x_n]$ is the integral of $f^{(n)}(u_0x_0 + \dots + u_nx_n)$ over the n -dimensional simplex. This **simplex** is the set S_n in $\mathbb{R}^{n+1}$ described as follows:

$$S_n = \left\{ u = (u_0, u_1, \dots, u_n) \in \mathbb{R}^{n+1} : u_i \geq 0, \sum_{i=0}^n u_i = 1 \right\}$$

We shall prove the Hermite-Genocchi formula by induction on n , and therefore begin with the case $n = 1$. We have

$$\begin{aligned} S_1 &= \{u = (u_0, u_1) \in \mathbb{R}^2 : u_0 \geq 0, u_1 \geq 0, u_0 + u_1 = 1\} \\ &= \{(1 - u_1, u_1) : 0 \leq u_1 \leq 1\} \end{aligned}$$

Hence, the integral in question can be written

$$\begin{aligned} \int_{S_1} f'(u_0x_0 + u_1x_1) du &= \int_0^1 f'((1 - u_1)x_0 + u_1x_1) du_1 \\ &= \int_0^1 f'(x_0 + u_1(x_1 - x_0)) du_1 \\ &= \int_0^1 \frac{d}{du_1} [f(x_0 + u_1(x_1 - x_0))] \frac{du_1}{x_1 - x_0} \\ &= \frac{1}{x_1 - x_0} f(x_0 + u_1(x_1 - x_0)) \Big|_{u_1=0}^{u_1=1} \\ &= \frac{1}{x_1 - x_0} \{f(x_1) - f(x_0)\} = f[x_0, x_1] \end{aligned}$$

For the inductive step, let us assume the correctness of the formula for $n - 1$ and prove it for n . Define

$$I(x_0, x_1, \dots, x_n) = \int_{S_n} f^{(n)}(u_0x_0 + \dots + u_nx_n) du$$

Since $\sum_{i=0}^n u_i = 1$, we have $u_0 = 1 - \sum_{i=1}^n u_i$, and consequently,

$$I(x_0, x_1, \dots, x_n) = \int_{S_n} f^{(n)}(x_0 + u_1(x_1 - x_0) + \dots + u_n(x_n - x_0)) du$$

$$\begin{aligned}
&= \int_0^1 \int_0^{1-u_1} \cdots \int_0^{1-u_1-\cdots-u_{n-1}} f^{(n)}(x_0 + u_1(x_1 - x_0) \\
&\quad + \cdots + u_n(x_n - x_0)) du_n \cdots du_2 du_1
\end{aligned}$$

The innermost integration is carried out as in the case $n = 1$; that is, we write

$$\begin{aligned}
&\int_0^{1-u_1-\cdots-u_{n-1}} \frac{d}{du_n} [f^{(n-1)}(x_0 + u_1(x_1 - x_0) + \cdots + u_n(x_n - x_0))] \frac{du_n}{x_n - x_0} \\
&= \frac{1}{x_n - x_0} [f^{(n-1)}(x_0 + u_1(x_1 - x_0) + \cdots + u_n(x_n - x_0))] \Big|_{u_n=0}^{u_n=1-u_1-\cdots-u_{n-1}} \\
&= \frac{1}{x_n - x_0} \left[f^{(n-1)} \left(x_0 + \sum_{i=1}^{n-1} u_i(x_i - x_0) + \left(1 - \sum_{i=1}^{n-1} u_i \right) (x_n - x_0) \right) \right. \\
&\quad \left. - f^{(n-1)} \left(x_0 + \sum_{i=1}^{n-1} u_i(x_i - x_0) \right) \right] \\
&= \frac{1}{x_n - x_0} \left[f^{(n-1)} \left(x_n + \sum_{i=1}^{n-1} u_i(x_i - x_n) \right) - f^{(n-1)} \left(x_0 + \sum_{i=1}^{n-1} u_i(x_i - x_0) \right) \right]
\end{aligned}$$

Thus, our integral I becomes

$$\begin{aligned}
&I(x_0, x_1, \dots, x_n) \\
&= \frac{1}{x_n - x_0} \int_0^1 \int_0^{1-u_1} \cdots \int_0^{1-u_1-\cdots-u_{n-2}} \left[f^{(n-1)} \left(x_n + \sum_{i=1}^{n-1} u_i(x_i - x_n) \right) \right. \\
&\quad \left. - f^{(n-1)} \left(x_0 + \sum_{i=1}^{n-1} u_i(x_i - x_0) \right) \right] du_{n-1} \cdots du_2 du_1 \\
&= \frac{1}{x_n - x_0} [I(x_n, x_1, \dots, x_{n-1}) - I(x_0, x_1, \dots, x_{n-1})]
\end{aligned}$$

By the induction hypothesis, this last expression is

$$\frac{1}{x_n - x_0} \{f[x_n, x_1, \dots, x_{n-1}] - f[x_0, x_1, \dots, x_{n-1}]\}$$

By the recursive formula for divided differences and the invariance of divided differences under permutation of the arguments, the preceding expression is none other than $f[x_0, x_1, \dots, x_n]$.

PROBLEMS 6.2

1. Prove Equation (12) by the process outlined in the text.
2. Prove that if f is continuous, then $f[x_0, x_1, \dots, x_n]$ is continuous on the open set in $\mathbb{R}^{n+1}$ where the components of the vector $(x_0, x_1, \dots, x_n)$ are distinct.

3. Let $f \in C^n[a, b]$. Prove that if $x_0 \in (a, b)$ and if $x_1, x_2, \dots, x_n$ all converge to x_0 , then $f[x_0, x_1, \dots, x_n]$ will converge to $f^{(n)}(x_0)/n!$.
4. Prove that if f is a polynomial of degree k , then for $n > k$,

$$f[x_0, x_1, \dots, x_n] = 0$$

5. Prove that if p is a polynomial of degree at most n , then

$$p(x) = \sum_{i=0}^n p[x_0, x_1, \dots, x_i] \prod_{j=0}^{i-1} (x - x_j)$$

6. Show that the divided differences are linear maps on functions. That is, prove the equation

$$(\alpha f + \beta g)[x_0, x_1, \dots, x_n] = \alpha f[x_0, x_1, \dots, x_n] + \beta g[x_0, x_1, \dots, x_n]$$

7. The divided difference $f[x_0, x_1]$ is analogous to a first derivative, as indicated in Theorem 4. Does it have a property analogous to $(fg)' = f'g + fg'$?
8. Using the functions ℓ_i defined in Section 6.1 (p. 312) and based on nodes $x_0, x_1, \dots, x_n$, show that for any f ,

$$\sum_{i=0}^n f(x_i) \ell_i(x) = \sum_{i=0}^n f[x_0, x_1, \dots, x_i] \prod_{j=0}^{i-1} (x - x_j)$$

9. (Continuation) Prove this formula:

$$f[x_0, x_1, \dots, x_n] = \sum_{i=0}^n f(x_i) \prod_{\substack{j=0 \\ j \neq i}}^n (x_i - x_j)^{-1}$$

10. Compare the efficiency of the divided difference algorithm to the procedure described in Section 6.1 (p. 311) for computing the coefficients in a Newton interpolating polynomial.
11. Use Cramer's rule in matrix theory to prove that

$$f[x_0, x_1, \dots, x_n] = \frac{\begin{vmatrix} 1 & x_0 & x_0^2 & \cdots & x_0^{n-1} & f(x_0) \\ 1 & x_1 & x_1^2 & \cdots & x_1^{n-1} & f(x_1) \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^{n-1} & f(x_n) \end{vmatrix}}{\begin{vmatrix} 1 & x_0 & x_0^2 & \cdots & x_0^n \\ 1 & x_1 & x_1^2 & \cdots & x_1^n \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^n \end{vmatrix}}$$

12. For the particular function $f(x) = x^m$, $m \in \mathbb{N}$, show that

$$f[x_0, x_1, \dots, x_n] = \begin{cases} 1 & \text{if } n = m \\ 0 & \text{if } n > m \end{cases}$$

13. Prove the **Leibniz formula**

$$(fg)[x_0, x_1, \dots, x_n] = \sum_{k=0}^n f[x_0, x_1, \dots, x_k] g[x_k, x_{k+1}, \dots, x_n]$$

Hint: See Problem 6.2.7 (above).

14. Write the equation in Problem 6.2.9 (above) in this form:

$$f[x_0, x_1, \dots, x_n] = \sum_{i=0}^n \alpha_i f(x_i) \quad \text{where} \quad \alpha_i = \prod_{\substack{j=0 \\ j \neq i}}^n (x_i - x_j)^{-1}$$

Prove that if the x_i 's are ordered thus:

$$x_0 < x_1 < x_2 < \cdots < x_n$$

then the α_i 's alternate in sign.

15. (Continuation) Prove that

$$\sum_{i=0}^n \alpha_i x_i^n = 1 \quad \text{and} \quad \sum_{i=0}^n \alpha_i = \begin{cases} 1 & \text{if } n = 0 \\ 0 & \text{if } n > 0 \end{cases}$$

16. Let $f(x) = 1/x$ and prove that

$$f[x_0, x_1, \dots, x_n] = (-1)^n \prod_{i=0}^n x_i^{-1}$$

17. Find the Newton interpolating polynomial for these data:

x	1	3/2	0	2
$f(x)$	3	13/4	3	5/3

18. Prove that if f is a polynomial, then the divided difference $f[x_0, x_1, \dots, x_n]$ is a polynomial in the variables $x_0, x_1, \dots, x_n$.

19. Show that if u is any function that interpolates f at $x_0, x_1, \dots, x_{n-1}$, and if v is a function that interpolates f at $x_1, x_2, \dots, x_n$, then the function

$$[(x_n - x)u(x) + (x - x_0)v(x)] / (x_n - x_0)$$

interpolates f at $x_0, x_1, \dots, x_n$.

20. (Continuation) Consider the array

$$\begin{array}{cc|c} x_0 & y_0 & a_0 \quad b_0 \quad c_0 \\ x_1 & y_1 & a_1 \quad b_1 \\ x_2 & y_2 & a_2 \\ x_3 & y_3 & \end{array}$$

in which, for some fixed x , the a_i, b_i , and c_i are computed by the formulas

$$\begin{aligned} a_i &= [(x_{i+1} - x)y_i + (x - x_i)y_{i+1}] / (x_{i+1} - x_i) \\ b_i &= [(x_{i+2} - x)a_i + (x - x_i)a_{i+1}] / (x_{i+2} - x_i) \\ c_i &= [(x_{i+3} - x)b_i + (x - x_i)b_{i+1}] / (x_{i+3} - x_i) \end{aligned}$$

Show that c_0 is the value of the cubic interpolating polynomial at x .

21. (Continuation) Generalize the algorithm suggested in the preceding problem to compute $p_n(x)$ for any n . This is known as **Neville's algorithm**.

22. Determine the Newton interpolating polynomial for this table:

x	0	1	2	7
y	51	3	1	201

23. The polynomial $p(x) = 2 - (x + 1) + x(x + 1) - 2x(x + 1)(x - 1)$ interpolates the first four points in the table:

x	-1	0	1	2	3
y	2	1	2	-7	10

By adding one additional term to p , find a polynomial that interpolates the whole table.

24. Write the Newton interpolating polynomial for these data:

x	4	2	0	3
$f(x)$	63	11	7	28

25. Establish an iteration method for solving $f(x) = 0$ as follows: Let q_2 be the interpolating quadratic polynomial for the following data and let x_{n+1} be the zero of q_2 closest to x_n :

$f(x_n)$	$f(x_{n-1})$	$f(x_{n-2})$
x_n	x_{n-1}	x_{n-2}

26. Prove that for $h > 0$,

$$f(x + 2h) - 2f(x + h) + f(x) = h^2 f''(\xi)$$

for some ξ in the interval $(x, x + 2h)$.

27. Prove

$$m! f[0, 1, \dots, m] = \sum_{j=0}^m (-1)^{m-j} \binom{m}{j} f(j)$$

Hint: Use induction. In the middle of your proof, you will have to stop and prove the identity that is the basis for Pascal's triangle:

$$\binom{m}{j-1} + \binom{m}{j} = \binom{m+1}{j}$$

28. Suppose that the Hermite-Genocchi formula were adopted as the *definition* of divided differences. On that basis, derive the recursive formula, Equation (11).
29. Extend the result in Problem 6.2.2 (p. 335) by proving that if $f^{(n)}$ is continuous, then $f[x_0, x_1, \dots, x_n]$ is continuous everywhere in $\mathbb{R}^{n+1}$.
30. In Theorem 4, does ξ depend continuously on $x_0, x_1, \dots, x_n$? Answer the same question for $f^{(n)}(\xi)$.

COMPUTER PROBLEMS 6.2

- For $n = 5, 10$, and 15 , find the Newton interpolating polynomial p_n for the function $f(x) = 1/(1 + x^2)$ on the interval $[-5, 5]$. Use equally spaced nodes. In each case, compute $f(x) - p_n(x)$ for 30 equally spaced points in $[-5, 5]$ in order to see the divergence of p_n from f .
- Program the algorithm described in Problem 6.2.25 (above) and test it.

6.3 Hermite Interpolation

The term **Hermite interpolation** refers to the interpolation of a function and some of its derivatives at a set of nodes. (A more precise definition will be given later.)

When a distinction is being made between this type of interpolation and the simpler type (in which no derivatives are interpolated), the latter is often called **Lagrange interpolation**.

Basic Concepts

An instructive and useful example of Hermite interpolation is the following: We require a polynomial of least degree that interpolates a function f and its *derivative* f' at two distinct points, say x_0 and x_1 . The polynomial sought will satisfy these four conditions:

$$p(x_i) = f(x_i) \quad p'(x_i) = f'(x_i) \quad (i = 0, 1)$$

Since there are four conditions, it seems reasonable to look for a solution in Π_3 , the linear space of all polynomials of degree at most 3. An element of Π_3 has four coefficients at our disposal. Rather than writing $p(x)$ in terms of $1, x, x^2, x^3$, however, let us write

$$p(x) = a + b(x - x_0) + c(x - x_0)^2 + d(x - x_0)^2(x - x_1)$$

because this will simplify the work. This leads to

$$p'(x) = b + 2c(x - x_0) + 2d(x - x_0)(x - x_1) + d(x - x_0)^2$$

The four conditions on p can now be written in the form

$$f(x_0) = a$$

$$f'(x_0) = b$$

$$f(x_1) = a + bh + ch^2 \quad (h = x_1 - x_0)$$

$$f'(x_1) = b + 2ch + dh^2$$

Obviously, a and b are obtained at once. Then c can be determined from the third equation, in which the terms involving a and b should be transferred to the left side. Finally, d can be determined from the fourth equation. Hence, the problem is solvable, no matter what the values of $f(x_i)$ and $f'(x_i)$ may be.

In general, if values of a function f and some of its derivatives are to be interpolated by a polynomial, we shall encounter some difficulties because the linear systems of equations (from which we expect to compute the coefficients in the polynomial) may be *singular*. A simple example will illustrate this.

EXAMPLE 1 Find a polynomial p that assumes these values: $p(0) = 0$, $p(1) = 1$, $p'(\frac{1}{2}) = 2$.

Solution Since there are three conditions, we try a quadratic:

$$p(x) = a + bx + cx^2$$

The condition $p(0) = 0$ leads to $a = 0$. The other two conditions lead to

$$1 = p(1) = b + c$$

$$2 = p'(\tfrac{1}{2}) = b + c$$

Thus, no quadratic solves our problem. Notice that the coefficient matrix is singular. If we now try a cubic polynomial for the same problem,

$$p(x) = a + bx + cx^2 + dx^3$$

we discover that a solution exists but is not unique. We notice that $a = 0$, as before. The remaining conditions are

$$1 = b + c + d$$

$$2 = b + c + \tfrac{3}{4}d$$

The solution of this system is $d = -4$ and $b + c = 5$. ■

The *general* problem of this type obviously has some intriguing difficulties associated with it. The topic is known as **Birkhoff interpolation**, and a large corpus of recent research is devoted to it. The reader wishing to explore further is referred to the monograph of Lorentz, Jetter, and Riemenschneider [1983].

A large class of interpolation problems having unique solutions will now be discussed. The problems in this restricted class are the ones generally known as **Hermite interpolation**. In a Hermite problem, it is assumed that whenever a derivative $p^{(j)}(x_i)$ is to be prescribed (at a node x_i), $p^{(j-1)}(x_i)$, $p^{(j-2)}(x_i)$, $\dots$, $p'(x_i)$, and $p(x_i)$ will also be prescribed. We choose our notation so that at node x_i , k_i interpolatory conditions are prescribed. Notice that k_i may vary with i . Let the nodes be $x_0, x_1, \dots, x_n$, and suppose that at node x_i these interpolation conditions are given:

$$p^{(j)}(x_i) = c_{ij} \quad (0 \leq j \leq k_i - 1, 0 \leq i \leq n) \quad (1)$$

The total number of conditions on p is denoted by $m + 1$, and therefore

$$m + 1 = k_0 + k_1 + \dots + k_n \quad (2)$$

THEOREM 1 Theorem on Hermite Interpolation

There exists a unique polynomial p in Π_m fulfilling the Hermite interpolation conditions in Equation (1).

Proof The polynomial p is sought in the space Π_m , and it therefore has $m + 1$ coefficients. The number of interpolatory conditions that are imposed on p by Equation (1) is also $m + 1$. Thus, we have a *square* system of $m + 1$ equations in $m + 1$ unknowns to solve, and we wish to be assured that the coefficient matrix is nonsingular. To prove that a square matrix A is nonsingular, it suffices to prove that the homogeneous equation $Au = 0$ has *only* the 0 solution ($u = 0$). In the interpolation problem under discussion, the **homogeneous problem** is to find $p \in \Pi_m$ such that

$$p^{(j)}(x_i) = 0 \quad (0 \leq j \leq k_i - 1, 0 \leq i \leq n)$$

Such a polynomial has a zero of multiplicity k_i at x_i ($0 \leq i \leq n$) and must therefore be a multiple of the polynomial q given by

$$q(x) = \prod_{i=0}^n (x - x_i)^{k_i}$$

Observe, however, that q is of degree

$$m + 1 = \sum_{i=0}^n k_i$$

whereas p is to be of degree at most m . We therefore conclude that $p = q = 0$. ■

EXAMPLE 2 What happens in Hermite interpolation when there is only one node?

Solution In this case, we require a polynomial p of degree k , say, for which

$$p^{(j)}(x_0) = c_{0j} \quad (0 \leq j \leq k)$$

The solution is the Taylor polynomial

$$p(x) = c_{00} + c_{01}(x - x_0) + \frac{c_{02}}{2!}(x - x_0)^2 + \cdots + \frac{c_{0k}}{k!}(x - x_0)^k \quad \blacksquare$$

Newton Divided Difference Method

Now let us explain how the Newton divided difference method can be extended to solve Hermite interpolation problems. We begin with a simple case in which a quadratic polynomial p is sought, taking prescribed values:

$$p(x_0) = c_{00} \quad p'(x_0) = c_{01} \quad p(x_1) = c_{10} \tag{3}$$

We write the divided difference table in this form:

x_0	c_{00}	c_{01}	?
x_0	c_{00}	?	
x_1	c_{10}		

The question marks stand for entries that are not yet computed. Observe that x_0 appears twice in the argument column since two conditions are being imposed on p at x_0 . Note further that the prescribed value of $p'(x_0)$ has been placed in the column of first-order divided differences. This is in accordance with the equation

$$\lim_{x \rightarrow x_0} f[x_0, x] = \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} = f'(x_0)$$

This equation justifies our defining

$$f[x_0, x_0] \equiv f'(x_0)$$

The remaining entries in the divided difference table can be computed in the usual way. The difficulty to be expected when the nodes are repeated occurs only at the entry c_{01} , and the value of c_{01} has been supplied by the *data* so it does not need to be computed. The entries denoted by question marks are then computed in the usual way:

$$p[x_0, x_1] = \frac{p(x_1) - p(x_0)}{x_1 - x_0} = \frac{c_{10} - c_{00}}{x_1 - x_0} \tag{4}$$

and

$$p[x_0, x_0, x_1] = \frac{p[x_0, x_1] - p[x_0, x_0]}{x_1 - x_0} = \frac{c_{10} - c_{00}}{(x_1 - x_0)^2} - \frac{c_{01}}{x_1 - x_0} \tag{5}$$

The interpolating polynomial is written in the usual way:

$$p(x) = p(x_0) + p[x_0, x_0](x - x_0) + p[x_0, x_0, x_1](x - x_0)^2 \tag{6}$$

A direct verification that this polynomial solves the problem in Equation (3) is suggested in Problem 6.3.5 (p. 348).

Returning to the example used at the beginning of this section, we can obtain the interpolation polynomial

$$\begin{aligned} p(x) = & f(x_0) + f'(x_0)(x - x_0) + f[x_0, x_0, x_1](x - x_0)^2 \\ & + f[x_0, x_0, x_1, x_1](x - x_0)^2(x - x_1) \end{aligned}$$

directly from the following divided difference table:

x_0	$f(x_0)$	$f'(x_0)$	$f[x_0, x_0, x_1]$	$f[x_0, x_0, x_1, x_1]$
x_0	$f(x_0)$	$f[x_0, x_1]$	$f[x_0, x_1, x_1]$	
x_1	$f(x_1)$	$f'(x_1)$		
x_1	$f(x_1)$			

Divided differences in which all arguments are identical are defined in accordance with Theorem 4 in Section 6.2 (p. 333). That theorem asserts the existence of a point ξ such that

$$f[x_0, x_1, \dots, x_k] = \frac{1}{k!} f^{(k)}(\xi)$$

Here it must be assumed that $f^{(k)}$ exists and is continuous in the smallest interval containing the nodes $x_0, x_1, \dots, x_k$. The point ξ will lie in the same interval. If the length of that interval shrinks to 0, we obtain in the limit

$$f[x_0, x_0, \dots, x_0] = \frac{1}{k!} f^{(k)}(x_0) \tag{7}$$

Notice that when $k \geq 2$, we must be careful to include the factor $1/k!$.

EXAMPLE 3 Use the extended Newton divided difference algorithm to determine a polynomial that takes these values:

$$p(1) = 2 \qquad p'(1) = 3 \qquad p(2) = 6 \qquad p'(2) = 7 \qquad p''(2) = 8$$

Solution We put the data in the divided difference array as follows, using question marks to signify that quantities are to be computed. The final result is on the right.

1	2	3	?	?	?	1	2	3	1	2	−1
1	2	?	?	?		1	2	4	3	1	
2	6	7	4			2	6	7	4		
2	6					2	6				
2	6					2	6				

Notice that in the third row, a second difference of 4 is inserted in accordance with Formula (7) in the case $k = 2$. When the array is completed, the numbers in the top row (excluding the node) are the coefficients in the interpolating polynomial:

$$p(x) = 2 + 3(x - 1) + (x - 1)^2 + 2(x - 1)^2(x - 2) - (x - 1)^2(x - 2)^2 \quad \blacksquare$$

Lagrange Form

In principle, formulas akin to the Lagrange interpolation formula can be developed for Hermite interpolation. We shall present one such formula that pertains to an important special case. As before, let the nodes be $x_0, x_1, \dots, x_n$, and let us assume that at each node a function value and the first derivative have been prescribed. The polynomial p that we seek must satisfy these equations:

$$p(x_i) = c_{i0} \qquad p'(x_i) = c_{i1} \qquad (0 \leq i \leq n) \tag{8}$$

In analogy with the Lagrange formula, we write

$$p(x) = \sum_{i=0}^n c_{i0} A_i(x) + \sum_{i=0}^n c_{i1} B_i(x) \tag{9}$$

in which A_i and B_i are to be polynomials with certain properties. A moment's reflection reveals that the following properties would perfectly serve our purpose:

$$\begin{cases} A_i(x_j) = \delta_{ij} \\ A'_i(x_j) = 0 \end{cases} \qquad \begin{cases} B_i(x_j) = 0 \\ B'_i(x_j) = \delta_{ij} \end{cases}$$

Indeed, if A_i and B_i have these properties, then it is elementary to prove that p , as given in Equation (9), will have the interpolation properties in Equation (8). It can now be verified that with the aid of the functions

$$\ell_i(x) = \prod_{\substack{j=0 \\ j \neq i}}^n \frac{x - x_j}{x_i - x_j} \qquad (0 \leq i \leq n)$$

A_i and B_i can be defined as follows:

$$\begin{cases} A_i(x) = [1 - 2(x - x_i)\ell'_i(x_i)]\ell_i^2(x) & (0 \leq i \leq n) \\ B_i(x) = (x - x_i)\ell_i^2(x) & (0 \leq i \leq n) \end{cases}$$

Notice that each ℓ_i is of degree n , and therefore A_i and B_i are of degree $2n + 1$. Hence, p will be of degree at most $2n + 1$. This is as it should be because $2n + 2$ conditions have been placed on p in Equation (8).

The Lagrange form of the interpolation polynomial in Example 1 is given by

$$p(x) = f(x_0)A_0(x) + f(x_1)A_1(x) + f'(x_0)B_0(x) + f'(x_1)B_1(x)$$

where

$$A_0(x) = [1 - 2(x - x_0)\ell'_0(x_0)]\ell_0^2(x)$$

$$A_1(x) = [1 - 2(x - x_1)\ell'_1(x_1)]\ell_1^2(x)$$

$$B_0(x) = (x - x_0)\ell_0^2(x)$$

$$B_1(x) = (x - x_1)\ell_1^2(x)$$

and

$$\ell_0(x) = \frac{x - x_1}{x_0 - x_1}$$

$$\ell_1(x) = \frac{x - x_0}{x_1 - x_0}$$

$$\ell'_0(x) = \frac{1}{x_0 - x_1}$$

$$\ell'_1(x) = \frac{1}{x_1 - x_0}$$

An error formula for this type of Hermite interpolation is given in the next theorem.

THEOREM 2 Theorem on Hermite Interpolation Error Estimate

Let $x_0, x_1, \dots, x_n$ be distinct nodes in $[a, b]$ and let $f \in C^{2n+2}[a, b]$. If p is the polynomial of degree at most $2n + 1$ such that

$$p(x_i) = f(x_i) \quad p'(x_i) = f'(x_i) \quad (0 \leq i \leq n)$$

then to each x in $[a, b]$ there corresponds a point ξ in (a, b) such that

$$f(x) - p(x) = \frac{f^{(2n+2)}(\xi)}{(2n+2)!} \prod_{i=0}^n (x - x_i)^2$$

Proof If x is a node, the formula is obviously correct. Now fix x (not a node). Define w and ϕ by

$$w(t) = \prod_{i=0}^n (t - x_i)^2 \quad \phi = f - p - \lambda w$$

where λ is chosen so that $\phi(x) = 0$. Notice that ϕ has at least $n + 2$ zeros in $[a, b]$; namely, $x, x_0, x_1, \dots, x_n$. By Rolle's Theorem, ϕ' has at least $n + 1$ zeros different from the points just enumerated. In addition, ϕ' vanishes at each node. Thus ϕ' has at least $2n + 2$ zeros in $[a, b]$. By Rolle's Theorem, ϕ'' has at least $2n + 1$ zeros in (a, b) , and by repeating the argument, we conclude that $\phi^{(2n+2)}$ has a zero, say ξ , in (a, b) . Thus,

$$0 = \phi^{(2n+2)}(\xi) = f^{(2n+2)}(\xi) - p^{(2n+2)}(\xi) - \lambda w^{(2n+2)}(\xi)$$

Since p is of degree at most $2n + 1$, we have $p^{(2n+2)} = 0$. Since $w(t)$ has the leading term t^{2n+2} , it follows that $w^{(2n+2)} = (2n + 2)!$. Using this information together with the value $\lambda = [f(x) - p(x)]/w(x)$ in the equation for $\phi^{(2n+2)}$ gives us

$$0 = f^{(2n+2)}(\xi) - [f(x) - p(x)](2n + 2)!/w(x)$$

which is another form of the equation to be proved. ■

Divided Differences with Repetitions

The remainder of this section is devoted to a discussion of divided differences when repetitions are permitted in the arguments. There are many ways to proceed, such as with a recursive definition (Braess [1984]), or by using a definition with determinants (Schumaker [1981]), or by simply extending the definition previously given for $f[x_0, x_1, \dots, x_n]$ in the case of distinct arguments. We follow the last course, as given by Conte and de Boor [1980].

Recall that if the points $x_0, x_1, \dots, x_n$ are distinct, then $f[x_0, x_1, \dots, x_n]$ has been defined in Section 6.2 (p. 329) as the coefficient of x^n in the polynomial from Π_n that interpolates f at $x_0, x_1, \dots, x_n$. Let us now define the meaning of interpolation when the list of points contains repetitions.

DEFINITION 1 **Definition of Interpolation with Repeated Notes**

We say that f interpolates 0 at $x_0, x_1, \dots, x_n$ if $f^{(k-1)}(\xi) = 0$ for each point ξ that occurs k or more times in the list $x_0, x_1, \dots, x_n$.

Thus, for example, we shall say that f interpolates 0 at 1, 3, 8, 1, 13, 1, 8 if

$$f(1) = f(3) = f(8) = f'(1) = f(13) = f''(1) = f'(8) = 0$$

It is easy to prove that a polynomial p interpolates 0 at $x_0, x_1, \dots, x_n$ if and only if $p(x)$ contains the factor

$$q(x) = \prod_{j=0}^n (x - x_j)$$

If f and g are two functions such that $f - g$ interpolates 0 at $x_0, x_1, \dots, x_n$, then we say that f interpolates g (or g interpolates f) at $x_0, x_1, \dots, x_n$.

Using the terminology just explained, we can restate the theorem proved earlier in this section in the following form.

THEOREM 3 Theorem on Uniqueness of Polynomial Interpolation

Let $x_0, x_1, \dots, x_m$ be a list of points in which no element is repeated more than k times. Let f belong to class C^{k-1} on an interval containing these points. Then a unique polynomial in Π_m exists that interpolates f at the given points.

The definition of $f[x_0, x_1, \dots, x_n]$ in the general case is as the coefficient of x^n in the polynomial from Π_n that interpolates f at $x_0, x_1, \dots, x_n$. If a number ξ occurs k times in the list, then this definition requires the existence of the derivative $f^{(k-1)}(\xi)$. In any contrary case, the divided difference does not exist (that is, is not defined). Having made this definition, we can now prove the validity of the **general Newton interpolation formula**

$$p(x) = \sum_{j=0}^n f[x_0, x_1, \dots, x_j] \prod_{i=0}^{j-1} (x - x_i) \quad (10)$$

Recall that $\prod_{i=0}^{-1} (x - x_i) = 1$ by definition.

THEOREM 4 Theorem on General Newton Interpolation Polynomial

If f is sufficiently differentiable so that the divided differences occurring in Equation (10) exist, then that equation gives the polynomial p in Π_n that interpolates f at $x_0, x_1, \dots, x_n$.

Proof We proceed by induction on n . For $n = 0$, it is asserted that $f[x_0]$ is the polynomial in Π_0 that interpolates f at x_0 . This is obviously true.

Suppose now that the polynomial q defined by

$$q(x) = \sum_{j=0}^{n-1} f[x_0, x_1, \dots, x_j] \prod_{i=0}^{j-1} (x - x_i)$$

interpolates f at $x_0, x_1, \dots, x_{n-1}$. Let p be the polynomial in Π_n that interpolates f at $x_0, x_1, \dots, x_n$. The previous theorems guarantee the existence and unicity of p . By the definition of the divided difference, the coefficient of x^n in p is $f[x_0, x_1, \dots, x_n]$. Hence, the polynomial

$$p(x) - f[x_0, x_1, \dots, x_n] \prod_{i=0}^{n-1} (x - x_i)$$

is of degree at most $n - 1$. It interpolates f at $x_0, x_1, \dots, x_{n-1}$, by Problems 6.3.8–9, 11 (p. 348). By the unicity of q , we therefore must have

$$p(x) - f[x_0, x_1, \dots, x_n] \prod_{i=0}^{n-1} (x - x_i) = q(x)$$

Hence, we have

$$\begin{aligned} p(x) &= q(x) + f[x_0, x_1, \dots, x_n] \prod_{i=0}^{n-1} (x - x_i) \\ &= \sum_{j=0}^n f[x_0, x_1, \dots, x_j] \prod_{i=0}^{j-1} (x - x_i) \end{aligned}$$

The final issue to be settled is whether the divided differences in their general form obey the recurrence relation

$$f[x_0, x_1, \dots, x_n] = \{f[x_1, x_2, \dots, x_n] - f[x_0, x_1, \dots, x_{n-1}]\} / (x_n - x_0)$$

Let us assume that $x_0 \leq x_1 \leq \dots \leq x_n$. This involves no loss of generality because the generalized divided differences are symmetric functions of their arguments. (The proof of this fact given in Section 6.2, p. 333, is valid in the general case.) The recurrence relation will obviously fail in the case $x_n = x_0$. But this equality implies that $x_0 = x_1 = \dots = x_n$, and so this is precisely the case in which we shall want to use the formula

$$f[x_0, x_0, \dots, x_0] = \frac{1}{n!} f^{(n)}(x_0)$$

In all other cases the recursive formula is valid. We now prove this formally.

THEOREM 5 Theorem on Divided Difference Recursive Formula

Let $x_0 \leq x_1 \leq \dots \leq x_n$. Then the divided differences obey this recursive formula:

$$f[x_0, x_1, \dots, x_n] = \begin{cases} \frac{f[x_1, x_2, \dots, x_n] - f[x_0, x_1, \dots, x_{n-1}]}{x_n - x_0} & \text{if } x_n \neq x_0 \\ f^{(n)}(x_0)/n! & \text{if } x_n = x_0 \end{cases} \quad (11)$$

Proof We proceed by induction on n . The polynomial in Π_1 that interpolates f at x_0 and x_1 is

$$p(x) = \begin{cases} (x - x_0)[f(x_1) - f(x_0)]/(x_1 - x_0) + f(x_0) & (x_0 \neq x_1) \\ f'(x_0)(x - x_0) + f(x_0) & (x_0 = x_1) \end{cases}$$

In this polynomial, the coefficient of x is

$$f[x_0, x_1] = \begin{cases} \{f[x_1] - f[x_0]\}/(x_1 - x_0) & (x_0 \neq x_1) \\ f'(x_0) & (x_0 = x_1) \end{cases}$$

This is in accordance with Equation (11).

Now assume that Equation (11) is valid for all n from 1 to $m - 1$, inclusive. Let $x_0 \leq x_1 \leq \dots \leq x_m$ be a list of nodes, and consider the polynomial p in Π_m that interpolates f at these points. If $x_m = x_0$, then all the points $x_0, x_1, \dots, x_m$ are the same. In this case, by Problem 6.3.7 (p. 348), p is the Taylor polynomial of f at x_0

given by

$$p(x) = \sum_{k=0}^m \frac{1}{k!} f^{(k)}(x_0)(x - x_0)^k$$

The coefficient of x^m in $p(x)$ is $f^{(m)}(x_0)/m!$, and this establishes Equation (11) when $n = m$ and $x_m = x_0$. In the other case, the argument used to prove Theorem 1 in Section 6.2 (p. 330) is valid. ■

PROBLEMS 6.3

1. Use the extended Newton divided difference method to obtain a quartic polynomial that takes these values:

x	0	1	2
$p(x)$	2	-4	44
$p'(x)$	-9	4	

2. (Continuation) Find a quintic polynomial that takes the values given in the preceding problem and, in addition, satisfies $p(3) = 2$. *Hint:* Add a suitable term to the polynomial found in the preceding problem.
3. Obtain a formula for the polynomial p of least degree that takes these values:

$$p(x_i) = y_i \quad p'(x_i) = 0 \quad (0 \leq i \leq n)$$

4. What condition will have to be placed on the nodes x_0 and x_1 if the interpolation problem

$$p(x_i) = c_{i0} \quad p''(x_i) = c_{i2} \quad (i = 0, 1)$$

is to be solvable by a cubic polynomial (for arbitrary c_{ij})?

5. Show that the polynomial given in Equation (6), with the divided differences in Equations (4) and (5), fulfills the conditions in Equation (3).
6. Verify the properties claimed for the functions A_i and B_i .
7. Prove that the Taylor polynomial

$$p(x) = \sum_{j=0}^{k-1} \frac{1}{j!} f^{(j)}(x_0)(x - x_0)^j$$

interpolates f at $x_0, x_0, \dots, x_0$ (k repetitions).

8. Prove that a polynomial interpolates 0 at $x_0, x_1, \dots, x_n$ (repetitions permitted) if and only if it contains the factor $\prod_{j=0}^n (x - x_j)$.
9. Prove that if f interpolates g at $x_0, x_1, \dots, x_n$ and if h interpolates 0 at these points, then $f \pm ch$ interpolates g at these points.
10. Fix $x_0, x_1, \dots, x_n$ and show that the set of functions that interpolate 0 at these points is an *algebra*; that is, it is closed under addition, multiplication, and multiplication by scalars.
11. Prove that if f interpolates 0 at the nodes $x_0, x_1, \dots, x_n$, then it interpolates 0 at the nodes $x_0, x_1, \dots, x_{n-1}$.
12. Let $x_0 < x_1 < \dots < x_n$, and let f be continuously differentiable. Show that

$$\frac{\partial}{\partial x_i} f[x_0, x_1, \dots, x_n] = f[x_0, x_1, \dots, x_i, x_i, x_{i+1}, \dots, x_n]$$

- 13. Write out in detail and then simplify the functions A_i and B_i occurring in Equation (9), taking $n = 2$.
- 14. Derive a formula for $\ell'_i(x)$.
- 15. (Use Rolle's Theorem.) Let $f \in C^n[\alpha, \beta]$. Suppose that f has a zero of multiplicity m at α and a root of multiplicity k at β , where $m \geq 1, k \geq 1$, and $m + k - 1 = n$. Prove that $f^{(n)}$ has at least one zero in (α, β) .
- 16. Consider the polynomial

$$p(t) = b - (b - a) \left[3 \left(\frac{b - t}{b - a} \right)^2 - 2 \left(\frac{b - t}{b - a} \right)^3 \right]$$

Show that $|p'(t)| \leq p'((a + b)/2) = 3/2, p(a) = a, p(b) = b, p'(a) = 0$, and $p'(b) = 0$.

6.4 Spline Interpolation

A **spline function** consists of polynomial pieces on subintervals joined together with certain continuity conditions. Formally, suppose that $n + 1$ points $t_0, t_1, \dots, t_n$ have been specified and satisfy $t_0 < t_1 < \dots < t_n$. These points are called **knots**. Suppose also that an integer $k \geq 0$ has been prescribed. A **spline function of degree k** having knots $t_0, t_1, \dots, t_n$ is a function S such that:

- 1. On each interval $[t_{i-1}, t_i)$, S is a polynomial of degree $\leq k$.
- 2. S has a continuous $(k - 1)$ st derivative on $[t_0, t_n]$.

Hence, S is a piecewise polynomial of degree at most k having continuous derivatives of all orders up to $k - 1$.

Splines of degree 0 are piecewise constants. A spline of degree 0 can be given explicitly in the form

$$S(x) = \begin{cases} S_0(x) = c_0 & x \in [t_0, t_1) \\ S_1(x) = c_1 & x \in [t_1, t_2) \\ \vdots & \vdots \\ S_{n-1}(x) = c_{n-1} & x \in [t_{n-1}, t_n] \end{cases}$$

The intervals $[t_{i-1}, t_i)$ do not intersect each other, and so no ambiguity arises in defining such a function at the knots. A typical spline of degree 0 with six knots is shown in Figure 6.3.

Figure 6.4 shows the graph of a typical spline function of degree 1 with nine knots. A function such as this can be defined explicitly by

$$S(x) = \begin{cases} S_0(x) = a_0x + b_0 & x \in [t_0, t_1) \\ S_1(x) = a_1x + b_1 & x \in [t_1, t_2) \\ \vdots & \vdots \\ S_{n-1}(x) = a_{n-1}x + b_{n-1} & x \in [t_{n-1}, t_n] \end{cases}$$

If the knots t_i and the coefficients a_i, b_i are all prescribed, then the value of S at x is obtained by first identifying the subinterval $[t_i, t_{i+1})$ that contains x . The spline

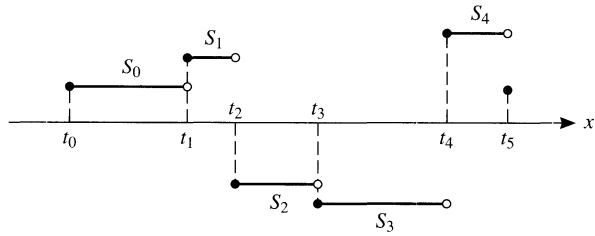


FIGURE 6.3
A spline of degree 0

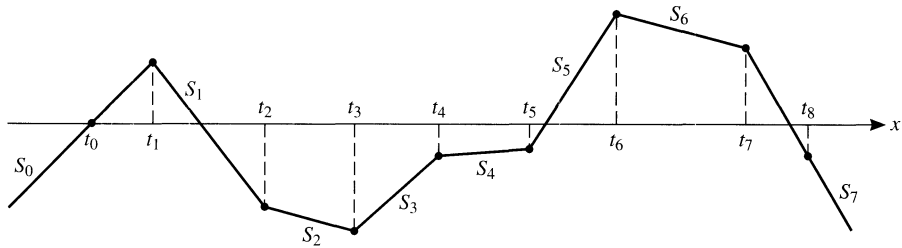


FIGURE 6.4
A spline of degree 1

function can be defined on the entire real line. For convenience, we can use the expression $a_0x + b_0$ on the interval $(-\infty, t_1)$ and the expression $a_{n-1}x + b_{n-1}$ on the interval $[t_{n-1}, \infty)$. The function S is continuous, and so the piecewise polynomials match up at the nodes; that is, $S_i(t_{i+1}) = S_{i+1}(t_{i+1})$. A pseudocode to evaluate the first-degree spline $S(x)$ follows:

```

input  $(t_i), (a_i), (b_i), x, n$ 
for  $i = 1$  to  $n - 1$  do
    if  $x < t_i$  then
         $S(x) = a_{i-1}x + b_{i-1}$ 
        output  $S(x)$ 
        exit loop
    end if
end do
 $S(x) = a_{n-1}x + b_{n-1}$ 
output  $S(x)$ 

```

Cubic Splines

We shall develop more thoroughly the theory and construction of the **cubic splines** ($k = 3$) since these are often used in practice. We assume that a table of values has been given:

x	t_0	t_1	$\cdots$	t_n
y	y_0	y_1	$\cdots$	y_n

(1)

and that a cubic spline S is to be constructed to interpolate the table. On each interval $[t_0, t_1], [t_1, t_2], \dots, [t_{n-1}, t_n]$, S is given by a different cubic polynomial. Let S_i be

the cubic polynomial that represents S on $[t_i, t_{i+1}]$. Thus,

$$S(x) = \begin{cases} S_0(x) & x \in [t_0, t_1] \\ S_1(x) & x \in [t_1, t_2] \\ \vdots & \vdots \\ S_{n-1}(x) & x \in [t_{n-1}, t_n] \end{cases} \quad (2)$$

The polynomials S_{i-1} and S_i interpolate the same value at the point t_i and therefore

$$S_{i-1}(t_i) = y_i = S_i(t_i) \quad (1 \leq i \leq n-1) \quad (3)$$

Hence, S is automatically continuous. Moreover, S' and S'' are assumed to be continuous, and these conditions will be used in the derivation of the cubic spline function.

Does the continuity of S , S' , and S'' provide enough conditions to define a cubic spline? There are $4n$ coefficients in the piecewise cubic polynomial since there are four coefficients in each of the n cubic polynomials. On each subinterval $[t_i, t_{i+1}]$, there are two interpolation conditions, $S(t_i) = y_i$ and $S(t_{i+1}) = y_{i+1}$, giving $2n$ conditions. The continuity of S gives no additional conditions since it has already been counted in the interpolation conditions. The continuity of S' gives one condition at each interior knot, $S'_{i-1}(t_i) = S'_i(t_i)$, accounting for $n-1$ additional conditions. Similarly, the continuity of S'' gives another $n-1$ conditions. Thus, there are altogether $4n-2$ conditions for determining $4n$ coefficients. Two *degrees of freedom* remain, and there are various ways of using them to advantage.

Now we derive the equation for $S_i(x)$ on the interval $[t_i, t_{i+1}]$. First we define the numbers $z_i = S''(t_i)$. Clearly, z_i exist for $0 \leq i \leq n$ and satisfy

$$\lim_{x \downarrow t_i} S''(x) = z_i = \lim_{x \uparrow t_i} S''(x) \quad (1 \leq i \leq n-1) \quad (4)$$

because S'' is continuous at each interior knot. Since S_i is a cubic polynomial on $[t_i, t_{i+1}]$, S'_i is a linear function satisfying $S'_i(t_i) = z_i$ and $S'_i(t_{i+1}) = z_{i+1}$ and therefore is given by the straight line between z_i and z_{i+1} :

$$S'_i(x) = \frac{z_i}{h_i}(t_{i+1} - x) + \frac{z_{i+1}}{h_i}(x - t_i) \quad (5)$$

where $h_i \equiv t_{i+1} - t_i$. If this is integrated twice, the result is S_i itself:

$$S_i(x) = \frac{z_i}{6h_i}(t_{i+1} - x)^3 + \frac{z_{i+1}}{6h_i}(x - t_i)^3 + C(x - t_i) + D(t_{i+1} - x) \quad (6)$$

where C and D are constants of integration. (To verify, differentiate (6) twice to obtain (5).) The interpolation conditions $S_i(t_i) = y_i$ and $S_i(t_{i+1}) = y_{i+1}$ can now be imposed on S_i to determine C and D . The result is

$$\begin{aligned} S_i(x) = & \frac{z_i}{6h_i}(t_{i+1} - x)^3 + \frac{z_{i+1}}{6h_i}(x - t_i)^3 \\ & + \left(\frac{y_{i+1}}{h_i} - \frac{z_{i+1}h_i}{6} \right)(x - t_i) + \left(\frac{y_i}{h_i} - \frac{z_ih_i}{6} \right)(t_{i+1} - x) \end{aligned} \quad (7)$$

Equation (7) is also easily verified. Simply let $x = t_i$ and $x = t_{i+1}$ to see that the interpolation conditions are fulfilled. Once the values of $z_0, z_1, \dots, z_n$ have been determined, Equations (2) and (7) can be used to evaluate $S(x)$ for any x in the interval $[t_0, t_n]$.

To determine $z_1, z_2, \dots, z_{n-1}$, we use the continuity conditions for S' . At the interior knots t_i , we must have $S'_{i-1}(t_i) = S'_i(t_i)$. Equation (7) gives us $S'_i(x)$ by differentiating. Then substitution of $x = t_i$ and simplification lead to

$$S'_i(t_i) = -\frac{h_i}{3}z_i - \frac{h_i}{6}z_{i+1} - \frac{y_i}{h_i} + \frac{y_{i+1}}{h_i} \tag{8}$$

Analogously, using Equation (7) to obtain S_{i-1} , we have

$$S'_{i-1}(t_i) = \frac{h_{i-1}}{6}z_{i-1} + \frac{h_{i-1}}{3}z_i - \frac{y_{i-1}}{h_{i-1}} + \frac{y_i}{h_{i-1}} \tag{9}$$

When the right-hand sides of Equations (8) and (9) are set equal to each other, the result can be written as

$$h_{i-1}z_{i-1} + 2(h_i + h_{i-1})z_i + h_iz_{i+1} = \frac{6}{h_i}(y_{i+1} - y_i) - \frac{6}{h_{i-1}}(y_i - y_{i-1}) \tag{10}$$

This equation is used only for $i = 1, 2, \dots, n - 1$. (Why?) It then gives a system of $n - 1$ linear equations for the $n + 1$ unknowns $z_0, z_1, \dots, z_n$. We can select z_0 and z_n arbitrarily and solve the resulting system of equations to obtain $z_1, z_2, \dots, z_{n-1}$. One excellent choice is $z_0 = z_n = 0$. The resulting spline function is called a **natural cubic spline**. (This choice of spline is supported by a theorem to be proved later.)

The linear system of equations (10) for $1 \leq i \leq n - 1$ with $z_0 = 0$ and $z_n = 0$ is symmetric, tridiagonal, diagonally dominant, and of the form

$$\begin{bmatrix} u_1 & h_1 & & & \\ h_1 & u_2 & h_2 & & \\ & h_2 & u_3 & h_3 & \\ & & \ddots & \ddots & \ddots \\ & & & h_{n-3} & u_{n-2} & h_{n-2} \\ & & & & h_{n-2} & u_{n-1} \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \\ \vdots \\ z_{n-2} \\ z_{n-1} \end{bmatrix} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ \vdots \\ v_{n-2} \\ v_{n-1} \end{bmatrix}$$

where

$$\begin{aligned} h_i &= t_{i+1} - t_i \\ u_i &= 2(h_i + h_{i-1}) \\ b_i &= \frac{6}{h_i}(y_{i+1} - y_i) \\ v_i &= b_i - b_{i-1} \end{aligned}$$

It can be solved by a special algorithm (Gaussian elimination without scaled row pivoting) as follows:

```

input  $n, (t_i), (y_i)$ 
for  $i = 0$  to  $n - 1$  do
     $h_i \leftarrow t_{i+1} - t_i$ 
     $b_i \leftarrow 6(y_{i+1} - y_i)/h_i$ 
end do
 $u_1 \leftarrow 2(h_0 + h_1)$ 
 $v_1 \leftarrow b_1 - b_0$ 
for  $i = 2$  to  $n - 1$  do
     $u_i \leftarrow 2(h_i + h_{i-1}) - h_{i-1}^2/u_{i-1}$ 
     $v_i \leftarrow b_i - b_{i-1} - h_{i-1}v_{i-1}/u_{i-1}$ 
end do
 $z_n \leftarrow 0$ 
for  $i = n - 1$  to  $1$  step  $-1$  do
     $z_i \leftarrow (v_i - h_i z_{i+1})/u_i$ 
end do
 $z_0 \leftarrow 0$ 
output  $(z_i)$ 

```

A subprogram or procedure can be easily written based on the preceding algorithm. It would accept as input an array of nodes (t_i) and an array of corresponding function values (y_i) . The routine would compute and return an array of the (z_i) values.

Since there are divisions by u_i in this algorithm, we need to prove that $u_i \neq 0$. We show by induction that $u_i > h_i > 0$. For $i = 1$, this is clear since $u_1 = 2(h_0 + h_1)$. If $u_{i-1} > h_{i-1}$, then $u_i > h_i$ since

$$u_i = 2(h_i + h_{i-1}) - \frac{h_{i-1}^2}{u_{i-1}} > 2(h_i + h_{i-1}) - h_{i-1} > h_i = t_{i+1} - t_i > 0$$

After the coefficients $z_0, z_1, \dots, z_n$ have been determined, any value of the cubic spline function (2) can be computed from Equation (7). Given any x , it is necessary first to determine which of the intervals

$$(-\infty, t_1), \quad [t_1, t_2), \quad \dots, \quad [t_{n-2}, t_{n-1}), \quad [t_{n-1}, \infty)$$

contains x . By convention, we use S_0 not only on $[t_0, t_1]$ but also on $(-\infty, t_0)$. Likewise, we use S_{n-1} on $[t_{n-1}, t_n]$ and on (t_n, ∞) . The interval containing x is determined by testing, in order,

$$x - t_{n-1}, \quad x - t_{n-2}, \quad \dots, \quad x - t_1$$

to see whether any are nonnegative. If one of these is nonnegative, we take the first such occurrence, say $x - t_i$. Then $x - t_i \geq 0$ but $x - t_{i+1} < 0$, so $t_i \leq x < t_{i+1}$. If all the tested terms are negative, then we have $x \in (-\infty, t_1]$. From the index i so determined, the desired polynomial S_i could be evaluated at the given x value using Equation (7). However, we can rewrite (7) in a more efficient, *nested* form by using Problem 6.4.4 (p. 361):

$$S_i(x) = y_i + (x - t_i) \left[C_i + (x - t_i) \left[B_i + (x - t_i) A_i \right] \right] \tag{11}$$

where

$$\begin{aligned} A_i &= \frac{1}{6h_i} (z_{i+1} - z_i) \\ B_i &= \frac{z_i}{2} \\ C_i &= -\frac{h_i}{6} z_{i+1} - \frac{h_i}{3} z_i + \frac{1}{h_i} (y_{i+1} - y_i) \end{aligned}$$

With these explanations in mind, the reader should have no trouble writing a sub-program or procedure that carries out the calculation in Equation (11). It would take as input the number n , the array of nodes (t_i) , the array of function values (y_i) , and the (z_i) that came from the previous routine. It also would take a single real number x as input and return the value of $S(x)$.

Here are some of the results from a simple computer test for two routines based on the algorithms just given. We sample the function $f(x) = \sqrt{x}$ at 10 knots equally spaced in the interval $[0, 2.25]$, interpolate with a cubic spline function S , and print the differences $E(x) \equiv S(x) - f(x)$ at 37 points:

x	$ E(x) $
0.0000	0.0
0.0625	1.07321×10^{-1}
0.1250	7.52666×10^{-2}
0.1875	3.32617×10^{-2}
0.2500	0.0
$\vdots$	$\vdots$
1.7500	0.0
1.8125	3.64780×10^{-5}
1.8750	6.36578×10^{-5}
1.9375	5.85318×10^{-5}
2.0000	0.0
2.0625	1.14083×10^{-4}
2.1250	2.12312×10^{-4}
2.1875	2.04682×10^{-4}
2.2500	0.0

The output shows that the error, E , is 0 at each knot. For points other than knots, $|E|$ is no greater than 0.11. The greatest values of $|E|$ occur in $[t_0, t_1]$. (Why?)

We now present a theorem to the effect that the natural cubic spline produces the *smoothest possible* interpolating function. The word **smooth** is given a technical meaning in the theorem.

THEOREM 1 Theorem on Optimality of Natural Cubic Splines

Let f'' be continuous in $[a, b]$ and let $a = t_0 < t_1 < \cdots < t_n = b$. If S is the natural cubic spline interpolating f at the knots t_i for $0 \leq i \leq n$, then

$$\int_a^b [S''(x)]^2 dx \leq \int_a^b [f''(x)]^2 dx$$

Proof Let $g \equiv f - S$. Then $g(t_i) = 0$ for $0 \leq i \leq n$ and

$$\int_a^b (f'')^2 dx = \int_a^b (S'')^2 dx + \int_a^b (g'')^2 dx + 2 \int_a^b S'' g'' dx$$

The proof will be complete if we can show that

$$\int_a^b S'' g'' dx \geq 0$$

We establish this in the following analysis, using integration by parts, the condition $S''(t_0) = S''(t_n) = 0$, and the fact that S''' is constant (say c_i) on $[t_{i-1}, t_i]$.

$$\begin{aligned} \int_a^b S'' g'' dx &= \sum_{i=1}^n \int_{t_{i-1}}^{t_i} S'' g'' dx \\ &= \sum_{i=1}^n \left\{ (S'' g')(t_i) - (S'' g')(t_{i-1}) - \int_{t_{i-1}}^{t_i} S''' g' dx \right\} \\ &= - \sum_{i=1}^n \int_{t_{i-1}}^{t_i} S''' g' dx = - \sum_{i=1}^n c_i \int_{t_{i-1}}^{t_i} g' dx \\ &= - \sum_{i=1}^n c_i [g(t_i) - g(t_{i-1})] = 0 \end{aligned}$$

Recall that the **curvature** of a curve described by the equation $y = f(x)$ is the quantity

$$|f''(x)| \left[1 + \{f'(x)\}^2 \right]^{-3/2}$$

If the nonlinear bracketed term is dropped, $|f''(x)|$ is left as an approximation to the curvature. In natural cubic spline interpolation, we are finding a curve with minimal (approximate) curvature over an interval because the quantity $\int_a^b [f''(x)]^2 dx$ is being minimized.

At one step in the above proof, we note that there is a “collapsing” sum:

$$\sum_{i=1}^n [(S'' g')(t_i) - (S'' g')(t_{i-1})] = (S'' g')(b) - (S'' g')(a)$$

Our proof will still be valid if this last expression is nonnegative. Putting in $g = f - S$, we arrive at the condition

$$S''(b)[f'(b) - S'(b)] \geq S''(a)[f'(a) - S'(a)]$$

For example, if instead of assuming that S is the *natural* cubic spline interpolant we assume the end conditions $S'(a) = f'(a)$ and $S'(b) = f'(b)$, then the conclusion is true.

Tension Splines

In some data-fitting problems, it is useful to have a parameter τ available called the **tension**. When τ is given a large value, the curve passing through the data points will have high tension. This can be interpreted as a force that stretches the curve tightly among the data points, as shown in Figures 6.5 (a) and (b).

When τ is given a low value, the curve will more nearly assume the shape of the interpolating cubic spline. When $\tau \rightarrow +\infty$, the curve approaches the piecewise linear function; that is, a spline of degree 1.

One mathematical model of a curve such as that just described is as follows. As before, we have knots

$$t_0 < t_1 < \cdots < t_n$$

and data y_i given at each t_i . The **tension spline** that we seek is a function f having these properties

- 1. The function $f \in C^2[t_0, t_n]$.
- 2. The interpolation conditions $f(t_i) = y_i$ hold for $0 \leq i \leq n$.
- 3. On each open interval (t_{i-1}, t_i) , f satisfies $f^{(4)} - \tau^2 f'' = 0$.

FIGURE 6.5(a)
 Spline curve with high
 tension ($\tau = 10$)

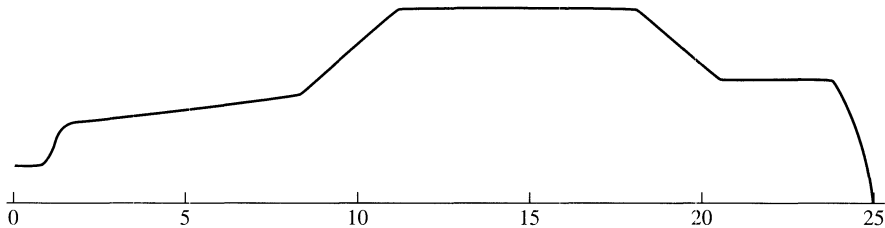
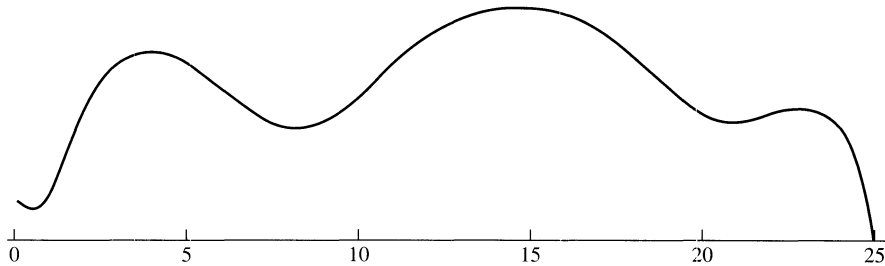


FIGURE 6.5(b)
 Spline curve with low
 tension ($\tau = 0.1$)



Thus, f has globally two continuous derivatives, interpolates the given data, and satisfies a certain differential equation in each subinterval. It is clear that this prescription yields a cubic spline when $\tau = 0$ because the solutions of the equation $f^{(4)} = 0$ are cubic polynomials.

To determine f , we proceed as in the case of the cubic splines. Hence, we set $z_i \equiv f''(t_i)$ and write down the conditions f must satisfy on the interval $[t_i, t_{i+1}]$:

$$\begin{aligned} f^{(4)} - \tau^2 f'' &= 0 \\ f(t_i) &= y_i & f(t_{i+1}) &= y_{i+1} \\ f''(t_i) &= z_i & f''(t_{i+1}) &= z_{i+1} \end{aligned}$$

One can verify that the solution of this **two-point boundary-value problem** is

$$\begin{aligned} f(x) &= \{z_i \sinh[\tau(t_{i+1} - x)] + z_{i+1} \sinh[\tau(x - t_i)]\} / [\tau^2 \sinh(\tau h_i)] \\ &\quad + (y_i - z_i/\tau^2)(t_{i+1} - x)/h_i + (y_{i+1} - z_{i+1}/\tau^2)(x - t_i)/h_i \end{aligned} \quad (12)$$

After the coefficients z_i have been determined, this equation will be used to compute values of f on the interval $[t_i, t_{i+1}]$. All of this is analogous to the case of cubic splines.

For f to have C^2 global smoothness, the conditions

$$\lim_{x \downarrow x_i} f'(x) = \lim_{x \uparrow x_i} f'(x) \quad (1 \leq i \leq n-1)$$

must be imposed at the interior knots. The tedious calculations involved in this are not given here. They proceed as in the case of the cubic splines. The result is a tridiagonal system of equations for the unknowns $z_0, z_1, \dots, z_n$ that can be written in the form

$$\alpha_{i-1} z_{i-1} + (\beta_{i-1} + \beta_i) z_i + \alpha_i z_{i+1} = \gamma_i - \gamma_{i-1} \quad (1 \leq i \leq n-1) \quad (13)$$

with these abbreviations:

$$\begin{aligned} \alpha_i &= 1/h_i - \tau/\sinh(\tau h_i) \\ \beta_i &= \tau \cosh(\tau h_i)/\sinh(\tau h_i) - 1/h_i \\ \gamma_i &= \tau^2(y_{i+1} - y_i)/h_i \end{aligned}$$

We will observe that two additional conditions are needed to determine the z -vector. As in the cubic spline case, one possibility is to specify $z_0 = z_n = 0$.

The steps in obtaining a tension spline f to fit data (t_i, y_i) are as follows:

1. Test to be sure that $t_0 < t_1 < \dots < t_n$.
2. Compute $h_i, \alpha_i, \beta_i, \gamma_i$ for $0 \leq i \leq n-1$.
3. Set $z_1 = z_n = 0$.
4. Solve the tridiagonal system (13) for z_i for $2 \leq i \leq n-1$.
5. Use Formula (12) to compute values of f on the interval $[t_i, t_{i+1}]$.

The unsophisticated approach we have outlined here can be used for interesting experimentation. For example, it led to the two curves shown in Figures 6.5 (a) and (b). In the first we let $\tau = 10$, and in the second $\tau = 0.1$. The data were the same and consisted of prescribed function values at the integer points $t_i = i$ for $0 \leq i \leq 25$.

Tension splines were introduced by Schweikert [1966]. Related papers are by Cline [1974a, b] and Pruess [1976, 1978]. Software for computing curves and surfaces with the aid of tension splines has been developed by Cline. An alternative to the *spline in tension* is the **taut spline** of de Boor [1984]. These functions are ordinary cubic splines with additional knots (and data) placed in regions where we desire the curve to change direction abruptly. The advantage of the taut splines is that no new computer programs are needed, and the computational burden involved in using hyperbolic functions is avoided.

Theory of Higher-Degree Natural Splines

In the final part of this section, we present some of the theory of higher-degree natural splines. Natural splines exist for odd degrees only, and the degree will be denoted here by $2m + 1$. When $m = 1$, one has the natural cubic splines, as developed previously. For the general theory, it is convenient to proceed in a slightly different manner. As before, a set of knots is given:

$$t_0 < t_1 < \cdots < t_n$$

A natural spline of degree $2m + 1$ is a function $S \in C^{2m}(\mathbb{R})$ that reduces to a polynomial of degree $\leq 2m + 1$ in each interval $[t_0, t_1], [t_1, t_2], \dots, [t_{n-1}, t_n]$ and to a polynomial of degree at most m in $(-\infty, t_0)$ and (t_n, ∞) . A natural cubic spline, under this definition, must reduce to linear polynomials in $(-\infty, t_0)$ and (t_n, ∞) .

The linear space of all natural splines of degree $2m + 1$ with $n + 1$ knots $t_0, t_1, \dots, t_n$ is denoted by $\mathcal{N}_n^{2m+1}(t_0, t_1, \dots, t_n)$ or, for brevity, $\mathcal{N}_n^{2m+1}$.

It is convenient to use the so-called **truncated power function**, denoted by x_+^n , which is defined to be x^n if $x \geq 0$, and defined to be 0 if $x < 0$. This function belongs to the continuity class C^{n-1} .

THEOREM 2 Theorem on Truncated Power Functions

Every member of $\mathcal{N}_n^{2m+1}$ has a representation

$$S(x) = \sum_{i=0}^m a_i x^i + \sum_{i=0}^n b_i (x - t_i)_+^{2m+1} \tag{14}$$

in which $\sum_{i=0}^n b_i t_i^j = 0$ for $0 \leq j \leq m$.

Proof In the interval $(-\infty, t_0)$, S is a polynomial p_0 of degree at most m . This polynomial determines the coefficients a_i . In the interval (t_0, t_1) , S reduces to a polynomial p_1 of degree $2m + 1$. The continuity conditions at t_0 are

$$p_0^{(j)}(t_0) = p_1^{(j)}(t_0) \quad (0 \leq j \leq 2m)$$

By Taylor's Theorem, p_1 can be written in the form

$$\begin{aligned} p_1(x) &= \sum_{j=0}^{2m+1} \frac{1}{j!} p_1^{(j)}(t_0)(x - t_0)^j \\ &= \sum_{j=0}^{2m} \frac{1}{j!} p_0^{(j)}(t_0)(x - t_0)^j + b_0(x - t_0)^{2m+1} \\ &= p_0(x) + b_0(x - t_0)^{2m+1} \end{aligned}$$

This equation shows that on $(-\infty, t_1)$, we have

$$S(x) = p_0(x) + b_0(x - t_0)_+^{2m+1}$$

(We note in passing that $S^{(j)}(t_0) = 0$ for $m < j \leq 2m$.) The argument used at t_0 is now repeated at each node to obtain the remaining terms $b_i(x - t_i)_+^{2m+1}$. On the interval (t_n, ∞) , S must reduce to a polynomial of degree $\leq m$. Hence, in this interval,

$$0 = S^{(m+1)}(x) = \sum_{i=0}^n b_i(2m+1)(2m)\cdots(m+1)(x - t_i)^m$$

From this equation, with the help of the Binomial Theorem, we obtain

$$0 = \sum_{i=0}^n b_i(x - t_i)^m = \sum_{i=0}^n b_i \sum_{j=0}^m \binom{m}{j} x^{m-j} (-t_i)^j$$

Here we have a polynomial of degree at most m in x . Its coefficients must vanish, and so $\sum_{i=0}^n b_i t_i^j = 0$ for $j = 0, 1, \dots, m$. ■

THEOREM 3 Theorem on Uniqueness of Natural Splines of Odd Degree

Let the knots $t_0 < t_1 < \cdots < t_n$ be given, and let $0 \leq m \leq n$. There is a unique natural spline of degree $2m + 1$ taking prescribed values at the knots.

Proof By Theorem 2, a natural spline has the form

$$S(x) = \sum_{j=0}^m a_j x^j + \sum_{j=0}^n b_j (x - t_j)_+^{2m+1}$$

If the values prescribed for S are λ_i , then (in accordance with Theorem 2) the interpolation problem requires us to solve the system

$$\begin{cases} S(t_i) \equiv \sum_{j=0}^m a_j t_i^j + \sum_{j=0}^n b_j (t_i - t_j)_+^{2m+1} = \lambda_i & (0 \leq i \leq n) \\ \sum_{j=0}^m b_j t_j^i = 0 & (0 \leq i \leq m) \end{cases}$$

This is a square system of $m + n + 2$ equations in $m + n + 2$ unknowns. To show that it is nonsingular, it suffices to prove that the corresponding homogeneous prob-

lem has only the 0-solution. Suppose therefore that $S(t_i) = 0$ for $0 \leq i \leq n$. We shall prove that

$$I \equiv \int_a^b [S^{(m+1)}(x)]^2 dx = 0 \quad (15)$$

where $a = t_0$ and $b = t_n$. Integration by parts yields

$$\begin{aligned} I &= S^{(m+1)}(x)S^{(m)}(x) \Big|_a^b - \int_a^b S^{(m)}(x)S^{(m+2)}(x)dx \\ &= - \int_a^b S^{(m)}(x)S^{(m+2)}(x)dx \end{aligned}$$

Here we exploited the fact that S is a polynomial of degree at most m in $(-\infty, a)$, and hence $S^{(m+1)}(a) = 0$. Likewise, $S^{(m+1)}(b) = 0$. This argument is repeated until we arrive at

$$I = (-1)^m \int_a^b S^{(1)}(x)S^{(2m+1)}(x)dx$$

Now $S^{(2m+1)}$ is piecewise constant, and so

$$I = (-1)^m \sum_{i=1}^n \int_{t_{i-1}}^{t_i} c_i S'(x)dx = (-1)^m \sum_{i=1}^n c_i [S(t_i) - S(t_{i-1})] = 0$$

This proves Equation (15). From it we infer that $S^{(m+1)} \equiv 0$. Hence, S is a polynomial of degree at most m . Since S has zeros at $t_0, t_1, \dots, t_n$ and $n + 1 > m$, we see that $S = 0$. ■

The analogue of the Theorem on Smoothness of Natural Cubic Splines follows.

THEOREM 4 Theorem on Optimality of Natural Splines of Odd Degree

Let $m \leq n$ and $f \in C^{m+1}[a, b]$. Let S be the natural spline of degree $2m + 1$ that interpolates f at knots $a = t_0 < t_1 < \dots < t_n = b$. Then

$$\int_a^b [S^{(m+1)}(x)]^2 dx \leq \int_a^b [f^{(m+1)}(x)]^2 dx$$

Proof Proceed as in Theorem 1. Let $g = f - S$. Then $g(t_i) = 0$ for $0 \leq i \leq n$. By repeated integration by parts, we can prove that

$$\int_a^b g^{(m+1)}(x)S^{(m+1)}(x)dx = 0$$

Then the proof is completed by writing

$$\begin{aligned} \int_a^b [f^{(m+1)}(x)]^2 dx &= \int_a^b [S^{(m+1)}(x) + g^{(m+1)}(x)]^2 dx \\ &= \int_a^b [S^{(m+1)}(x)]^2 dx + 2 \int_a^b S^{(m+1)}(x)g^{(m+1)}(x)dx + \int_a^b [g^{(m+1)}(x)]^2 dx \\ &= \int_a^b [S^{(m+1)}(x)]^2 dx + \int_a^b [g^{(m+1)}(x)]^2 dx \\ &\geq \int_a^b [S^{(m+1)}(x)]^2 dx \end{aligned}$$

PROBLEMS 6.4

- 1. Refer to the algorithm for determining the values of z_i , and prove that for all $i = 1, 2, \dots, n - 1$, we have $u_i z_i + h_i z_{i+1} - v_i = 0$.
- 2. (Continuation) Denote the left side of the equation in the preceding problem by E_i . Then $(h_i/u_i)E_i + E_{i+1} = 0$. Show that the latter equation can be reduced to Equation (10) by using the formulas in the algorithms. This will establish that the algorithm produces a solution to Equation (10).
- 3. Show that if we replace t_{i+1} by $t_i + h_i$ in Equation (6), the result is

$$S_i(x) = \frac{z_{i+1}}{6h_i}(x - t_i)^3 - \frac{z_i}{6h_i}(x - t_i - h_i)^3 + C(x - t_i) - D(x - t_i - h_i)$$

- 4. (Continuation) By expanding the term $(x - t_i - h_i)^3$ in the preceding equation and by using the correct values of C and D , show that Equation (11) in the text is correct.
- 5. Determine whether this is a quadratic spline function:

$$f(x) = \begin{cases} x & x \in (-\infty, 1] \\ -\frac{1}{2}(2 - x)^2 + \frac{3}{2} & x \in [1, 2] \\ \frac{3}{2} & x \in [2, \infty) \end{cases}$$

- 6. (Continuation) Is the function in the preceding problem a cubic spline function?
- 7. Determine all the values of a, b, c, d, e for which the following function is a cubic spline:

$$f(x) = \begin{cases} a(x - 2)^2 + b(x - 1)^3 & x \in (-\infty, 1] \\ c(x - 2)^2 & x \in [1, 3] \\ d(x - 2)^2 + e(x - 3)^3 & x \in [3, \infty) \end{cases}$$

Next, determine the values of the parameters so that the cubic spline interpolates this table:

x	0	1	4
y	26	7	25

- 8. Verify that Equations (7), (9), and (10) in the text are correct.
- 9. Using the development of the cubic splines as a model, derive the appropriate equations and algorithm to provide a quadratic spline interpolant to data (t_i, y_i) for $0 \leq i \leq n$, where $t_0 < t_1 < \dots < t_n$. If Q is the spline interpolant, then the numbers $z_i = Q'(t_i)$ are well defined. Find the equations governing $z_0, z_1, \dots, z_n$. You should discover that one of the z points can be arbitrary, say $z_0 = 0$.
- 10. Show that in the algorithm for solving the system (10), $u_i > h_i + h_{i-1}$ for all $i = 1, 2, \dots, n - 1$.

11. Determine the values of a , b , and c so that this is a cubic spline having knots 0, 1, and 2:

$$f(x) = \begin{cases} 3 + x - 9x^2 & x \in [0, 1] \\ a + b(x - 1) + c(x - 1)^2 + d(x - 1)^3 & x \in [1, 2] \end{cases}$$

Next, determine d so that $\int_0^2 [f''(x)]^2 dx$ is a minimum. Finally, find the value of d that makes $f''(2) = 0$ and explain why this value is different from the one previously determined.

12. Determine whether this is a cubic spline:

$$f(x) = \begin{cases} x^3 + x & x \leq 0 \\ x^3 - x & x \geq 0 \end{cases}$$

Show that

$$\lim_{x \uparrow 0} f''(x) = \lim_{x \downarrow 0} f''(x)$$

13. Determine whether the natural cubic spline that interpolates the table

x	0	1	2	3
y	1	1	0	10

is or is not the function

$$f(x) = \begin{cases} 1 + x - x^3 & x \in [0, 1] \\ 1 - 2(x - 1) - 3(x - 1)^2 + 4(x - 1)^3 & x \in [1, 2] \\ 4(x - 2) + 9(x - 2)^2 - 3(x - 2)^3 & x \in [2, 3] \end{cases}$$

14. Determine whether this function is a natural cubic spline:

$$f(x) = \begin{cases} 2(x + 1) + (x + 1)^3 & x \in [-1, 0] \\ 3 + 5x + 3x^2 & x \in [0, 1] \\ 11 + 11(x - 1) + 3(x - 1)^2 - (x - 1)^3 & x \in [1, 2] \end{cases}$$

15. A theorem in calculus asserts that if a function is differentiable at a point, then it is necessarily continuous at that point. The reason for this is elementary: If the limit defining $f'(x)$ exists,

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x + h) - f(x)}{h}$$

then the limit of the numerator alone must be 0, and this implies the continuity of f at x . Suppose now that for a certain function f and a certain point x_0 , we have

$$\lim_{x \downarrow x_0} f'(x) = \lim_{x \uparrow x_0} f'(x)$$

Can we conclude that f' is continuous at x_0 ?

16. (Continuation) In checking to determine whether a piecewise cubic function f is a cubic spline, does it suffice to verify the equation

$$\lim_{x \downarrow t} f''(x) = \lim_{x \uparrow t} f''(x)$$

for each knot t ?

17. Find the natural cubic spline function whose knots are -1 , 0 , and 1 and that takes the values $S(-1) = 13$, $S(0) = 7$, and $S(1) = 9$.

18. Which properties of a natural cubic spline does the following function possess, and which properties does it not possess?

$$f(x) = \begin{cases} (x+1) + (x+1)^3 & x \in [-1, 0] \\ 4 + (x-1) + (x-1)^3 & x \in (0, 1] \end{cases}$$

19. Find a natural cubic spline function whose knots are $-1, 0$, and 1 and that takes these values:

x	-1	0	1
y	5	7	9

20. Determine whether the coefficients a, b, c , and d exist so that the function

$$S(x) = \begin{cases} 1 - 2x & x \in (-\infty, -3] \\ a + bx + cx^2 + dx^3 & x \in [-3, 4] \\ 157 - 32x & x \in [4, +\infty) \end{cases}$$

is a natural cubic spline for the interval $[-3, 4]$.

21. Is the following function a natural cubic spline?

$$S(x) = \begin{cases} x^3 - 1 & x \in [-1, \frac{1}{2}] \\ 3x^3 - 1 & x \in [\frac{1}{2}, 1] \end{cases}$$

22. (Continuation) Repeat the preceding problem for the function

$$S(x) = \begin{cases} x^3 - 1 & x \in [-1, 0] \\ 3x^3 - 1 & x \in [0, 1] \end{cases}$$

23. Can a and b be defined so that the function

$$S(x) = \begin{cases} (x-2)^3 + a(x-1)^2 & x \in (-\infty, 2] \\ (x-2)^3 - (x-3)^2 & x \in [2, 3] \\ (x-3)^3 + b(x-2)^2 & x \in [3, +\infty) \end{cases}$$

is a natural cubic spline? Why or why not?

24. If S is a first-degree spline function that interpolates f at a sequence of knots $0 = t_0 < t_1 < \cdots < t_n = 1$, what is $\int_0^1 S(x)dx$?
25. What value of (a, b, c, d) makes this a cubic spline?

$$f(x) = \begin{cases} x^3 & x \in [-1, 0] \\ a + bx + cx^2 + dx^3 & x \in [0, 1] \end{cases}$$

26. Determine the value of (a, b, c) that makes the function

$$f(x) = \begin{cases} x^3 & x \in [0, 1] \\ \frac{1}{2}(x-1)^3 + a(x-1)^2 + b(x-1) + c & x \in [1, 3] \end{cases}$$

a cubic spline. Is it a natural cubic spline?

27. Let $t_0 < t_1 < \cdots < t_n$ and $-\infty < x < \infty$. What is the output value of k from the following algorithm?

```
for  $i = 1$  to  $n$  do
  if  $x < t_i$  then
     $k \leftarrow i$ 
```

```

        exit loop
    end if
end do

```

28. How many conditions are needed to define a quadratic spline interpolation function $S(x)$ over $a = t_0 < t_1 < \cdots < t_{20} = b$? Does the continuity of $S'(x)$ provide all the needed conditions?
29. Prove Theorem 1 when the end conditions on S are changed to $S'(a) = f'(a)$ and $S'(b) = f'(b)$.
30. Develop a suitable procedure for finding the cubic spline interpolant when the end conditions specify values of $S'(t_0)$ and $S'(t_n)$.
31. Give the simplified version of natural cubic spline interpolation that pertains to the case of equally spaced knots.
32. Show that a spline function of degree 1 having knots $t_0 < t_1 < \cdots < t_n$ can be expressed in the form

$$S(x) = ax + b + \sum_{i=1}^{n-1} c_i |x - t_i|$$

33. Show that the function in Equation (12) solves the two-point boundary-value problem preceding Equation (12).
34. Show that Equation (13) follows from the global smoothness conditions, as asserted in the text.
35. Prove that the coefficient matrix in Equation (13) is diagonally dominant.
36. Using the notation and hypotheses of Theorem 4, prove that the inequality

$$\|f^{(m+1)}\| \geq \|f^{(m+1)} - S^{(m+1)}\|$$

is valid.

COMPUTER
PROBLEMS 6.4

1. Draw a curve, such as an oval or spiral, on a sheet of graph paper. Select points in a more or less regular distribution along the curve and label them $t_0 = 1.0$, $t_1 = 2.0$, and so on. Read the x - and y -coordinates at each selected point to obtain a table of $x(t)$ and $y(t)$. Fit these functions by spline functions S and S^* . Then the formulas $x = S(t)$ and $y = S^*(t)$ give an approximate parametric representation of the curve. Plot the resulting curves for several test cases using an automatic plotter.
2. In the development of Problem 6.4.9 (p. 361), find a way to solve for the array $(z_0, z_1, \dots, z_n)$ so that $\sum_{i=0}^n z_i^2$ is a minimum. That is, use this condition to remove the arbitrariness referred to in the problem. Incorporate this feature in your algorithm and test it on a computer.
3. Prove this formula:

$$\int_{t_i}^{t_{i+1}} S_i(x)dx = \frac{h_i}{2}(y_i + y_{i+1}) - \frac{h_i^3}{24}(z_i + z_{i+1})$$

Then write and test a program to compute

$$\int_{t_0}^{t_n} S(x)dx$$

4. Write and test a computer program to compute the cubic spline function S having pre-scribed knots $t_0 < t_1 < \dots < t_n$ and satisfying these conditions:

$$\begin{cases} S(t_i) = y_i & (0 \leq i \leq n) \\ S''(t_0) = \alpha \\ S''(t_n) = \beta \end{cases}$$

5. Program the algorithm for tension splines and test it with a variety of values of the tension parameter τ .
6. Start with the silhouette of a car body as shown in Figure 6.6. Prepare a table of abscissas and ordinates for 10 to 20 points. Using values of the tension $\tau = 0.25, 4, \text{ and } 10$, generate and plot values of the interpolating tension spline to see which one produces the most pleasing appearance.

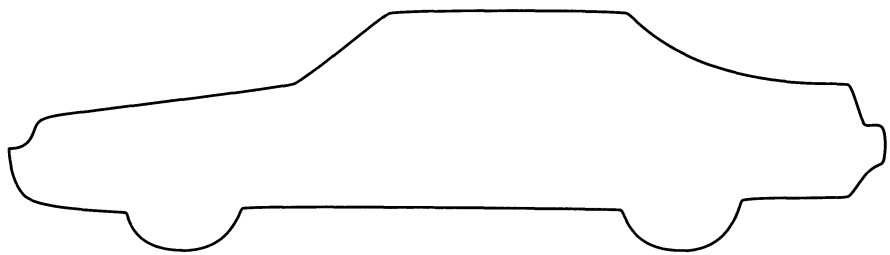


FIGURE 6.6
Car silhouette

7. Draw a script letter, such as the one shown in Figure 6.7. Then reproduce it with the aid of cubic splines and a plotter. Proceed as follows: Select a modest number of points on the curve, say $n = 11$. Label these $t = 1, 2, \dots, n$. For each point, obtain the corresponding x - and y -coordinates. Then fit $x = S_x(t)$ and $y = S_y(t)$, using cubic spline interpolating functions S_x and S_y . This will produce a parametric representation of the original curve. Compute a large number of values of $S_x(t)$ and $S_y(t)$ to give to the plotter. To learn more about how spline curves are used in designing typefaces, the reader should consult Knuth [1979].

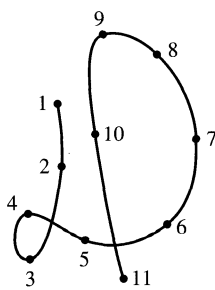


FIGURE 6.7
Script letter from 11 knots

8. Interpret the results of the following numerical experiment and draw some conclusions.
- a. Define p to be the polynomial of degree 20 that interpolates the function $f(x) = (1 + 6x^2)^{-1}$ at 21 equally spaced nodes in the interval $[-1, 1]$. Include the endpoints as nodes. Print a table of $f(x)$, $p(x)$, and $f(x) - p(x)$ at 41 equally spaced points on the interval.

- b. Repeat the experiment using the **Chebyshev nodes** given by

$$x_i = \cos[(i-1)\pi/20] \quad (1 \leq i \leq 21)$$

- c. With 21 equally spaced knots, repeat the experiment using a cubic interpolating spline.

6.5 B-Splines: Basic Theory

This section is devoted to a system of spline functions from which all other spline functions can be obtained by forming linear combinations. These splines provide bases for certain spline spaces and are therefore called **B-splines**. Once the knots are known, the B -splines are easily generated by recurrence relations and the algorithm is relatively simple. The B -splines are distinguished by their elegant theory and their model behavior in numerical calculations. Moreover, B -splines can be generalized.

We begin with a system of knots, t_i , on the real line. For practical purposes, only a finite set of knots is ever needed, but for the theoretical development it is much easier to suppose that the knots form an infinite set extending to $+\infty$ on the right and to $-\infty$ on the left:

$$\cdots < t_{-2} < t_{-1} < t_0 < t_1 < t_2 < \cdots$$

This knot sequence is assumed to be fixed throughout this section, and all of our splines will be based on it.

B-Splines of Degree 0

The B -splines of degree 0 are denoted by B_i^0 and have the appearance shown in Figure 6.8. The index i ranges over all the integers. The heavy dots on the graph indicate that we define $B_i^0(t_i) = 1$ and $B_i^0(t_{i+1}) = 0$. The formal definition is

$$B_i^0(x) = \begin{cases} 1 & \text{if } t_i \leq x < t_{i+1} \\ 0 & \text{otherwise} \end{cases}$$

These B -splines form an infinite sequence, $\{B_i^0 : i \in \mathbb{Z}\}$. (Here $\mathbb{Z}$ denotes the set of all integers, positive, negative, or 0.) We observe some of their salient properties:

1. The **support** of B_i^0 , defined as the set of x where $B_i^0(x) \neq 0$, is the interval $[t_i, t_{i+1})$.

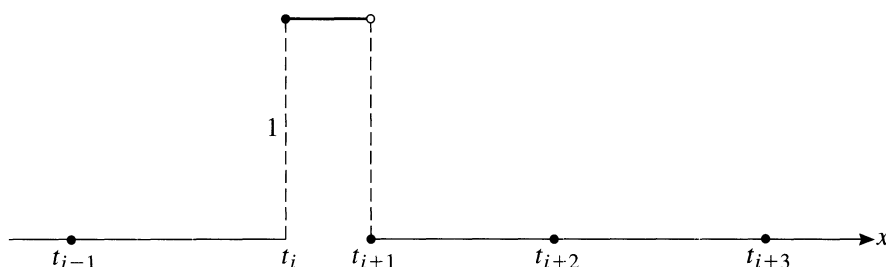


FIGURE 6.8
The B -spline B_i^0

2. $B_i^0(x) \geq 0$ for all i and all x .
3. B_i^0 is continuous from the right on the entire real line.
4. $\sum_{i=-\infty}^{\infty} B_i^0(x) = 1$ for all x .

This last equation is verified by selecting any $x \in \mathbb{R}$ and then determining the knot interval in which x lies, say $t_j \leq x < t_{j+1}$; then $\sum_{i=-\infty}^{\infty} B_i^0(x) = B_j^0(x) = 1$.

A final remark about the splines B_i^0 is that they do form a basis for all splines of degree 0 based on the given knot sequence, provided that we standardize such splines to be continuous from the right. To verify this assertion, suppose that S is such a spline function. Then it is *piecewise constant* and is defined by a set of rules of the form

$$S(x) = c_i \quad \text{if} \quad t_i \leq x < t_{i+1} \quad (i \in \mathbb{Z})$$

It is apparent that $S(x) = \sum_{i=-\infty}^{\infty} c_i B_i^0(x)$. (Thus, we have a basis in the sense of Schauder: Each vector in the space has a *unique* representation as an *infinite* series $\sum_{i=-\infty}^{\infty} c_i B_i^0$.)

The functions B_i^0 are the starting point for a recursive definition of all the higher-degree B-splines. The basic recurrence relation is

$$B_i^k(x) = \left(\frac{x - t_i}{t_{i+k} - t_i} \right) B_i^{k-1}(x) + \left(\frac{t_{i+k+1} - x}{t_{i+k+1} - t_{i+1}} \right) B_{i+1}^{k-1}(x) \quad (k \geq 1) \quad (1)$$

All the properties of the higher-order B-splines will follow from this recursive definition. By introducing some special linear functions

$$V_i^k(x) = \frac{x - t_i}{t_{i+k} - t_i} \quad (2)$$

we can write the recurrence relation in the following more elegant form:

$$B_i^k = V_i^k B_i^{k-1} + (1 - V_{i+1}^k) B_{i+1}^{k-1} \quad (3)$$

Since B_i^0 is a piecewise polynomial of degree 0, and since V_i^k is linear, B_i^1 is a piecewise polynomial of degree ≤ 1 . The same reasoning shows that, in general, B_i^k will be a piecewise polynomial of degree $\leq k$.

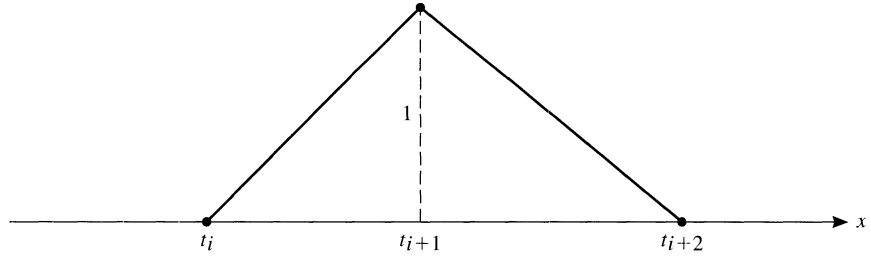
B-Splines of Degree 1

With the aid of Equation (1), we can give an explicit formula for $B_i^1(x)$ as follows:

$$\begin{aligned} B_i^1(x) &= \left(\frac{x - t_i}{t_{i+1} - t_i} \right) B_i^0(x) + \left(\frac{t_{i+2} - x}{t_{i+2} - t_{i+1}} \right) B_{i+1}^0(x) \\ &= \begin{cases} 0 & \text{if } x < t_i \text{ or } x \geq t_{i+2} \\ \frac{x - t_i}{t_{i+1} - t_i} & \text{if } t_i \leq x < t_{i+1} \\ \frac{t_{i+2} - x}{t_{i+2} - t_{i+1}} & \text{if } t_{i+1} \leq x < t_{i+2} \end{cases} \end{aligned}$$

The graph of B_i^1 is shown in Figure 6.9.

FIGURE 6.9
The B -spline B_i^1



Again, some observations can be made about the functions B_i^1 :

1. The support of B_i^1 is (t_i, t_{i+2}) .
2. $B_i^1(x) \geq 0$ for all i and all x .
3. B_i^1 is continuous and is differentiable at every point except t_i, t_{i+1} , and t_{i+2} .
4. $\sum_{i=-\infty}^{\infty} B_i^1(x) = 1$ for all x .

To verify this last equation, take any $x \in \mathbb{R}$. Since t_j converges to $+\infty$ when i increases, and it converges to $-\infty$ when i decreases, we can find an index j such that $t_j \leq x < t_{j+1}$. Then $B_i^1(x) = 0$ for all i with the possible exception of $i = j$ or $i = j - 1$. Hence for this x ,

$$\begin{aligned}\sum_{i=-\infty}^{\infty} B_i^1(x) &= B_{j-1}^1(x) + B_j^1(x) \\ &= \frac{t_{j+1} - x}{t_{j+1} - t_j} + \frac{x - t_j}{t_{j+1} - t_j} = 1\end{aligned}$$

Properties of B -Splines

Now, in a sequence of lemmas, we shall develop the important properties of the family B_i^k ($i \in \mathbb{Z}, k \in \mathbb{N}$).

LEMMA 1 Lemma on Support of B -Splines

If $k \geq 1$ and $x \notin (t_i, t_{i+k+1})$, then $B_i^k(x) = 0$.

Proof We have already observed that this is true for $k = 1$, but not for $k = 0$. If it is true for a certain index $k - 1$, then it is true for k by the following reasoning: If $x \notin (t_i, t_{i+k+1})$, then $x \notin (t_i, t_{i+k})$ and $x \notin (t_{i+1}, t_{i+k+1})$. By the induction hypothesis, $B_i^{k-1}(x) = 0$ and $B_{i+1}^{k-1}(x) = 0$. From Equation (3), it follows that $B_i^k(x) = 0$. ■

LEMMA 2 Lemma on Positivity of B -Splines

Let $k \geq 0$. If $x \in (t_i, t_{i+k+1})$, then $B_i^k(x) > 0$.

Proof We observed earlier that the assertion of Lemma 2 is true when $k = 0$ or $k = 1$. (It is true for $k = 1$ by the explicit formulas given previously for B_i^1 .) Now assume that

the assertion is true for an index $k - 1$, with $k \geq 2$. This assumption and Lemma 1 imply that $B_i^{k-1}(x) \geq 0$ for all x and for all i . Let $t_i < x < t_{i+k+1}$. Then the linear factors on the right side of Equation (1) are positive. By the induction hypothesis, $B_i^{k-1}(x) > 0$ in (t_i, t_{i+k}) , and $B_{i+1}^{k-1}(x) > 0$ in (t_{i+1}, t_{i+k+1}) . Since $k \geq 2$, these two intervals overlap, and by Equation (1) we see that $B_i^k(x) > 0$. ■

Since we expect to use the B -splines B_i^k as a basis for all splines of degree k , we shall be interested in linear combinations of the form $\sum_{i=-\infty}^{\infty} c_i B_i^k(x)$.

LEMMA 3 Lemma on Recurrence Relation of B -Splines

For all $k \geq 0$, we have

$$\sum_{i=-\infty}^{\infty} c_i B_i^k = \sum_{i=-\infty}^{\infty} [c_i V_i^k + c_{i-1}(1 - V_i^k)] B_i^{k-1}$$

Proof We use Equation (3) and elementary series manipulations as follows:

$$\begin{aligned} \sum_{i=-\infty}^{\infty} c_i B_i^k &= \sum_{i=-\infty}^{\infty} c_i [V_i^k B_i^{k-1} + (1 - V_{i+1}^k) B_{i+1}^{k-1}] \\ &= \sum_{i=-\infty}^{\infty} c_i V_i^k B_i^{k-1} + \sum_{i=-\infty}^{\infty} c_{i-1}(1 - V_i^k) B_i^{k-1} \end{aligned}$$

Numerical Procedure

In Lemma 3, the coefficients c_i can be constants or functions. The lemma therefore provides a means for evaluating a function given in the form

$$f(x) = \sum_{i=-\infty}^{\infty} C_i^k(x) B_i^k(x)$$

We suppose that the functions C_i^k are given; they, of course, may be constants. We define now

$$C_i^{k-1}(x) = C_i^k(x) V_i^k(x) + C_{i-1}^k(x) [1 - V_i^k(x)] \quad (4)$$

Then by Lemma 3 and Formula (4), we have

$$\sum_{i=-\infty}^{\infty} C_i^k(x) B_i^k(x) = \sum_{i=-\infty}^{\infty} C_i^{k-1}(x) B_i^{k-1}(x)$$

By repeating the argument for $k - 1, k - 2, \dots, 0$, we eventually obtain

$$\sum_{i=-\infty}^{\infty} C_i^k(x) B_i^k(x) = \sum_{i=-\infty}^{\infty} C_i^0(x) B_i^0(x)$$

As we know, the expression on the right is easily evaluated: For $t_j \leq x < t_{j+1}$, its value is $C_j^0(x)$. We write Equation (4) in detail, using Equation (2), as follows:

$$C_i^{j-1}(x) = [(x - t_i)C_i^j(x) + (t_{i+j} - x)C_{i-1}^j(x)] / (t_{i+j} - t_i) \tag{5}$$

The preceding remarks lead to the following numerical procedure.

ALGORITHM 1 Algorithm: *B*-Spline Coefficients

If coefficients C_i^k are given, the spline function $S(x) = \sum_{i=-\infty}^{\infty} C_i^k B_i^k(x)$ for a given x can be computed as follows: Determine the index m such that $t_m \leq x < t_{m+1}$. Compute the triangular array

$$\begin{array}{ccccccc} C_m^k & C_m^{k-1} & \cdots & C_m^1 & C_m^0 \\ C_{m-1}^k & C_{m-1}^{k-1} & \cdots & C_{m-1}^1 & \\ \vdots & \vdots & \ddots & & \\ C_{m-k+1}^k & C_{m-k+1}^{k-1} & & & \\ C_{m-k}^k & & & & \end{array}$$

using Equation (5). Then $S(x) = C_m^0$.

LEMMA 4 Lemma on Partition of Unity for *B*-Splines

For all k , we have

$$\sum_{i=-\infty}^{\infty} B_i^k(x) = 1$$

Proof We can use the algorithm just described. We start with $\sum_{i=-\infty}^{\infty} C_i^k B_i^k(x)$, in which $C_i^k = 1$ for all i . We then fix x and apply Equation (4) to compute

$$\begin{aligned} C_i^{k-1} &= C_i^k V_i^k + C_{i-1}^k [1 - V_i^k] \\ &= V_i^k + 1 - V_i^k \\ &= 1 \end{aligned}$$

By repeating the argument, we discover that $C_i^j = 1$ for all i and for $j = k, k - 1, \dots, 0$. Hence, we have

$$\sum_{i=-\infty}^{\infty} B_i^k(x) = \sum_{i=-\infty}^{\infty} B_i^0(x) = 1 \quad \blacksquare$$

Derivative and Integral of *B*-Splines

The next lemma gives an important formula for the derivative of B_i^k . It is convenient to use Formula (2) for V_i^k and to set

$$\alpha_i^k = \frac{1}{t_{i+k} - t_i} \quad (6)$$

We note that, with this notation,

$$\frac{d}{dx} V_i^k(x) = \alpha_i^k \quad (7)$$

Other useful formulas, which are trivial to verify, are these:

$$\alpha_i^k V_i^{k+1} = \alpha_i^{k+1} V_i^k \quad (8)$$

$$\alpha_{i+1}^k (1 - V_i^{k+1}) = \alpha_{i+1}^{k+1} (1 - V_{i+1}^k) \quad (9)$$

In both cases, the verification consists simply in using the definitions of α_i^k and V_i^k .

LEMMA 5 Lemma on Derivative of B-Splines

The derivatives of B-spline functions are calculated as follows, for $k \geq 2$:

$$\frac{d}{dx} B_i^k(x) = \left(\frac{k}{t_{i+k} - t_i} \right) B_i^{k-1}(x) - \left(\frac{k}{t_{i+k+1} - t_{i+1}} \right) B_{i+1}^{k-1}(x) \quad (10)$$

When $k = 1$, the equation is true for all x except $x = t_i, t_{i+1}, t_{i+2}$.

Proof The proof is by induction, with the cases $k = 1$ and $k = 2$ left to the reader. We assume that the formula is true for a fixed k , and with that hypothesis, we establish the next case. The induction hypothesis is Equation (10), which is now rewritten in a compact form using notation previously explained:

$$\frac{d}{dx} B_i^k = k\alpha_i^k B_i^{k-1} - k\alpha_{i+1}^k B_{i+1}^{k-1} \quad (11)$$

From the basic recurrence relation, Equation (3), we have

$$\begin{aligned} \frac{d}{dx} B_i^{k+1} &= \frac{d}{dx} [V_i^{k+1} B_i^k + (1 - V_{i+1}^{k+1}) B_{i+1}^k] \\ &= V_i^{k+1} \frac{d}{dx} B_i^k + \alpha_i^{k+1} B_i^k + (1 - V_{i+1}^{k+1}) \frac{d}{dx} B_{i+1}^k - \alpha_{i+1}^{k+1} B_{i+1}^k \end{aligned} \quad (12)$$

Here we used Equation (7) twice. Next we replace the derivatives that appear on the right side of (12) by using the induction hypothesis, Equation (11). The result is

$$\begin{aligned} \frac{d}{dx} B_i^{k+1} &= V_i^{k+1} (k\alpha_i^k B_i^{k-1} - k\alpha_{i+1}^k B_{i+1}^{k-1}) + \alpha_i^{k+1} B_i^k \\ &\quad + (1 - V_{i+1}^{k+1}) (k\alpha_{i+1}^k B_{i+1}^{k-1} - k\alpha_{i+2}^k B_{i+2}^{k-1}) - \alpha_{i+1}^{k+1} B_{i+1}^k \end{aligned}$$

By straightforward regrouping, we write this in the form

$$\frac{d}{dx} B_i^{k+1} = \alpha_i^{k+1} B_i^k + k\alpha_i^k V_i^{k+1} B_i^{k-1} - \alpha_{i+1}^{k+1} B_{i+1}^k$$

$$\begin{aligned}
 & -k\alpha_{i+2}^k(1 - V_{i+1}^{k+1})B_{i+2}^{k-1} - k\alpha_{i+1}^k V_i^{k+1} B_{i+1}^{k-1} \\
 & + k\alpha_{i+1}^k(1 - V_{i+1}^{k+1})B_{i+1}^{k-1}
 \end{aligned} \tag{13}$$

In this equation we shall make the following replacements—all justified by Equations (8) and (9):

$$\begin{aligned}
 k\alpha_i^k V_i^{k+1} B_i^{k-1} &= k\alpha_i^{k+1} V_i^k B_i^{k-1} \\
 -k\alpha_{i+2}^k(1 - V_{i+1}^{k+1})B_{i+2}^{k-1} &= -k\alpha_{i+1}^{k+1}(1 - V_{i+2}^k)B_{i+2}^{k-1} \\
 &\quad - k\alpha_{i+1}^k V_{i+1}^{k+1} B_{i+1}^{k-1} + k\alpha_{i+1}^k(1 - V_{i+1}^{k+1})B_{i+1}^{k-1} \\
 &= k\alpha_{i+1}^k(1 - V_i^{k+1})B_{i+1}^{k-1} - k\alpha_{i+1}^k V_{i+1}^{k+1} B_{i+1}^{k-1} \\
 &= k\alpha_i^{k+1}(1 - V_{i+1}^k)B_{i+1}^{k-1} - k\alpha_{i+1}^{k+1} V_{i+1}^k B_{i+1}^{k-1}
 \end{aligned}$$

The transformed version of Equation (13) is now

$$\begin{aligned}
 \frac{d}{dx} B_i^{k+1} &= \alpha_i^{k+1} B_i^k + k[\alpha_i^{k+1} V_i^k B_i^{k-1} + \alpha_i^{k+1}(1 - V_{i+1}^k)B_{i+1}^{k-1}] \\
 &\quad - \alpha_{i+1}^{k+1} B_{i+1}^k - k[\alpha_{i+1}^{k+1} V_{i+1}^k B_{i+1}^{k-1} + \alpha_{i+1}^{k+1}(1 - V_{i+2}^k)B_{i+2}^{k-1}]
 \end{aligned} \tag{14}$$

By means of the basic recurrence relation (3), the bracketed terms in Equation (14) can be simplified, with the result being

$$\begin{aligned}
 \frac{d}{dx} B_i^{k+1} &= \alpha_i^{k+1} B_i^k + k\alpha_i^{k+1} B_i^k - \alpha_{i+1}^{k+1} B_{i+1}^k - k\alpha_{i+1}^{k+1} B_{i+1}^k \\
 &= (k+1)\alpha_i^{k+1} B_i^k - (k+1)\alpha_{i+1}^{k+1} B_{i+1}^k
 \end{aligned} \tag{15}$$

Because this is Equation (11) with k replaced by $k+1$, the induction step is complete. ■

LEMMA 6 Lemma on Smoothness of B -Splines

For $k \geq 1$, the B -splines B_i^k belong to continuity class $C^{k-1}(\mathbb{R})$.

Proof It is clear that B_i^1 is continuous: $B_i^1 \in C^0(\mathbb{R})$. Now let us assume that $B_i^k \in C^{k-1}(\mathbb{R})$. By Lemma 5, $(d/dx)B_i^{k+1} \in C^{k-1}(\mathbb{R})$ because this derivative is a linear combination of B_i^k and B_{i+1}^k . Hence, $B_i^{k+1} \in C^k(\mathbb{R})$. By the principle of induction, this completes the proof. ■

From the lemma concerning derivatives of B -splines, we obtain a useful formula:

$$\frac{d}{dx} \sum_{i=-\infty}^{\infty} c_i B_i^k(x) = k \sum_{i=-\infty}^{\infty} \left(\frac{c_i - c_{i-1}}{t_{i+k} - t_i} \right) B_i^{k-1}(x) \quad (k \geq 2) \tag{16}$$

This formula can be used for numerical differentiation, although numerical differentiation of noisy data is a very dubious undertaking. For $k = 1$, Equation (16) is true for all x except the knots.

LEMMA 7 Lemma on Integral of B-Splines

The integrals of B-spline functions are calculated as follows:

$$\int_{-\infty}^x B_i^k(s) ds = \left(\frac{t_{i+k+1} - t_i}{k+1} \right) \sum_{j=i}^{\infty} B_j^{k+1}(x)$$

Proof We can verify that the derivatives of both sides of the equation are equal by using Equation (16) with

$$c_j = \begin{cases} 0 & \text{if } j < i \\ 1 & \text{if } j \geq i \end{cases}$$

Then $c_j - c_{j-1}$ is 0 except when $j = i$, and we obtain

$$B_i^k(x) = \left(\frac{t_{i+k+1} - t_i}{k+1} \right) (k+1) \left(\frac{1}{t_{i+k+1} - t_i} \right) B_i^k(x)$$

To be certain that the functions on the two sides of the equation do not differ by a constant, note that they both reduce to 0 at $x = t_i$. ■

Additional Properties

If f is a function and K is a subset of its domain, then $f|K$ denotes the **restriction** of f to K . Thus,

$$(f|K)(x) = f(x) \quad (x \in K)$$

This concept is useful in working with splines because each function $B_i^k| (t_j, t_{j+1})$ is a polynomial (more precisely, the restriction of a polynomial). When it is said that a set of functions f_i is **linearly independent on a set K** , this means that the set of restricted functions $f_i|K$ is linearly independent in the usual sense.

Now consider the B-splines $B_0^k, B_1^k, \dots, B_k^k$. When these are restricted to any single interval between knots (t_v, t_{v+1}) , the result is a set of polynomials of degree $\leq k$. A surprising and useful fact is that these restrictions form a basis for the polynomial space Π_k if the interval is (t_k, t_{k+1}) .

LEMMA 8 Lemma on Linear Independence of B-Splines

The set of B-splines $\{B_j^k, B_{j+1}^k, \dots, B_{j+k}^k\}$ is linearly independent on (t_{k+j}, t_{k+j+1}) .

Proof First consider the case $k = 0$. The lemma asserts that $\{B_j^0\}$ is linearly independent on the interval (t_j, t_{j+1}) . This is obviously true. For the purposes of an inductive proof, let $k \geq 1$, and assume that the lemma is correct for index $k - 1$. On the basis of this assumption, we shall prove the lemma for the index k . Let $S = \sum_{i=0}^k c_{j+i} B_{j+i}^k$, and suppose that $S| (t_{k+j}, t_{k+j+1}) = 0$. By Equation (16),

$$0 = S'| (t_{k+j}, t_{k+j+1}) = k \sum_{i=1}^k \frac{c_{j+i} - c_{j+i-1}}{t_{j+i+k} - t_{j+i}} B_{j+i}^{k-1} | (t_{k+j}, t_{k+j+1})$$

To arrive at this equation, we used $B_{j+k+1}^{k-1} = 0$ and $B_j^{k-1} = 0$ on (t_{k+j}, t_{k+j+1}) . By applying the induction hypothesis to $\{B_{j+1}^{k-1}, B_{j+2}^{k-1}, \dots, B_{j+k}^{k-1}\}$, we conclude that this set is linearly independent on the interval (t_{k+j}, t_{k+j+1}) . Therefore, in Equation (16) all the coefficients must be 0, and thus we have $c_0 = c_1 = \dots = c_k$. If this common value is denoted by λ , we have $S(x) = \lambda$ on (t_{k+j}, t_{k+j+1}) by Lemma 4. (Observe that in Lemma 4, the only terms that are nonzero on the interval (t_{k+j}, t_{k+j+1}) are $B_j^k, B_{j+1}^k, \dots, B_{j+k}^k$.) Since it has been assumed that S vanished on (t_{k+j}, t_{k+j+1}) , we conclude that $\lambda = 0$. ■

LEMMA 9 Lemma on Linear Independence of B -Splines

The set of B -splines $\{B_{-k}^k, B_{-k+1}^k, \dots, B_{n-1}^k\}$ is linearly independent on (t_0, t_n) .

Proof Let $S = \sum_{i=-k}^{n-1} c_i B_i^k$, and suppose that $S \mid (t_0, t_n) = 0$. On the interval (t_0, t_1) only $B_{-k}^k, B_{-k+1}^k, \dots, B_0^k$ are nonzero, and therefore

$$0 = S \mid (t_0, t_1) = \sum_{i=-k}^0 c_i B_i^k \mid (t_0, t_1) \tag{17}$$

By Lemma 8, the set $\{B_{-k}^k, B_{-k+1}^k, \dots, B_0^k\}$ is linearly independent on (t_0, t_1) . Hence, from Equation (17), we infer that $c_i = 0$ when $-k \leq i \leq 0$. If all the c_i 's are 0, we have the desired conclusion. If not all the c_i 's are 0, let j be the first index for which $c_j \neq 0$. By the prior work, $j \geq 1$. Hence, $(t_j, t_{j+1}) \subseteq (t_0, t_n)$. For any $x \in (t_j, t_{j+1})$, we obtain the contradiction

$$0 = S(x) = \sum_{i=j}^{n-1} c_i B_i^k(x) = c_j B_j^k(x) \neq 0$$

Hence, all the c_i 's are 0. ■

PROBLEMS 6.5

1. Verify this formula for $B_j^2(t_i)$:

$$B_j^2(t_i) = \left(\frac{t_i - t_{i-1}}{t_{i+1} - t_{i-1}}\right) \delta_{i,j+1} + \left(\frac{t_{i+1} - t_i}{t_{i+1} - t_{i-1}}\right) \delta_{i,j+2}$$

2. Prove that if $t_m \leq x < t_{m+1}$, then

$$\sum_{i=-\infty}^{\infty} c_i B_i^k(x) = \sum_{i=m-k}^m c_i B_i^k(x)$$

3. Let $h_i = t_{i+1} - t_i$. Show that if

$$c_{i-1}h_{i-1} + c_{i-2}h_i = y_i(h_i + h_{i-1}) \qquad (i \in \mathbb{Z})$$

then the spline function $S = \sum_{i=-\infty}^{\infty} c_i B_i^2$ will have the interpolation property $S(t_j) = y_j$ for all j .

4. Suppose that the knots are taken to be all the integers: $t_i = i$ ($i \in \mathbb{Z}$). Show that $B_i^k(x) = B_0^k(x - t_i)$.
5. Count the number of multiplications, divisions, and additions/subtractions involved in the algorithm for computing $\sum_{i=-\infty}^{\infty} C_i^k B_i^k(x)$.

6. Verify Formula (16) for derivatives of splines.

7. Prove that

$$\int_{-\infty}^{\infty} B_i^k(x) dx = \frac{t_{i+k+1} - t_i}{k+1}$$

8. Prove that if $\sum_{i=-\infty}^{\infty} c_i B_i^k(x) = 0$ for all x , then $c_i = 0$ for all i .

9. Define functions of s by the formula

$$U_i^k(s) = (t_{i+1} - s)(t_{i+2} - s) \cdots (t_{i+k} - s)$$

For $k = 0$, let $U_i^0(s) = 1$. Prove that

$$U_i^k(s) V_i^k(x) + U_{i-1}^k(s) [1 - V_i^k(x)] = (x - s) U_i^{k-1}(s)$$

10. (Continuation) Prove that

$$\sum_{i=-\infty}^{\infty} U_i^k(s) B_i^k(x) = (x - s) \sum_{i=-\infty}^{\infty} U_i^{k-1}(s) B_i^{k-1}(x)$$

11. (Continuation) Prove **Marsden's Identity**:

$$\sum_{i=-\infty}^{\infty} U_i^k(s) B_i^k(x) = (x - s)^k$$

12. (Continuation) Prove that every polynomial of degree $\leq k$ is expressible in the form $\sum_{i=-\infty}^{\infty} c_i B_i^k$.

13. Using the notation in the text, prove that B_i^2 is given by the formulas

$$B_i^2(x) = \begin{cases} V_i^2 V_i^1 & x \in [t_i, t_{i+1}) \\ V_i^k - V_{i+1}^1 (V_i^2 - 1 + V_{i+1}^2) & x \in [t_{i+1}, t_{i+2}) \\ (1 - V_{i+1}^2)(1 - V_{i+2}^1) & x \in [t_{i+2}, t_{i+3}) \\ 0 & \text{elsewhere} \end{cases}$$

14. Verify these two equations:

$$\frac{d}{dx} B_i^1 = \frac{B_i^0}{t_{i+1} - t_i} - \frac{B_{i+1}^0}{t_{i+2} - t_{i+1}}$$

$$\frac{d}{dx} B_i^2 = \frac{2B_i^1}{t_{i+2} - t_i} - \frac{2B_{i+1}^1}{t_{i+3} - t_{i+1}}$$

15. Prove that

$$\sup_{-\infty < x < \infty} \left| \sum_{i=-\infty}^{\infty} c_i B_i^k(x) \right| \leq \sup_{-\infty < i < \infty} |c_i|$$

16. Prove that if $\sup_i |t_{i+1} - t_i| \leq m$, then

$$\int_{-\infty}^{\infty} \left| \sum_{i=-\infty}^{\infty} c_i B_i^k(x) \right| dx \leq m \sum_{i=-\infty}^{\infty} |c_i|$$

17. Find an upper bound for

$$\int_{-\infty}^{\infty} \left[\sum_{i=-\infty}^{\infty} c_i B_i^k(x) \right]^2 dx$$

18. Prove that for $k \geq 3$,

$$\frac{d^2}{dx^2} \sum_{i=-\infty}^{\infty} c_i B_i^k(x) = k(k-1) \sum_{i=-\infty}^{\infty} \left(\frac{c_i - c_{i-1}}{t_{i+k} - t_i} - \frac{c_{i-1} - c_{i-2}}{t_{i+k-1} - t_{i-1}} \right) \frac{B_i^{k-2}(x)}{t_{i+k-1} - t_i}$$

19. Prove that if $a_i = k^{-1}(t_{i+1} + t_{i+2} + \cdots + t_{i+k})$, then

$$\sum_{i=-\infty}^{\infty} a_i B_i^k(x) = x \quad (k \geq 1)$$

20. Give an example of a spline function $\sum_{i=-\infty}^{\infty} c_i B_i^k$ that is not the 0 function yet vanishes at all the knots.

21. Suppose that in the sequence of constants $C_{m-k}^k, C_{m-k+1}^k, \dots, C_m^k$, two adjacent elements are the same. Show that the spline function

$$\sum_{i=-\infty}^{\infty} C_i^k B_i^k(x)$$

will be a polynomial of degree less than k on the interval (t_m, t_{m+1}) .

22. Use the algorithm that follows Lemma 3 to prove that $\sum_{i=-\infty}^{\infty} c_i B_i^k$ is a polynomial of degree $\leq k$ on (t_m, t_{m+1}) .

23. Use the algorithm following Lemma 3 to give new proofs of Lemmas 1 and 2.

24. Prove formally that $B_i^k(x) > 0$ on (t_j, t_{j+1}) if and only if $j - k \leq i \leq j$.

25. Prove that Formula (10) in Lemma 5 is true for $k = 1$ except at the three knots t_i, t_{i+1}, t_{i+2} .

26. Find α and β in the formula

$$\text{support}\left(\prod_{i=0}^r B_i^k\right) = \bigcap_{i=0}^r \text{support}(B_i^k) = (\alpha, \beta)$$

27. What is the smallest interval on which $[B_j^k, B_{j+1}^k, \dots, B_{n+k+j-1}^k]$ is linearly independent?

28. Prove that

$$\sum_{i=0}^n B_i^k(x) = 1 \quad (t_k \leq x \leq t_{k+n})$$

29. Prove that

$$\sum_{i=0}^n B_i^k(x) > 0 \quad (t_1 < x < t_{n+k+1})$$

COMPUTER PROBLEMS 6.5

1. Write and test a subroutine for computing approximate values of $g(x) = \int_a^x f(t) dt$ on an interval $[a, b]$. The function f is furnished by the user in a separate function subprogram. The method to be used is first to interpolate f on $[a, b]$ by a natural cubic spline, S , using n equally spaced knots, and then to use $g(x) \approx \int_a^x S(t) dt$. The user will specify a, b , and n .
2. Let knots $t_1, t_2, \dots, t_{n+k+1}$ be specified, as well as n and k . Let coefficients $c_1, c_2, \dots, c_n$ be given, and write $f(x) = \sum_{i=1}^n c_i B_i^k(x)$. Write a subroutine that gives the value of $f(x)$ for any real value of x .
3. (Challenging problem) Let the knots be the set of all integers. Write a program to compute the supremum norm of B_0^k for $1 \leq k \leq 100$. A short program will suffice. Test cases: For $k = 1, 5$, and 10 , the values are $1, 0.55$, and 0.410963 , respectively.

6.6 B-Splines: Applications

In this section, we adopt the notation of the preceding section. An initial sequence of knots is prescribed

$$\cdots < t_{-2} < t_{-1} < t_0 < t_1 < t_2 < \cdots \quad (1)$$

and a system of B -splines, $B_i^k(x)$, is defined, based on the given knots. We want to relate these B -splines to the spline functions introduced at the beginning of Section 6.4 (p. 349). There, we considered functions that are globally of class C^{k-1} and are piecewise polynomials of degree $\leq k$ on the n intervals $[t_0, t_1]$, $[t_1, t_2]$, $[t_2, t_3]$, $\dots$, $[t_{n-1}, t_n]$. Let us denote the family of all such splines by $\mathcal{S}_n^k$. The notation does not display the knots, which are fixed at the beginning. We regard the spline functions in $\mathcal{S}_n^k$ as having domain $[t_0, t_n]$. How are these functions related to the B -splines B_i^k ? Throughout, we assume that $n \geq 1$.

Basis for the Space $\mathcal{S}_n^k$

We restrict the functions B_i^k to the interval $[t_0, t_n]$ so that the domains of these functions are the same. The notation for the restricted functions is $B_i^k \mid [t_0, t_n]$.

THEOREM 1 Theorem on Basis for Space $\mathcal{S}_n^k$

A basis for the space $\mathcal{S}_n^k$ is

$$\{B_i^k \mid [t_0, t_n] : -k \leq i \leq n-1\} \quad (2)$$

Consequently, the dimension of $\mathcal{S}_n^k$ is $k+n$.

Proof

First, it is obvious that the functions in Equation (2) belong to $\mathcal{S}_n^k$ because they are splines of degree k based on the given knot sequence in (1).

Next, we prove that the dimension of $\mathcal{S}_n^k$ is at most $k+n$. This will be accomplished by exhibiting a set of $k+n$ functions that spans $\mathcal{S}_n^k$. In fact, each element of $\mathcal{S}_n^k$ can be expressed in the form

$$S(x) = \sum_{i=0}^k a_i x^i + \sum_{i=1}^{n-1} b_i (x - t_i)_+^k \quad (3)$$

In this equation, the truncated powers are used:

$$(x - t_i)_+^k = \begin{cases} (x - t_i)^k & \text{if } x \geq t_i \\ 0 & \text{if } x < t_i \end{cases}$$

To prove Equation (3), we start with the interval $[t_0, t_1]$, in which all the truncated powers are 0. On $[t_0, t_1]$, $S(x)$ is a polynomial of degree k , say p_0 . Then we have $p_0(x) = \sum_{i=0}^k a_i x^i$. This determines all the coefficients a_i . In the interval $[t_1, t_2]$, $S(x)$ is another polynomial, say p_1 . By the continuity conditions at t_1 , we have

$$(p_1 - p_0)^{(r)}(t_1) = 0 \quad (0 \leq r \leq k-1)$$

Since $p_1 - p_0$ is of degree at most k , we conclude that $(p_1 - p_0)(x) = b_1(x - t_1)^k$ for some b_1 . Therefore, we can write

$$S(x) = \sum_{i=0}^k a_i x^i + b_1(x - t_1)_+^k \quad (t_0 \leq x \leq t_2)$$

This same argument is repeated at $t_2, t_3, \dots, t_{n-1}$ to justify the remaining terms in Equation (3).

The third part of the proof is to apply Lemma 9 of Section 6.5 (p. 374) and conclude that the set in Equation (2) is linearly independent. It is therefore a basis for $\mathcal{S}_n^k$. ■

The proof of Theorem 1 shows that the functions

$$1, x, x^2, \dots, x^k, (x - t_1)_+^k, (x - t_2)_+^k, \dots, (x - t_{n-1})_+^k$$

form a basis for $\mathcal{S}_n^k$. We should be cautious about using this basis for numerical work because it is very badly conditioned. Instead, one should use the basis of B -splines given in the statement of the theorem.

Interpolation Matrix

Spline functions can be used for interpolation at points other than the knots. Let a set of **nodes** be given: $x_1 < x_2 < \dots < x_n$. We would like to interpolate arbitrary data given at the nodes by a spline function of the form $\sum_{j=1}^n c_j B_j^k$. For this to be possible, the **interpolation matrix** A given by

$$A_{ij} = B_j^k(x_i) \quad (1 \leq i, j \leq n) \quad (4)$$

must be nonsingular. A beautiful theorem by Schoenberg and Whitney reveals the necessary and sufficient conditions for this nonsingularity; namely, that the diagonal of A should contain no 0 elements.

For the proof of this result, we follow de Boor [1976 and personal communications]. Several parts will be considered separately.

LEMMA 1 Lemma 1 on Interpolation Matrix

If the matrix in Equation (4) is nonsingular, then $A_{ii} \neq 0$ for $1 \leq i \leq n$.

Proof Suppose that $A_{rr} = 0$ for some r . Then $B_r^k(x_r) = 0$, and by Lemma 2 in Section 6.5 (p. 368), $x_r \notin (t_r, t_{r+k+1})$. Suppose first that $x_r \leq t_r$. If $i \leq r \leq j$, then $x_i \leq x_r \leq t_r \leq t_j$ and x_i is not in the support of B_j . Consequently, $A_{ij} = B_j^k(x_i) = 0$. The first r rows in A can be interpreted as vectors in $\mathbb{R}^{r-1}$ because $A_{ij} = 0$ for $j = r, r + 1, \dots, n$. This set of rows is therefore linearly dependent, and A is singular.

In the other case, $x_r \geq t_{r+k+1}$. If $i \geq r \geq j$, then $x_i \geq x_r \geq t_{r+k+1} \geq t_{j+k+1}$, and $A_{ij} = B_j^k(x_i) = 0$. The columns numbered 1 to r form a linearly dependent set because their components numbered $r, r + 1, \dots, n$ are all 0. Again we conclude that A is singular. ■

LEMMA 2 Lemma 2 on Interpolation Matrix

If $k = 1$ and $t_i < x_i < t_{i+2}$ for $1 \leq i \leq n$, then A is nonsingular.

Proof We use induction on n to establish this. If $n = 1$, the matrix A is a 1×1 matrix with sole element $A_{11} = B_1^1(x_1) \neq 0$.

Now suppose that Lemma 2 has been proved whenever the number of nodes is less than n . Consider then the case of n nodes, with $n \geq 2$. If there is an index r for which $1 \leq r \leq n - 1$ and $x_r \leq t_{r+1}$, then for any pair (i, j) satisfying $i \leq r < j$, we shall have $x_i \leq x_r \leq t_{r+1} \leq t_j$ and $A_{ij} = B_j^1(x_i) = 0$. The matrix A will therefore have the form

$$A = \begin{bmatrix} C & 0 \\ E & D \end{bmatrix}$$

in which C is $r \times r$ and D is $(n - r) \times (n - r)$. The matrix A is invertible if and only if C and D are invertible. (See Problem 6.6.11, p. 387.) The matrices C and D are just like A except smaller (because $r < n$ and $n - r < n$). Therefore, they are invertible by the induction hypothesis.

A similar argument can be used if $x_r \geq t_{r+1}$ for some index r in the range $2 \leq r \leq n$. In this case, if $j < r \leq i$, then $A_{ij} = 0$. The structure of A is

$$A = \begin{bmatrix} C & E \\ 0 & D \end{bmatrix}$$

with C being $(r - 1) \times (r - 1)$ and D being $(n - r + 1) \times (n - r + 1)$.

The only case remaining is that in which $x_i > t_{i+1}$ for $1 \leq i \leq n - 1$ and $x_i < t_{i+1}$ for $2 \leq i \leq n$. This can occur only if $n = 1$ (a case already considered) or if $n = 2$. In this special case, $n = 2$, $k = 1$, $x_1 > t_2$, and $x_2 < t_3$. Hence, $t_2 < x_1 < x_2 < t_3$. On the interval (t_2, t_3) , we have $B_1^1(x) + B_2^1(x) = 1$. Thus the matrix A has the form

$$\begin{bmatrix} B_1^1(x_1) & B_2^1(x_1) \\ B_1^1(x_2) & B_2^1(x_2) \end{bmatrix} = \begin{bmatrix} \lambda & 1 - \lambda \\ \mu & 1 - \mu \end{bmatrix}$$

The determinant is $\lambda - \mu \equiv B_1^1(x_1) - B_1^1(x_2) > 0$. ■

The Schoenberg-Whitney Theorem follows.

THEOREM 2 Schoenberg-Whitney Theorem

Assume that $x_1 < x_2 < \cdots < x_n$. For the matrix A , given by $A_{ij} = B_j^k(x_i)$, to be nonsingular, it is necessary and sufficient that there is no 0 element on its diagonal.

Proof The necessity has been proved in Lemma 1. The sufficiency, when $k = 0$, is trivial because the condition $B_i^0(x_i) \neq 0$ is equivalent to $t_i \leq x_i < t_{i+1}$. Because the supports of the B -splines are disjoint, $B_j^0(x_i) = \delta_{ij}$.

For $k = 1$, the proof of sufficiency has been given in Lemma 2. Now we proceed by induction on k . Assume that the theorem has been proved for B -splines of degree

less than k . On that hypothesis, we prove it for splines of degree k . Here $k \geq 2$ because of Lemma 2. Our proof will require another induction on n , as in the proof of Lemma 2, but we shall be less formal about it here. The point of the argument is that if $x_r \leq t_{r+1}$ for some $r \in \{1, 2, \dots, n-1\}$ or if $x_r \geq t_{r+k}$ for some $r \in \{2, 3, \dots, n\}$, then the matrix A will have a block structure such as in Lemma 2 and will be invertible by the induction hypothesis applied to two submatrices of A . We can therefore assume that

$$t_{i+1} < x_i < x_{i+1} < t_{i+k+1} \quad (1 \leq i \leq n-1) \quad (5)$$

Suppose that A is singular. Then $Au = 0$ for some $u \neq 0$. Let $f = \sum_1^n u_j B_j^k$. We note that f vanishes at the $n+2$ points $t_1, x_1, x_2, \dots, x_n, t_{n+k+1}$. Since $k \geq 2$, f' exists and is continuous. Hence Rolle's Theorem is applicable, and we infer the existence of $n+1$ zeros, ξ_i , of f' , arranged as follows:

$$t_1 < \xi_1 < x_1 < \xi_2 < x_2 < \dots < x_{n-1} < \xi_n < x_n < \xi_{n+1} < t_{n+k+1}$$

Now f' is of the form $\sum_{j=1}^{n+1} v_j B_j^{k-1}$, as shown in Equation (16) of Section 6.5 (p. 372). We also have

$$\sum_{j=1}^{n+1} v_j B_j^{k-1}(\xi_i) = f'(\xi_i) = 0 \quad (1 \leq i \leq n+1) \quad (6)$$

By using Equation (5), we see that ξ_i belongs to the support of B_i^{k-1} for $2 \leq i \leq n$.

Now several cases arise. In Case 1, assume that $\xi_1 < t_{k+1}$ and that $\xi_{n+1} > t_{n+1}$. Then ξ_1 is in the support of B_1^{k-1} and ξ_{n+1} is in the support of B_{n+1}^{k-1} . Hence, by the induction hypothesis, the $(n+1) \times (n+1)$ matrix $(B_j^{k-1}(\xi_i))$ is nonsingular. The coefficients, v_j , in Equation (6) must therefore all be 0. The actual formula for v_j , as in Equation (16) of Section 6.5 (p. 372), is

$$v_j = k(u_j - u_{j-1}) / (t_{j+k} - t_j)$$

We can use this formula for $1 \leq j \leq n+1$ if we agree to set $u_0 = u_{n+1} = 0$. The vanishing of the coefficients v_j implies that

$$u_1 - u_0 = u_2 - u_1 = \dots = u_{n+1} - u_n = 0$$

Since $u_0 = u_{n+1} = 0$, we conclude that all $u_i = 0$.

In Case 2, assume that $\xi_1 \geq t_{k+1}$ and that $\xi_{n+1} > t_{n+1}$. Then $B_1^{k-1}(\xi_i) = 0$ for $1 \leq i \leq n+1$ because ξ_i is not in the support of B_1^{k-1} . In this case, we infer from Equation (6) that

$$\sum_{j=2}^{n+1} v_j B_j^{k-1}(\xi_i) = 0 \quad (2 \leq i \leq n+1)$$

But ξ_i is in the support of B_i^{k-1} for $2 \leq i \leq n+1$, and so by the induction hypothesis we conclude that $v_j = 0$ for $2 \leq j \leq n+1$. As before, this leads to

$$u_2 - u_1 = u_3 - u_2 = \cdots = u_{n+1} - u_n = 0$$

and to $u_i = 0$ for $1 \leq i \leq n$ because $u_{n+1} = 0$.

In Case 3, assume that $\xi_1 < t_{k+1}$ and that $\xi_n \leq t_{n+1}$. This is similar to Case 2. Now ξ_i is in the support of B_i^{k-1} for $1 \leq i \leq n$, and the same arguments as before lead to $u_i = 0$ for $1 \leq i \leq n$.

In Case 4, assume that $\xi_1 \geq t_{k+1}$ and $t_{n+1} \geq \xi_{n+1}$. Now we have $B_1^{k-1}(\xi_i) = 0$ and $B_{n+1}^{k-1}(\xi_i) = 0$ for $1 \leq i \leq n+1$. Equation (6) gives us

$$\sum_{j=2}^n v_j B_j^{k-1}(\xi_i) = 0 \quad (2 \leq i \leq n)$$

But ξ_i is in the support of B_i^{k-1} for $2 \leq i \leq n$, and by familiar reasoning, we conclude that

$$u_2 - u_1 = u_3 - u_2 = \cdots = u_n - u_{n-1} = 0$$

This shows that all the u_j are equal, say to λ . Hence, $f = \lambda \sum_1^n B_j^k$. Since $B_j^k \geq 0$ and $B_1^k(x_1) > 0$, the equation $f(x_1) = 0$ shows that $\lambda = 0$. ■

EXAMPLE 1 Let the knots be all the integers. If we wish to interpolate by linear combinations of $B_1^2, B_2^2, \dots, B_5^2$, can we use the set of nodes $\{3.1, 3.5, 3.6, 6.1, 6.6\}$?

Solution In this case, the condition given by the Schoenberg-Whitney Theorem can be written in the form

$$i < x_i < i + 3 \quad (1 \leq i \leq 5)$$

One easily verifies that this condition is satisfied by the given nodes. ■

Existence

In Theorem 1, we saw that the spline space $\mathcal{S}_n^k$ has dimension $n + k$. The functions that make up $\mathcal{S}_n^k$ are regarded as having the domain $[t_0, t_n]$. If nodes $x_1, x_2, \dots, x_{n+k}$ are chosen in $[t_0, t_n]$, what conditions must they satisfy so that interpolation by $\mathcal{S}_n^k$ will be possible?

THEOREM 3 Theorem on Interpolation by B-Splines

If nodes $x_1, x_2, \dots, x_{n+k}$ are chosen in $[t_0, t_n]$ and satisfy

$$t_{i-k-1} < x_i < t_i \quad (1 \leq i \leq n+k)$$

then arbitrary data at these nodes can be interpolated by the spline space $\mathcal{S}_n^k$.

Proof By Theorem 1, a basis for $\mathcal{S}_n^k$ is given by the set of functions $B_j^k \mid [t_0, t_n]$, with $-k \leq j \leq n-1$. Relabel the nodes by putting $y_i = x_{i+1+k}$ for $-k \leq i \leq n-1$. Then $y_i \in \text{supp}(B_i^k)$, and by the Schoenberg-Whitney Theorem, the $B_j^k(y_i)$ is nonsingular. ■

Now we return to the interpolation problem for nodes $x_1, x_2, \dots, x_n$ using the space S_n^k . Our interpolation matrix is $A_{ij} = B_j^k(x_i)$, as in Equation (4). By the Schoenberg-Whitney Theorem, this matrix is nonsingular if we assume that $B_i^k(x_i) \neq 0$ for $1 \leq i \leq n$ (which is the same as the assumption $t_i < x_i < t_{i+k+1}$). Then our interpolating function can be taken to be

$$S(x) = \sum_{j=1}^n c_j B_j^k(x)$$

The coefficients in this equation are obtained by solving the following linear system of equations:

$$\sum_{j=1}^n B_j^k(x_i) c_j = f(x_i) \quad (1 \leq i \leq n) \quad (7)$$

This solution to the interpolation problem is not necessarily the only one. We shall now see this *nonuniqueness* in considering the special case of interpolation at the knots. Suppose that $x_i = t_i$. The interpolation problem for these nodes requires the solution of the system

$$\sum_{j=-\infty}^{\infty} c_j B_j^k(t_i) = f(t_i) \quad (1 \leq i \leq n) \quad (8)$$

The simplest case, $k = 0$, is immediately solved by using $B_j^0(t_i) = \delta_{ij}$. Hence, we have $c_j = f(t_j)$ for $1 \leq j \leq n$. The case $k = 1$ is solved in the same way because $B_j^1(t_i) = \delta_{i-1,j}$. Thus, $c_{i-1} = f(t_i)$ for $1 \leq i \leq n$.

To solve this interpolation problem when $k = 2$, we require the following fact (see Problem 6.5.1, p. 374):

$$B_j^2(t_i) = \left(\frac{t_i - t_{i-1}}{t_{i+1} - t_{i-1}} \right) \delta_{i-1,j} + \left(\frac{t_{i+1} - t_i}{t_{i+1} - t_{i-1}} \right) \delta_{i-2,j} \quad (9)$$

Equation (9) implies immediately that each linear equation in System (8) will contain only two of the unknowns. Indeed, Equation (8) now reads as follows:

$$\left(\frac{t_{i+1} - t_i}{t_{i+1} - t_{i-1}} \right) c_{i-2} + \left(\frac{t_i - t_{i-1}}{t_{i+1} - t_{i-1}} \right) c_{i-1} = f(t_i) \quad (1 \leq i \leq n) \quad (10)$$

This system contains n equations and $n + 1$ unknowns. We can solve it by assigning any value to c_{-1} and recursively computing $c_0, c_1, \dots, c_{n-1}$ by means of Equation (10). Hence, there are *many* solutions to this interpolation problem.

If the case $k = 3$ is treated in the same manner, we obtain n equations with $n + 2$ unknowns. Here is system (8) for $k = 3$:

$$c_{i-3} B_{i-3}^3(t_i) + c_{i-2} B_{i-2}^3(t_i) + c_{i-1} B_{i-1}^3(t_i) = f(t_i) \quad (1 \leq i \leq n) \quad (11)$$

The values $B_j^3(t_i)$ needed here can be computed by means of Equation (1) in Section 6.5 (p. 367). The results are

$$\begin{aligned}
B_{i-3}^3(t_i) &= \frac{(t_{i+1} - t_i)^2}{(t_{i+1} - t_{i-2})(t_{i+1} - t_{i-1})} \\
B_{i-2}^3(t_i) &= \frac{(t_{i+2} - t_i)(t_i - t_{i-1})}{(t_{i+2} - t_{i-1})(t_{i+1} - t_{i-1})} + \frac{(t_i - t_{i-2})(t_{i+1} - t_i)}{(t_{i+1} - t_{i-2})(t_{i+1} - t_{i-1})} \\
B_{i-1}^3(t_i) &= \frac{(t_i - t_{i-1})^2}{(t_{i+2} - t_{i-1})(t_{i+1} - t_{i-1})}
\end{aligned}$$

As in the case $k = 2$, an easy solution to the system of equations is possible; namely, assign arbitrary values to c_{-2} and c_{-1} , then solve recursively for $c_0, c_1, \dots, c_{n-1}$, using Equation (11). This manner of solving system (11) is not recommended because the two extra parameters c_{-2} and c_{-1} are not used to advantage. It is usual to impose one additional condition on the spline at the knots t_1 and t_n —that is, at *each end* of the interval of interest. For example, the *natural* spline conditions are defined to be $S''(t_1) = S''(t_n) = 0$.

The system of equations to be solved for the natural spline interpolant consists of Equation (11) together with a pair of equations given in Problems 6.6.8–10 (p. 387). For advice on spline interpolation and the arrangement of computations, the reader should consult de Boor [1984]. (In that book, B_j^k denotes a spline of degree $k - 1$.)

Noninterpolatory Approximation Methods

To illustrate *noninterpolatory* approximation methods, we turn to an elegant procedure introduced by Schoenberg [1967]. Given a function f , we define a spline function Sf by the equation

$$Sf = \sum_{i=-\infty}^{\infty} f(x_i) B_i^k \quad x_i = \frac{1}{k}(t_{i+1} + \dots + t_{i+k}) \quad (12)$$

When $k = 0$, we let $x_i = t_i$. In this case we have nothing new because Equation (12) is the interpolation scheme discussed previously. The same remark pertains to the case $k = 1$. But for higher values of k , the spline function Sf defined by Equation (12) does not interpolate f at any prescribed set of nodes. (Such operators have been termed **quasi-interpolation** operators.) The salient properties of this approximation scheme are these:

1. If f is a linear function, then $Sf = f$.
2. For any linear function ℓ , $Sf - \ell$ has no more variations in sign than $f - \ell$.
3. If $f \geq 0$, then $Sf \geq 0$.
4. If $|f| \leq M$, then $|Sf| \leq M$.
5. S is a linear operator: $S(\alpha f + \beta g) = \alpha Sf + \beta Sg$.

The interested reader should consult either Marsden [1970] or Schoenberg [1967].

Our next task is to investigate whether continuous functions can be approximated to arbitrary precision by splines. In situations like this, we expect to hold the degree k fixed while increasing the number of knots in order to obtain higher precision. The question addressed is whether any desired precision can be reached by increasing the

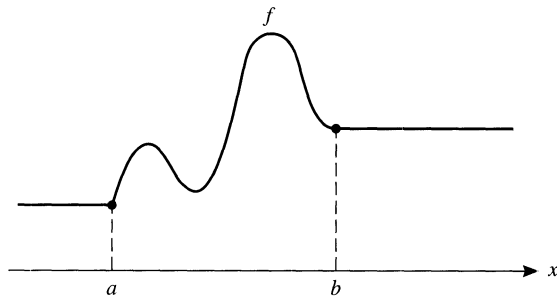


FIGURE 6.10
Extension of f

density of knots; the theorem sought will be the spline analogue of the Weierstrass Approximation Theorem.

As usual, a set of knots is prescribed:

$$\cdots < t_{-2} < t_{-1} < t_0 < t_1 < t_2 < \cdots$$

The interval $[t_0, t_n]$ is denoted also by $[a, b]$, and we suppose that a function f is given on $[a, b]$. The function f is extended as shown in Figure 6.10. Thus, if f is continuous on $[a, b]$, its extension will also be continuous.

Whether f is continuous or not, we define its **modulus of continuity** by the equation

$$\omega(f; \delta) = \max_{|s-t| \leq \delta} |f(s) - f(t)|$$

If f is a continuous function defined on an interval $[a, b]$, then it is uniformly continuous. This means that for any $\varepsilon > 0$, there is a $\delta > 0$ such that for all s and t in $[a, b]$,

$$|s - t| < \delta \Rightarrow |f(s) - f(t)| < \varepsilon$$

Hence, $\omega(f; \delta) \leq \varepsilon$. In other words, for a continuous function f on a closed and bounded interval, the modulus of continuity, $\omega(f; \delta)$, converges to 0 as δ converges to 0.

If f' exists, is continuous, and satisfies $|f'(x)| \leq M$, then by the Mean-Value Theorem,

$$|f(s) - f(t)| = |f'(\xi)| |s - t| \leq M |s - t|$$

Hence, $\omega(f, \delta) \leq M\delta$.

The next step is to introduce a spline function that approximates f in a simple manner. For this purpose we choose

$$g = \sum_{i=-\infty}^{\infty} f(t_{i+2}) B_i^k \tag{13}$$

With the aid of this function, we can now prove the following result.

THEOREM 4 Theorem on Spline Function Approximation

If f is a function on $[t_0, t_n]$, then the spline function, g , in Equation (13) satisfies

$$\max_{t_0 \leq x \leq t_n} |f(x) - g(x)| \leq k\omega(f; \delta)$$

where $\delta = \max_{-k \leq i \leq n+1} |t_i - t_{i-1}|$ and $\omega(f; \cdot)$ is the modulus of continuity of f .

Proof Recall that $B_i^k \geq 0$ and $\sum_{i=-\infty}^{\infty} B_i^k = 1$. Then with g as in Equation (13),

$$\begin{aligned} |g(x) - f(x)| &= \left| \sum_{i=-\infty}^{\infty} f(t_{i+2}) B_i^k(x) - f(x) \sum_{i=-\infty}^{\infty} B_i^k(x) \right| \\ &= \left| \sum_{i=-\infty}^{\infty} [f(t_{i+2}) - f(x)] B_i^k(x) \right| \\ &\leq \sum_{i=-\infty}^{\infty} |f(t_{i+2}) - f(x)| B_i^k(x) \end{aligned}$$

Let $x \in [t_j, t_{j+1}] \subseteq [a, b]$. On the interval $[t_j, t_{j+1}]$, only $B_{j-k}^k, B_{j-k+1}^k, \dots, B_j^k$ are active. Hence,

$$\begin{aligned} |g(x) - f(x)| &\leq \sum_{i=j-k}^j |f(t_{i+2}) - f(x)| B_i^k(x) \\ &\leq \max_{j-k \leq i \leq j} |f(t_{i+2}) - f(x)| \end{aligned}$$

For i in the range $j - k \leq i \leq j$, we have

$$\begin{aligned} t_{i+2} - x &\leq t_{j+2} - t_j = (t_{j+2} - t_{j+1}) + (t_{j+1} - t_j) \leq 2\delta \\ x - t_{i+2} &\leq t_{j+1} - t_{j-k+2} = (t_{j+1} - t_j) + \dots + (t_{j-k+3} - t_{j-k+2}) \leq k\delta \end{aligned}$$

By the definition of the modulus of continuity and by Problem 6.6.20 (p. 387), we have

$$|f(t_{i+2}) - f(x)| \leq \omega(f; k\delta) \leq k\omega(f; \delta) \quad \blacksquare$$

Distance from a Function to a Spline Space

The preceding result can be stated in terms of the distance from f to the space $\mathcal{S}_n^k$. The distance from a function f to a subspace G in a normed space is defined by

$$\text{dist}(f, G) = \inf_{g \in G} \|f - g\|$$

Let us use the norm defined by

$$\|f\| = \max_{a \leq x \leq b} |f(x)|$$

Theorem 4 asserts that

$$\text{dist}(f, \mathcal{S}_n^k) \leq k\omega(f; \delta) \tag{14}$$

If f is continuous, then

$$\lim_{\delta \downarrow 0} \omega(f; \delta) = 0$$

Hence, as the density of the knots is increased, the upper bound in Equation (14) will approach 0.

For functions possessing some derivatives, more can be said.

THEOREM 5 Theorem on Distance from a Function to a Spline Space

Let $r < k < n$. If $f \in C^r[t_0, t_n]$, then (with δ as in the Theorem on Spline Function Approximation)

$$\text{dist}(f, \mathcal{S}_n^k) \leq k^r \delta^r \|f^{(r)}\|$$

Proof Let g be any element of $\mathcal{S}_n^k$. By Theorem 4, we have

$$\text{dist}(f, \mathcal{S}_n^k) = \text{dist}(f - g, \mathcal{S}_n^k) \leq k\omega(f - g; \delta) \leq k\delta \|f' - g'\|$$

As g runs over $\mathcal{S}_n^k$, g' runs over all of $\mathcal{S}_n^{k-1}$. (See Problem 6.6.13, p. 387.) Thus, by taking an infimum on g in the previous inequality, we obtain

$$\text{dist}(f, \mathcal{S}_n^k) \leq k\delta \text{dist}(f', \mathcal{S}_n^{k-1})$$

This argument can be repeated $r - 2$ times, yielding

$$\begin{aligned} \text{dist}(f, \mathcal{S}_n^k) &\leq k^{r-1} \delta^{r-1} \text{dist}(f^{(r-1)}, \mathcal{S}_n^{k+1-r}) \\ &\leq k^r \delta^{r-1} \omega(f^{(r-1)}; \delta) \\ &\leq k^r \delta^r \|f^{(r)}\| \end{aligned}$$

PROBLEMS 6.6

1. Show that the matrix $B_j^k(x_i)$ that arises in B -spline interpolation is *banded*. Specifically, if $B_j^k(x_j) \neq 0$ and $x_j < x_{j+1}$ for each j , then each row and column of the matrix contains at most $2k + 1$ nonzero elements.
2. Let $k = 2$ and $f(x) = x$ in Equation (12). Show that $Sf = f$.
3. Prove that

$$x^2 = \sum_{i=-\infty}^{\infty} t_{i+1} t_{i+2} B_i^2(x)$$

4. Let $k = 2$ and $f(x) = 1$ in Equation (12). Show that $Sf = f$.
5. Prove Property 5 of the Schoenberg approximation process.

6. Prove Property 1 of the Schoenberg process in the case $k = 2$. *Hint:* Problems 6.6.2, 4–5 (p. 386) are helpful.
7. Prove Properties 3 and 4 of the Schoenberg process.
8. Prove that if $S = \sum_{j=-\infty}^{\infty} c_j B_j^3$, then $S'' = \sum_{j=-\infty}^{\infty} e_j B_j^1$ with

$$e_j = \frac{6}{t_{j+2} - t_j} \left(\frac{c_j - c_{j-1}}{t_{j+3} - t_j} - \frac{c_{j-1} - c_{j-2}}{t_{j+2} - t_{j-1}} \right)$$

9. (Continuation) Prove that if $S = \sum_{j=-\infty}^{\infty} c_j B_j^3$, then $S''(t_i) = e_{i-1}$, where e_j is defined in the preceding problem.
10. (Continuation) Prove that the natural cubic spline interpolant for the function f is $\sum_{j=-2}^{n-1} c_j B_j^3$, where the coefficients satisfy Equation (11) in the text and these *natural* end conditions corresponding to $i = 1$ and $i = n$:

$$(t_{i+2} - t_{i-1})c_{i-3} - (t_{i+2} + t_{i+1} - t_{i-1} - t_{i-2})c_{i-2} + (t_{i+1} - t_{i-2})c_{i-1} = 0$$

11. Let C and D be square matrices. Prove that

$$\begin{bmatrix} C & 0 \\ E & D \end{bmatrix}$$

is nonsingular if and only if C and D are nonsingular.

12. Let K be any subset of $\mathbb{R}$. Prove that in order for $\{B_1^k, \dots, B_n^k\}$ to be linearly independent on K , it is necessary and sufficient that $K \cap (t_i, t_{i+k+1})$ be nonempty for $1 \leq i \leq n$.
13. Use Theorem 1 to prove that the derivative operator is surjective from S_n^k to S_n^{k-1} .
14. Prove Lemma 8 of Section 6.5 (p. 373) as a corollary of the Schoenberg-Whitney Theorem.
15. Prove Lemma 9 of Section 6.5 (p. 374) as a corollary of the Schoenberg-Whitney Theorem.
16. Is the following a valid version of the Schoenberg-Whitney Theorem? If the nodes x_i are not necessarily ordered, then the nonsingularity of the matrix $(B_j^k(x_i))$ is equivalent to the assertion that each interval (t_i, t_{i+k+1}) for $1 \leq i \leq n$ contains at least one node.
17. Use the method of proof in Theorem 4 to prove a similar result for the Schoenberg operator.
18. (Continuation) Refine the result obtained in the preceding problem for the case of equally spaced knots.
19. Prove Theorem 1 by showing that only $n + k$ elements of the sequence B_i^k have nonzero values on $[t_0, t_n]$.
20. Prove that $\omega(f; k\delta) \leq k\omega(f; \delta)$.
21. Prove Property 1 of the Schoenberg process for all values of k . It suffices, because of linearity, to prove that $Sf = f$ if $f(x) = 1$ or if $f(x) = x$. The first of these is very easy. To prove the second, show that $(Sf)' = 1$ by using Equation (16) in Section 6.5 (p. 372).

COMPUTER PROBLEM 6.6

1. Let $k = 3$ and take knots to be the integers. For $n = 5, 10, 15$, and 20 , estimate the conditioning of the matrix $(f_j(x_i))$ when f_j represents the basis for S_n^k consisting of powers and truncated powers (as in the proof of Theorem 1). Take x_i to be equally spaced points in $[t_0, t_n]$.

6.7 Taylor Series

In this short section, we want to illustrate (and emphasize) the utility of Taylor series as a mechanism for approximation. Taylor's Theorem is, of course, useful for functions that possess a certain number of continuous derivatives. However, it is not very useful for empirical data or for functions that have only a few derivatives. For functions to which Taylor's Theorem is applicable, one should not overlook the possibility of representing them efficiently by a *Taylor polynomial*.

Recall that if f is a function having a continuous $(n+1)$ st derivative on an interval $(c - \delta, c + \delta)$, then

$$f(x) = p_n(x) + E_n(x)$$

where p_n is a polynomial of degree $\leq n$ and E_n is a **remainder function**. These are given by

$$p_n(x) = \sum_{k=0}^n \frac{1}{k!} f^{(k)}(c)(x-c)^k$$

$$E_n(x) = \frac{1}{(n+1)!} f^{(n+1)}(\xi_x)(x-c)^{n+1} \quad |\xi_x - c| < \delta$$

An important special case occurs when $c = 0$; this is the **Maclaurin series**.

By using Taylor's Theorem together with an analysis of $E_n(x)$ as $n \rightarrow \infty$, we obtain Taylor series for many important functions, such as

$$\cos x = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k}}{(2k)!} \quad (-\infty < x < \infty) \quad (1)$$

$$\frac{1}{x} = \sum_{k=0}^{\infty} (-1)^k (x-1)^k \quad (0 < x < 2) \quad (2)$$

The series that occur in these illustrations are **power series**. Here is a theorem about the convergence of a power series.

THEOREM 1 Theorem on Convergence of Power Series

For every power series

$$\sum_{k=0}^{\infty} a_k (x-c)^k$$

there is a number r in the range $[0, \infty]$ such that the series diverges for $|x-c| > r$ and converges for $|x-c| < r$.

The number r (it may be $+\infty$) is called the **radius of convergence** of the series. It can often be computed by the **ratio test**. (See Problems 6.7.2–3, p. 390.) The radius of convergence of the cosine series, Equation (1), is $+\infty$; that of the series in Equation (2) is $+1$.

For applications, the following theorem is important.

THEOREM 2 Theorem on Radius of Convergence

Let r be the radius of convergence of $\sum_{k=0}^{\infty} a_k(x - c)^k$. Then the equation

$$f(x) = \sum_{k=0}^{\infty} a_k(x - c)^k$$

defines a function that is continuously differentiable in the interval $|x - c| < r$. Furthermore,

$$f'(x) = \sum_{k=0}^{\infty} k a_k(x - c)^{k-1}$$

and this series has r for its radius of convergence. Finally, if $|b - c| < r$ and $|x - c| < r$, then

$$\int_b^x f(t) dt$$

can be obtained by integrating the f -series term by term. The resulting series has r for its radius of convergence.

In summary, this theorem says that we can integrate and differentiate a power series term by term within its interval of convergence.

As an illustration of how this theorem works, we consider one of the **higher transcendental functions**, the **sine integral**. This is defined by the formula

$$S(x) \equiv \int_0^x \frac{\sin t}{t} dt$$

This integration is not amenable to the usual techniques of elementary calculus. Instead, one can proceed as follows:

$$\sin t = \sum_{k=0}^{\infty} (-1)^k \frac{t^{2k+1}}{(2k+1)!}$$

$$\frac{\sin t}{t} = \sum_{k=0}^{\infty} (-1)^k \frac{t^{2k}}{(2k+1)!}$$

$$\int_0^x \frac{\sin t}{t} dt = \sum_{k=0}^{\infty} (-1)^k \frac{1}{(2k+1)!} \int_0^x t^{2k} dt$$

$$S(x) = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k+1}}{(2k+1)!(2k+1)}$$

The series we have obtained for $S(x)$ is rapidly convergent for small x . Thus, $S(1)$ can be computed to 20 decimal places of accuracy by taking only ten terms. (See Problem 6.7.36, p. 392.)

Another similar example is

$$\int_0^x e^{t^2} dt = \sum_{k=0}^{\infty} \frac{x^{2k+1}}{(2k+1)k!} \quad (3)$$

PROBLEMS 6.7

1. Show that $\sum_{k=0}^{\infty} a_k x^k$ and $\sum_{k=0}^{\infty} a_k (x-c)^k$ have the same radius of convergence. For p as a natural number, what about $\sum_{k=0}^{\infty} a_k x^{k+p}$?
2. (**Ratio test**) If $\lim_{n \rightarrow \infty} |A_{n+1}/A_n| < 1$, then $\sum_{k=0}^{\infty} A_k$ converges. Use the test to show that the cosine series in Equation (1) converges for all real x .
3. (**Ratio test**, continuation) If $\lim_{n \rightarrow \infty} |A_{n+1}/A_n| > 1$, then $\sum_{k=0}^{\infty} A_k$ diverges. Use this fact together with the preceding problem to find the radius of convergence of

$$\sum_{k=0}^{\infty} (-1)^k (x-1)^k$$

4. Find the radius of convergence of

$$\sum_{k=0}^{\infty} k! x^k$$

5. If $f(x) = \sum_{k=0}^{\infty} a_k (x-c)^k$ and if the radius of convergence is r , then f possesses derivatives of all orders in the interval $|x-c| < r$. Furthermore,

$$f^{(n)}(x) = \sum_{k=n}^{\infty} \frac{a_k k!}{(k-n)!} (x-c)^{k-n} \quad (|x-c| < r)$$

Use Theorem 2 on radius of convergence to prove this.

6. Find a power series for

$$\frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$$

Note: This function is known as the **error function** and is denoted by $\text{erf}(x)$. It plays a role in statistics.

7. Find a power series for the function

$$f(x) = \int_0^x \frac{e^t - 1}{t} dt$$

Use the series you have obtained to compute $f(1)$ with three significant figures. (In this example, that means two decimal places.) Finally, prove that the sum of all the terms you did not include is so small that it will not affect the answer. *Cultural note:* The function f is an important one in applied mathematics. For further information, consult Abramowitz and Stegun [1964, chap. 5].

8. Show that the function $f(x) = \sum_{k=0}^{\infty} x^{2k}/(k!2^k)$ solves the differential equation $y' = xy$.
9. Let $a_0 = 1$ and $a_n = a_{n-1}/[2(n+1)]$ for $n \geq 1$. What is the radius of convergence of $\sum_{k=0}^{\infty} a_k x^k$?
10. (Continuation) Let $f(x)$ be the function whose series is in the preceding problem. What is the power series for $f'(x)$? A recurrence relation for the coefficients suffices.

11. Using the ratio test (see Problems 6.7.2–3, p. 390), prove that if $\lim_{n \rightarrow \infty} |a_n/a_{n+1}|$ exists or is $+\infty$, then it is the radius of convergence of $\sum_{k=0}^{\infty} a_k(x-c)^k$.
12. Let $a_0 = 1$ and $a_{n+1} = [2 + (-1)^n]a_n$ for $n \geq 1$. What is the radius of convergence of $\sum_{k=0}^{\infty} a_k(x-c)^k$?
13. If r and r' are the radii of convergence of $\sum_{k=0}^{\infty} a_k x^k$ and $\sum_{k=0}^{\infty} b_k x^k$, respectively, what is the radius of convergence of $\sum_{k=0}^{\infty} (a_k + b_k)x^k$?
14. A function related to the **dilogarithm** is defined by

$$f(x) = -\int_0^x \frac{\ln(1-t)}{t} dt \quad (-\infty < x \leq 1)$$

Find the Maclaurin series for f and determine its radius of convergence. How would you compute $f(-2)$? What about $f(0.001)$?

15. Obtain a series for $\ln x$ by integrating the series in Equation (2).
16. Show how the series for $\sin x$ can be obtained in two ways by differentiating or by integrating the series in Equation (1).
17. Obtain a power series for $\tan^{-1} x$ as follows: Start with $(1+x)^{-1} = \sum_{k=0}^{\infty} (-x)^k$, replace x by x^2 , and then integrate the terms in the resulting equation. Compare this method to the alternative procedure of computing the successive derivatives of $\tan^{-1} x$ and obtaining the Taylor series.
18. Criticize the following analysis and correct it:

$$\begin{aligned} \int_0^x \frac{1 - \cos t}{t} dt &= \int_0^x \left[\frac{1}{t} - \frac{\cos t}{t} \right] dt = \int_0^x \left[\frac{1}{t} - \sum_{k=0}^{\infty} \frac{(-1)^k t^{2k-1}}{(2k)!} \right] dt \\ &= \ln x - \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k}}{2k(2k)!} \end{aligned}$$

Hint: There are several errors.

19. Find the Maclaurin series for the function

$$f(x) = \int_0^x \frac{1 - \cos t}{t} dt$$

Cultural note: This function is sometimes called the **cosine integral** and is denoted by $\text{Cin}(x)$. The nomenclature has not been standardized.

20. Find the Maclaurin series for the **Fresnel integral**:

$$\varphi(x) = \int_0^x \sin t^2 dt$$

21. (**Reciprocal function**) Suppose $f(x) = \sum_{k=0}^{\infty} a_k x^k$, with $a_0 = 1$. Then, in some neighborhood of 0, $1/f(x)$ is well defined. Assume a series $\sum_{k=0}^{\infty} b_k x^k$ for the reciprocal function, and determine recursively the b_k 's from the fact that the product of the two series is 1.
22. (Continuation) Use the result of the preceding problem to find the Maclaurin series for $x/(e^x - 1)$.

23. (Continuation) Show that the coefficients, b_k , of Problem 6.7.21 (p. 391) have the property $b_3 = b_5 = b_7 = \dots = 0$. *Cultural note:* The coefficients $k!b_k$ are called the **Bernoulli numbers** and are often denoted by $B_0, B_1, \dots, B_k$.
24. (Continuation) Show that the Bernoulli numbers, B_k , in Problem 6.7.23 (above) have the property

$$\sum_{k=0}^{n-1} \binom{n}{k} B_k = 0 \quad (n > 1)$$

- Show that with $B_0 = 1$, this equation can be used to compute $B_1, B_2, \dots$ and that $B_0 = 1, B_1 = -\frac{1}{2}, B_2 = \frac{1}{6}, B_3 = 0$, and $B_4 = -\frac{1}{30}$.
25. Prove that if r is the radius of convergence of $\sum_{k=0}^{\infty} a_k x^k$, then the radius of the derivative series, $\sum_{k=0}^{\infty} k a_k x^{k-1}$, is also r .
26. Show that

$$\frac{x+1}{x-1} = 1 - 2 \sum_{k=0}^{\infty} x^k \quad (|x| < 1)$$

27. Find a power series about 0 for the function $\csc x - 1/x$.
28. Find the Maclaurin series for the function f defined by the equation

$$-\ln(1-x)f(x) = \frac{x}{1-x}$$

29. Find a series in powers of $(x-1)$ for the function

$$f(x) = \int_1^x \frac{e^t - e}{t-1} dt$$

- Use the resulting series to compute $f(0)$ correct to three significant figures.
30. Establish Equation (3).
31. What is the radius of convergence of the series $\sum_{k=0}^{\infty} k(2x)^k$?
32. What is the radius of convergence of $\sum_{k=0}^{\infty} c_k x^k$ if $c_0 = 1$ and $c_n = (n/3)c_{n-1}$ for $n > 1$?
33. What is the radius of convergence of the series $\sum_{k=0}^{\infty} c_k x^k$, where $c_0 = 1$ and $c_{n+1} = (3n+3)c_n/(n+5)$?
34. Find the first three terms in the power series in x for $e^{\cos x}$. *Hint:* Let $z = 1 - \cos x$.
35. Find a power series for

$$\int_0^x e^{-t} dt$$

36. Show that only ten terms in the series for the sine integral $S(1)$ are needed to obtain 20 decimal places of accuracy. How many are needed for 25 decimal places of accuracy?

COMPUTER
PROBLEM 6.7

1. Write a subprogram or procedure that computes the sine integral with ten decimal places of precision for any x in $[-1, 1]$. Let the number of terms in the series depend on x . Program an error return for x out of range.

6.8 Best Approximation: Least-Squares Theory

One of the classic problems of best approximation can be stated as follows: A continuous function f is defined on an interval $[a, b]$. For a fixed n , we ask for a polynomial p of degree at most n that deviates as little as possible from f . The **deviation**

can be measured by the expression

$$\max_{a \leq x \leq b} |f(x) - p(x)|$$

One recognizes immediately that this is quite different from simply interpolating f at some fixed set of nodes, and it is different from simply truncating the Taylor series of f , if it has one. In fact, we have here an **extremum problem** whose solution is not at all evident.

Existence

The general linear problem of best approximation encompasses the one just described. Here we consider any normed linear space E and a subspace G in E . For any $f \in E$, the distance from f to G is defined to be the quantity

$$\text{dist}(f, G) = \inf_{g \in G} \|f - g\|$$

This measures the absolute minimum deviation that we can hope to achieve in approximating the vector f by an element of G . If an element g of G has the property

$$\|f - g\| = \text{dist}(f, G)$$

then g achieves this minimum deviation and is termed a **best approximation** of f from G . The meaning of *best approximation* thus depends on the norm chosen for the problem.

In the classic problem mentioned above, the normed space E is the space $C[a, b]$ of all continuous functions defined on $[a, b]$, and the norm is

$$\|f\| = \max_{a \leq x \leq b} |f(x)| \quad f \in C[a, b] \tag{1}$$

The subspace G would then be the space Π_n of all polynomials of degree $\leq n$, interpreted as a subspace of $C[a, b]$. Section 6.9 (p. 405) is devoted to this particular type of approximation.

In the general problem of best approximation, one of the main issues is whether a best approximation to f exists. Here is an important existence theorem.

THEOREM 1 Theorem on Existence of Best Approximation

If G is a finite-dimensional subspace in a normed linear space E , then each point of E possesses at least one best approximation in G .

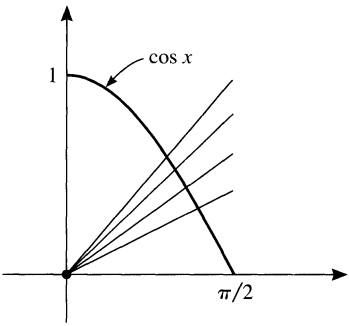
Proof Let f be an element of E . Any candidate, g , for a best approximation to f must compete with the 0 element of G ; hence,

$$\|f - g\| \leq \|f - 0\| = \|f\|$$

We can therefore confine our search to the set

$$K = \{g \in G : \|g - f\| \leq \|f\|\}$$

FIGURE 6.11
Approximating $\cos x$
by λx



The set K is closed and bounded. Because G is finite dimensional, K is compact. Because the function $g \mapsto \|f - g\|$ is continuous, we can invoke the theorem that a continuous real-valued function on a compact set attains its infimum. ■

In general, best approximations are not unique. An example that easily shows this occurs in approximating $f(x) = \cos x$ on the interval $[0, \pi/2]$ by a function $g(x) = \lambda x$, where λ is a constant at our disposal. Figure 6.11 shows several best approximations when the norm in Equation (1) is adopted.

Inner-Product Spaces

Obtaining best approximations for specific E , G , and f can be a difficult task. It usually entails the solution of a *nonlinear* system of equations. There is one important case, however, when we need only to solve some *linear* equations. This is the case in which E is an inner-product space.

Recall that a real inner-product space is a linear space E in which an inner product and a norm have been introduced with the following axioms.

AXIOM 1 Inner Product Axioms

- a. $\langle f, h \rangle = \langle h, f \rangle$
- b. $\langle f, \alpha h + \beta g \rangle = \alpha \langle f, h \rangle + \beta \langle f, g \rangle$
- c. $\langle f, f \rangle > 0$ if $f \neq 0$
- d. $\|f\| = \sqrt{\langle f, f \rangle}$

Two important inner-product spaces are $\mathbb{R}^n$, with

$$\langle x, y \rangle = \sum_{i=1}^n x_i y_i$$

and $C_w[a, b]$, the space of continuous functions on $[a, b]$, with

$$\langle f, g \rangle = \int_a^b f(x)g(x)w(x)dx$$

where w is a fixed continuous positive function. In any inner-product space, we write $f \perp g$ if $\langle f, g \rangle = 0$. We write $f \perp G$ if $f \perp g$ for all $g \in G$.

LEMMA 1 Lemma on Inner-Product Space Properties

In an inner-product space, we have

1. $\langle \sum_{i=1}^n a_i f_i, g \rangle = \sum_{i=1}^n a_i \langle f_i, g \rangle$
2. $\|f + g\|^2 = \|f\|^2 + 2\langle f, g \rangle + \|g\|^2$
3. If $f \perp g$, then $\|f + g\|^2 = \|f\|^2 + \|g\|^2$
4. $|\langle f, g \rangle| \leq \|f\| \|g\|$
5. $\|f + g\|^2 + \|f - g\|^2 = 2\|f\|^2 + 2\|g\|^2$

Proof Part 1 is proved by induction, using Axioms **a** and **b**. Part 2 uses Axioms **a** and **b** and the definition in Part 4:

$$\begin{aligned} \|f + g\|^2 &= \langle f + g, f + g \rangle = \langle f, f \rangle + \langle f, g \rangle + \langle g, f \rangle + \langle g, g \rangle \\ &= \|f\|^2 + 2\langle f, g \rangle + \|g\|^2 \end{aligned}$$

Part 3, the **Pythagorean law**, follows at once from Part 2. Part 4 is the **Schwarz inequality**. To prove it, suppose that it fails for the pair (f, g) :

$$|\langle f, g \rangle| > \|f\| \|g\|$$

Since $\langle f, 0 \rangle = 0$, it is clear that $g \neq 0$. By homogeneity, we can assume $\|g\| = 1$, so that $|\langle f, g \rangle| > \|f\|$. By homogeneity, we can assume $\langle f, g \rangle = 1$, so that $\|f\| < 1$. Then a contradiction arises:

$$0 \leq \|f - g\|^2 = \|f\|^2 - 2\langle f, g \rangle + \|g\|^2 = \|f\|^2 - 1$$

Part 5 is a straightforward computation based on Part 2. ■

Once the Schwarz inequality is available, the triangle inequality for the norm is proved by writing

$$\begin{aligned} \|f + g\|^2 &= \|f\|^2 + 2\langle f, g \rangle + \|g\|^2 \\ &\leq \|f\|^2 + 2\|f\| \|g\| + \|g\|^2 \\ &= (\|f\| + \|g\|)^2 \end{aligned}$$

THEOREM 2 Theorem on Characterizing Best Approximation

Let G be a subspace in an inner-product space E . For $f \in E$ and $g \in G$, these properties are equivalent:

1. g is a best approximation to f in G
2. $f - g \perp G$

Proof If $f - g \perp G$, then for any $h \in G$ we have, by the Pythagorean law,

$$\|f - h\|^2 = \|(f - g) + (g - h)\|^2 = \|f - g\|^2 + \|g - h\|^2 \geq \|f - g\|^2$$

For the converse, suppose that g is a best approximation to f . Let $h \in G$ and $\lambda > 0$. Then

$$\begin{aligned} 0 &\leq \|f - g + \lambda h\|^2 - \|f - g\|^2 \\ &= \|f - g\|^2 + 2\lambda \langle f - g, h \rangle + \lambda^2 \|h\|^2 - \|f - g\|^2 \\ &= \lambda \{2\langle f - g, h \rangle + \lambda \|h\|^2\} \end{aligned}$$

Letting $\lambda \downarrow 0$, we obtain $\langle f - g, h \rangle \geq 0$. The same inequality must be satisfied by $-h$, and therefore $\langle f - g, h \rangle \leq 0$. Hence, $\langle f - g, h \rangle = 0$. Because h was arbitrary in G , $f - g \perp G$. ■

Notice that a consequence of the proof is the *unicity* of a best approximation in the special setting.

Normal Equations

EXAMPLE 1 Use Theorem 2 to determine the best approximation of the function $f(x) = \sin x$ by a polynomial $g(x) = c_1x + c_2x^3 + c_3x^5$ on the interval $[-1, 1]$, using the norm

$$\|f\| = \left\{ \int_{-1}^1 [f(x)]^2 dx \right\}^{1/2}$$

Solution The optimal function g has the property $f - g \perp G$, where G is the space generated by $g_1(x) = x$, $g_2(x) = x^3$, and $g_3(x) = x^5$. Thus, $\langle g - f, g_i \rangle = 0$ is required for $i = 1, 2, 3$. These equations can be written

$$c_1 \langle g_1, g_i \rangle + c_2 \langle g_2, g_i \rangle + c_3 \langle g_3, g_i \rangle = \langle f, g_i \rangle \quad (i = 1, 2, 3)$$

and are called the **normal equations** in this problem. Putting in the details, we have

$$\begin{cases} c_1 \int_{-1}^1 x^2 dx + c_2 \int_{-1}^1 x^4 dx + c_3 \int_{-1}^1 x^6 dx = \int_{-1}^1 x \cdot \sin x dx \\ c_1 \int_{-1}^1 x^4 dx + c_2 \int_{-1}^1 x^6 dx + c_3 \int_{-1}^1 x^8 dx = \int_{-1}^1 x^3 \sin x dx \\ c_1 \int_{-1}^1 x^6 dx + c_2 \int_{-1}^1 x^8 dx + c_3 \int_{-1}^1 x^{10} dx = \int_{-1}^1 x^5 \sin x dx \end{cases}$$

After all the integrals have been worked out numerically, we have this 3×3 system of equations to solve:

$$\begin{bmatrix} \frac{1}{3} & \frac{1}{5} & \frac{1}{7} \\ \frac{1}{5} & \frac{1}{7} & \frac{1}{9} \\ \frac{1}{7} & \frac{1}{9} & \frac{1}{11} \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} \alpha - \beta \\ -3\alpha + 5\beta \\ 65\alpha - 101\beta \end{bmatrix}$$

where $\alpha = \sin 1$ and $\beta = \cos 1$. Using numerical values for α and β , we solve, obtaining $c_1 \approx -0.99998$, $c_2 \approx -0.16652$, and $c_3 \approx 0.00802$. Recall that we discussed the normal equations and numerical difficulties associated with them in Section 5.3 (p. 279). Also, this coefficient matrix is an example of the ill-conditioned Hilbert matrix that was encountered in Section 2.3 (p. 68). Thus, the basis chosen for G is a poor one for numerical work. ■

Orthonormal Systems

Another way of approaching approximation problems in an inner-product space is by means of orthonormal systems. We say that a finite or infinite sequence of vectors $f_1, f_2, \dots$ in an inner-product space is **orthogonal** if

$$\langle f_i, f_j \rangle = 0 \quad (i \neq j)$$

The set is said to be **orthonormal** if for all i and j ,

$$\langle f_i, f_j \rangle = \delta_{ij}$$

Such systems are very well adapted to approximation because of the following result.

THEOREM 3 Theorem on Constructing Best Approximation

Let $\{g_1, g_2, \dots, g_n\}$ be an orthonormal system in an inner-product space E . The best approximation of f by an element $\sum_{i=1}^n c_i g_i$ is obtained if and only if $c_i = \langle f, g_i \rangle$.

Proof Let G be the subspace generated by $g_1, g_2, \dots, g_n$. Notice that the best approximation, $\sum_{i=1}^n c_i g_i$, is characterized, as in Theorem 2, by the condition

$$f - \sum_{i=1}^n c_i g_i \perp G$$

Thus, we need only verify that this condition is equivalent to the condition $c_i = \langle f, g_i \rangle$. Now, $f - \sum_{i=1}^n c_i g_i$ will be orthogonal to G if and only if it is orthogonal to each basis vector g_j . Computing the required inner products, we have (for $1 \leq j \leq n$):

$$\begin{aligned} \left\langle f - \sum_{i=1}^n c_i g_i, g_j \right\rangle &= \langle f, g_j \rangle - \sum_{i=1}^n c_i \langle g_i, g_j \rangle \\ &= \langle f, g_j \rangle - c_j = 0 \end{aligned}$$

The following is the strategy suggested by these considerations: If it is desired to approximate elements of E by elements of a subspace G , first obtain an orthonormal base $\{g_1, g_2, \dots, g_n\}$ for G . Then the best approximation of f is $\sum_{i=1}^n \langle f, g_i \rangle g_i$.

As an illustration of this strategy, we reconsider the previous example

$$\sin x \approx c_1 x + c_2 x^3 + c_3 x^5$$

It is known that an orthonormal base for our three-dimensional subspace is provided by three Legendre polynomials as follows:

$$g_1(x) = x/\sqrt{2/3}$$

$$g_2(x) = (5x^3 - 3x)/(2\sqrt{2/7})$$

$$g_3(x) = (63x^5 - 70x^3 + 15x)/(8\sqrt{2/11})$$

The solution to our problem is then the polynomial $\sum_{i=1}^3 c_i g_i$, where $c_i = \langle f, g_i \rangle$. Thus,

$$c_1 = \sqrt{3/2} \int_{-1}^1 x \cdot \sin x dx = 2\sqrt{3/2} (\alpha - \beta)$$

$$c_2 = \frac{1}{2}\sqrt{7/2} \int_{-1}^1 \sin x (5x^3 - 3x) dx = \sqrt{7/2} (-18\alpha + 28\beta)$$

$$c_3 = \frac{1}{8}\sqrt{11/2} \int_{-1}^1 \sin x (63x^5 - 70x^3 + 15x) dx = \frac{1}{4}\sqrt{11/2} (4320\alpha - 6728\beta)$$

where $\alpha = \sin 1$ and $\beta = \cos 1$. Using numerical values for α and β , we have the approximate solution $c_1 \approx 0.738$, $c_2 \approx -3.37 \times 10^{-2}$, and $c_3 \approx 4.34 \times 10^{-4}$. Since the orthonormal base is already available, these results can be obtained with less effort using this second method. Our basis, being orthonormal, is well conditioned (indeed, optimally so).

Generalized Pythagorean Law and Bessel’s Inequality

The next result is a generalized Pythagorean law.

LEMMA 2 Lemma on Generalized Pythagorean Law

If $[g_1, g_2, \dots, g_n]$ is orthogonal, then

$$\left\| \sum_{i=1}^n a_i g_i \right\|^2 = \sum_{i=1}^n a_i^2 \|g_i\|^2$$

Proof If $n = 1$, it is obvious. If the equation is correct for n , then it is correct for $n + 1$ because $a_{n+1}g_{n+1}$ is orthogonal to $\sum_{i=1}^n a_i g_i$, and the elementary case of the Pythagorean law applies:

$$\begin{aligned} \left\| \sum_{i=1}^{n+1} a_i g_i \right\|^2 &= \left\| \sum_{i=1}^n a_i g_i \right\|^2 + \|a_{n+1} g_{n+1}\|^2 \\ &= \sum_{i=1}^n a_i^2 \|g_i\|^2 + a_{n+1}^2 \|g_{n+1}\|^2 \end{aligned}$$

■

The following result is **Bessel’s inequality**.

LEMMA 3 Lemma on Bessel's Inequality

If $[g_1, g_2, \dots, g_n]$ is orthonormal, then

$$\sum_{i=1}^n |\langle f, g_i \rangle|^2 \leq \|f\|^2$$

Proof Let $g^* = \sum_{i=1}^n \langle f, g_i \rangle g_i$. By Theorem 3, g^* is the best approximation to f in the space G generated by the g_i . By Theorem 2, we have $f - g^* \perp G$. By the Pythagorean law (applied twice),

$$\|f\|^2 = \|f - g^*\|^2 + \|g^*\|^2 \geq \|g^*\|^2 = \sum_{i=1}^n |\langle f, g_i \rangle|^2 \quad \blacksquare$$

The Gram-Schmidt Process

We now turn our attention to the question of how to obtain orthonormal bases. The **Gram-Schmidt process** is described in the following theorem. (Recall that the Gram-Schmidt process was discussed in Chapter 5 for the matrix case.)

THEOREM 4 Theorem on Gram-Schmidt Process

Let $\{v_1, v_2, \dots, v_n\}$ be a basis for a subspace U in an inner-product space. Define recursively

$$u_i = \left\| v_i - \sum_{j=1}^{i-1} \langle v_i, u_j \rangle u_j \right\|^{-1} \left(v_i - \sum_{j=1}^{i-1} \langle v_i, u_j \rangle u_j \right) \quad (i = 1, 2, \dots, n)$$

Then $\{u_1, u_2, \dots, u_n\}$ is an orthonormal base for U .

Proof Denote by U_i the space spanned by $\{v_1, v_2, \dots, v_i\}$. We shall prove by induction on i that:

1. $\{u_1, u_2, \dots, u_i\} \subseteq U_i$
2. $\{u_1, u_2, \dots, u_i\}$ is orthonormal

For $i = 1$, we see at once that $u_1 \in U_1$ and that $\|u_1\| = 1$. Now suppose that Properties 1 and 2 are true for the index $i - 1$. Then $\{u_1, u_2, \dots, u_{i-1}\} \subseteq U_{i-1}$ and $v_i \notin U_{i-1}$. Thus v_i is not a linear combination of $u_1, u_2, \dots, u_{i-1}$, and the denominator in the definition of u_i is not 0. It follows that u_i is well defined and belongs to U_i . Hence, $\{u_1, u_2, \dots, u_i\} \subseteq U_i$. Since $\sum_{j=1}^{i-1} \langle v_i, u_j \rangle u_j$ is the best approximation of v_i in U_{i-1} (by Theorem 3), we have $u_i \perp U_{i-1}$, by Theorem 2. Hence, we have $u_i \perp \{u_1, u_2, \dots, u_{i-1}\}$. Obviously, $\|u_i\| = 1$. ■

A striking simplification occurs when the Gram-Schmidt process is applied to the monomial functions $1, x, x^2, \dots$ (in their natural order). In the following theorem, any inner product can be used as long as it has the property that for any three functions,

$$\langle fg, h \rangle = \langle f, gh \rangle$$

This is obviously valid for the commonly used inner product:

$$\langle f, g \rangle = \int_a^b f(x)g(x)w(x)dx$$

THEOREM 5 Theorem on Orthogonal Polynomials

The sequence of polynomials defined inductively as follows is orthogonal:

$$p_n(x) = (x - a_n)p_{n-1}(x) - b_np_{n-2}(x) \quad (n \geq 2)$$

with $p_0(x) = 1$, $p_1(x) = x - a_1$, and

$$a_n = \langle xp_{n-1}, p_{n-1} \rangle / \langle p_{n-1}, p_{n-1} \rangle$$

$$b_n = \langle xp_{n-1}, p_{n-2} \rangle / \langle p_{n-2}, p_{n-2} \rangle$$

Proof It is clear from the formulas that each p_n is a monic polynomial of degree n and is therefore not 0. Hence, the denominators in the formulas are not 0. Now we show by induction on n that $\langle p_n, p_i \rangle = 0$ for $i = 0, 1, \dots, n-1$. For $n = 0$, there is nothing to prove. For $n = 1$, we note that by the definition of a_1 ,

$$\langle p_1, p_0 \rangle = \langle (x - a_1)p_0, p_0 \rangle = \langle xp_0, p_0 \rangle - a_1 \langle p_0, p_0 \rangle = 0$$

Now assume the validity of our assertion for an index $n-1$, where $n \geq 2$. Then

$$\langle p_n, p_{n-1} \rangle = \langle xp_{n-1}, p_{n-1} \rangle - a_n \langle p_{n-1}, p_{n-1} \rangle - b_n \langle p_{n-2}, p_{n-1} \rangle = 0$$

$$\langle p_n, p_{n-2} \rangle = \langle xp_{n-1}, p_{n-2} \rangle - a_n \langle p_{n-1}, p_{n-2} \rangle - b_n \langle p_{n-2}, p_{n-2} \rangle = 0$$

For any $i = 0, 1, \dots, n-3$, we have

$$\begin{aligned} \langle p_n, p_i \rangle &= \langle xp_{n-1}, p_i \rangle - a_n \langle p_{n-1}, p_i \rangle - b_n \langle p_{n-2}, p_{n-i} \rangle \\ &= \langle p_{n-1}, xp_i \rangle \\ &= \langle p_{n-1}, p_{i+1} + a_{i+1}p_i + b_{i+1}p_{i-1} \rangle = 0 \end{aligned}$$

In the last step, we used the recurrence formula to represent xp_i . If $i = 0$, we should write instead $xp_0 = p_1 + a_1p_0$. ■

EXAMPLE 2 Use Theorem 5 with the inner product $\int_{-1}^1 f(x)g(x)dx$ to obtain the **Legendre polynomials**.

Solution The first few calculations proceed as follows:

$$p_0(x) = 1$$

$$a_1 = \langle xp_0, p_0 \rangle / \langle p_0, p_0 \rangle = 0$$

$$p_1(x) = x$$

$$a_2 = \langle xp_1, p_1 \rangle / \langle p_1, p_1 \rangle = 0$$

$$b_2 = \langle xp_1, p_0 \rangle / \langle p_0, p_0 \rangle = \frac{1}{3}$$

$$p_2(x) = x^2 - \frac{1}{3}$$

The next three Legendre polynomials are

$$p_3(x) = x^3 - \frac{3}{5}x$$

$$p_4(x) = x^4 - \frac{6}{7}x^2 + \frac{3}{35}$$

$$p_5(x) = x^5 - \frac{10}{9}x^3 + \frac{5}{21}x$$

EXAMPLE 3 Show that the **Chebyshev polynomials** form an orthogonal system on $[-1, 1]$ when the following inner product is used:

$$\langle f, g \rangle = \int_{-1}^1 f(x)g(x) \frac{dx}{\sqrt{1-x^2}}$$

Solution By making the change of variable $x = \cos \theta$, we put the inner product into the form

$$\langle f, g \rangle = \int_0^\pi f(\cos \theta)g(\cos \theta)d\theta$$

Since $T_n(x) = \cos(n \cos^{-1} x)$, we have (if $n \neq m$)

$$\begin{aligned} \langle T_n, T_m \rangle &= \int_0^\pi \cos n\theta \cos m\theta d\theta \\ &= \frac{1}{2} \int_0^\pi [\cos(n+m)\theta + \cos(n-m)\theta] d\theta \\ &= \frac{1}{2} \left[\frac{\sin(n+m)\theta}{n+m} + \frac{\sin(n-m)\theta}{n-m} \right]_0^\pi = 0 \end{aligned}$$

Algorithm

If a polynomial is given in the form $u = \sum_{i=0}^n c_i p_i$, where the polynomials p_i are as described in Theorem 5, then the evaluation of $u(x)$ can be accomplished efficiently, as follows:

```

 $d_{n+2} \leftarrow 0; d_{n+1} \leftarrow 0$ 
for  $k = n$  to 0 step  $-1$  do
     $d_k \leftarrow c_k + (x - a_{k+1})d_{k+1} - b_{k+2}d_{k+2}$ 
end do
    
```

Then, $u(x) = d_0$.

Here is the proof that the algorithm is valid:

$$\begin{aligned}
 u(x) &= \sum_{k=0}^n c_k p_k(x) \\
 &= \sum_{k=0}^n [d_k - (x - a_{k+1})d_{k+1} + b_{k+2}d_{k+2}] p_k(x) \\
 &= d_0 p_0(x) + d_1 [p_1(x) - (x - a_1)p_0(x)] \\
 &\quad + \sum_{k=2}^n d_k [p_k(x) - (x - a_k)p_{k-1}(x) + b_k p_{k-2}(x)] \\
 &= d_0
 \end{aligned}$$

THEOREM 6 Theorem on Extremal Property

The polynomials, p_n , described in Theorem 5 have this property: p_n is the monic polynomial of degree n for which $\|p_n\|$ is a minimum.

Proof An arbitrary monic polynomial of degree n can be represented as $p_n - \sum_{i=0}^{n-1} c_i p_i$. The norm of this function will be a minimum if

$$p_n - \sum_{i=0}^{n-1} c_i p_i \perp \Pi_{n-1}$$

This orthogonality relation is achieved by taking all $c_i = 0$ because $p_n \perp \Pi_{n-1}$. ■

If $[u_1, u_2, \dots]$ is an orthonormal system in an inner-product space E , then a family of projection operators, P_n , is defined by the equation

$$P_n f = \sum_{i=1}^n \langle f, u_i \rangle u_i$$

THEOREM 7 Theorem on Orthogonal Projections

The operators, P_n , have these properties:

1. P_n maps E linearly onto the subspace U_n generated by $u_1, u_2, \dots, u_n$.
2. Each P_n is a projection; that is, $P_n^2 = P_n$.
3. Each P_n is an orthogonal map in the sense that $f - P_n f \perp U_n$.
4. $P_n f$ is the best approximation of f in U_n .
5. Each P_n is self-adjoint: $\langle P_n f, g \rangle = \langle f, P_n g \rangle$.

Proof The proof is left to the reader as Problem 6.8.18 (p. 405). ■

The Gram Matrix

Approximation problems in inner-product spaces can be solved in an elementary fashion without using orthonormal bases. If $\{u_1, u_2, \dots, u_n\}$ is any basis for a subspace U , and if we wish to compute a best approximation in U for an element f , we can proceed as follows: In order that an element $u \in U$ be the best approximation to f , it is necessary and sufficient that $u - f \perp U$ by Theorem 2. An equivalent condition is that $\langle u - f, u_i \rangle = 0$ for $1 \leq i \leq n$. On setting $u = \sum_{j=1}^n c_j u_j$, we find the condition

$$\sum_{j=1}^n c_j \langle u_j, u_i \rangle = \langle f, u_i \rangle \quad (1 \leq i \leq n)$$

These are the **normal equations** for the problem. They constitute a system of n linear equations in the n unknowns $c_1, c_2, \dots, c_n$. The coefficient matrix is called a **Gram matrix**; its elements are $G_{ij} = \langle u_i, u_j \rangle$.

LEMMA 4 Lemma on Gram Matrix

If $[u_1, u_2, \dots, u_n]$ is linearly independent, then its Gram matrix is nonsingular.

Proof By the Gram-Schmidt process, we can find an $n \times n$ matrix B such that the vectors

$$v_i = \sum_{s=1}^n B_{is} u_s$$

form an orthonormal set. Then

$$\delta_{ij} = \langle v_i, v_j \rangle = \left\langle \sum_{s=1}^n B_{is} u_s, \sum_{r=1}^n B_{jr} u_r \right\rangle$$

This equation is of the form $I = B G B^T$. It follows that G is nonsingular. ■

It is impossible to resist giving another proof of this result. The approximation problem described above has a unique solution, u , by the remark following Theorem 2. The expression of u in terms of the basis, $u = \sum c_i u_i$, is also unique because this is a property of any basis. It follows that our problem has a unique solution, $(c_1, c_2, \dots, c_n)$. Since this is true for any f , the coefficient matrix G must be nonsingular.

Lemma 4 assures us that the normal equations have a unique solution. Thus, in principle, any basis for the subspace U can be used to solve the approximation problem. In practice, the condition number of the Gram matrix must be considered. Recall the remarks made in Section 2.3 about the Hilbert matrix (p. 68). This is the Gram matrix for the functions $u_j(x) = x^{j-1}$ when the inner product $\langle f, g \rangle = \int_0^1 f(x)g(x)dx$ is used. From the standpoint of conditioning, the most satisfactory bases are the orthonormal ones because their Gram matrices are identity matrices.

PROBLEMS 6.8

- Find the best approximation to $\sin x$ by a function $u(x) = \lambda x$ on the interval $[0, \pi/2]$ using the supremum norm. *Hint:* Draw a sketch. The function $\sin t - \lambda t$ should have a maximum at a point ξ in $(0, \pi/2)$, and it should have a minimum of the same magnitude at $\pi/2$. These conditions should determine ξ and λ .
- (Continuation) Solve the preceding problem using the usual quadratic norm.
- Suppose that we wish to approximate an even function by a polynomial of degree $\leq n$ using the norm $\|f\| = \left\{ \int_{-1}^1 |f(x)|^2 dx \right\}^{1/2}$. Prove that the best approximation is also even. Generalize.
- Let $p_0, p_1, p_2, \dots$ be a sequence of polynomials such that (for each n) p_n has exact degree n . Show that the sequence is linearly independent.
- Prove the **Parseval identity**:

$$\langle f, g \rangle = \sum_{i=1}^n \langle f, u_i \rangle \langle g, u_i \rangle$$

which is valid if f and g are in the span of the orthonormal set $\{u_1, u_2, \dots, u_n\}$.

- Show that the notorious **Hilbert matrix**, with elements

$$a_{ij} = (1 + i + j)^{-1} \quad (0 \leq i, j \leq n)$$

is a Gram matrix for the functions $1, x, x^2, \dots, x^{n-1}$.

- In a vector space with basis $\{v_1, v_2, \dots, v_n\}$, any other basis is obtained by a linear transformation

$$u_j = \sum_{i=1}^n a_{ij} v_i \quad (1 \leq j \leq n)$$

in which the coefficient matrix is nonsingular. Show that the matrix that arises in this way from the Gram-Schmidt process is upper triangular.

- In the three-term recurrence relation for the orthogonal polynomials, assume that the inner product is $\langle f, g \rangle = \int_{-a}^a f(x)g(x)w(x)dx$, where w is an even function. Prove that $a_n = 0$ for all n . Prove that p_n is even if n is even and that p_n is odd if n is odd.
- Let $\{v_1, v_2, \dots, v_n\}$ be an orthogonal set of vectors in an inner-product space. What choice of coefficients produces a minimum value in $\|f - \sum_{i=1}^n c_i v_i\|$? Don't overlook the possibility that some of the v 's may be 0.
- In the algorithm for computing a linear combination of orthogonal polynomials, show that at most $2n - 1$ multiplications are required. In the case of the Chebyshev polynomials, n multiplications suffice.
- In the three-term recurrence formula for orthogonal polynomials, prove that b_n is positive by establishing that $b_n = \|p_{n-1}\|^2 / \|p_{n-2}\|^2$.
- Prove that for the Legendre polynomials, the coefficients in the three-term recurrence are $a_n = 0$ and $b_n = (n-1)^2 / [(2n-1)(2n-3)]$.
- How would the three-term recurrence formula for orthogonal polynomials have to be changed if we desired to produce an *orthonormal* system?
- Using the inner product $\langle u, v \rangle = \int_{-1}^1 u(x)v(x)dx$, let the Gram-Schmidt process be applied to the sequence of functions $x \mapsto (x^2 - 1)x^k$, $k = 0, 1, 2, \dots$. Prove that if the resulting orthonormal sequence is renormalized to form a sequence of *monic* polynomials, then the latter satisfy a three-term recurrence relation of the form

$$q_{n+1}(x) = xq_n(x) - b_nq_{n-1}(x)$$

Give a formula for b_n . Determine the first three q -polynomials.

15. Prove that an orthogonal set of nonzero elements is necessarily linearly independent.
16. Let A be a linear transformation on an inner-product space. Assume that A is **self-adjoint**, which means that $\langle Af, g \rangle = \langle f, Ag \rangle$ for all f and g . Prove that the solutions of the equation $Af = \lambda f$, corresponding to different values of λ , are mutually orthogonal.
17. Let $[u_1, u_2, \dots]$ be an orthonormal sequence in an inner-product space. Prove that for any f in the space, the Fourier coefficients $\langle f, u_n \rangle$ are **square summable**:

$$\sum_{n=1}^{\infty} \langle f, u_n \rangle^2 < \infty.$$

18. Prove Theorem 7. *Suggestion:* Ordering the properties **1, 4, 3, 2, 5** is quite efficient.
19. Let $\tilde{T}_n$ be the monic multiple of T_n . Find the three-term recurrence relation satisfied by $\tilde{T}_0, \tilde{T}_1, \dots$.
20. Find a formula for $\text{dist}(f, G)$, where G is the subspace spanned by an orthonormal set $[g_1, g_2, \dots, g_n]$.
21. Derive these Legendre polynomials:

$$p_3(x) = x^3 - \frac{3}{5}x$$

$$p_4(x) = x^4 - \frac{6}{7}x^2 + \frac{3}{35}$$

$$p_5(x) = x^5 - \frac{10}{9}x^3 + \frac{5}{21}x$$

22. Using Theorem 5 directly, find p_0, p_1, p_2, p_3 for $[a, b] = [0, 1]$ and $w(x) = 1$.
23. (Continuation) Determine p_3 in the form $p_3 = x^3 + Bx^2 + Cx + D$ by making p_3 orthogonal to Π_2 . Verify your results by the use of the preceding problem.
24. Devise an algorithm for computing $\sum_{i=0}^n c_i p_i$ that computes, in order, each partial sum, $\sum_{i=0}^k c_i p_i$, for $k = 0, 1, 2, \dots, n$. Assume that the coefficients a_k and b_k in Theorem 5 are known.

6.9 Best Approximation: Chebyshev Theory

In this section, we work with the space $C(X)$ of all continuous real-valued functions defined on a given topological space X . We assume that X is a compact Hausdorff space. The reader who wishes to avoid considerations of general topology may take X to be a closed and bounded set in the real space $\mathbb{R}^k$, for example, an interval $[a, b]$ in $\mathbb{R}$.

The space $C(X)$ becomes a normed space (indeed, a *Banach* space) if we define the norm to be

$$\|f\| = \max_{x \in X} |f(x)|$$

This norm is used throughout this section.

An important problem of best approximation in the space $C(X)$ is as follows: An element f is given in $C(X)$, and a finite-dimensional subspace G is given in $C(X)$. We want to approximate f as well as possible by an element of G . Hence (as in the preceding section) we define

$$\text{dist}(f, G) = \inf_{g \in G} \|f - g\|$$

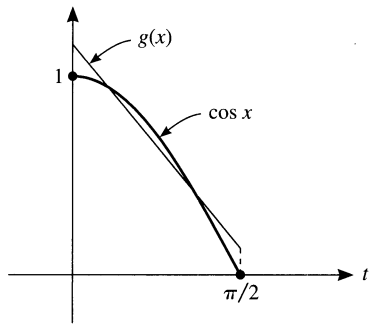


FIGURE 6.12
Approximating $\cos x$ on
the interval $[0, \pi/2]$

and ask whether a *best* approximation exists—that is, an element $g \in G$ such that

$$\|f - g\| = \text{dist}(f, G)$$

This question was answered affirmatively in Section 6.8 (Theorem 1, p. 393). We can therefore address the problem of determining best approximations in this setting. To give some idea of what to expect, let us consider an example.

EXAMPLE 1 Describe the best approximation of the function $f(x) = \cos x$ on the interval $[0, \frac{\pi}{2}]$ by an element of Π_1 .

Solution Elements of Π_1 are linear functions, whose graphs are straight lines. A graph of $\cos x$ and a typical linear function not too far from $\cos x$ is shown in Figure 6.12. Is the linear function shown a *best* approximation? No, it could be lowered, thus reducing the maximum deviation. Also, its slope could be adjusted. For the best approximation, there will be *three* points where the maximum deviation occurs. These will apparently be the points 0 , $\frac{\pi}{2}$, and a point ξ interior to the interval. Denoting the maximum deviation by δ and the linear function by $g(x)$, we have

$$\begin{cases} g(0) - f(0) = \delta \\ g(\xi) - f(\xi) = -\delta \\ g(\pi/2) - f(\pi/2) = \delta \\ g'(\xi) - f'(\xi) = 0 \end{cases}$$

These four equations serve to determine δ , ξ , and the two coefficients in the equation $g(x) = c_1 + c_2x$. ■

Characterizing Best Approximations

Before going on to the *characterization* of best approximations, we make an observation.

LEMMA 1 **Lemma on Best Approximations Properties**
For an element $g \in G$, these properties are equivalent:

1. g is a best approximation to f , from the subspace G .
2. 0 is a best approximation to $f - g$, from the subspace G .

Proof If Property 1 is true, then for all $h \in G$,

$$\|f - g\| \leq \|f - g - h\|$$

This implies that 0 is a best approximation of $f - g$. Conversely, if Property 2 is true, then the preceding inequality is valid for all $h \in G$, and (since $g + h$ can be any element of G) Property 1 is true. ■

Thus, we need to understand exactly which elements f of $C(X)$ have 0 for a best approximation in G ; that is, have the property $\|f\| = \text{dist}(f, G)$. For $f \in C(X)$, we define its **critical set** to be

$$\text{crit}(f) = \{x \in X : |f(x)| = \|f\|\}$$

THEOREM 1 Theorem on Kolmogorov's Characterization

For an element f and a subspace G in $C(X)$, these properties are equivalent:

1. $\|f\| = \text{dist}(f, G)$
2. No element of G has the same signs as f on $\text{crit}(f)$.

Proof Suppose that Property 1 is false. Then for some $g \in G$, $\|f - g\| < \|f\|$. For $x \in \text{crit}(f)$ we put $\sigma(x) = \text{sign } f(x)$ and write

$$\sigma(x)[f(x) - g(x)] \leq |f(x) - g(x)| \leq \|f - g\| < \|f\| = |f(x)| = \sigma(x)f(x)$$

Thus, $\sigma(x)g(x) > 0$, and $g(x)$ has the same sign as $f(x)$ on $\text{crit}(f)$.

Now assume that Property 2 is false. Let $g(x)f(x) > 0$ on $\text{crit}(f)$. There is no loss of generality in supposing that $\|g\| = 1$. Since $\text{crit}(f)$ is a compact set and gf is a continuous function, there is a positive ε such that $g(x)f(x) > \varepsilon$ on $\text{crit}(f)$. Put

$$\mathcal{O} = \{x \in X : g(x)f(x) > \varepsilon\}$$

Then $\mathcal{O}$ is an open set containing $\text{crit}(f)$. Its complement is a compact set *disjoint* from $\text{crit}(f)$. Hence, we have

$$\rho \equiv \max\{|f(x)| : x \in X \setminus \mathcal{O}\} < \|f\|$$

Now we shall try to approximate f by λg , where the coefficient λ is to be judiciously chosen. Let us see what is required. First, for the points in $\mathcal{O}$, we shall want this inequality to be valid pointwise:

$$(f - \lambda g)^2 = f^2 - 2\lambda gf + \lambda^2 g^2 \leq \|f\|^2 - 2\lambda\varepsilon + \lambda^2 = \|f\|^2 - \lambda(2\varepsilon - \lambda) < \|f\|^2$$

This will be true if $0 < \lambda < 2\varepsilon$, as is easily verified. For the remaining points—that is, those in $X \setminus \mathcal{O}$ —we shall want

$$|f - \lambda g| \leq |f| + \lambda|g| \leq \rho + \lambda < \|f\|$$

This is true if $0 < \lambda < \|f\| - \rho$. Thus, if λ is correctly chosen, $\|f - \lambda g\| < \|f\|$, showing that Property 1 is false. ■

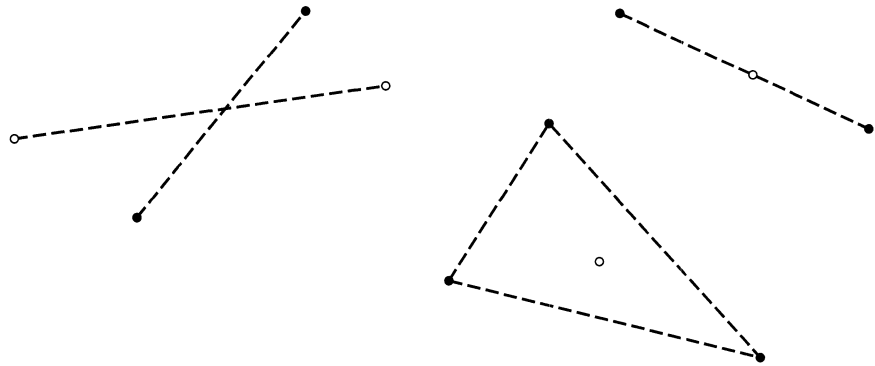


FIGURE 6.13
Critical points in a best
approximation

By using Lemma 1 and the Theorem on Kolmogorov’s Characterization, we obtain the next corollary.

COROLLARY 1 **Corollary 1 on Best Approximation, Necessary and Sufficient Condition**
Let f be an element of $C(X)$, G a subspace of $C(X)$, and g^ an element of G . In order that g^* be a best approximation of f in G , it is necessary and sufficient that no element g in G satisfy $g(x)[f(x) - g^*(x)] > 0$ on the set $\{x : |f(x) - g^*(x)| = \|f - g^*\|\}$.*

COROLLARY 2 **Corollary 2 on Best Approximation, Necessary and Sufficient Condition**
In order that an element $g \in \Pi_1$ be a best approximation to an $f \in C[a, b]$, it is necessary and sufficient that the function $f - g$ assume the value $\pm\|f - g\|$, with alternating sign in at least three points of $[a, b]$.

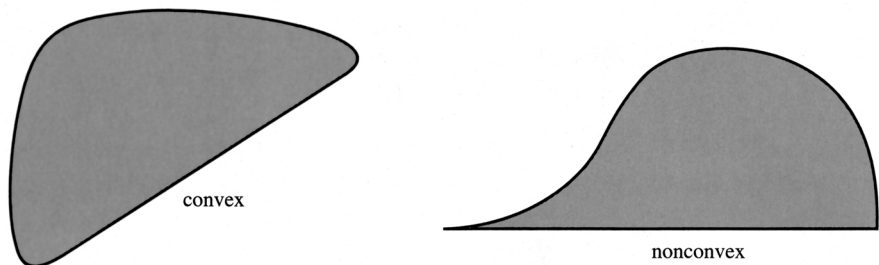
Proof By the Theorem on Kolmogorov’s Characterization applied to $f - g$, the characteristic property is that no element of Π_1 can have the same signs as $f - g$ on $\text{crit}(f - g)$. At least three points in $\text{crit}(f - g)$ must exist at which the values of $f - g$ alternate in sign, because otherwise there would be a point ξ in $[a, b]$ such that the points satisfying $f(x) - g(x) = \|f - x\|$ would be on one side of ξ and the points satisfying $f(x) - g(x) = -\|f - g\|$ would be on the other. Then a linear function vanishing at ξ would exist having the same signs as $f - g$ on $\text{crit}(f - g)$. ■

COROLLARY 3 **Corollary 3 on Best Approximation, Necessary and Sufficient Condition**
Let X be a closed and bounded set in $\mathbb{R}^2$. Let G be the subspace of $C(X)$ consisting of linear functions

$$g(x, y) = a + bx + cy$$

In order that g be a best approximation to an element $f \in C(X)$, it is necessary and sufficient that the critical set of $f - g$ contain one of the three patterns in Figure 6.13.

FIGURE 6.14
A convex set and a
nonconvex set



Convexity

A set K in a linear space is said to be **convex** if it contains every line segment connecting two points of K . Formally, this is expressed by the implication

$$\left. \begin{array}{l} u, v \in K \\ 0 \leq \theta \leq 1 \end{array} \right\} \implies \theta u + (1 - \theta)v \in K$$

A convex set and a nonconvex set in $\mathbb{R}^2$ are shown in Figure 6.14.

Linear combinations of vectors of the form $\sum_{i=1}^k \theta_i u_i$ are called **convex combinations** when $\sum_{i=1}^k \theta_i = 1$ and $\theta_i \geq 0$. The set of all convex combinations of points selected from a given set S is called the **convex hull** of S . Thus, we have

$$\text{co}(S) = \left\{ \sum_{i=1}^k \theta_i u_i : k \in \mathbb{N}, u_i \in S, \theta_i \geq 0, \sum_{i=1}^k \theta_i = 1 \right\}$$

LEMMA 2 Lemma on Closed Convex Set Properties

Let K be a closed convex set in $\mathbb{R}^n$ (or any Hilbert space). Then K contains a unique point of minimum norm. Furthermore, these properties of K are equivalent:

1. $0 \notin K$
2. A vector v exists such that $\langle v, u \rangle > 0$ for all $u \in K$.

Proof Let $\rho = \inf\{\|u\| : u \in K\}$, and select $u_j \in K$ so that $\|u_j\| \rightarrow \rho$. The sequence $[u_j]$ is a Cauchy sequence because the parallelogram law yields

$$\begin{aligned} \|u_i - u_j\|^2 &= 2\|u_i\|^2 + 2\|u_j\|^2 - \|u_i + u_j\|^2 \\ &= 2\|u_i\|^2 + 2\|u_j\|^2 - 4\|(u_i + u_j)/2\|^2 \\ &\leq 2\|u_i\|^2 + 2\|u_j\|^2 - 4\rho^2 \rightarrow 2\rho^2 + 2\rho^2 - 4\rho^2 = 0 \end{aligned}$$

By the completeness of the space, the Cauchy sequence converges, say, $u_j \rightarrow u$. Then $u \in K$ because K is closed. Also $\|u\| = \rho$, by continuity. The uniqueness is proved by an argument like the one above:

If u and u' are two elements of norm ρ in K , then

$$\|u - u'\|^2 \leq 2\|u\|^2 + 2\|u'\|^2 - 4\rho^2 = 0$$

If Property 2 is true, then Property 1 follows at once. For the converse, suppose that Property 1 is true. Let v be the point of least norm in K . We have, for arbitrary $u \in K$ and $\theta \in (0, 1)$,

$$\begin{aligned} 0 &\leq \|\theta u + (1 - \theta)v\|^2 - \|v\|^2 = \|\theta(u - v) + v\|^2 - \|v\|^2 \\ &= \theta^2\|u - v\|^2 + 2\theta\langle u - v, v \rangle \end{aligned}$$

This shows that

$$0 \leq \theta\|u - v\|^2 + 2\langle u - v, v \rangle$$

Letting $\theta \downarrow 0$, we conclude that $\langle u - v, v \rangle \geq 0$, whence $\langle v, u \rangle \geq \langle v, v \rangle > 0$. ■

THEOREM 2 Carathéodory's Theorem

Let S be a subset of an n -dimensional linear space. Each point of the convex hull of S is a convex combination of at most $n + 1$ points of S .

Proof Let p be a point in the convex hull of S . There is no loss of generality in taking p to be 0. Then we have $0 = \sum_{i=1}^k \theta_i u_i$ for appropriate $\theta_i \in [0, 1]$, $u_i \in S$, and $\sum_{i=1}^k \theta_i = 1$. We can assume that a minimal such representation of 0 has been chosen (i.e., k is as small as possible). It follows that $\theta_i > 0$ for $i = 1, \dots, k$. If $k \leq n + 1$, the proof is finished. If $k > n + 1$, then find a nontrivial linear dependence $\sum_{i=2}^k \lambda_i u_i = 0$. Put $\lambda_1 = 0$. Observe that for all real t , $\sum_{i=1}^k (\theta_i + t\lambda_i) u_i = 0$. Let

$$\phi(t) = \min_{1 \leq i \leq k} (\theta_i + t\lambda_i)$$

The function ϕ is continuous and $\phi(0) > 0$. For some t , $\phi(t) < 0$. Hence, there is a particular t_0 for which $\phi(t_0) = 0$. Putting $\theta'_i = \theta_i + t_0\lambda_i$, we have $\theta'_i \geq 0$, $\min \theta'_i = 0$, and $\theta'_1 = \theta_1 > 0$. Hence in the expression $\sum_{i=1}^k \theta'_i u_i = 0$, one term has $\theta'_i = 0$. Dividing by $\sum \theta'_i$, we express 0 as a convex combination of $k - 1$ elements of S , thus contradicting the assumed minimality of k . ■

LEMMA 3 Lemma on Convex Hull of Compact Sets

The convex hull of a compact set in a finite-dimensional normed linear space is compact.

Proof Let S be a compact set in an n -dimensional space X . Consider the set V of all $(2n + 2)$ -tuples

$$v \equiv (\theta_0, \theta_1, \dots, \theta_n, u_0, u_1, \dots, u_n)$$

in which $\theta_i \in \mathbb{R}$, $u_i \in S$, $\theta_i \geq 0$, and $\sum_{i=0}^n \theta_i = 1$. These $(2n + 2)$ -tuples are points in $\mathbb{R}^{n+1} \times X \times X \times \dots \times X$. This space has dimension $n + 1 + (n + 1)n = (n + 1)^2$, and our set V is closed and bounded; hence, V is compact. Define $f(v) = \sum_{i=0}^n \theta_i u_i$. Then f maps V into $\text{co}(S)$. By the Carathéodory's Theorem, f is surjective (onto). It is also continuous. Because the continuous image of a compact set is compact, we can conclude that $\text{co}(S)$ is compact. ■

THEOREM 3 Theorem on Linear Inequalities

For a compact set S in $\mathbb{R}^n$, these properties are equivalent:

1. A point v exists such that $\langle v, u \rangle > 0$ for all $u \in S$.
2. 0 is not in the convex hull of S .

Proof If Property 2 is false, then we can write $0 = \sum_{i=1}^k \theta_i u_i$ for appropriate $\theta_i > 0$ and $u_i \in S$. For any $v \in \mathbb{R}^n$, we have

$$0 = \langle v, 0 \rangle = \sum_{i=1}^k \theta_i \langle v, u_i \rangle$$

It is clear that not all the numbers $\langle v, u_i \rangle$ can be positive. Hence, Property 1 is false.

For the converse, assume that Property 2 is true. By Lemma 3, $\text{co}(S)$ is compact. Lemma 2 can then be applied to conclude that Property 1 is true. ■

Chebyshev Solution of Linear Equations

COROLLARY 4 Corollary 4 on Best Approximation, Necessary and Sufficient Condition

Let A be an $m \times n$ matrix, $b \in \mathbb{R}^m$, $x \in \mathbb{R}^n$, $\sigma_i = \text{sign}(Ax - b)_i$, and $I = \{i : |(Ax - b)_i| = \|Ax - b\|_\infty\}$. In order that x minimize the norm $\|Ax - b\|_\infty$, it is necessary and sufficient that 0 lie in the convex hull of the set $\{\sigma_i A_i : i \in I\}$, where A_i denotes the i th row of A .

Proof In seeking to minimize $\|Ax - b\|_\infty$, we are looking for a linear combination of the columns of A that is as close as possible to b . We interpret all column vectors as functions on the set $T = \{1, 2, \dots, m\}$. The element being approximated is b , and the subspace G of approximants is the column space of A . By the Theorem on Kolmogorov's Characterization, the solution x is characterized by the fact that no element of G has the signs σ_i on I . Equivalently, the system $\sigma_i (Av)_i > 0$, $i \in I$, is inconsistent. Since this can be written $\langle \sigma_i A_i, v \rangle > 0$, $i \in I$, Theorem 3 gives an equivalent condition; namely, $0 \in \text{co}\{\sigma_i A_i : i \in I\}$. ■

EXAMPLE 2 Use Corollary 4 to determine whether $x = (2, 3)^T$ is the Chebyshev solution of the system

$$\begin{cases} 5x_1 - 7x_2 = 14 \\ 3x_1 + x_2 = 8 \\ x_1 - 9x_2 = -23 \\ 3x_1 - 1x_2 = 6 \\ 6x_1 - x_2 = 6 \end{cases}$$

Solution The residuals, $r_i = \langle A_i, x \rangle - b_i$, are 3, 1, -2, -3, 3. Hence, the σ_i are +1, +1, -1, -1, +1. The critical set is $I = \{1, 4, 5\}$.

The corollary asserts that x is a Chebyshev solution if and only if

$$0 \in \text{co}\{\sigma_1 A_1, \sigma_4 A_4, \sigma_5 A_5\}$$

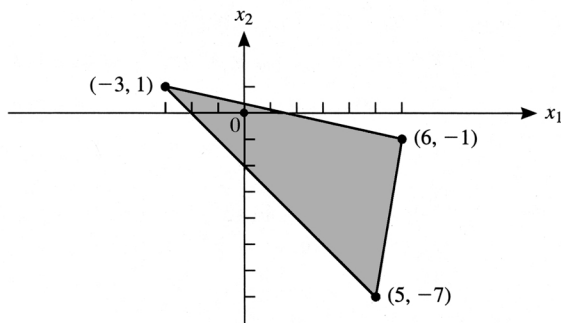


FIGURE 6.15
Zero in the hull of
three points

These three vectors can be quickly plotted to show that this circumstance is true. See Figure 6.15. ■

Further Characterization Theorems

In the next theorems, we use some special notation. If G is an n -dimensional subspace in $C(X)$, and if $\{g_1, g_2, \dots, g_n\}$ is a basis for G , we put

$$\vec{g}(x) = (g_1(x), g_2(x), \dots, g_n(x)) \quad (x \in X)$$

This is interpreted as a point in $\mathbb{R}^n$.

For any $x \in X$, we define a linear functional $\hat{x}$ on $C(X)$ by means of the equation

$$\hat{x}(f) = f(x) \quad (f \in C(X))$$

The linearity of $\hat{x}$ is a consequence of a simple calculation:

$$\hat{x}(\alpha f + \beta g) = (\alpha f + \beta g)(x) = \alpha f(x) + \beta g(x) = \alpha \hat{x}(f) + \beta \hat{x}(g)$$

THEOREM 4 Characterization Theorem

Let G be an n -dimensional subspace in $C(X)$, and let f be an element of $C(X)$. These statements are equivalent:

1. $\|f\| = \text{dist}(f, G)$.
2. No element of G has the same signs as $f(x)$ on $\text{crit}(f)$.
3. 0 lies in the convex hull of the set $\{f(x)\vec{g}(x) : x \in \text{crit}(f)\}$.
4. A functional $\sum_{i=1}^k \lambda_i \hat{x}_i$ exists that annihilates G and satisfies the conditions $x_i \in \text{crit}(f)$, $\lambda_i f(x_i) > 0$, and $k \leq n + 1$.

Proof The implication Property 1 $\Rightarrow$ Property 2 is part of the Theorem on Kolmogorov's Characterization proved earlier.

For the implication Property 2 $\Rightarrow$ Property 3, assume Property 2. Let $\{g_1, g_2, \dots, g_n\}$ be any basis for G . Then we *cannot* find $c_1, c_2, \dots, c_n$ such that

$$\sum_{i=1}^n c_i f(x) g_i(x) > 0$$

on $\text{crit}(f)$. Since this inequality can be written as $\langle c, f(x)\vec{g}(x) \rangle > 0$, the Theorem on Linear Inequalities implies that 0 lies in the convex hull of the point set

$$\{f(x)\vec{g}(x) : x \in \text{crit}(f)\}$$

For the implication Property 3 $\Rightarrow$ Property 4, assume Property 3. By Carathéodory's Theorem, we can write

$$0 = \sum_{i=1}^k \theta_i f(x_i) \vec{g}(x_i)$$

where $k \leq n+1$, $\theta_i > 0$, and $x_i \in \text{crit}(f)$. Putting $\lambda_i = \theta_i f(x_i)$, we have $\lambda_i f(x_i) > 0$ and $0 = \sum_{i=1}^k \lambda_i \vec{g}(x_i)$. From the definition of $\vec{g}$, this last equation shows that for $j = 1, \dots, n$, $0 = \sum_{i=1}^k \lambda_i g_j(x_i) = (\sum_{i=1}^k \lambda_i \hat{x}_i)(g_j)$. Since the functions g_j span G , we see that the functional $\sum_{i=1}^k \lambda_i \hat{x}_i$ annihilates G .

For the implication Property 4 $\Rightarrow$ Property 1, assume Property 4. If $h \in G$, then

$$\|f\| \sum_{i=1}^k |\lambda_i| = \sum_{i=1}^k \lambda_i f(x_i) = \sum_{i=1}^k \lambda_i [f(x_i) - h(x_i)] \leq \|f - h\| \sum_{i=1}^k |\lambda_i|$$

Hence, $\|f\| \leq \|f - h\|$, and Property 1 is true. ■

EXAMPLE 3 Let $X = [0, 1]$ and let G be generated by $g_1(x) = 1 - 2x^2$ and $g_2(x) = x - x^2$. Show that any $f \in C[0, 1]$ that satisfies $\|f\| = f(0) = f(1)$ must have the property $\|f\| = \text{dist}(f, G)$.

Solution Use Part 4 of Theorem 4. For $g \in G$, we notice that $g(0) + g(1) = 0$. (Simply test the two basis functions for this property.) Thus we can take $x_1 = 0$, $x_2 = 1$, $\lambda_1 = 1$, and $\lambda_2 = 1$ in the theorem. ■

Haar Subspaces

DEFINITION 1 Definition of Haar Subspace

An n -dimensional subspace G in $C(X)$ is called a **Haar subspace** if no element of G (except 0) has n or more zeros in X .

EXAMPLE 4 The subspace Π_{n-1} of polynomials having degree $< n$ is a Haar subspace in $C(X)$ whenever X is a subset of $\mathbb{R}$.

The next result shows that Haar subspaces are ideally suited for interpolation problems.

LEMMA 4 Lemma

An n -dimensional subspace G in $C(X)$ is a Haar subspace if and only if for arbitrary real numbers $\lambda_1, \lambda_2, \dots, \lambda_n$ and for arbitrary distinct points $x_1, x_2, \dots, x_n$ in X there is a unique element g in G such that $g(x_i) = \lambda_i$ ($1 \leq i \leq n$).

Proof Select a basis $\{g_1, g_2, \dots, g_n\}$ for G . Then the interpolation problem is to determine $c_1, c_2, \dots, c_n$ so that

$$\sum_{j=1}^n c_j g_j(x_i) = \lambda_i \quad (1 \leq i \leq n)$$

If this is solvable for all choices of λ_i , then the matrix having elements $g_j(x_i)$ is nonsingular, and the corresponding homogeneous problem (with all $\lambda_i = 0$) can have only the 0 solution ($c_i = 0$ for all i). Thus, no linear combination of $g_1, g_2, \dots, g_n$ can vanish at the points $x_1, x_2, \dots, x_n$. This is the Haar property of G because the points $x_1, x_2, \dots, x_n$ are arbitrary. Since the argument is reversible, the proof is complete. ■

For Haar subspaces, best approximations can be described in the following manner.

THEOREM 5 Theorem on Best Approximations in Haar Subspaces

Let G be an n -dimensional Haar subspace in $C(X)$, and let $f \in C(X)$. A necessary and sufficient condition for $\|f\| = \text{dist}(f, G)$ is that a functional of the form $\sum_{i=1}^{n+1} \lambda_i \widehat{x}_i$ exists that annihilates G and satisfies $x_i \in \text{crit}(f)$ and $\lambda_i f(x_i) > 0$.

Proof Theorem 4 gives this same necessary and sufficient condition with $n + 1$ replaced by k . The equation $\sum_{i=1}^k \lambda_i g(x_i) = 0$ cannot be valid nontrivially unless $k \geq n + 1$. To verify this, let $k \leq n$ and use the interpolating property of G (Lemma 4) to select $g \in G$ such that $g(x_i) = \lambda_i$. Consequently, we would have $0 = \sum_{i=1}^k \lambda_i g(x_i) = \sum_{i=1}^k \lambda_i^2$. ■

Unicity of Best Approximations

THEOREM 6 Theorem on Strong Unicity

Let G be a finite-dimensional Haar subspace in $C(X)$, and let f be an element of $C(X)$ such that $\|f\| = \text{dist}(f, G)$. Then a positive constant γ (depending on f) exists such that for all $g \in G$, $\|f + g\| \geq \|f\| + \gamma \|g\|$.

Proof Let n be the dimension of G . By Theorem 5 there exist points $x_0, x_1, \dots, x_n$ in $\text{crit}(f)$ and positive coefficients θ_i such that $\sum_{i=0}^n \theta_i \sigma_i g(x_i) = 0$ for all $g \in G$, and $\sigma_i = \text{sign } f(x_i)$. Let h be an element of norm 1 in G . Since $\sum_{i=0}^n \theta_i \sigma_i h(x_i) = 0$, at least one of the numbers $\sigma_i h(x_i)$ is positive. Thus, $\max_i \sigma_i h(x_i) > 0$. Since this expression is continuous and the surface of the unit cell of G is compact, we infer that

$$\gamma \equiv \inf_h \max_i \sigma_i h(x_i) > 0$$

If $g \in G$ and $g \neq 0$, then for some i , $\sigma_i g(x_i)/\|g\| \geq \gamma$. Hence, we have

$$\|f + g\| \geq \sigma_i f(x_i) + \sigma_i g(x_i) \geq \|f\| + \gamma \|g\|$$
 ■

COROLLARY 5 Corollary

If G is a finite-dimensional Haar subspace in $C(X)$, then each element of $C(X)$ has a unique best approximation in G .

Proof Let g be a best approximation to an element f in $C(X)$. Then $f - g$ has 0 for a best approximation in G . If $h \in G$ and $h \neq 0$, then

$$\|f - g + h\| \geq \|f - g\| + \gamma\|h\| > \|f - g\|$$

by the Theorem on Strong Unicity. ■

THEOREM 7 Theorem on Continuity

Let G be a finite-dimensional Haar subspace in $C(X)$. Let $A : C(X) \rightarrow G$ be the map such that $\|f - Af\| = \text{dist}(f, G)$. Then for each f there is a positive number $\lambda(f)$ such that

$$\|Af - Ah\| \leq \lambda(f)\|f - h\| \quad h \in C(X)$$

Proof By Theorem 6 on strong unicity, a positive number $\gamma(f)$ exists such that for all $g \in G$,

$$\|f - g\| \geq \|f - Af\| + \gamma(f)\|Af - g\|$$

Letting $g = Ah$, we then have

$$\begin{aligned} \gamma(f)\|Af - Ah\| &\leq \|f - Ah\| - \|f - Af\| \\ &\leq \|f - h\| + \|h - Ah\| - \|f - Af\| \\ &\leq \|f - h\| + \|h - Af\| - \|f - Af\| \\ &\leq \|f - h\| + \|h - f\| + \|f - Af\| - \|f - Af\| \\ &= 2\|f - h\| \end{aligned}$$

■

Chebyshev's Alternation Theorem**LEMMA 5** Lemma

Let G be an n -dimensional Haar subspace in $C[a, b]$. Suppose that we have $n + 1$ points such that $a \leq x_0 < x_1 < \cdots < x_n \leq b$ and that $\sum_{i=0}^n \lambda_i g(x_i) = 0$ for every $g \in G$. If $\sum_{i=0}^n |\lambda_i| \neq 0$, then the λ 's alternate in sign: $\lambda_i \lambda_{i-1} < 0$ for $i = 1, 2, \dots, n$.

Proof For any $j \in 1, 2, \dots, n$ we carry out the following argument: By Lemma 2, a unique element g_j exists in G having this interpolation property:

$$\begin{cases} g_j(x_i) = 0 & \text{for } 0 \leq i \leq j-2 \text{ and for } j+1 \leq i \leq n \\ g_j(x_j) = 1 \end{cases}$$

Then we have

$$0 = \sum_{i=0}^n \lambda_i g_j(x_i) = \lambda_{j-1} g_j(x_{j-1}) + \lambda_j \quad (1)$$

Now $g_j(x_{j-1}) > 0$ because if $g_j(x_{j-1}) = 0$ or if $g_j(x_{j-1}) < 0$, we can conclude that g_j has n zeros, contrary to the Haar property. Equation (1) now shows that if any λ_j is 0, then all are 0, contrary to the hypothesis. Then Equation (1) shows that λ_j and λ_{j-1} must have opposite signs. ■

THEOREM 8 Theorem on Chebyshev's Alternation

Let G be an n -dimensional Haar subspace in $C[a, b]$, and let X be a closed set in $[a, b]$. For an element $f \in C(X)$, these properties are equivalent:

1. $\|f\| = \text{dist}(f, G)$; that is, f has 0 as a best approximation in G .
2. Points $x_0 < x_1 < \cdots < x_n$ in X exist such that $f(x_{i-1})f(x_i) = -\|f\|^2$ for $1 \leq i \leq n$.

Proof By Theorem 7, Property 1 is equivalent to

3. There exist points $x_0, \dots, x_n$ in $\text{crit}(f)$ and coefficients $\lambda_0, \dots, \lambda_n$ such that

$$x_0 < x_1 < \cdots < x_n, \quad \sum_{i=0}^n \lambda_i g(x_i) = 0 \text{ for all } g \in G, \text{ and } \lambda_i f(x_i) > 0$$

If Property 3 is true, then by Lemma 5 the coefficients λ_i must alternate in sign. Since $\lambda_i f(x_i) > 0$, the numbers $f(x_i)$ must alternate in sign. Hence, Property 2 is true.

Conversely, if Property 2 is true, it is clear that each x_i is a critical point of f , and the signs of $f(x_i)$ alternate. Suppose for definiteness that $\text{sign } f(x_i) = (-1)^i$. If 1 is false, then for some $g \in G$, $\|f - g\| < \|f\|$. Hence, we have

$$(-1)^i [f(x_i) - g(x_i)] \leq \|f - g\| < \|f\| = (-1)^i f(x_i)$$

Thus $(-1)^i g(x_i) > 0$, and consequently g must have at least n zeros. This contradicts the Haar property of G . Notice that these zeros of g are certainly in $[a, b]$ but not necessarily in X . That is why the Haar condition is needed on $[a, b]$. ■

COROLLARY 6 Corollary on Chebyshev's Alternation

Let G be an n -dimensional Haar subspace in $C[a, b]$. Let $f \in C[a, b]$ and $g \in G$. In order that g be the best approximation of f in G , it is necessary and sufficient that there exist points $x_0 < x_1 < \cdots < x_n$ in $[a, b]$ such that

$$f(x_i) - g(x_i) = (-1)^i c \|f - g\| \quad (0 \leq i \leq n, \quad |c| = 1)$$

Algorithms

In the remainder of this section, we discuss the numerical problems of Chebyshev approximation and outline some of the algorithms available. We take as a basic

problem the minimization of an expression

$$\Delta(c) = \left\| f - \sum_{i=1}^n c_i g_i \right\|_{\infty} = \sup_{x \in X} \left| f(x) - \sum_{i=1}^n c_i g_i(x) \right|$$

In this problem, the functions $f, g_1, g_2, \dots, g_n$ have been prescribed in $C(X)$, and we wish to determine the coefficient vectors $c = (c_1, c_2, \dots, c_n)$ that make $\Delta(c)$ as small as possible.

One of the basic algorithms for this problem is an iterative one that requires in each step the solution of a similar problem of more elementary nature. This algorithm is known as the **Remez first algorithm**; we describe it now.

Let us denote by G the subspace generated by $\{g_1, g_2, \dots, g_n\}$. At the k th step of the algorithm, a finite subset X_k is given in X . This finite set is used to define a seminorm in $C(X)$ via the equation

$$\|f\|_k = \max_{x \in X_k} |f(x)|$$

At this step it is necessary to determine (by methods to be discussed later) an element h_k in G to minimize the seminorm

$$\|f - h_k\|_k$$

This is another Chebyshev approximation problem, but it involves only the finite point set X_k . Next, a point x_k is selected in X so that

$$|f(x_k) - h_k(x_k)| = \|f - h_k\|$$

This point is now adjoined to the set X_k to form the next set X_{k+1} , and the process is repeated. The vector h_k obtained at the k th step is usually a good starting point in the search for h_{k+1} in the next step.

THEOREM 9 Theorem on Remez Algorithm

If the initial seminorm, $\|\cdot\|_1$, in the Remez first algorithm is a true norm on G , then $\lim_{k \rightarrow \infty} \|f - h_k\| = \text{dist}(f, G)$. The sequence $[h_k]$ has cluster points, and each of these is a best approximation to f .

Proof Directly from the definitions of the norms and the inclusions $X_1 \subseteq X_2 \subseteq \dots$, we have, for $1 \leq k \leq i$ and $g \in G$,

$$\|f - g\|_1 \leq \|f - g\|_k \leq \|f - g\|_i \leq \|f - g\|$$

From the preceding inequality we infer that

$$\|f - h_k\|_1 \leq \|f - h_k\|_k \leq \|f - h_i\|_i \leq \text{dist}(f, G)$$

This shows that the sequence $\|h_k\|_1$ is bounded. Since all norms are equivalent on the finite-dimensional space G , the sequence $\|h_k\|$ is bounded. Hence, some subsequence of the h_k 's converges to a point h^* . Given $\varepsilon > 0$, let k be chosen so that $\|h_k - h^*\| < \varepsilon$. Select $i > k$ so that $\|h_i - h^*\| < \varepsilon$. Then we have

$$\begin{aligned} \text{dist}(f, G) &\leq \|f - h^*\| \leq \|f - h_k\| + \|h_k - h^*\| \\ &\leq |f(x_k) - h_k(x_k)| + \varepsilon \end{aligned}$$

$$\begin{aligned}
&\leq \|f - h_k\|_i + \varepsilon \\
&\leq \|f - h_i\|_i + \|h_i - h^*\|_i + \|h^* - h_k\|_i + \varepsilon \\
&\leq \text{dist}(f, G) + 3\varepsilon
\end{aligned}$$

Since ε was arbitrary, we conclude that $\|f - h^*\| = \text{dist}(f, G)$. Thus any cluster point of the sequence $[h_k]$ is a best approximation to f in G .

It remains to be proved that the sequence $d_k = \|f - h_k\|$ converges to $\text{dist}(f, G)$. This sequence is bounded and has convergent subsequences. Let $d_{k_i} \rightarrow d^*$. Let h' now denote a cluster point of the subsequence $[h_{k_i}]$. By the first part of our proof, $d^* = \|f - h'\| = \text{dist}(f, G)$. This shows that the sequence $[d_k]$ has only one cluster point; namely, $\text{dist}(f, G)$. Consequently, we have the convergence $d_k \rightarrow \text{dist}(f, G)$. ■

It is useful to note that at each step of the Remez first algorithm, an upper and a lower bound are easily computed for the unknown number $d^* = \text{dist}(f, G)$. In fact, we have

$$\|f - h_k\|_k \leq d^* \leq \min_{1 \leq i \leq k} \|f - h_i\|$$

The lower bound converges monotonically upward to d^* , whereas the upper bound converges monotonically downward to d^* . The simpler upper bound $\|f - h_k\|$ also converges to d^* , but not always monotonically.

Next we discuss a variation of this algorithm, called the **exchange method**, in which the sets X_k are restrained from growing larger by deleting one *old* element whenever a *new* element is added. In practical problems, this method is usually used, although its validity depends on further assumptions being made concerning the basic functions $g_1, g_2, \dots, g_n$. The process of deleting one element and adding another is called an **exchange**. If the initial subset X_1 contains $n + 1$ elements, and if an exchange occurs in every step of the algorithm, then each X_k will contain exactly $n + 1$ elements.

From the characterization theorem for best approximations, we know that at the k th step, the origin of $\mathbb{R}^n$ will lie in the convex hull of the $n + 1$ vectors

$$e_k(x)(g_1(x), g_2(x), \dots, g_n(x)) \quad x \in X_k \quad (2)$$

where $e_k(x) = f(x) - h_k(x)$. With h_k in hand, the point previously referred to as x_k is chosen. That is, x_k is a critical point of the function $f - h_k$. Now one of the points in X_k is *replaced* by x_k . This exchange is done in such a way that the origin of $\mathbb{R}^n$ remains in the convex hull of the vectors described in (2). Thus, the vector

$$e_k(x_k)(g_1(x_k), g_2(x_k), \dots, g_n(x_k))$$

will replace one of the previous set of $n + 1$ vectors. The theorem that governs the exchange and reveals how to do it is presented next.

THEOREM 10 Exchange Theorem

Let $\{u_0, u_1, \dots, u_{n+1}\}$ be a set of $n + 2$ points in $\mathbb{R}^n$. If 0 lies in the convex hull of $\{u_0, u_1, \dots, u_n\}$, then for some $k \leq n$ this assertion remains true when u_k is replaced by u_{n+1} .

Proof If it is possible to write $0 = \sum_{i=0}^n \theta_i u_i$ with $\theta_i \geq 0$, $\sum_{i=0}^n \theta_i = 1$, and $\min_i \theta_i = 0$, then do so and select k so that $\theta_k = 0$. Clearly we can replace u_k by u_{n+1} in the equation $0 = \sum_{i=0}^n \theta_i u_i$.

Suppose that it is *not* possible to write $0 = \sum_{i=0}^n \theta_i u_i$ with $\theta_i \geq 0$, $\sum_{i=0}^n \theta_i = 1$, and $\min_i \theta_i = 0$. By Carathéodory's Theorem, the vectors $u_0, u_1, \dots, u_n$ must span an n -dimensional space, which of course must be $\mathbb{R}^n$. Hence, we can find $\lambda_0, \dots, \lambda_n$ such that $u_{n+1} = \sum_{i=0}^n \lambda_i u_i$. Also, there exist $\theta_i > 0$ such that $0 = \sum_{i=0}^n \theta_i u_i$. Select k so that $\lambda_k/\theta_k \geq \lambda_i/\theta_i$ for all i . For $0 \leq i \leq n$, put $\theta'_i = \lambda_k \theta_i - \lambda_i \theta_k$. Also, let $\theta'_{n+1} = \theta_k$. Observe that $\theta'_k = 0$. Now we have

$$\begin{aligned} \sum_{i=0}^{n+1} \theta'_i u_i &= \sum_{i=0}^n \theta'_i u_i + \theta'_{n+1} u_{n+1} = \sum_{i=0}^n (\lambda_k \theta_i - \lambda_i \theta_k) u_i + \theta_k u_{n+1} \\ &= \lambda_k \sum_{i=0}^n \theta_i u_i - \theta_k \sum_{i=0}^n \lambda_i u_i + \theta_k u_{n+1} \\ &= \lambda_k (-\theta_k u_k) - \theta_k (u_{n+1} - \lambda_k u_k) + \theta_k u_{n+1} = 0 \end{aligned}$$

Furthermore, $\theta'_i \geq 0$ because $\theta'_{n+1} = \theta_k > 0$, and for $i \leq n$,

$$\theta'_i = \theta_i \theta_k (\lambda_k/\theta_k - \lambda_i/\theta_i) \geq 0$$

If we divide the equation $\sum_{i=0}^n \theta'_i u_i = 0$ by $\sum_{i=0}^n \theta'_i$, we see that 0 is expressed as a convex combination of $u_0, \dots, u_{n+1}$, with u_k omitted. ■

PROBLEMS 6.9

- 1. Solve for c_1, c_2, δ , and ξ in Example 1 of this section.
- 2. Find the best approximation of $\sqrt{x}$ by a first-degree polynomial on the interval $[0, 1]$.
- 3. Show that the subspaces in $C[0, 1]$ spanned by these sets are Haar subspaces:
a. $\{1, x^2, x^3\}$ b. $\{1, e^x, e^{2x}\}$ c. $\{(x+2)^{-1}, (x+3)^{-1}, (x+4)^{-1}\}$
- 4. Show that the subspaces in $C[-1, 1]$ spanned by these sets are not Haar subspaces:
a. $\{1, x^2, x^3\}$ b. $\{|x|, |x-1|\}$ c. $\{e^x, x+1\}$
- 5. In the space $C[0, 1]$, consider the subspace Π_0 of polynomials of degree 0 (i.e., constant functions). Using the quantities

$$M(f) = \max_{0 \leq x \leq 1} f(x) \quad \text{and} \quad m(f) = \min_{0 \leq x \leq 1} f(x)$$

describe the best approximation of an element f in $C[0, 1]$ by an element of Π_0 .

- 6. Let $u_0, u_1, \dots, u_n$ be $n+1$ points in $\mathbb{R}^n$ such that it is not possible to write $0 = \sum_{i=0}^n \theta_i u_i$ with $\theta \geq 0$, $\theta \neq 0$, and $\min_i \theta_i = 0$. Show that each set of n vectors chosen from $\{u_0, u_1, \dots, u_n\}$ is a basis for $\mathbb{R}^n$ or give a counterexample.
- 7. Let A be an $n \times (n+2)$ matrix. Show that if the system

$$Ax = 0 \quad x \geq 0 \quad x \neq 0 \quad x_{n+2} = 0 \quad x \in \mathbb{R}^{n+2}$$

is consistent, then for some $k < n + 2$, so is the system

$$Ax = 0 \quad x \geq 0 \quad x \neq 0 \quad x_k = 0 \quad x \in \mathbb{R}^{n+2}$$

8. Prove that the quadratic polynomial of best approximation to the function $\cosh x$ on the interval $[-1, 1]$ is $a + bx^2$, where $b = \cosh 1 - 1$ and a is obtained by solving this pair of equations simultaneously for a and t :

$$2a = 1 + \cosh t - t^2b$$

$$\sinh t = 2tb$$

9. Prove that the convex hull of a set is convex and that it is the smallest convex set containing the original set.
10. Prove that in a normed linear space every closed ball $\{f : \|f - g\| \leq r\}$ is convex.
11. Give an example of a convex set whose complement is bounded.
12. Given a line in the plane, $ax + by + c = 0$, with $a^2 + b^2 > 0$, provide a complete description of the set of points on the line whose distance from the origin is a minimum, using the ℓ_∞ -norm to define distance.
13. Let f be a continuous function of (x, y) in the square $0 \leq x \leq 1, 0 \leq y \leq 1$. Describe the best approximation of f by a continuous function of x alone.
14. Do the three functions

$$g_0(x, y) = 1 \quad g_1(x, y) = x \quad g_2(x, y) = y$$

generate a Haar subspace in $C(\mathbb{R}^2)$?

15. Prove that the set of functions $\{1, x, x^2, \dots, x^{n-1}, f\}$ generates a Haar subspace on $[a, b]$ if $f^{(n)}(x) > 0$ on $[a, b]$.
16. Let $f = a_0T_0 + a_1T_1 + \dots + a_{n+1}T_{n+1}$, where the T_k are Chebyshev polynomials. Prove that the best approximation of f in the sup-norm on $[-1, 1]$ in the space Π_n is $a_0T_0 + a_1T_1 + \dots + a_nT_n$.

6.10 Interpolation in Higher Dimensions

The problem of finding smooth interpolants for functions of several variables is a difficult one that has attracted much attention, both in the past and currently. The multivariate case shows some unusual features that are not present in the univariate case, and these features are already apparent when the number of variables is only two. Therefore, very little is lost in restricting the discussion to the *bivariate* case (two independent variables), at least at the beginning.

Interpolation Problem

The central problem that we shall discuss is as follows: A set of interpolation points (or *nodes*) is given in the xy -plane. These can be denoted by

$$(x_1, y_1) \quad (x_2, y_2) \quad \dots \quad (x_n, y_n) \tag{1}$$

We assume that these n points are *distinct*. With each point (x_i, y_i) there is associated a real number, c_i , and our objective is to find a smooth and easily computed function

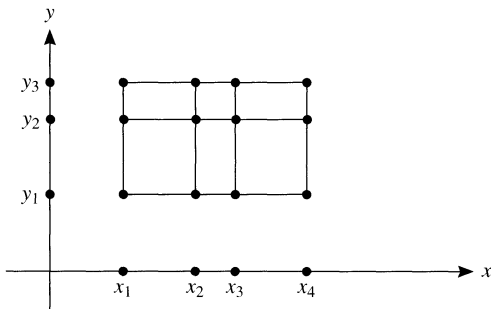


FIGURE 6.16
A Cartesian grid of nodes

F such that

$$F(x_i, y_i) = c_i \quad (1 \leq i \leq n)$$

It is understood that F will be defined on all of $\mathbb{R}^2$, or at least on some large domain that includes the nodes. In the preceding description, the terms *smooth* and *easily computed* have only informal or intuitive meanings.

Cartesian Product and Grid

The interpolation problem just described can sometimes be solved by a *tensor product* of univariate interpolation methods. Let us dispose of this case first. The procedure is limited to the situation in which the set of nodes, denoted here by $\mathcal{N}$, is a **Cartesian product**:

$$\mathcal{N} = \{x_1, x_2, \dots, x_p\} \times \{y_1, y_2, \dots, y_q\}$$

Thus, $\mathcal{N}$ is the set of *all* pairs (x_i, y_j) , where x_i is chosen from the first set and y_j from the second. In other words,

$$\mathcal{N} = \{(x_i, y_j) : 1 \leq i \leq p, 1 \leq j \leq q\} \quad (2)$$

Observe that for this case, the notation in Equation (2) is more convenient than the notation in Equation (1). An example of Equation (2), in which $p = 4$ and $q = 3$, is shown in Figure 6.16.

Such an array of nodes is often called a **Cartesian grid**. For convenience, we have numbered the x -points from left to right and the y -points from bottom to top, although this is not necessary.

Suppose that we have a linear interpolation scheme for nodes $x_1, x_2, \dots, x_p$. This will be a **univariate process**. We want to think of this as a linear operator P of the form

$$(Pf)(x) = \sum_{i=1}^p f(x_i)u_i(x) \quad (3)$$

in which the functions u_i have the **cardinal property**

$$u_i(x_j) = \delta_{ij} \quad (1 \leq i, j \leq p) \quad (4)$$

For example, in ordinary polynomial interpolation, these functions u_i are given by a familiar formula from Section 6.1 (p. 312):

$$u_i(x) = \prod_{\substack{j=1 \\ j \neq i}}^p \frac{x - x_j}{x_i - x_j} \quad (1 \leq i \leq p) \quad (5)$$

Notice that the operator P can be extended in a trivial manner to operate on functions of *two* or more variables. Thus, if f is a function of (x, y) , we can write

$$(\overline{P}f)(x, y) = \sum_{i=1}^p f(x_i, y)u_i(x) \quad (6)$$

One can see immediately that $\overline{P}f$ is a function of two variables that interpolates f on the *vertical* lines

$$L_i \equiv \{(x_i, y) : -\infty < y < \infty\} \quad (1 \leq i \leq p) \quad (7)$$

Suppose that another operator is available for interpolation at the nodes $y_1, y_2, \dots, y_q$. We write

$$(Qf)(y) = \sum_{i=1}^q f(y_i)v_i(y) \quad (8)$$

where the functions v_i are any convenient ones having the cardinal property

$$v_i(y_j) = \delta_{ij} \quad (1 \leq i, j \leq q) \quad (9)$$

Again, Q can be extended to operate on bivariate functions by use of the equation

$$(\overline{Q}f)(x, y) = \sum_{i=1}^q f(x, y_i)v_i(y) \quad (10)$$

The function $\overline{Q}f$ interpolates f on all the *horizontal* lines

$$L^i \equiv \{(x, y_i) : -\infty < x < \infty\} \quad (1 \leq i \leq q) \quad (11)$$

Boolean Sum

Two useful bivariate interpolation operators can now be constructed from $\overline{P}$ and $\overline{Q}$; they are the **product** $\overline{P}\overline{Q}$ and the **Boolean sum** $\overline{P} \oplus \overline{Q}$, defined by

$$\overline{P} \oplus \overline{Q} = \overline{P} + \overline{Q} - \overline{P}\overline{Q} \quad (12)$$

More detailed formulas for these operators are easily derived from the definitions of $\overline{P}$ and $\overline{Q}$. Thus,

$$\begin{aligned} (\overline{P}\overline{Q}f)(x, y) &= \overline{P}(\overline{Q}f)(x, y) = \sum_{i=1}^p (\overline{Q}f)(x_i, y)u_i(x) \\ &= \sum_{i=1}^p \sum_{j=1}^q f(x_i, y_j)v_j(y)u_i(x) \end{aligned} \quad (13)$$

Since $v_j(y_k)u_i(x_\ell) = \delta_{jk}\delta_{i\ell}$, we see without difficulty that $\overline{PQ}f$ is a function that interpolates f at all the nodes (x_i, y_j) . The **tensor product** notation $P \otimes Q$ is also used for the operator $\overline{PQ}$.

In the same way, a formula for $\overline{P} \oplus \overline{Q}$ is

$$\begin{aligned} [(\overline{P} \oplus \overline{Q})f](x, y) &= (\overline{P}f)(x, y) + (\overline{Q}f)(x, y) - (\overline{PQ}f)(x, y) \tag{14} \\ &= \sum_{i=1}^p f(x_i, y)u_i(x) + \sum_{j=1}^q f(x, y_j)v_j(y) \\ &\quad - \sum_{i=1}^p \sum_{j=1}^q f(x_i, y_j)u_i(x)v_j(y) \end{aligned}$$

It is left as Problem 6.10.6 (p. 437) to prove that the function $(\overline{P} \oplus \overline{Q})f$ interpolates f on all the horizontal and vertical lines L_i ($1 \leq i \leq p$) and L^j ($1 \leq j \leq q$).

EXAMPLE 1 Give a formula for a polynomial in two variables that assumes the following values:

(x, y)	$\parallel$	(1,1)	(2,1)	(4,1)	(5,1)	(1,3)	(2,3)	(4,3)(5,3)	(1,4)	(2,4)	(4,4)	(5,4)
$f(x, y)$	$\parallel$	1.7	−4.1	−3.2	4.9	6.1	−4.2	2.3 7.5	−5.9	3.8	−1.7	2.5

Solution Observe first that the nodes form a Cartesian grid, and the tensor product method is applicable. The functions u_i and v_i are given by Equation (5). In this example, they are as follows:

$$\begin{aligned} u_1(x) &= \frac{x-2}{1-2} \cdot \frac{x-4}{1-4} \cdot \frac{x-5}{1-5} = -\frac{1}{12}(x-2)(x-4)(x-5) \\ u_2(x) &= \frac{1}{6}(x-1)(x-4)(x-5) \\ u_3(x) &= -\frac{1}{6}(x-1)(x-2)(x-5) \\ u_4(x) &= \frac{1}{12}(x-1)(x-2)(x-4) \\ v_1(y) &= \frac{y-3}{1-3} \cdot \frac{y-4}{1-4} = \frac{1}{6}(y-3)(y-4) \\ v_2(y) &= -\frac{1}{2}(y-1)(y-4) \\ v_3(y) &= \frac{1}{3}(y-1)(y-3) \end{aligned}$$

The polynomial interpolant is then

$$\begin{aligned} F(x, y) &= u_1(x)[1.7v_1(y) + 6.1v_2(y) - 5.9v_3(y)] \\ &\quad + u_2(x)[-4.1v_1(y) - 4.2v_2(y) + 3.8v_3(y)] \\ &\quad + u_3(x)[-3.2v_1(y) + 2.3v_2(y) - 1.7v_3(y)] \\ &\quad + u_4(x)[4.9v_1(y) + 7.5v_2(y) + 2.5v_3(y)] \end{aligned} \tag{15}$$



Tensor Product

If the function F in Example 1 is written as a sum of terms $x^i y^j$, the following twelve terms appear:

$$1, x, x^2, x^3, y, xy, x^2y, x^3y, y^2, xy^2, x^2y^2, x^3y^2 \quad (16)$$

Thus, we are interpolating by means of a 12-dimensional subspace of bivariate polynomials. The proper notation for this subspace is $\Pi_3 \otimes \Pi_2$. This is the **tensor product** of two linear spaces, and it consists of all functions of the form

$$(x, y) \mapsto \sum_{i=1}^m a_i(x)b_i(y)$$

in which $a_i \in \Pi_3$ and $b_i \in \Pi_2$. (The sum can have any number of terms.) It is not difficult to prove that a basis for this space consists of the functions in (16).

It is to be emphasized that the theory just outlined applies to general functions u_i and v_i , not just to polynomials. All that is needed is the cardinality property. (In an abstract theory, one works directly with the operators P and Q ; their detailed structure does not enter the analysis.)

In the tensor product method of polynomial interpolation, the general case will involve bivariate polynomials from the space $\Pi_{p-1} \otimes \Pi_{q-1}$, where p and q are the numbers of points that figure in Equation (2). A basis for this space is given by the functions

$$(x, y) \mapsto x^i y^j \quad (0 \leq i \leq p-1, 0 \leq j \leq q-1) \quad (17)$$

A generic element of the space is then of the form

$$(x, y) \mapsto \sum_{i=0}^{p-1} \sum_{j=0}^{q-1} c_{ij} x^i y^j$$

The **degree** of a term $x^i y^j$ is defined to be $i + j$. Thus, the space $\Pi_{p-1} \otimes \Pi_{q-1}$ will contain one basis element of degree $p + q - 2$; namely, $x^{p-1} y^{q-1}$. But it will not contain all terms of degree $p + q - 2$. For example, a term such as $x^p y^{q-2}$ will not be present. The degree of a polynomial in (x, y) is defined to be the largest degree of the terms present in the polynomial. The space of all bivariate polynomials of degree at most k will be denoted here by $\Pi_k(\mathbb{R}^2)$. A typical element of $\Pi_k(\mathbb{R}^2)$ is a function of the form

$$(x, y) \mapsto \sum_{i=0}^k \sum_{j=0}^{k-i} c_{ij} x^i y^j = \sum_{0 \leq i+j \leq k} c_{ij} x^i y^j \quad (18)$$

THEOREM 1 Theorem on Basis Functions of Bivariate Polynomials

A basis for $\Pi_k(\mathbb{R}^2)$ is the set of functions

$$(x, y) \mapsto x^i y^j \quad (0 \leq i + j \leq k)$$

Proof It is clear that this set spans $\Pi_k(\mathbb{R}^2)$, and it is necessary to prove only its linear independence. Suppose, therefore, that the function in Equation (18) is 0. If y is assigned a fixed value, say $y = y_0$, then the equation

$$\sum_{i=0}^k \left(\sum_{j=0}^{k-i} c_{ij} y_0^j \right) x^i = 0$$

exhibits an apparent linear dependence among the functions $x \mapsto x^i$. Since this set of functions is linearly independent, we conclude that

$$\sum_{j=0}^{k-i} c_{ij} y_0^j = 0 \quad (0 \leq i \leq k)$$

In this equation, y_0 can be *any* point. By the linear independence of the set of functions

$$y \mapsto y^j \quad (0 \leq j \leq k)$$

we conclude that $c_{ij} = 0$ for all i and j . ■

COROLLARY 1 Corollary on Bivariate Polynomials

The dimension of $\Pi_k(\mathbb{R}^2)$ is $\frac{1}{2}(k+1)(k+2)$.

Proof The basis elements of $\Pi_k(\mathbb{R}^2)$ given in Theorem 1 can be arrayed as follows:

$$\begin{array}{ccccccc} & & & & & & x^k \\ & & & & & & x^{k-1} \\ x^{k-1} & & & & & & x^{k-1}y \\ & & & & & & x^{k-2}y^2 \\ x^{k-2} & & & & & & x^{k-2}y \\ & & & & & & \vdots \\ \vdots & & & & & & \ddots \\ x^0 & & & & & & x^0y \\ & & & & & & x^0y^2 \\ & & & & & & \dots \\ & & & & & & x^0y^k \end{array}$$

The number of basis elements is thus

$$1 + 2 + 3 + \cdots + (k+1) = \frac{1}{2}(k+1)(k+2) \quad \blacksquare$$

Recall that in the one-variable case, Π_k can be used for interpolation at *any* set of $k+1$ nodes in $\mathbb{R}$. It is natural to expect that for two variables, $\Pi_k(\mathbb{R}^2)$ can be used to interpolate at any set of $n \equiv \frac{1}{2}(k+1)(k+2)$ nodes. This expectation is not fulfilled, however, and a simple example will show this. Suppose that $k = 1$, so that $n = 3$. A generic element of $\Pi_1(\mathbb{R}^2)$ has the form

$$c_0 + c_1x + c_2y$$

If we attempt to solve an interpolation problem with three nodes (x_i, y_i) , we are led to a linear system whose coefficient determinant is

$$\begin{vmatrix} 1 & x_1 & y_1 \\ 1 & x_2 & y_2 \\ 1 & x_3 & y_3 \end{vmatrix}$$

Since this determinant represents twice the area of a triangle whose vertices are the given nodes, the determinant will be zero when the nodes are colinear. In that case the interpolation problem will be (in general) insoluble.

Geometry

The preceding considerations indicate that the *geometry* of the node set $\mathcal{N}$ will determine whether interpolation by $\Pi_k(\mathbb{R}^2)$ is possible on $\mathcal{N}$. Of course, the number of nodes should be $n = \frac{1}{2}(k+1)(k+2)$. Some theorems concerning this question will be given here to illustrate what is known. The following theorem is due to Gasca and Maeztu [1982].

THEOREM 2 Theorem of Gasca and Maeztu

Interpolation of arbitrary data by the subspace $\Pi_k(\mathbb{R}^2)$ is possible on a set of $\frac{1}{2}(k+1)(k+2)$ nodes if the nodes lie on lines $L_0, L_1, \dots, L_k$ in such a way that (for each i) L_i contains exactly $i+1$ nodes.

Proof Let $\mathcal{N}$ denote the set of nodes. Assuming the hypothesis of the theorem, we have $\#(\mathcal{N} \cap L_i) = i+1$, in which the symbol $\#$ is used to signify the number of elements in a set. The sets $\mathcal{N} \cap L_i$ must be pairwise disjoint; if they were not, the following contradiction would arise:

$$\#\mathcal{N} < \sum_{i=0}^k \#(\mathcal{N} \cap L_i) = \sum_{i=0}^k (i+1) = \frac{1}{2}(k+1)(k+2)$$

Because the number of nodes is equal to the dimension of the space $\Pi_k(\mathbb{R}^2)$, it suffices to prove that the homogeneous interpolation problem has only the 0 solution. Accordingly, let $p \in \Pi_k(\mathbb{R}^2)$, and suppose that $p(x, y) = 0$ for each point (x, y) in $\mathcal{N}$. For each i , let ℓ_i be a linear function describing L_i :

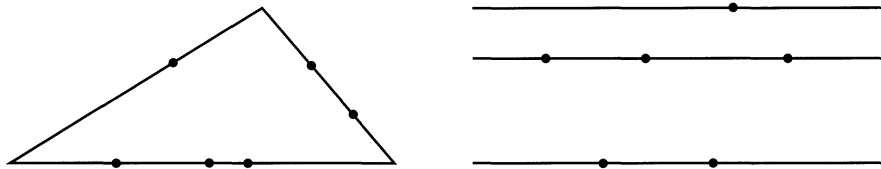
$$L_i = \{(x, y) : \ell_i(x, y) = 0\} \qquad (0 \leq i \leq k)$$

Notice that $p^2 + \ell_k^2$ has at least $k+1$ zeros; namely, the points of $\mathcal{N} \cap L_k$. Bézout’s Theorem (given below) allows us to conclude that ℓ_k is a divisor of p . This argument can be repeated because $(p/\ell_k)^2 + \ell_{k-1}^2$ has at least k zeros, and ℓ_{k-1} must be a divisor of p/ℓ_k . After k steps in this argument, we draw the conclusion that p is divisible by $\ell_1 \ell_2 \cdots \ell_k$. Thus p is a *scalar* multiple of $\ell_1 \ell_2 \cdots \ell_k$ because p is of degree at most k . Since p vanishes on $\mathcal{N} \cap L_0$ while $\ell_1 \ell_2 \cdots \ell_k$ does not, p must be 0. ■

The **Bézout’s Theorem** states that if $p \in \Pi_k(\mathbb{R}^2)$, if $q \in \Pi_m(\mathbb{R}^2)$, and if $p^2 + q^2$ has more than km zeros, then p and q must have a common nonconstant factor. Because this theorem is limited to $\mathbb{R}^2$, the same is true of Theorem 2. An algorithmic proof of Theorem 2, not requiring Bézout’s Theorem, is given later.

EXAMPLE 2 The sets of nodes in Figure 6.17 are suitable for interpolation by the space $\Pi_2(\mathbb{R}^2)$.

FIGURE 6.17
Node sets for interpolation
by $\Pi_2(\mathbb{R}^2)$



A theorem closely related to Theorem 2 can be given in higher-dimensional spaces, $\mathbb{R}^d$. We appeal to Problem 6.10.5 (p. 437), according to which the dimension of $\Pi_k(\mathbb{R}^d)$ is $\binom{d+k}{k}$. The following result is from Chung and Yao [1977].

THEOREM 3 Theorem of Chung and Yao

Let k and d be given, and set $n = \binom{d+k}{k}$. Let a set of n nodes $z_1, z_2, \dots, z_n$ be given in $\mathbb{R}^d$. If there exist hyperplanes H_{ij} in $\mathbb{R}^d$, with $1 \leq i \leq n$ and $1 \leq j \leq k$, such that

$$z_j \in \bigcup_{v=1}^k H_{iv} \iff j \neq i \quad (1 \leq i, j \leq n) \quad (19)$$

then arbitrary data on the node set can be interpolated by polynomials in $\Pi_k(\mathbb{R}^d)$.

Proof Each hyperplane is the zero set of a nonzero linear function, and we write

$$H_{ij} = \{z \in \mathbb{R}^d : \ell_{ij}(z) = 0\}$$

where $\ell_{ij} \in \Pi_1(\mathbb{R}^d)$. Define the functions

$$q_i(z) = \prod_{j=1}^k \ell_{ij}(z) \quad (1 \leq i \leq n)$$

Now z_i does not belong to any of the hyperplanes $H_{i1}, H_{i2}, \dots, H_{ik}$ by Condition (19), and therefore $\ell_{ij}(z_i) \neq 0$ for $1 \leq j \leq k$. This proves that $q_i(z_i) \neq 0$.

Again, by Condition (19), if $j \neq i$, then $z_j \in H_{iv}$ for some v , and consequently $\ell_{iv}(z_j)$ and $q_i(z_j)$ are 0. Now if we set $p_i(z) = q_i(z)/q_i(z_i)$, we shall have the cardinality property, $p_i(z_j) = \delta_{ij}$. Since $p_i \in \Pi_k(\mathbb{R}^n)$, we have a Lagrangian formula for interpolating a function f at the nodes by a polynomial of degree k :

$$P(z) = \sum_{i=1}^n f(z_i) p_i(z) \quad \blacksquare$$

A node configuration satisfying the hypotheses of Theorem 3 is shown in Figure 6.18. Here the dimension is $d = 2$, the degree is $k = 2$, and the number of nodes is $n = 6$.

A very weak result concerning polynomial interpolation at arbitrary sets of nodes follows.

THEOREM 4 Theorem on Capability of Interpolating Arbitrary Data

The space $\Pi_k(\mathbb{R}^2)$ is capable of interpolating arbitrary data on any set of $k + 1$ distinct nodes in $\mathbb{R}^2$.

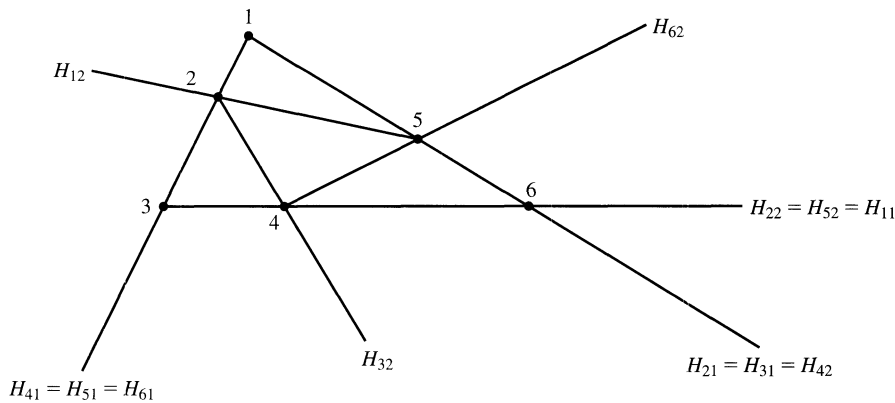


FIGURE 6.18
Illustration of Theorem 3

Proof If the nodes are (x_i, y_i) , with $0 \leq i \leq k$, we select a linear function

$$\ell(x, y) = ax + by + c$$

such that the $k + 1$ numbers $t_i = \ell(x_i, y_i)$ are all different. (See Problem 6.10.16, p. 438.) If f is the function to be interpolated, we find $p \in \Pi_k(\mathbb{R})$, such that $p(t_i) = f(x_i, y_i)$. Then $p \circ \ell \in \Pi_k(\mathbb{R}^2)$ and

$$(p \circ \ell)(x_i, y_i) = p(\ell(x_i, y_i)) = p(t_i) = f(x_i, y_i) \quad \blacksquare$$

A function of the form $f \circ \ell$, with $\ell \in \Pi_1(\mathbb{R}^2)$, has been termed a **ridge function**. Its graph is a ruled surface because $f \circ \ell$ remains constant on each line $\ell(x, y) = \lambda$. Theorem 4 has an obvious extension to $\mathbb{R}^d$.

A Newtonian Scheme

For the practical implementation of any interpolation method, it is advantageous to have an algorithm like Newton's procedure in univariate polynomial interpolation. Recall that one feature of the Newton scheme is that from a polynomial p interpolating f at nodes $x_1, x_2, \dots, x_n$, we can easily obtain a polynomial p^* interpolating f at nodes $x_1, x_2, \dots, x_{n+1}$ by adding one term to p . Indeed, we put

$$\begin{aligned} q(x) &= (x - x_1)(x - x_2) \cdots (x - x_n) \\ p^*(x) &= p(x) + cq(x) \\ c &= [f(x_{n+1}) - p(x_{n+1})]/q(x_{n+1}) \end{aligned}$$

The advantage of this algorithm is that an interpolating polynomial can be constructed step by step, adding one new interpolation node and one new term to p in each stage.

The abstract form of this procedure is as follows: Let X be a set and f a real-valued function defined on X . Let $\mathcal{N}$ be a set of nodes. If p is any function that interpolates f on $\mathcal{N}$, and if q is any function that vanishes on $\mathcal{N}$, then a function p^*

interpolating f on $\mathcal{N} \cup \{\xi\}$ can be obtained in the form $p^* = p + cq$, provided that $q(\xi) \neq 0$.

A more general version of this strategy deals with sets of nodes. Let q be a function from X to $\mathbb{R}$ and let Z be its 0 set. If p interpolates f on $\mathcal{N} \cap Z$ and r interpolates $(f - p)/q$ on $\mathcal{N} \setminus Z$, then $p + qr$ interpolates f on $\mathcal{N}$.

The procedure just outlined can be used to give an algorithmic proof of Theorem 2. To begin, select $p_k \in \Pi_k(\mathbb{R}^2)$ that interpolates f on $\mathcal{N} \cap L_k$. (Use Theorem 4.) Proceeding inductively downward, suppose that p_i has been found in $\Pi_k(\mathbb{R}^2)$ and interpolates f on all the nodes in $L_k \cup L_{k-1} \cup \cdots \cup L_i$. We shall attempt to construct p_{i-1} in the **Newton form**

$$p_{i-1} = p_i + r\ell_k\ell_{k-1} \cdots \ell_i$$

It is clear that p_{i-1} will still interpolate f at the nodes in $L_k \cup L_{k-1} \cup \cdots \cup L_i$, because the term added to p_i vanishes on this set. To make p_{i-1} interpolate f on the nodes in L_{i-1} , we write

$$f(x) = p_i(x) + r(x)(\ell_k\ell_{k-1} \cdots \ell_i)(x) \quad x \in \mathcal{N} \cap L_{i-1}$$

from which we infer that r should interpolate $(f - p_i)/(\ell_k\ell_{k-1} \cdots \ell_i)$ on $\mathcal{N} \cap L_{i-1}$. By Theorem 4, there is an $r \in \Pi_{i-1}(\mathbb{R}^2)$ that does so. Finally, observe that $p_{i-1} \in \Pi_k(\mathbb{R}^2)$ because r is of degree $i - 1$ and $(\ell_k\ell_{k-1} \cdots \ell_i)$ is of degree $k - i + 1$. This algorithm was given by Micchelli [1986a].

It is an interesting fact that *no* n -dimensional subspace in $C(\mathbb{R}^2)$ can serve for interpolation at arbitrary sets of n nodes (except in the trivial case of $n = 1$). This was probably first noticed by Haar in 1918, and his argument goes like this. Suppose that n functions $u_1, u_2, \dots, u_n$ are given in $C(\mathbb{R}^2)$. Let n nodes in $\mathbb{R}^2$ be given, say $p_i = (x_i, y_i)$. If we wish to interpolate at these nodes using the base functions u_i , we shall have to solve a linear system whose determinant is

$$D = \begin{vmatrix} u_1(p_1) & u_2(p_1) & \cdots & u_n(p_1) \\ u_1(p_2) & u_2(p_2) & \cdots & u_n(p_2) \\ \vdots & \vdots & \ddots & \vdots \\ u_1(p_n) & u_2(p_n) & \cdots & u_n(p_n) \end{vmatrix}$$

This determinant may be nonzero for the given set of nodes. However, let the first two of the nodes undergo a continuous motion in $\mathbb{R}^2$ in such a way that during the motion these two points never coincide, nor do they coincide with any of the other nodes, yet at the end of the motion they have exchanged their original positions. By the rules of determinants, the determinant D will have changed sign (because rows 1 and 2 are interchanged). By continuity, D assumed the value 0 during the continuous motion described. Hence, D will *sometimes* be 0, even for distinct nodes. The fact that two nodes can move in $\mathbb{R}^2$ and exchange places without ever being coincident is characteristic of $\mathbb{R}^2, \mathbb{R}^3, \dots$ but not of $\mathbb{R}^1$. This explains why interpolation in $\mathbb{R}^2, \mathbb{R}^3, \dots$ must be approached somewhat differently from $\mathbb{R}^1$. What is usually done is to fix the nodes *first* and *then* to ask what subspaces of interpolating functions are suitable.

Shepard Interpolation

A very general method of the type just mentioned (in which the subspace depends on the nodes) is known as **Shepard interpolation**, after its originator, D. Shepard [1968]. Let the (distinct) nodes be listed as

$$p_i = (x_i, y_i) \qquad (1 \leq i \leq n) \tag{20}$$

We shall use p and q to denote generic elements in $\mathbb{R}^2$ because this will make the extension to $\mathbb{R}^3, \mathbb{R}^4, \dots$ conceptually transparent. Next, we select a real-valued function ϕ on $\mathbb{R}^2 \times \mathbb{R}^2$, subject to the sole condition that

$$\phi(p, q) = 0 \quad \text{if and only if} \quad p = q \tag{21}$$

Examples that come to mind are $\phi(p, q) = \|p - q\|$ and $\phi(p, q) = \|p - q\|^2$. Next, we set up some cardinal functions in exact analogy with the Lagrange formulas in univariate approximation. This is done as follows:

$$u_i(p) = \prod_{\substack{j=1 \\ j \neq i}}^n \frac{\phi(p, p_j)}{\phi(p_i, p_j)} \qquad (1 \leq i \leq n) \tag{22}$$

It is easy to see that these functions have the cardinality property

$$u_i(p_j) = \delta_{ij} \qquad (1 \leq i, j \leq n)$$

This is a consequence of the hypothesis in (21). It follows that an interpolant to f at the given nodes is provided by the function

$$F = \sum_{i=1}^n f(p_i)u_i \tag{23}$$

EXAMPLE 3 What are the formulas for Shepard interpolation when $\|p - q\|^2$ is used for $\phi(p, q)$?

Solution Let $p_i = (x_i, y_i)$, $p = (x, y)$, and $\phi(p, p_j) = \|p - p_j\|^2 = (x - x_j)^2 + (y - y_j)^2$. Then

$$F(x, y) = \sum_{i=1}^n f(x_i, y_i) \prod_{\substack{j=1 \\ j \neq i}}^n \frac{(x - x_j)^2 + (y - y_j)^2}{(x_i - x_j)^2 + (y_i - y_j)^2} \quad \blacksquare$$

An algorithm to compute $F(x, y)$ in Example 3 goes as follows:

```
input n, x, y
input (x_i, y_i, c_i) (1 ≤ i ≤ n)
for i = 1 to n do
    d_i = (x - x_i)^2 + (y - y_i)^2
    for j = 1 to n do
        d_ij = (x_i - x_j)^2 + (y_i - y_j)^2
    end do
end do
```

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Then

$$F(x, y) = \sum_{i=1}^n c_i \prod_{\substack{j=1 \\ j \neq i}}^n d_j / d_{ij}$$

(No attempt has been made to make the algorithm efficient. It contains much duplication.)

Another version of Shepard's method starts with the additional assumption on ϕ that it is a *nonnegative* function. Next, let

$$v_i(p) = \prod_{\substack{j=1 \\ j \neq i}}^n \phi(p, p_j) \quad v(p) = \sum_{i=1}^n v_i(p) \quad w_i(p) = v_i(p)/v(p) \quad (24)$$

By our assumptions on ϕ , we have $v_i(p_j) = 0$ if $i \neq j$ and $v_i(p) > 0$ for all points except $p_1, \dots, p_{i-1}, p_{i+1}, \dots, p_n$. It follows that $v(p) > 0$ and that w_i is well defined. By the construction, $w_i(p_j) = \delta_{ij}$ and $0 \leq w_i(p) \leq 1$. Furthermore, $\sum_{i=1}^n w_i(p) = 1$. The interpolation process is given by the equation

$$F = \sum_{i=1}^n f(p_i)w_i = \sum_{i=1}^n f(p_i)v_i/v \quad (25)$$

This process has two favorable properties not possessed by the previous version; namely, if the data are nonnegative, then the interpolant F will be a nonnegative function, and if f is a constant function, then $F = f$. These two properties give evidence that the interpolant F inherits certain characteristics of the function being interpolated. On the other hand, if ϕ is differentiable, then F will exhibit a flat spot at each node. This is because $0 \leq w_i \leq 1$ and $w_i(p_j) = \delta_{ij}$, so that the nodes are extrema (maximum points or minimum points) of each w_i . Thus the partial derivatives of w_i are 0 at each node, and consequently the same is true of F .

An important case of Shepard interpolation arises when the function ϕ is given by a power of the Euclidean distance:

$$\phi(x, y) = \|x - y\|^\mu \quad (\mu > 0)$$

Here x and y can be points in $\mathbb{R}^s$. It will be seen that ϕ is differentiable if $\mu > 1$ but not if $0 < \mu \leq 1$. It suffices to examine the simpler function $g(x) = \|x\|^\mu$ at the questionable point $x = 0$. The directional derivative of g at 0 is obtained by differentiating the function $G(t) = \|tu\|^\mu$, where u is a unit vector defining the direction. Because $G(t) = |t|^\mu$, the derivative at $t = 0$ does not exist when $0 < \mu \leq 1$, but for $\mu > 1$, $G'(0) = 0$.

The formula for w_i can be given in two ways:

$$w_i(x) = \prod_{\substack{j=1 \\ j \neq i}}^n \|x - x_j\|^\mu / \sum_{k=1}^n \prod_{\substack{j=1 \\ j \neq k}}^n \|x - x_j\|$$

$$w_i(x) = \|x - x_i\|^{-\mu} / \sum_{j=1}^n \|x - x_j\|^{-\mu}$$

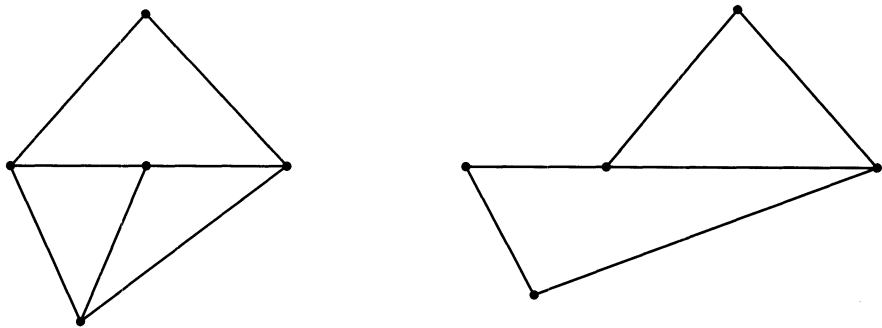


FIGURE 6.19
Illegal triangulations

The second equation must be used with care because the right side assumes the indeterminate form ∞/∞ at x_i .

A *local* multivariate interpolation method of Franke and Little is designed so that the datum at one node will have a very small influence on the interpolating function at points far from that node. Given nodes (x_i, y_i) , $1 \leq i \leq n$, we introduce functions

$$g_i(x, y) = \left(1 - r_i^{-1} \sqrt{(x - x_i)^2 + (y - y_i)^2}\right)_+^\mu$$

The subscript $+$ indicates that when the quantity inside the parentheses is negative, it is replaced by 0. This will occur if (x, y) is far from the node (x_i, y_i) . The parameter μ influences the smoothness of the function. The parameter r_i controls the support of g_i . Thus, $g_i(x, y) = 0$ if (x, y) is more than r units distant from (x_i, y_i) .

If r_i is chosen to be the distance from (x_i, y_i) to the nearest neighboring node, then $g_i(x_j, y_j) = \delta_{ij}$. In this case, we interpolate an arbitrary function f by means of the function

$$\sum_{i=1}^n f(x_i, y_i) g_i(x, y)$$

Triangulation

Another general strategy for interpolating functions given on $\mathbb{R}^2$ begins by creating a **triangulation**. Informally, this means that triangles are drawn by joining nodes. In the end, we shall have a family of triangles, $T_1, T_2, \dots, T_m$. We consider this collection of triangles to be the triangulation. These rules must be satisfied:

1. Each interpolation node must be the vertex of some triangle T_i .
2. Each vertex of a triangle in the collection must be a node.
3. If a node belongs to a triangle, it must be a vertex of that triangle.

The effect of rule 3 is to disallow the construction shown in Figure 6.19.

The simplest type of interpolation on a triangulation is the piecewise linear function that interpolates a function f at all the vertices of all triangles. In any triangle,

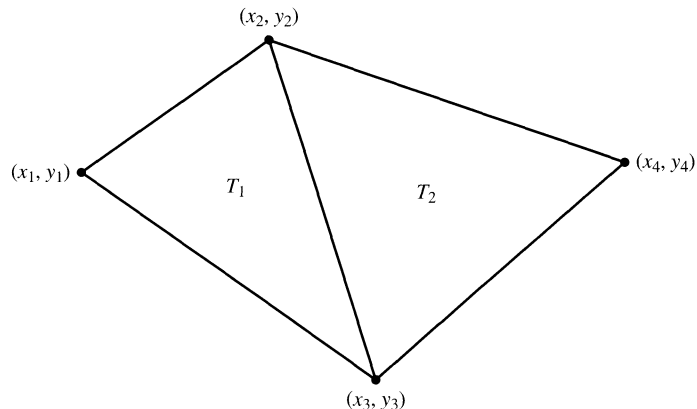


FIGURE 6.20
Triangulation

T_i , a linear function will be prescribed:

$$\ell_i(x, y) = a_i x + b_i y + c_i \quad (x, y) \in T_i$$

The coefficients in ℓ_i are uniquely determined by the prescribed function values at the vertices of T_i . This can be seen as an application of Theorem 2 because L_1 in that theorem can be taken to be one side of the triangle, and L_0 can be a line parallel to L_1 containing the vertex not on L_1 .

Let us consider the situation shown in Figure 6.20. The line segment joining (x_2, y_2) to (x_3, y_3) is common to both triangles. This line segment can be represented as

$$\{t(x_2, y_2) + (1 - t)(x_3, y_3) : 0 \leq t \leq 1\}$$

The variable t can be considered the coordinate for the points on the line segment. The linear function ℓ_1 , when restricted to this line segment, will be a linear function of the single variable t , namely,

$$a_1(t x_2 + (1 - t)x_3) + b_1(t y_2 + (1 - t)y_3) + c_1$$

or

$$(a_1 x_2 - a_1 x_3 + b_1 y_2 - b_1 y_3)t + (a_1 x_3 + b_1 y_3 + c_1)$$

This linear function of t is completely determined by the interpolation conditions at (x_2, y_2) and (x_3, y_3) . The same remarks pertain to the linear function ℓ_2 . Thus ℓ_1 and ℓ_2 agree on this line segment, and the piecewise linear function defined on $T_1 \cup T_2$ is continuous. This proves the following result.

THEOREM 5 Theorem

Let $\{T_1, T_2, \dots, T_m\}$ be a triangulation in the plane. The piecewise linear function taking prescribed values at all the vertices of all the triangles, T_i , is continuous.

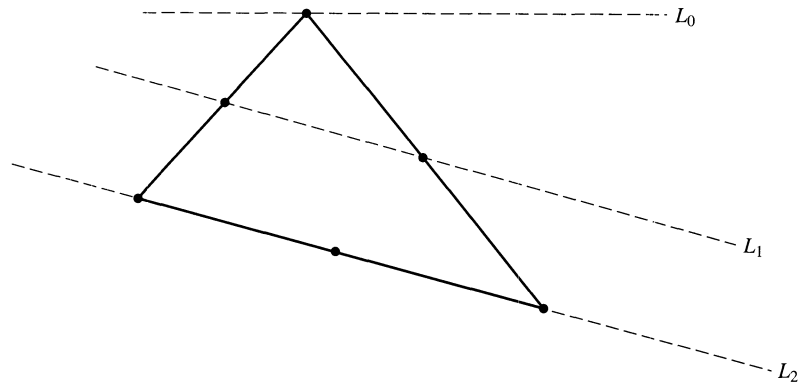


FIGURE 6.21
Applying Theorem 2

Consider next the use of piecewise quadratic functions on a triangulation. In each triangle, T_i , a quadratic polynomial will be prescribed:

$$q_i(x, y) = a_1x^2 + a_2xy + a_3y^2 + a_4x + a_5y + a_6$$

Six conditions will be needed to determine the six coefficients. One such set of conditions consists of values at the vertices of the triangle and the midpoints of the sides. Again, an application of Theorem 2 shows that this interpolation is always uniquely possible. Indeed, in that theorem, L_2 can be one side of the triangle, L_1 can be the line passing through two midpoints not on L_2 , and L_0 can be a line containing the remaining vertex but no other node. (See Figure 6.21.) Reasoning as before, we see that the global piecewise quadratic function will be continuous because the three prescribed function values on the side of a triangle determine the quadratic function of one variable on that side.

Moving Least Squares

Another versatile method of smoothing and interpolating multivariate functions is called **moving least squares**. First, we explain it in a general setting, and then some specific examples will be given.

We start with a set X that is the domain of the functions involved. For example, X can be $\mathbb{R}$, $\mathbb{R}^2$, or a subset of either. Next, a set of nodes $\{x_1, x_2, \dots, x_n\}$ is given. These are the points at which a certain function f has been sampled. Thus, the values $f(x_i)$ are known for $1 \leq i \leq n$. For purposes of approximation, we select a set of functions $u_1, u_2, \dots, u_m$. These are real-valued functions defined on X . The number m will usually be very small relative to n .

In the familiar least-squares method, a set of nonnegative weights $w_i \geq 0$ is given. We try to find coefficients $c_1, c_2, \dots, c_m$ to minimize the expression

$$\sum_{i=1}^n \left[f(x_i) - \sum_{j=1}^m c_j u_j(x_i) \right]^2 w_i$$

This is the sum of the squares of the residuals. If we write

$$\langle f, g \rangle = \sum_{i=1}^n f(x_i)g(x_i)w_i \quad \|f\| = \sqrt{\langle f, f \rangle}$$

then the theory of approximation in an inner-product space is applicable, and the solution to the minimization problem is characterized by the orthogonality condition

$$f - \sum_{j=1}^m c_j u_j \perp u_i \quad (1 \leq i \leq m)$$

This leads to the **normal equations**

$$\sum_{j=1}^m c_j \langle u_j, u_i \rangle = \langle f, u_i \rangle \quad (1 \leq i \leq m)$$

How does the moving-least-squares method differ from the procedure just outlined? The weights w_i are now allowed to be functions of x . The formalism of the usual least-squares method can be retained, although the following notation may be better:

$$\langle f, g \rangle_x = \sum_{i=1}^n f(x_i)g(x_i)w_i(x)$$

The normal equations should now be written in the form

$$\sum_{j=1}^m c_j(x) \langle u_j, u_i \rangle_x = \langle f, u_i \rangle_x$$

and the final approximating function will be

$$g(x) = \sum_{j=1}^m c_j(x)u_j(x)$$

The computations necessary to produce this function will be quite formidable if m is large, because the normal equations change with x . For this reason, m is usually no greater than 10.

The weight functions can be used to achieve several desirable effects. First, if $w_i(x)$ is “strong” at x_i , the function g will nearly interpolate f at x_i . In the limiting case, $w_i(x_i) = +\infty$ and $g(x_i) = f(x_i)$. If $w_i(x)$ decreases rapidly to 0 when x moves away from x_i , then the nodes far from x_i will have little effect on $g(x_i)$.

A choice for w_i that achieves these two objectives in a space $\mathbb{R}^d$ is

$$w_i(x) = \|x - x_i\|^{-2}$$

where any norm can be used, although the Euclidean norm is usual.

If the moving-least-squares procedure is used with a single function, $u_1(x) \equiv 1$, and with weight functions like the one just mentioned, then Shepard’s method will

result. To see that this is so, write the normal equation for this case, with $c_1(x) = c(x)$, $u_1(x) = u(x) = 1$:

$$c(x)\langle u, u \rangle_x = \langle f, u \rangle_x$$

The approximating function will be

$$\begin{aligned} g(x) &= c(x)u(x) = c(x) = \langle f, u \rangle_x / \langle u, u \rangle_x \\ &= \sum_{i=1}^n f(x_i)w_i(x) / \sum_{j=1}^n w_j(x) \end{aligned}$$

If $w_i(x) = \|x - x_i\|^{-2}$, then after the singularities are removed, $w_i / \sum_{j=1}^n w_j$ has the cardinal property: It takes the value 1 at x_i and the value 0 at all other nodes.

Interpolation by Multiquadrics

Another multivariate interpolation process is one proposed by R. L. Hardy [1971] that uses as its basic functions these so-called **multiquadrics**:

$$z_i(p) = \{\|p - p_i\|^2 + c^2\}^{1/2} \quad (1 \leq i \leq n)$$

Here the norm is Euclidean, and c is a parameter that Hardy suggested to be set equal to 0.8 times the average distance between nodes. In interpolating with these functions, we need to know that the coefficient matrix $(z_i(p_j))$ is not singular. This has been proved by Micchelli [1986b].

Some further references on multivariate interpolation are Chui [1988], Hartley [1976], Micchelli [1986a], Franke [1982], and Lancaster and Salkauskas [1986].

References on Shepard's method of interpolation are Shepard [1968], Gordon and Wixom [1978], Newman and Rivlin [1983], Barnhill, Dube, and Little [1983], Farwing [1986], and Cheney and Light [1999].

PROBLEMS 6.10

1. Give an algorithm for determining whether a set $\mathcal{N}$ of points in $\mathbb{R}^2$ can be expressed as a Cartesian product, as in Equation (2). If the factorization is possible, the algorithm should deliver the factors.
2. For each of these inclusions, either prove that it is true or prove that it is false:
 - a. $\Pi_n \otimes \Pi_m \subseteq \Pi_{n+m}(\mathbb{R}^2)$
 - b. $\Pi_k(\mathbb{R}^2) \subseteq \Pi_n \otimes \Pi_m$, where $k = \max(n, m)$
 - c. $\Pi_k(\mathbb{R}^2) \subseteq \Pi_n \otimes \Pi_m$, where $k = \min(n, m)$
3. Prove that $\dim \Pi_k(\mathbb{R}^3) = \frac{1}{6}(k+1)(k+2)(k+3)$. Here the space consists of all polynomials of degree at most k in three variables.
4. This problem and the next deal with polynomials in d variables. Label the variables with subscripts and put them into a vector called x :

$$x = (x_1, x_2, \dots, x_d) \in \mathbb{R}^d$$

Now we require the concept of a **multi-index**. This is a d -tuple of nonnegative integers:

$$\alpha = (\alpha_1, \alpha_2, \dots, \alpha_d) \in \mathbb{Z}_{+}^d$$

We define

$$x^\alpha = x_1^{\alpha_1} x_2^{\alpha_2} x_3^{\alpha_3} \cdots x_d^{\alpha_d}$$

This is called a **multinomial** and is the basic building block for polynomials in d variables. The **order** of the multi-index α is defined by the equation

$$|\alpha| = \alpha_1 + \alpha_2 + \cdots + \alpha_d$$

Notice that what is called the **degree** of the multinomial x^α is nothing but $|\alpha|$. Prove that $x^\alpha x^\beta = x^{\alpha+\beta}$. Prove that any polynomial in d variables having degree $\leq k$ can be written in the form

$$\sum_{|\alpha| \leq k} c_\alpha x^\alpha$$

- 5. Let $\Pi_k(\mathbb{R}^d)$ be the set of all polynomials in d variables having degree at most k . Show that the dimension of this space is $\binom{d+k}{k}$.
- 6. Prove the assertion in the text (preceding Example 1) concerning the interpolation property of the operator $\overline{P} \oplus \overline{Q}$.
- 7. Show that interpolation by $\Pi_2(\mathbb{R}^2)$ is not generally possible at a set of six nodes if these nodes lie on an ellipse, a parabola, a hyperbola, or a pair of straight lines.
- 8. What type of bivariate polynomial would be suitable for interpolation on the sets of nodes in Figure 6.22?

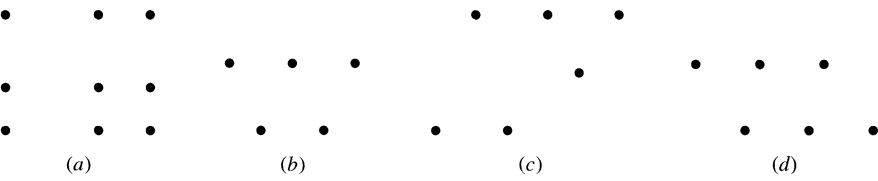


FIGURE 6.22

- 9. Prove that a basis for $\Pi_n \otimes \Pi_m$ is given by the functions

$$(x, y) \longmapsto x^i y^j \quad (0 \leq i \leq n, \ 0 \leq j \leq m)$$

- 10. Give the formulas for the second version of Shepard interpolation using $\phi(p, q) = \|p - q\|_\infty^2$. Recall that this norm is $\|(x, y)\| = \max\{|x|, |y|\}$.
- 11. A polynomial in two variables having the form

$$F(x, y) = \sum_{0 \leq i+j \leq k} c_{ij} x^i y^j$$

is said to be of **degree at most** k . If we have a coefficient $c_{ij} \neq 0$ with $i + j = k$, we say that F is of **degree** k . Prove that the product of a polynomial in two variables of degree k with a polynomial in two variables of degree m is a polynomial of degree $m + k$.

- 12. Consider Shepard interpolation in two variables with n nodes and let the function ϕ be given by $\phi(p, q) = \|p - q\|^2$. What polynomial space $\Pi_k(\mathbb{R}^2)$ contains all the cardinal functions? (Give the least k .)
- 13. Let $f \in \Pi_k(\mathbb{R})$ and $\ell \in \Pi_1(\mathbb{R}^2)$. Prove that $f \circ \ell \in \Pi_k(\mathbb{R}^2)$. Is every element of $\Pi_k(\mathbb{R}^2)$ obtained in this way?

14. Prove that interpolation of arbitrary data at a set of four nodes is possible by a polynomial

$$p(x, y) = a + bx + cy + dxy$$

provided that the nodes are vertices of a rectangle whose sides are parallel to the coordinate axes.

15. Prove that interpolation by a linear polynomial

$$p(x, y) = ax + by + c$$

is always possible at the vertices of a triangle. Give two proofs, one of which is based on Theorem 3.

16. Prove that if n distinct points x_i are given in $\mathbb{R}^d$, then a vector b exists such that the n numbers $t_i = \langle x_i, b \rangle$ are all different. (This is needed for Theorem 4.)

17. If U and V are vector spaces of functions on X and Y , respectively, then $U \otimes V$ consists of all functions w that have a representation as a finite sum

$$w(x, y) = \sum_{i=1}^n u_i(x)v_i(y) \quad u_i \in U, \ v_i \in V$$

Prove that if U and V are finite dimensional, then $\dim(U \otimes V) = \dim U \cdot \dim V$.

18. Prove that for an operator L of the form

$$Lf = \sum_{i=1}^n f(x_i)w_i$$

these properties are equivalent

i. For all f , $\min f(x_i) \leq Lf \leq \max f(x_i)$

ii. $w_i \geq 0$ and $\sum_{i=1}^n w_i = 1$

19. Is the following an accurate summary of the rules for a triangulation?: The set of nodes is exactly the same as the set of vertices of the triangles.

COMPUTER
PROBLEM 6.10

1. Write and test an efficient code to carry out Shepard's method. The user will specify nodes (x_i, y_i) , ordinates c_i , and a list of points at which the Shepard interpolant is to be evaluated. If possible, obtain graphic output.

6.11 Continued Fractions

Many of the special functions that occur in the applications of mathematics are defined by infinite processes, such as series, integrals, and iterations. The continued fraction is one of these infinite processes. An example of a continued fraction is this one, due to Lambert in 1770:

$$\tan^{-1} x = \frac{x}{1 + \frac{x^2}{3 + \frac{4x^2}{5 + \frac{9x^2}{7 + \frac{16x^2}{9 + \dots}}}}} \quad (|x| < 1) \tag{1}$$

This can also be written as

$$\tan^{-1} x = \frac{x}{1+} \frac{x^2}{3+} \frac{4x^2}{5+} \frac{9x^2}{7+} \frac{16x^2}{9+} \cdots \quad (|x| < 1) \tag{2}$$

The right side of this equation represents a limit in the following way: The expression in Equation (2) is terminated after n terms to give a well-defined function f_n :

$$f_n(x) = \frac{x}{1+} \frac{x^2}{3+} \frac{4x^2}{5+} \cdots \frac{(n-1)^2x^2}{2n-1} \quad (n \geq 2) \tag{3}$$

This is called the n th *convergent* of the continued fraction. Equation (2) is defined to mean

$$\tan^{-1} x = \lim_{n \rightarrow \infty} f_n(x) \quad (|x| < 1) \tag{4}$$

If we assume the validity of this equation, we have an alternative method of computing values of the arctangent function. Whether the method is practical depends on the *rapidity* of convergence in Equation (4). To judge this numerically, let us compute $\tan^{-1}(1/\sqrt{3}) = \pi/6 \approx 0.5235987756$ by means of the sequence $f_n(1/\sqrt{3})$ for $n \geq 2$. Here are the results:

n	$f_n(1/\sqrt{3})$
2	0.519615
3	0.523892
4	0.523577
5	0.523600
6	0.523599
7	0.523599

This list shows that six decimal places of precision have been obtained at the sixth convergent.

Recursive Formulas

The task of evaluating a continued fraction is not as easy as evaluating a series. In the case of an infinite series, say $\sum_{k=1}^\infty a_k$, we compute the partial sums $S_n = \sum_{k=1}^n a_k$ by means of the formula $S_{n+1} = S_n + a_{n+1}$. Let us consider the analogous problem for a continued fraction:

$$C = \frac{a_1}{b_1+} \frac{a_2}{b_2+} \frac{a_3}{b_3+} \cdots \tag{5}$$

We want to discover a recursive algorithm for computing the successive convergents:

$$C_n = \frac{a_1}{b_1+} \frac{a_2}{b_2+} \frac{a_3}{b_3+} \cdots \frac{a_{n-1}}{b_{n-1}+} \frac{a_n}{b_n} \tag{6}$$

We introduce for temporary use the functions

$$f_n(x) = \frac{a_1}{b_1+} \frac{a_2}{b_2+} \frac{a_3}{b_3+} \cdots \frac{a_{n-1}}{b_{n-1}+} \frac{a_n}{b_n + x} \tag{7}$$

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Obviously then, we obtain

$$C_n = f_n(0)$$

To obtain $f_n(x)$ from $f_{n-1}(x)$, we observe first

$$f_n(x) = f_{n-1}\left(\frac{a_n}{b_n + x}\right)$$

By direct calculation, we have

$$\begin{aligned} f_1(x) &= \frac{a_1}{b_1 + x} \\ f_2(x) &= \frac{a_1 b_2 + a_1 x}{b_1 b_2 + a_2 + b_1 x} \\ f_3(x) &= \frac{a_1 b_2 b_3 + a_1 a_3 + (a_1 b_2)x}{b_1 b_2 b_3 + a_2 b_3 + b_1 a_3 + (b_1 b_2 + a_2)x} \end{aligned}$$

The pattern that emerges here suggests that

$$f_n(x) = \frac{A_n + A_{n-1}x}{B_n + B_{n-1}x} \quad (n \geq 1) \quad (8)$$

where

$$\begin{cases} A_0 = 0, & A_1 = a_1 \\ A_n = b_n A_{n-1} + a_n A_{n-2} \end{cases} \quad (n \geq 2) \quad (9)$$

and

$$\begin{cases} B_0 = 1, & B_1 = b_1 \\ B_n = b_n B_{n-1} + a_n B_{n-2} \end{cases} \quad (n \geq 2) \quad (10)$$

THEOREM 1 Theorem on Continued Fraction

If the sequences $\{a_n\}_{n=1}^{\infty}$ and $\{b_n\}_{n=1}^{\infty}$ are given, and if the sequences $\{A_n\}_{n=0}^{\infty}$ and $\{B_n\}_{n=0}^{\infty}$ are defined by Formulas (9) and (10), then

$$C_n = \frac{A_n}{B_n} \quad (n \geq 1) \quad (11)$$

Proof The validity of Equation (8) for indices 1, 2, and 3 has already been verified. Now suppose that Equation (8) is true for indices 1, 2, $\dots$, $n-1$. Then

$$\begin{aligned}
 f_n(x) &= f_{n-1}\left(\frac{a_n}{b_n+x}\right) \\
 &= \frac{A_{n-1} + A_{n-2}a_n/(b_n+x)}{B_{n-1} + B_{n-2}a_n/(b_n+x)} \\
 &= \frac{A_{n-1}(b_n+x) + A_{n-2}a_n}{B_{n-1}(b_n+x) + B_{n-2}a_n} \\
 &= \frac{A_{n-1}b_n + A_{n-2}a_n + A_{n-1}x}{B_{n-1}b_n + B_{n-2}a_n + B_{n-1}x} \\
 &= \frac{A_n + A_{n-1}x}{B_n + B_{n-1}x}
 \end{aligned}$$

This establishes Equation (8) by induction. Then, of course,

$$C_n = f_n(0) = \frac{A_n}{B_n} \quad \blacksquare$$

The recursive formulas developed here form the basis for an efficient algorithm. For example, the numerical results for $\tan^{-1}(1/\sqrt{3})$, at the beginning of this section, were computed using this recursive relation with $a_n = (n-1)^2x^2$ and $b_n = 2n-1$, starting with $a_1 = x$ and $b_1 = 1$.

Conversion of Series to Continued Fractions

Many important special functions that arise in applied mathematics have expansions in continued fractions. Sources of information are Abramowitz and Stegun [1964] and the textbooks by Khovanskii [1963], Perron [1929], and Wall [1948]. We shall indicate here one procedure by which a continued fraction can be obtained from a series.

THEOREM 2 Theorem on Series to Continued Fractions

$$\sum_{k=1}^{\infty} \frac{1}{x_k} = \frac{1}{x_1 -} \frac{x_1^2}{x_1 + x_2 -} \frac{x_2^2}{x_2 + x_3 -} \cdots \frac{x_{n-1}^2}{x_{n-1} + x_n -} \cdots \quad (12)$$

Proof This can be proved by induction and is left as Problem 6.11.17 (p. 443) for the reader. ■

To illustrate how this theorem can be used, we construct a continued fraction from the Maclaurin series for $\arctan x$:

$$\begin{aligned}
 \arctan x &= x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots \\
 &= \frac{1}{x^{-1}} + \frac{1}{-3x^{-3}} + \frac{1}{5x^{-5}} + \frac{1}{-7x^{-7}} + \cdots
 \end{aligned}$$

$$\begin{aligned} &= \frac{1}{x^{-1} - \frac{(x^{-1})^2}{x^{-1} + (-3x^{-3})}} - \frac{(-3x^{-3})^2}{(-3x^{-3}) + \frac{(5x^{-5})^2}{(5x^{-5}) + (-7x^{-7})}} - \dots \\ &= \frac{x}{1 + \frac{-x^2}{x^2 - 3 + \frac{-9x^2}{-3x^2 + 5 + \frac{-25x^2}{5x^2 - 7 - \dots}}}} \end{aligned}$$

Notice that if both the numerator and denominator of one of the component fractions in a continued fraction are multiplied by the same quantity, then the numerator of the next component fraction is affected as well. (Why?)

Equations (9), (10), and (11) can be used to approximate $\arctan x$ for a given value of x when

$$a_n = -(2n - 3)^2 x^2 \quad \text{and} \quad b_n = (-1)^n [(2n - 3)x^2 - (2n - 1)]$$

for $n \geq 2$. However, the algorithm for computing $\arctan x$ via this continued fraction is far more complicated than one based on computing the partial sums in the Maclaurin series, and it produces the same sequence of numbers. Observe that the Lambert continued fraction for $\arctan x$, in Equation (1), is not the same as the one derived from the Maclaurin series. To investigate this further, we compute $\arctan(1/\sqrt{3})$ using the partial sums in the series:

n	$f_n(1/\sqrt{3})$
1	0.577350
2	0.513200
3	0.526030
4	0.522976
5	0.523767
6	0.523551
7	0.523612
8	0.523595
9	0.523600
10	0.523598
11	0.523599
12	0.523599

A comparison of this table with the one produced from the continued fraction (1) shows that the continued fraction converges faster than the series, for this example.

PROBLEMS 6.11

1. Show that

$$\frac{1}{1+} \frac{1}{1+} \frac{1}{1+} \dots = \frac{1}{2}(\sqrt{5} - 1)$$

Hint: Set x equal to the continued fraction, and look at $1/x$.

2. (Continuation) For the continued fraction in the preceding problem, 6.11.1, show that

$$C_n = \frac{A_n}{A_{n+1}}$$

3. Show that

$$\sqrt{x} = 1 + 2 \left(\frac{v}{1+} \frac{v}{1+} \frac{v}{1+} \cdots \right)$$

where $v = \frac{1}{4}(x - 1)$.

4. (Continuation) For the continued fraction in the preceding problem, show that

$$C_n = \frac{vA_n}{A_{n+1}}$$

5. Show that

$$\sqrt{b^2 + a} = b + \frac{a}{2b+} \frac{a}{2b+} \frac{a}{2b+} \cdots$$

6. Compare two methods for computing values of the function

$$f(x) = \int_0^x e^{-t^2} dt$$

namely, the Taylor series expansion given in Problem 6.7.6 (p. 390) and the continued fraction

$$f(x) = \frac{\sqrt{\pi}}{2} - \frac{1}{2}e^{-x^2} \left(\frac{1}{x+} \frac{1}{2x+} \frac{2}{x+} \frac{3}{2x+} \frac{4}{x+} \cdots \right)$$

7. Show that every real number in the interval $0 < x < 1$ can be expressed as a continued fraction (possibly terminating) in the form

$$x = \frac{1}{b_1+} \frac{1}{b_2+} \frac{1}{b_3+} \cdots$$

where each b_i is a positive integer.

8. Prove that if

$$x = \frac{a_1}{b_1+} \frac{a_2}{b_2+} \frac{a_3}{b_3+} \cdots$$

with the a_i and b_i all positive, then the convergents of the fraction are alternately greater than and less than x .

9. Show that f_n is increasing if and only if $A_{n-1}B_n > A_nB_{n-1}$, using the notation of Equation (8).
10. Count the number of long operations involved in computing C_n from A_n/B_n by the use of Formulas (9) and (10).
11. Show that any two successive convergents in the continued fraction of Equation (11) obey this equation:

$$\frac{A_n}{B_n} - \frac{A_{n-1}}{B_{n-1}} = (-1)^{n-1} \frac{a_1 a_2 \cdots a_n}{B_n B_{n-1}}$$

12. (Continuation) Use the result of the preceding problem to prove Problem 6.11.8 (above).
13. Prove that if the numbers b_i are all positive and $a_i = 1$, then $B_n > \min(1, b_1)$ for all n .
14. Prove that if $b_i > 0$ and $a_i = 1$, then $B_n B_{n-1}$ is increasing as a function of n .
15. Prove that if $b_i \geq 1 = a_i$, then the continued fraction in Equation (11) converges.
16. Find the recursive formula for B_n if

$$\frac{a_1}{b_1+} \frac{a_2}{b_2+} \cdots = \frac{1}{B_1+} \frac{1}{B_2+} \cdots$$

17. Prove Theorem 2 in this section. *Hint:* Use induction.

18. Show that

$$e^x = \frac{1}{1-} \frac{x}{x+1-} \frac{x}{x+2-} \frac{2x}{x+3-} \frac{3x}{x+4-} \dots$$

19. (Continuation) Show that

$$\frac{1}{1+} \frac{2}{2+} \frac{3}{3+} \dots = \frac{1}{e-1}$$

Hint: Use the preceding problem.

20. Show that the functions

$$g_n(x) = \frac{1}{x+} \frac{2}{x+} \frac{3}{x+} \dots \frac{n}{x}$$

can be generated recursively by this algorithm:

$$g_n(x) = \frac{p_n(x)}{q_n(x)}$$

where

$$\begin{cases} p_0(x) = 0, & p_1(x) = 1 \\ p_{n+1}(x) = xp_n(x) + (n+1)p_{n-1}(x) \end{cases}$$

and

$$\begin{cases} q_0(x) = 1, & q_1(x) = x \\ q_{n+1}(x) = xq_n(x) + (n+1)q_{n-1}(x) \end{cases}$$

21. Assuming that the following continued fraction converges, find its value:

$$\frac{1}{6+} \frac{1}{6+} \frac{1}{6+} \frac{1}{6+} \dots$$

22. Find the value of x :

$$x = 1 + \frac{1}{1+} \frac{1}{2+} \frac{1}{1+} \frac{1}{2+} \dots$$

23. Find the value of x :

$$x = 2 + \frac{1}{4+} \frac{1}{4+} \frac{1}{4+} \dots$$

24. If

$$f_n(x) = \frac{2}{1+} \frac{4}{2+} \frac{6}{3+} \dots \frac{2n}{n+x}$$

how would you obtain f_{n+1} from f_n ?

25. What is the value of $\sqrt{6 + \sqrt{6 + \sqrt{6 + \dots}}}$?

26. Assume that the continued fraction

$$\frac{1}{2x+} \frac{1}{2x+} \frac{1}{2x+} \dots \quad (x > 0)$$

converges. Determine a closed-form expression for it in terms of x .

COMPUTER PROBLEMS 6.11

- 1. Write a program for computing $\sqrt{x}$ by means of the equation in Problem 6.11.3 (p. 443). Compute $\sqrt{10}$, $\sqrt{100}$, $\sqrt{1000}$, and $\sqrt{10000}$ by printing a table of the first 50 convergent values for each.
- 2. Write a program for computing $\arctan(1/\sqrt{3})$ without using subscripted variables, and compare to the results given in the text.

3. Write a computer program to evaluate the continued fraction for $\arctan x$, as given in Equation (1). Use the following elementary approach: Given n , compute $f_n(x)$ in Equation (3) by starting at the right side and forming the appropriate fractions. Test your program by computing $\pi^{-1} \arctan(\sqrt{3})$ with $n = 5, 10, 15$, and 20 .

6.12 Trigonometric Interpolation

First, we recall the salient facts about interpolation by ordinary algebraic polynomials. If $n + 1$ function values are given in a table

x	x_0	x_1	x_2	$\cdots$	x_n
y	y_0	y_1	y_2	$\cdots$	y_n

(1)

then a unique polynomial p of degree $\leq n$ exists that *interpolates* the data. In other words,

$$p(x_j) = y_j \qquad (0 \leq j \leq n) \tag{2}$$

It is assumed that the points $x_0, x_1, \dots, x_n$ are all different from each other, but no restriction is placed on the data y_j . The polynomial p is chosen from the linear space Π_n consisting of all polynomials of degree $\leq n$. One basis for Π_n is given by the functions $b_k(x) = x^k$ for $0 \leq k \leq n$.

Fourier Series

For representing *periodic* phenomena, the algebraic polynomial space Π_n is, of course, not appropriate. Trigonometric functions will be much more suitable. One must know the period of the function in question before selecting the basic trigonometric functions. For convenience, let us assume that the function being interpolated is periodic with period 2π . Then the functions $1, \cos x, \cos 2x, \dots$ and $\sin x, \sin 2x, \sin 3x, \dots$ are appropriate. One of the basic theorems from Fourier analysis states that if f is 2π -periodic and has a continuous first derivative, then its Fourier series

$$\frac{a_0}{2} + \sum_{k=1}^{\infty} (a_k \cos kx + b_k \sin kx) \tag{3}$$

converges uniformly to f . The Fourier coefficients in the series are computed from the formulas

$$a_k = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \cos kt \, dt \tag{4}$$

$$b_k = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \sin kt \, dt \tag{5}$$

The cited theorem assures us that it is reasonable to approximate 2π -periodic functions with the sines and cosines listed above.

Complex Fourier Series

Much of this theory can be expressed in a more elegant form by using complex exponentials. We recall **Euler’s formula**:

e^{i\theta} = \cos \theta + i \sin \theta \tag{6}

in which i^2 = -1. The Fourier series is given by

f(x) \sim \sum_{k=-\infty}^{\infty} \widehat{f}(k)e^{ikx} \tag{7}

where

\widehat{f}(k) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(t)e^{-ikt} dt \tag{8}

If f is a real function, then its Fourier series, as given in Equation (3), is the real part of the complex Fourier series given in Equation (7). Indeed, we have, from Equations (4), (5), (6), and (8),

\widehat{f}(k) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(t)[\cos kt - i \sin kt] dt = \frac{1}{2}(a_k - ib_k) \quad (k \geq 0) \tag{9}

Now we invoke the following theorem.

THEOREM 1 Theorem on Fourier Series

Given the real sequences [a_k]_{k=0}^{\infty} and [b_k]_{k=1}^{\infty}, define

b_0 = 0 \quad a_{-k} = a_k \quad b_{-k} = -b_k \quad c_k = \frac{1}{2}(a_k - ib_k)

Then

\frac{a_0}{2} + \sum_{k=1}^n (a_k \cos kx + b_k \sin kx) = \sum_{k=-n}^n c_k e^{ikx} \tag{10}

Proof The series on the right can be written as

\frac{1}{2} \sum_{k=-n}^n (a_k - ib_k)(\cos kx + i \sin kx) \tag{11}

The imaginary part of this series is 0, as shown by the following calculation:

\begin{aligned} &\frac{1}{2} \sum_{k=-n}^n [a_k \sin kx - b_k \cos kx] \\ &= \frac{1}{2} \sum_{k=1}^n [a_{-k} \sin(-kx) - b_{-k} \cos(-kx)] - \frac{b_0}{2} + \frac{1}{2} \sum_{k=1}^n [a_k \sin kx - b_k \cos kx] \\ &= \frac{1}{2} \sum_{k=1}^n [-a_k \sin kx + b_k \cos kx] + \frac{1}{2} \sum_{k=1}^n [a_k \sin kx - b_k \cos kx] = 0 \end{aligned}

The real part of the series in (11) is

$$\begin{aligned} & \frac{1}{2} \sum_{k=-n}^n [a_k \cos kx + b_k \sin kx] \\ &= \frac{1}{2} \sum_{k=1}^n [a_{-k} \cos(-kx) + b_{-k} \sin(-kx)] + \frac{a_0}{2} + \frac{1}{2} \sum_{k=1}^n [a_k \cos kx + b_k \sin kx] \\ &= \frac{a_0}{2} + \sum_{k=1}^n (a_k \cos kx + b_k \sin kx) \end{aligned} \quad \blacksquare$$

Inner Product, Pseudo–Inner Product, and Pseudonorm

The functions E_k defined by $E_k(x) = e^{ikx}$ (with $k = 0, \pm 1, \pm 2, \dots$) form an orthonormal system of functions in the complex Hilbert space $L_2[-\pi, \pi]$ provided that we define the **inner product** to be

$$\langle f, g \rangle = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) \overline{g(x)} dx$$

This means that $\langle E_k, E_n \rangle = 0$ when $n \neq k$ and that $\langle E_k, E_k \rangle = 1$. The calculation supporting this statement when $n \neq k$ is

$$\begin{aligned} \langle E_k, E_n \rangle &= \frac{1}{2\pi} \int_{-\pi}^{\pi} E_k(x) \overline{E_n(x)} dx = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{ikx} e^{-inx} dx \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{i(k-n)x} dx = \frac{1}{2\pi} \frac{e^{i(k-n)x}}{i(k-n)} \Big|_{x=-\pi}^{x=\pi} = 0 \end{aligned}$$

Obviously, $\langle E_k, E_k \rangle = 1$ and the functions E_k form an orthonormal sequence.

It will be convenient to use the following inner-product notation:

$$\langle f, g \rangle_N = \frac{1}{N} \sum_{j=0}^{N-1} f(2\pi j/N) \overline{g(2\pi j/N)} \tag{12}$$

This function is not a true inner product because the condition $\langle f, f \rangle_N = 0$ does not imply that $f = 0$. It implies only that $f(x)$ takes the value 0 at each node $2\pi j/N$. The other properties of a (complex) inner product are valid. They are:

Complex Inner Product Properties

1. $\langle f, f \rangle_N \geq 0$
2. $\langle f, g \rangle_N = \overline{\langle g, f \rangle_N}$
3. $\langle \alpha f + \beta g, h \rangle_N = \alpha \langle f, h \rangle_N + \beta \langle g, h \rangle_N$

Along with this **pseudo-inner product**, there is a **pseudonorm** defined by

$$\|f\|_N = \sqrt{\langle f, f \rangle_N}$$

We have $\|f\|_N = 0$ if and only if $f(2\pi j/N) = 0$ for $0 \leq j \leq N-1$.

For interpolation, the following theorem plays a crucial role.

THEOREM 2 Theorem on Pseudo-Inner Product

For any $N \geq 1$, we have

$$\langle E_k, E_m \rangle_N = \begin{cases} 1 & \text{if } k - m \text{ is divisible by } N \\ 0 & \text{otherwise} \end{cases} \tag{13}$$

Proof The expression in question can be written as

$$\frac{1}{N} \sum_{j=0}^{N-1} E_k \left(\frac{2\pi j}{N} \right) \overline{E_m \left(\frac{2\pi j}{N} \right)} = \frac{1}{N} \sum_{j=0}^{N-1} [e^{2\pi i(k-m)/N}]^j$$

If $k - m$ is divisible by N , then $(k - m)/N$ is an integer and $e^{2\pi i(k-m)/N} = 1$. Hence, each summand is 1 and their average is 1. On the other hand, if $k - m$ is not divisible by N , then $e^{2\pi i(k-m)/N} \neq 1$, and in this case we can apply the standard formula for the sum of a geometric series:

$$\sum_{j=0}^{N-1} \lambda^j = \frac{\lambda^N - 1}{\lambda - 1} \quad (\lambda \neq 1)$$

The result is

$$\frac{e^{2\pi i(k-m)} - 1}{e^{2\pi i(k-m)/N} - 1} = 0 \quad \blacksquare$$

Exponential Polynomials

An **exponential polynomial** of degree at most n is any function of the form

$$P(x) = \sum_{k=0}^n c_k e^{ikx} = \sum_{k=0}^n c_k E_k(x) = \sum_{k=0}^n c_k (e^{ix})^k$$

The last expression in this equation explains the source of the terminology because it shows P to be a polynomial of degree $\leq n$ in the variable e^{ix} . Interpolation by exponential polynomials is summarized in the next two results.

THEOREM 3 Theorem on Orthonormal Functions, E_k

Let E_k be the function $E_k(x) = e^{ikx}$. Then $\{E_0, E_1, \dots, E_{N-1}\}$ is orthonormal with respect to the inner product $\langle \cdot, \cdot \rangle_N$ defined in Equation (12).

COROLLARY 1 Corollary on Exponential Polynomials

The exponential polynomial that interpolates a prescribed function f at equally spaced nodes $x_j = 2\pi j/N$ is given by the equations

$$P = \sum_{k=0}^{N-1} c_k E_k \quad \text{with} \quad c_k = \langle f, E_k \rangle_N \quad (14)$$

Proof Using the given formula for c_k , we compute the value of the exponential polynomial at an arbitrary node, say x_v . The result is

$$\begin{aligned} \sum_{k=0}^{N-1} c_k E_k(x_v) &= \sum_{k=0}^{N-1} \langle f, E_k \rangle_N E_k(x_v) \\ &= \sum_{k=0}^{N-1} \frac{1}{N} \sum_{j=0}^{N-1} f(x_j) \overline{E_k(x_j)} E_k(x_v) \\ &= \sum_{j=0}^{N-1} f(x_j) \frac{1}{N} \sum_{k=0}^{N-1} \overline{E_k(x_j)} E_k(x_v) \\ &= \sum_{j=0}^{N-1} f(x_j) \langle E_v, E_j \rangle_N \\ &= f(x_v) \end{aligned}$$

EXAMPLE 1 Use Corollary 1 to obtain an explicit formula for the interpolant when $N = 2$.

Solution In this case, P is the exponential polynomial of degree 1 that interpolates f at the nodes 0 and π . Formula (14) gives us

$$\begin{aligned} P(x) &= \frac{1}{2} [f(0) + f(\pi)] + \frac{1}{2} [f(0) + f(\pi)e^{-i\pi}] e^{ix} \\ &= \frac{1}{2} [f(0) + f(\pi)] + \frac{1}{2} [f(0) - f(\pi)] e^{ix} \end{aligned}$$

COROLLARY 2 Corollary on Exponential Polynomials

If $n < N$, then the exponential polynomial $\sum_{k=0}^n c_k E_k$ that best approximates f in the least-squares sense on the finite set

$$x_j = 2\pi j/N \quad (0 \leq j \leq N-1)$$

is obtained with $c_k = \langle f, E_k \rangle_N$.

Proof By Theorem 3, the functions $x \mapsto e^{ikx}$ form an orthonormal system with respect to the inner product defined in Equation (12). Now use Theorem 3 of Section 6.8 (p. 397) to complete the proof. ■

In Corollary 1, the choice of words suggests that the exponential polynomial described there is unique. To verify this, suppose that $\sum_{k=0}^{N-1} a_k E_k$ is an exponential polynomial that interpolates f at $x_0, x_1, \dots, x_{N-1}$ (where $x_j = 2\pi j/N$). Then

$$\sum_{k=0}^{N-1} a_k E_k(x_j) = f(x_j) \quad (0 \leq j \leq N-1)$$

If we multiply both sides of this equation by $E_n(-x_j)$ and sum with respect to j , the result is

$$\sum_{k=0}^{N-1} a_k \sum_{j=0}^{N-1} E_k(x_j) E_n(-x_j) = \sum_{j=0}^{N-1} f(x_j) E_n(-x_j)$$

By Equation (12), this implies that

$$\sum_{k=0}^{N-1} a_k \langle E_k, E_n \rangle_N = \langle f, E_n \rangle_N$$

Since $\langle E_k, E_n \rangle_N = \delta_{kn}$, we conclude that

$$a_n = \langle f, E_n \rangle_N = c_n$$

PROBLEMS 6.12

1. Using the notation of Theorem 3 and Corollary 1, show that if an exponential polynomial $g(x) = \sum_{k=0}^{N-1} a_k E_k(x)$ assumes the value 0 at each node x_j , then the coefficients a_k are all 0.
2. (Continuation) Use the result of the preceding problem to give another proof that the interpolating function in Corollary 1 is unique.
3. Prove that $E_k E_n = E_{k+n}$ and that $\overline{E_k} = E_{-k}$.
4. Prove that if f and g are functions such that

$$f(x_j) = \langle g, E_j \rangle_n \quad (x_j = 2\pi j/n)$$

then $g(x_j) = n \langle f, E_j \rangle_n$.

5. By taking real and imaginary parts in a suitable exponential equation, prove that

$$\frac{1}{n} \sum_{j=0}^{n-1} \cos \frac{2\pi jk}{n} = \begin{cases} 1 & \text{if } k \text{ divides } n \\ 0 & \text{otherwise} \end{cases}$$

$$\frac{1}{n} \sum_{j=0}^{n-1} \sin \frac{2\pi jk}{n} = 0$$

6. Show that the inner product defined in Equation (12) satisfies the three Properties 1, 2, and 3 following that equation. Why is $\| \cdot \|_N$ not a norm?

6.13 Fast Fourier Transform

Fourier transforms are used to decompose a signal into its constituent frequencies. This is analogous to a prism separating white light into its component bands of colored light. In the same way that sunglasses reduce the glare of white light by permitting only the passage of the softer green light, the Fourier transform can be used to modify a signal to achieve a desired effect. Analyzing the component frequencies of a signal or a system, Fourier series and transforms find use in a wide variety of applications, such as aircraft and spacecraft guidance, digital signal processing, medical imaging, oil and gas exploration, and the solution of differential equations. See, for example, Briggs and Henson [1995] or Walker [1992].

This section is devoted to the computational aspects of trigonometric interpolation. In particular, an algorithm called the **fast Fourier transform** will be developed for the efficient determination of the coefficients in Equation (14) of Section 6.12 (p. 449). We follow the exposition of Stoer and Bulirsch [1980].

Suppose that the coefficients $c_0, c_1, \dots, c_{N-1}$ are defined as in Corollary 1 of Section 6.12 (p. 449). We write

$$c_k = \frac{1}{N} \sum_{j=0}^{N-1} f(x_j)(\lambda^k)^j \quad (\lambda = e^{-2\pi i/N})$$

Thus, c_k is the result of evaluating a polynomial of degree $N - 1$ at the point λ^k ; it can be computed at a cost of approximately N multiplications and N additions. Since there are N coefficients c_k to be computed, the total cost of constructing the interpolating exponential polynomial is $\mathcal{O}(N^2)$ operations, in this straightforward approach.

The fast Fourier transform will bring this computing cost down to a more reasonable amount, namely, $N \log_2 N$. The table shows what this means for the large values of N that are commonly encountered in signal processing:

N	N^2	$N \log_2 N$
1024	1048576	10240
4096	16777216	49152
16384	268435456	229375

THEOREM 1 Theorem on Exponential Polynomial

Let p and q be exponential polynomials of degree $\leq n - 1$ such that, for the points $x_j = \pi j/n$, we have

$$p(x_{2j}) = f(x_{2j}) \quad q(x_{2j}) = f(x_{2j+1}) \quad (0 \leq j \leq n - 1) \tag{1}$$

Then the exponential polynomial of degree $\leq 2n - 1$ that interpolates f at the points $x_0, x_1, \dots, x_{2n-1}$ is given by

$$P(x) = \frac{1}{2}(1 + e^{inx})p(x) + \frac{1}{2}(1 - e^{inx})q(x - \pi/n) \tag{2}$$

Proof Because p and q have degrees $\leq n - 1$, whereas e^{inx} is of degree n , it is clear that P has degree $\leq 2n - 1$. It remains to be shown that P interpolates f at the nodes. We

have, for $0 \leq j \leq 2n - 1$,

$$P(x_j) = \frac{1}{2}[1 + E_n(x_j)]p(x_j) + \frac{1}{2}[1 - E_n(x_j)]q(x_j - \pi/n)$$

Notice that

$$E_n(x_j) = e^{\pi i n j / n} = e^{\pi i j} = \begin{cases} +1 & j \text{ even} \\ -1 & j \text{ odd} \end{cases}$$

Thus for even j , we infer that $P(x_j) = p(x_j) = f(x_j)$, whereas for odd j , we have

$$P(x_j) = q(x_j - \pi/n) = q(x_{j-1}) = f(x_j) \quad \blacksquare$$

THEOREM 2 Theorem on Coefficients of Exponential Polynomial

Let the coefficients of the polynomials described in Theorem 1 be as follows:

$$p = \sum_{j=0}^{n-1} \alpha_j E_j \quad q = \sum_{j=0}^{n-1} \beta_j E_j \quad P = \sum_{j=0}^{2n-1} \gamma_j E_j$$

Then, for $0 \leq j \leq n - 1$,

$$\gamma_j = \frac{1}{2}\alpha_j + \frac{1}{2}e^{-ij\pi/n}\beta_j \quad (3)$$

$$\gamma_{j+n} = \frac{1}{2}\alpha_j - \frac{1}{2}e^{-ij\pi/n}\beta_j \quad (4)$$

Proof We shall use Equation (2) and shall require a formula for $q(x - \pi/n)$:

$$\begin{aligned} q\left(x - \frac{\pi}{n}\right) &= \sum_{j=0}^{n-1} \beta_j E_j\left(x - \frac{\pi}{n}\right) \\ &= \sum_{j=0}^{n-1} \beta_j e^{ij(x-\pi/n)} = \sum_{j=0}^{n-1} \beta_j e^{-i\pi j/n} E_j(x) \end{aligned}$$

Thus, from Equation (2),

$$\begin{aligned} P(x) &= \frac{1}{2}[1 + E_n(x)]p(x) + \frac{1}{2}[1 - E_n(x)]q\left(x - \frac{\pi}{n}\right) \\ P &= \frac{1}{2} \sum_{j=0}^{n-1} \{(1 + E_n)\alpha_j E_j + (1 - E_n)\beta_j e^{-i\pi j/n} E_j\} \\ &= \frac{1}{2} \sum_{j=0}^{n-1} \{(\alpha_j + \beta_j e^{-i\pi j/n})E_j + (\alpha_j - \beta_j e^{-i\pi j/n})E_{n+j}\} \end{aligned}$$

The formulas for the coefficients γ_j can now be read from this equation. \blacksquare

EXAMPLE 1 Use Theorem 2 to find P when $n = 1$.

Solution Formulas (3) and (4) give us

$$\gamma_0 = \frac{1}{2}(\alpha_0 + \beta_0) \quad \gamma_1 = \frac{1}{2}(\alpha_0 - \beta_0)$$

Hence, P is given by

$$P = \gamma_0 E_0 + \gamma_1 E_1 = \frac{1}{2}(\alpha_0 + \beta_0) + \frac{1}{2}(\alpha_0 - \beta_0)E_1$$

Putting in the values of α_0 and β_0 , we obtain the result of Example 1 in Section 6.12 (p. 449):

$$P(x) = \frac{1}{2}[f(0) + f(\pi)] + \frac{1}{2}[f(0) - f(\pi)]e^{ix} \quad \blacksquare$$

Analysis

For further analysis, let $R(n)$ denote the minimum number of multiplications necessary to compute the coefficients in an interpolating exponential polynomial for the set of points $\{2\pi j/n : 0 \leq j \leq n-1\}$.

THEOREM 3 Theorem on Function R Inequality

The function R obeys the inequality

$$R(2^m) \leq m2^m$$

Proof We begin by establishing the inequality

$$R(2n) \leq 2R(n) + 2n \quad (5)$$

By Theorem 2, the coefficients γ_j needed in the polynomial P can be obtained from the coefficients in p and q at the cost of $2n$ multiplications. Indeed, we require n multiplications to compute $\frac{1}{2}\alpha_j$ for $0 \leq j \leq n-1$ and another n multiplications to compute $(\frac{1}{2}e^{-ij\pi/n})\beta_j$ for $0 \leq j \leq n-1$. (In the latter, we assume that the factors $\frac{1}{2}e^{-ij\pi/n}$ have already been made available.) Since the coefficients $\alpha_0, \alpha_1, \dots, \alpha_{n-1}$ cost $R(n)$ multiplications, and since the same is true of $\beta_0, \beta_1, \dots, \beta_{n-1}$, we obtain a total cost for P of at most $2R(n) + 2n$ multiplications.

The inequality in the theorem is proved by induction. Consider the case $m = 0$. We wish to interpolate f at the point $x_0 = 0$ by an exponential polynomial of degree 0. The solution is the constant $f(0)$, and no multiplications are required. Thus the assertion of the theorem is true for $m = 0$. Now proceed by induction, using Equation (5). The calculation for the inductive step (m to $m+1$) is

$$\begin{aligned} R(2^{m+1}) &= R(2 \cdot 2^m) \leq 2R(2^m) + 2 \cdot 2^m \\ &\leq 2(m2^m) + 2^{m+1} = (m+1)2^{m+1} \quad \blacksquare \end{aligned}$$

As a consequence of Theorem 3, we see that if N is a power of 2, say 2^m , then the cost of computing the interpolating exponential polynomial obeys the inequality $R(N) \leq N \log_2 N$. The algorithm that repeatedly carries out the procedure in Theorem 1 is the fast Fourier transform.

The content of Theorem 1 can be interpreted in terms of two linear operators, L_n and T_h . For any f , let $L_n f$ denote the exponential polynomial of degree $n-1$

that interpolates f at the nodes $2\pi j/n$ for $0 \leq j \leq n-1$. Let T_h be a **translation operator** defined by

$$(T_h f)(x) = f(x + h)$$

We know from Corollary 1 in Section 6.12 (p. 449) that

$$L_n f = \sum_{k=0}^{n-1} \langle f, E_k \rangle_n E_k$$

Furthermore, in Theorem 1, we have

$$\begin{aligned} P &= L_{2n} f \\ p &= L_n f \\ q &= L_n T_{\pi/n} f \end{aligned}$$

The conclusion of Theorems 1 and 2 is that $L_{2n} f$ can be obtained efficiently from $L_n f$ and $L_n T_{\pi/n} f$. Of course, this is true for $n = 1, 2, \dots$

The notation just introduced allows Theorem 1 to be expressed in this nice form:

$$L_{2n} f = \frac{1}{2}(1 + E_n)L_n f + \frac{1}{2}(1 - E_n)T_{-\pi/n}L_n T_{\pi/n} f$$

Our goal now is to establish one version of the fast Fourier transform algorithm for computing $L_N f$, where $N = 2^m$.

DEFINITION 2 Definition of $P_k^{(n)}$

Having fixed the function f and the index $N = 2^m$, we define

$$P_k^{(n)} = L_{2^n} T_{2k\pi/N} f \quad (0 \leq n \leq m, 0 \leq k \leq 2^{m-n} - 1) \tag{6}$$

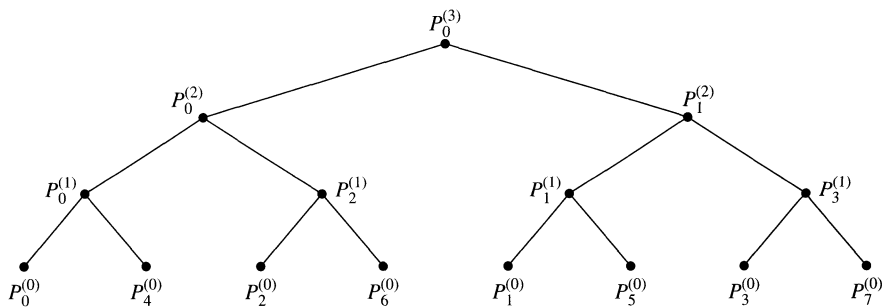
An alternative description of $P_k^{(n)}$ is as the exponential polynomial of degree $2^n - 1$ that interpolates f in the following way:

$$P_k^{(n)}\left(\frac{2\pi j}{2^n}\right) = f\left(\frac{2\pi k}{N} + \frac{2\pi j}{2^n}\right) \quad (0 \leq j \leq 2^n - 1) \tag{7}$$

By Problem 6.13.4 (p. 459), the set of nodes $(2\pi k/N) + (2\pi j/2^n)$ corresponding to one value of k is disjoint from the set corresponding to another value of k . A straightforward application of Theorem 1 shows that

$$P_k^{(n+1)}(x) = \frac{1}{2}(1 + e^{i2^n x})P_k^{(n)}(x) + \frac{1}{2}(1 - e^{i2^n x})P_{k+2^{m-n-1}}^{(n)}\left(x - \frac{\pi}{2^n}\right) \tag{8}$$

We can illustrate in a **tree diagram** how the exponential polynomials $P_k^{(n)}$ are related. Suppose that our objective is to compute $P_0^{(3)}$. In accordance with Equation (8), this function can be easily obtained from $P_0^{(2)}$ and $P_1^{(2)}$. Each of these, in turn, can be easily obtained from four polynomials of lower order, and so on. Figure 6.23 shows the connections.

FIGURE 6.23
Tree diagram

Algorithm

Denote the coefficients of $P_k^{(n)}$ by $A_{kj}^{(n)}$. Here $0 \leq n \leq m$, $0 \leq k \leq 2^{m-n} - 1$, and $0 \leq j \leq 2^n - 1$. We have

$$P_k^{(n)}(x) = \sum_{j=0}^{2^n-1} A_{kj}^{(n)} E_j(x) = \sum_{j=0}^{2^n-1} A_{kj}^{(n)} e^{ijx}$$

By Theorem 2, the following equations are valid:

$$\begin{aligned} A_{kj}^{(n+1)} &= \frac{1}{2} [A_{kj}^{(n)} + e^{-ij\pi/2^n} A_{k+2^{m-n-1},j}^{(n)}] \\ A_{k,j+2^n}^{(n+1)} &= \frac{1}{2} [A_{kj}^{(n)} - e^{-ij\pi/2^n} A_{k+2^{m-n-1},j}^{(n)}] \end{aligned}$$

For a fixed n , the array $A^{(n)}$ requires N storage locations in memory because $0 \leq k \leq 2^{m-n} - 1$ and $0 \leq j \leq 2^n - 1$. One way to carry out the computations is to use two linear arrays of length N , one to hold $A^{(n)}$ and the other to hold $A^{(n+1)}$. At the next stage, one array will contain $A^{(n+1)}$ and the other $A^{(n+2)}$. Let us call these arrays C and D . The two-dimensional array $A^{(n)}$ is stored in C by the rule

$$C(2^n k + j) \leftarrow A_{kj}^{(n)} \quad (0 \leq k \leq 2^{m-n} - 1, 0 \leq j \leq 2^n - 1)$$

Likewise, the array $A^{(n+1)}$ is stored in D by the rule

$$D(2^{n+1} k + j) \leftarrow A_{kj}^{(n+1)} \quad (0 \leq k \leq 2^{m-n-1} - 1, 0 \leq j \leq 2^{n+1} - 1)$$

The factors $Z(j) = e^{-2\pi i j/N}$ are computed at the beginning and stored. Then we use the fact that $e^{-ij\pi/2^n} = Z(j2^{m-n-1})$. Here is the **fast Fourier transform algorithm**:

```

input  $m$ 
 $N \leftarrow 2^m$ 
 $w \leftarrow e^{-2\pi i/N}$ 
for  $k = 0$  to  $N - 1$  do
     $Z(k) \leftarrow w^k$ 
     $C(k) \leftarrow f(2\pi k/N)$ 
end do
for  $n = 0$  to  $m - 1$  do

```



```

for  $k = 0$  to  $2^{m-n-1} - 1$  do
  for  $j = 0$  to  $2^n - 1$  do
     $u \leftarrow C(2^n k + j)$ 
     $v \leftarrow Z(j 2^{m-n-1}) C(2^n k + 2^{m-1} + j)$ 
     $D(2^{n+1} k + j) \leftarrow (u + v)/2$ 
     $D(2^{n+1} k + j + 2^n) \leftarrow (u - v)/2$ 
  end do
end do
for  $j = 0$  to  $N - 1$  do
   $C(j) \leftarrow D(j)$ 
end do
end do
output  $C(0), C(1), \dots, C(N)$ 

```

By scrutinizing the pseudocode, we can verify the bound $m2^m$ for the number of multiplications involved, as given by Theorem 3. Notice that in the nested loop of the code, n takes on m values; then k takes on 2^{m-n-1} values, and j takes on 2^n values. In this part of the code, just one command involving a multiplication occurs; namely, the one in which v is computed. This command will be encountered a number of times, equal to the product $m \times 2^{m-n-1} \times 2^n = m2^{m-1}$. At an earlier point in the code, the computation of the Z -array involves $2^m - 1$ multiplications. On any binary computer, a multiplication by $\frac{1}{2}$ need not be counted because it is accomplished by subtracting 1 from the exponent of the floating-point number.

EXAMPLE 2 Let $f = \sum_{k=0}^7 (k+1)E_k$, and reconstruct f by means of the fast Fourier transform, using samples of f at eight equally spaced points.

Solution The given algorithm can be used with $m = 3$. A subroutine or procedure for computing values of f directly from the formula goes as follows:

```

input  $x$ 
 $z \leftarrow \cos x + i \sin x$ 
 $d \leftarrow 8$ 
for  $k = 1$  to  $7$  do
   $d \leftarrow dz + 8 - k$ 
end do
output  $d$ 

```

When the programs were run on a 32-bit machine, the initial content of the C -array was as follows (rounded to five figures):

$C_0 = 36.000$		$C_4 = -4.0000$
$C_1 = -4.0000 - 9.6569i$		$C_5 = -4.0000 + 1.6568i$
$C_2 = -4.0000 - 4.0000i$		$C_6 = -4.0000 + 4.0000i$
$C_3 = -4.0000 - 1.6569i$		$C_7 = -4.0000 + 9.6569i$

The final content of the C -array was

(1.0000, 2.0000, 3.0000, 4.0000, 5.0000, 6.0000, 7.0000, 8.0000) ■

The preceding algorithm serves only a didactic purpose. Codes to be used in *production* computation should have further refinements. For example, with additional programming, we can dispense with the D -array. Furthermore, whenever the factor $Z(j)$ is $+1$ or -1 , the product $Z(j)C(k)$ should be programmed simply as $C(k)$ or $-C(k)$. Finally, a more versatile code can be written that can cope with values of N other than powers of 2.

Special codes can be written for functions f taking only real values at real arguments. Other codes can be written to produce cosine series or sine series alone. Consult the references at the end of this section for various ramifications of this topic.

Aliasing and Nyquist Frequency

As we attempt to reconstruct a function f from *samples* $f(x_j)$, the question arises of what limitations may be in the process. Obviously, a finite number of samples cannot convey all the information contained in a function that can be an arbitrary element in an infinite-dimensional linear space. Thus, two different continuous functions may appear to be the same from the evidence obtained from a finite number of samples. This phenomenon is known as **aliasing**. To investigate this, let us suppose that f is represented by its Fourier series:

$$f = \sum_{k=-\infty}^{\infty} \langle f, E_k \rangle E_k \quad (9)$$

Here we have used the following notation:

$$E_k(x) = e^{ikx} \quad \langle f, g \rangle = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) \overline{g(x)} dx$$

By contrast, the interpolating exponential polynomial of degree $\leq N-1$ is expressed as

$$P = \sum_{k=0}^{N-1} \langle f, E_k \rangle_N E_k \quad (10)$$

Here the notation is

$$\langle f, g \rangle_N = \frac{1}{N} \sum_{j=0}^{N-1} f(x_j) \overline{g(x_j)} \quad x_j = 2\pi j/N$$

To compare the coefficients in the two equations, (9) and (10), we write

$$\langle f, E_m \rangle_N = \left\langle \sum_{k=-\infty}^{\infty} \langle f, E_k \rangle E_k, E_m \right\rangle_N = \sum_{k=-\infty}^{\infty} \langle f, E_k \rangle \langle E_k, E_m \rangle_N$$

By Theorem 2 of Section 6.12 (p. 448), $\langle E_k, E_m \rangle_N$ is 0 except in those cases when $k-m$ is a multiple of N . Thus, if we let $k-m = \nu N$, with $\nu = 0, \pm 1, \pm 2, \dots$, then we obtain

$$\langle f, E_m \rangle_N = \sum_{v=-\infty}^{\infty} \langle f, E_{m+vN} \rangle$$

Writing these terms in order of decreasing importance, we have

$$\langle f, E_m \rangle_N = \langle f, E_m \rangle + \langle f, E_{m+N} \rangle + \langle f, E_{m-N} \rangle + \cdots$$

This shows that the coefficient of E_m in the interpolating function (10) has not only the wanted term $\langle f, E_m \rangle$ from Equation (9) but also terms that properly belong to higher frequencies, $\langle f, E_{m \pm vN} \rangle$ for $v = 1, 2, 3, \dots$. These terms are unwanted and create distortions in the reconstructed signal. The first of these unwanted components is $\langle f, E_N \rangle$, which belongs to E_N in the Fourier series. But E_N is not present in the interpolating function of Equation (10). The term E_N represents the lowest frequency component that is *present* in the Fourier series but *absent* in the interpolating function. This lowest frequency is known as the **Nyquist frequency**.

Computing Values of an Exponential Polynomial

The fast Fourier transform can also be used to evaluate an exponential polynomial at a set of equally spaced points. Suppose that such a polynomial is given in the form

$$p(x) = \sum_{j=0}^{n-1} a_j E_j(x)$$

To evaluate p at the points

$$t - \frac{2k\pi}{n} \quad (0 \leq k \leq n-1)$$

we write $x_k = 2k\pi/n$ and

$$\begin{aligned} p(t - x_k) &= \sum_{j=0}^{n-1} a_j E_j(t - x_k) = \sum_{j=0}^{n-1} a_j e^{ij(t-x_k)} \\ &= \sum_{j=0}^{n-1} a_j e^{ij t} \overline{E_k(x_j)} = n \langle g, E_k \rangle_n \end{aligned}$$

where g is a function such that

$$g(x_j) = a_j e^{ij t} \quad (0 \leq j \leq n-1)$$

Thus, we apply the fast Fourier transform to g , obtaining the coefficients $\langle g, E_k \rangle_n$. When these are multiplied by n , we have the values $p(t - x_k)$.

Some sources of information on the fast Fourier transform are Davis and Rabinowitz [1984], Cooley, Lewis, and Welch [1967], Bloomfield [1976], Briggs and Henson [1995], Brigham [1974], Conte and de Boor [1980], Kahaner [1970, 1978], Lanczos [1966], Kahaner, Moler, and Nash [1989], Elliott and Rao [1982], Nussbaumer [1982], and Scheid [1988].

PROBLEMS 6.13

1. Prove that $T_h E_j$ is a scalar multiple of E_j .
2. Show that f and $f + \lambda E_k$ are indistinguishable if they are interpolated at points $x_j = 2\pi j/N$, with k a multiple of N . Thus, to detect all components of f having frequencies $\leq \omega$, the sampling frequency should be greater than ω . This is another aspect of the aliasing effect.
3. (**Aliasing**) Two functions can appear to be the same if we know only samples of the functions taken at discrete points. Show that these two functions appear to be identical when sampled at the integer values of x :

$$f(x) = \cos[(n - \alpha)\pi x + \beta] \quad g(x) = \cos[(n + \alpha)\pi x - \beta]$$

4. Prove that if $k \neq r$, then the node sets on the right side of Equation (7) are disjoint.
5. Establish the validity of Equation (8).
6. Prove this equivalent form of Theorem 2:

$$\langle f, E_j \rangle_{2n} = \frac{1}{2}(u_j + v_j)$$

$$\langle f, E_{j+n} \rangle_{2n} = \frac{1}{2}(u_j - v_j)$$

where

$$u_j = \langle f, E_j \rangle_n \quad v_j = e^{-ij\pi/n} \langle T_{\pi/n} E_j \rangle_n$$

Here $T_{\pi/n}$ is the translation operator defined by $(T_{\pi/n} f)(x) = f(x + \pi/n)$.

7. This problem and the next three give a matrix formulation of the fast Fourier transform. Fix an index n and a function f . Let u and v be column vectors having components

$$u_j = f\left(\frac{2\pi j}{n}\right) \quad v_j = \langle f, E_j \rangle_n \quad (0 \leq j \leq n-1)$$

Prove that $v = Au$, where A is the matrix with components $A_{jk} = n^{-1}e^{-2\pi ijk/n}$ ($0 \leq j, k \leq n-1$).

8. (Continuation) Prove that the matrix A contains only n different elements. Prove that $n^{1/2}A$ is a unitary matrix. Prove that A is symmetric.
9. (Continuation) Using the Euclidean norm on $\mathbb{C}^n$, show that $\|v\| = n^{-1/2}\|u\|$.
10. (Continuation) Retain the notation of the preceding three problems and write $A = A^{(n)}$ to show the dependence on n . If $u \in \mathbb{C}^{2n}$, put $u' = (u_0, u_2, \dots, u_{2n-2})$ and $u'' = (u_1, u_3, \dots, u_{2n-1})$. Prove that for $0 \leq k \leq n-1$,

$$(A^{(2n)}u)_k = \frac{1}{2}(A^{(n)}u')_k + \frac{1}{2}e^{-ik\pi/n}(A^{(n)}u'')_k$$

$$(A^{(2n)}u)_{n+k} = \frac{1}{2}(A^{(n)}u')_k - \frac{1}{2}e^{-ik\pi/n}(A^{(n)}u'')_k$$

COMPUTER
PROBLEM 6.13

1. Write and test a fast Fourier transform code to compute the values of

$$p(x) = \sum_{j=0}^{N-1} a_j E_j(x)$$

assuming that $N = 2^m$ and that the coefficients a_j have been prescribed.

6.14 Adaptive Approximation

The distinguishing feature of *adaptive* approximation is that the domain of the function is repeatedly subdivided to obtain more accurate approximations on smaller subdomains. The resulting global approximation is then defined *piecewise* by patching together many local approximations. It is clear that spline functions with free knots are well suited for this type of approximation.

First-Degree Spline

We shall illustrate the principles with an algorithm that finds a first-degree spline function as an approximation to a given function f on a fixed interval $[a, b]$. It is assumed that an error tolerance, $\varepsilon > 0$, has been prescribed and that the result of our work is to be a first-degree spline function S that satisfies the inequality

$$|f(x) - S(x)| \leq \varepsilon \quad \text{for } a \leq x \leq b \quad (1)$$

The spline function S is piecewise linear and is chosen to interpolate f at the knots. The linear function that interpolates f at two points α and β is given by

$$\ell(f, \alpha, \beta; x) = \frac{f(\alpha)}{\beta - \alpha}(\beta - x) + \frac{f(\beta)}{\beta - \alpha}(x - \alpha)$$

If a set of knots is given, say

$$a = t_0 < t_1 < t_2 < \cdots < t_n = b$$

then the first-degree interpolating spline for f is defined by the formula

$$S(x) = \begin{cases} \ell(f, t_0, t_1; x) & t_0 \leq x \leq t_1 \\ \ell(f, t_1, t_2; x) & t_1 \leq x \leq t_2 \\ \vdots & \vdots \\ \ell(f, t_{n-1}, t_n; x) & t_{n-1} \leq x \leq t_n \end{cases}$$

If S meets the error criterion in Inequality (1), then our objective has been met. If, however,

$$\|f - S\|_\infty = \max_{a \leq x \leq b} |f(x) - S(x)| > \varepsilon$$

then the error tolerance ε has been exceeded at one or more points in the interval. Let ξ be a point where the error is worst; that is,

$$|f(\xi) - S(\xi)| = \|f - S\|_\infty$$

Let i be the index for which $t_{i-1} \leq \xi \leq t_i$. On this interval, the current spline function S is unsatisfactory. Thus, the difference $|f(x) - \ell(f, t_{i-1}, t_i; x)|$ exceeds ε at some points of the interval $[t_{i-1}, t_i]$. We therefore subdivide this interval by introducing ξ as a new knot. Instead of the single linear function $\ell(f, t_{i-1}, t_i; x)$ on

$[t_{i-1}, t_i]$, we shall now have two linear functions; namely, $\ell(f, t_{i-1}, \xi; x)$ on $[t_{i-1}, \xi]$ and $\ell(f, \xi, t_i; x)$ on $[\xi, t_i]$. In this way we pass from one set of knots to another set and from one spline approximation to a better one. The process is repeated until the spline function meets the error criterion.

Algorithm

In writing an algorithm for this process, we want to give high priority to efficiency. Thus at each stage of the process, we should store not only the current **knot array** $t = [t_0, t_1, \dots, t_n]$, but also the corresponding **ordinate array** $y = [y_0, y_1, \dots, y_n]$, where $y_i = f(t_i)$. Also, we should store the **deviation array** $d = [d_1, d_2, \dots, d_n]$, with $d_i = \max_{t_{i-1} \leq x \leq t_i} |f(x) - \ell(f, t_{i-1}, t_i; x)|$, and the array of **maximum deviation points** $c = [c_1, c_2, \dots, c_n]$, consisting of the points where these maximum deviations occur. Here d_i is the maximum deviation in the interval $[t_{i-1}, t_i]$, and c_i is a point in that interval where the maximum occurs. In terms of these stored quantities, we have

$$\begin{aligned}\ell(f, t_{i-1}, t_i; x) &= [y_{i-1}(t_i - x) + y_i(x - t_{i-1})] / (t_i - t_{i-1}) \\ \|f - S\|_\infty &= \max\{d_1, d_2, \dots, d_n\} \\ d_i &= |f(c_i) - \ell(f, t_{i-1}, t_i; c_i)| \quad t_{i-1} \leq c_i \leq t_i\end{aligned}$$

At the n th step, the arrays t , y , d , and c will be available from prior work. Now a test is performed to see whether the present knot sequence is satisfactory. If it is not, an index i is selected for which $d_i = \max\{d_1, d_2, \dots, d_n\}$. Since c_i is a point in $[t_{i-1}, t_i]$ at which the deviation $|f(x) - S(x)|$ is a maximum, we can take c_i to be the new knot (denoted previously by ξ). In the next step, two new intervals will be present in our partition of $[a, b]$; namely, $[t_{i-1}, c_i]$ and $[c_i, t_i]$. All the stored data with indices $i, i + 1, \dots, n$ are shifted *to the right* in memory, and t_i becomes c_i . In other words, the knots to the right of the new knot are relabeled, as well as the ordinates, deviations, and maximum deviation points, and the new knot is inserted into the array of knots. Then the interval containing the new knot is split at the new knot, and new points of maximum deviation over each of these subintervals are computed. In this way, a set of knots is generated with the maximum deviation over each subinterval within the specified tolerance, if possible. Here is an algorithm to do all this, starting with $t_0 = a$ and $t_1 = b$:

```

input  $a, b, \varepsilon, M$ 
 $t_0 \leftarrow a; t_1 \leftarrow b; y_0 \leftarrow f(t_0); y_1 \leftarrow f(t_1)$ 
call  $\text{MAX}(f, t_0, t_1, c_1, d_1)$ 
for  $n = 1$  to  $M - 1$  do
    select  $i$  so that  $d_i = \max\{d_1, d_2, \dots, d_n\}$ 
    if  $d_i \leq \varepsilon$  exit loop
    for  $j = n$  to  $i + 1$  step  $-1$  do
         $t_{j+1} \leftarrow t_j$ 
         $y_{j+1} \leftarrow y_j$ 

```

```

 $d_{j+1} \leftarrow d_j$ 
 $c_{j+1} \leftarrow c_j$ 
end do
 $t_{i+1} \leftarrow t_i$ 
 $y_{i+1} \leftarrow y_i$ 
 $t_i \leftarrow c_i$ 
 $y_i \leftarrow f(c_i)$ 
call Max( $f, t_{i-1}, t_i, c_i, d_i$ )
call Max( $f, t_i, t_{i+1}, c_{i+1}, d_{i+1}$ )
end do
output  $n, (t_0, t_1, \dots, t_n), (y_0, y_1, \dots, y_n), (d_1, d_2, \dots, d_n)$ 

```

The algorithm starts with just two knots, $t_0 = a$ and $t_1 = b$. Next, we determine the maximum deviation d_1 on the interval $[t_0, t_1]$. If d_1 is larger than the allowable tolerance ε , then the point c_1 where the maximum deviation occurred becomes a knot. The knot t_1 is relabeled t_2 , and c_1 becomes t_1 . The maximum deviations are then computed on $[t_0, t_1]$ and $[t_1, t_2]$. If either exceeds ε , the process is continued. Eventually, either all deviations fall below ε (success) or the upper limit on the number of steps (M) is exceeded (failure). Enough storage must be provided to allow M components in the arrays t, y, d , and c .

In this algorithm, three references to a subroutine or procedure Max occur. The purpose of Max(f, α, β, c, d) is to compute the maximum value of $|f(x) - \ell(f, \alpha, \beta; x)|$ on the interval $[\alpha, \beta]$, storing the maximum value in d and storing the appropriate point at which this occurs in c . In practice, the values of c and d need not be computed to high precision. A rough way to do this is to sample the values of $|f(x) - \ell(f, \alpha, \beta; x)|$ at, say, 11 points across the interval, and to accept one of these as the desired point. Here is an algorithm to do just this:

```

procedure Max( $f, \alpha, \beta, c, d$ )
 $k \leftarrow 10$ 
 $h \leftarrow (\beta - \alpha)/k$ 
for  $i = 0$  to  $k$  do
 $z_i \leftarrow f(\alpha + ih)$ 
end do
for  $i = 1$  to  $k - 1$  do
 $z_i \leftarrow |z_i - (iz_k + (k - i)z_0)|/k$ 
end do
 $d \leftarrow 0$ 
for  $i = 1$  to  $k - 1$  do
if  $z_i > d$  then
 $d \leftarrow z_i$ 
 $c \leftarrow \alpha + ih$ 
end if
end do
return

```

In the foregoing algorithm for Max, we have first stored $f(\alpha + ih)$ in z_i . After $z_0, z_1, \dots, z_k$ have been computed, we have this equation:

$$\ell(f, \alpha, \beta; x) = \frac{z_0}{kh}(\beta - x) + \frac{z_k}{kh}(x - \alpha)$$

where $h = (\beta - \alpha)/k$. For $x = \alpha + ih$, we have (because $\beta = \alpha + kh$)

$$\ell(f, \alpha, \beta; \alpha + ih) = \frac{z_0}{kh}(k - i)h + \frac{z_k}{kh}ih = (iz_k + (k - i)z_0)/k$$

Then the memory location z_i is used (line 8 in the algorithm) to store

$$|f(\alpha + ih) - \ell(f, \alpha, \beta; \alpha + ih)|$$

EXAMPLE 1 A computer program incorporating the adaptive approximation method described above was tested using the function $f(x) = \sqrt{x}$ on the interval $[0, 1]$ and with $\varepsilon = 10^{-2}$. As a result, ten computed knots were generated, clustered near the singularity at 0 (i.e., a point with an infinite derivative). See Figure 6.24.

Solution The ten knots, the values of f , and the maximum deviations were:

knots	values of f	deviations
0.00	0.000	
0.000729	0.027	0.007
0.00243	0.049	0.002
0.0081	0.09	0.003
0.027	0.16	0.005
0.09	0.3	0.01
0.174	0.417	0.005
0.3	0.548	0.004
0.58	0.762	0.009
1.0	1.000	0.008

If ε is reduced to 10^{-3} , then 32 knots are generated by this adaptive procedure. ■

General Case

The principles illustrated by the first-degree adaptive spline algorithm can be abstracted so that they apply to more general cases. The essential ingredient in the process is a local approximation operator A that operates on a function f and an interval $[\alpha, \beta]$ and produces an approximation to f on the given interval. We write $A(f, \alpha, \beta; x)$ for the value at x of this approximating function. The operation $\ell(f, \alpha, \beta; x)$ used in the first-degree spline procedure is such an operator.

At each step in the adaptive process, a set of subintervals of $[a, b]$ will be given. These intervals are nonoverlapping and cover $[a, b]$. If $[\alpha, \beta]$ is one of these subintervals, we say that it is *satisfactory* if

$$|f(x) - A(f, \alpha, \beta; x)| \leq \varepsilon \quad \text{on } [\alpha, \beta]$$

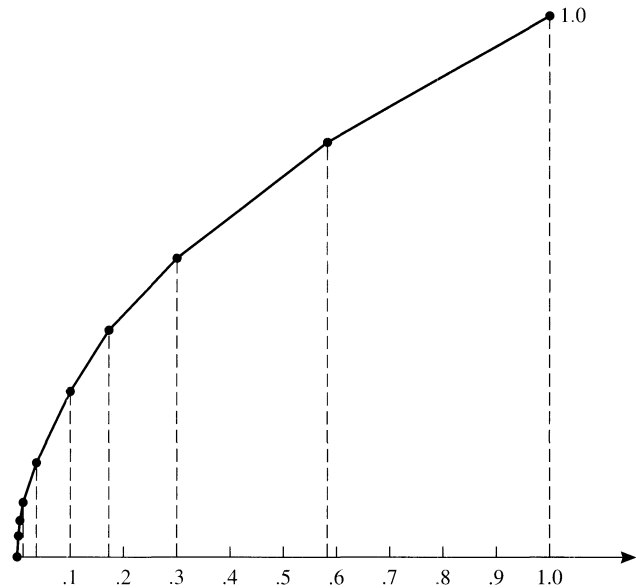


FIGURE 6.24
Adaptive spline
approximation of degree 1

If all the subintervals are satisfactory, the objective is met by a global approximating function G that is made up from the functions $A(f, \alpha, \beta; x)$ on various subintervals. (Notice that there is no guarantee that G will be continuous.) If a subinterval $[\alpha, \beta]$ is not satisfactory, it is subdivided in some standard way. For example, the midpoint $\xi = \frac{1}{2}(\alpha + \beta)$ can be added as a knot. An alternative is to take as the new knot a point ξ where the error is worst on $[\alpha, \beta]$. In any case, the interval $[\alpha, \beta]$ is replaced by two new intervals $[\alpha, \xi]$ and $[\xi, \beta]$. The process is then repeated.

PROBLEMS 6.14

1. If the procedure of Computer Problem 6.14.1 below is used on a monotone increasing function, what lower bound can be given on the number of subintervals required? The answer will be in terms of $f(b) - f(a)$ and ε .
2. For the function $f(x) = \sqrt{x}$, show that the maximum of $|f(x) - \ell(f, \alpha, \beta; x)|$ occurs at $x = \frac{1}{4}(\sqrt{\beta} + \sqrt{\alpha})^2$.
3. What will happen if the adaptive algorithm is applied (exactly as described in the text) to the function $f(x) = \sin 10x$ on the interval $[0, \pi]$?

COMPUTER
PROBLEMS 6.14

1. Write and test an adaptive routine that uses piecewise constant functions.
2. Write a computer routine for the adaptive approximation procedure, and apply it to the example in this section. Produce a computer plot of the results.
3. Modify the adaptive algorithm for use with cubic splines.
4. Modify the procedure Max so that it uses the y-array. Experiment with some other procedure for Max.